FRACTAL BASED FRAMEWORK FOR TIME SERIES VOLATILITY PREDICTION

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“So limited is our knowledge that we resort, not to science, but to shamans. We place control of the world’s largest economy in the hands of a few elderly men, the central bankers..”

–––Benoit Mandlebrot

Abstract

In this thesis, a novel mathematical framework is presented for pricing financial derivatives and modelling asset behaviour by bringing together fractional Brownian motion (fBm), fuzzy logic, and jump processes, all aligned with the no–arbitrage principle. In particular, our mathematical developments include fBm defined through Mandelbrot–Van Ness kernels, and advanced mathematical tools such Molchan martingale and BDG inequalities ensuring rigorous theoretical validity. These different concepts are combined to model uncertainties such as sudden market shocks and investor sentiment, providing a fresh perspective in financial mathematics and derivatives pricing. Using fuzzy logic to incorporate subject factors such as market optimism or pessimism, adjusting volatility dynamically according to the current market environment. Fractal mathematics with the Hurst exponent close to zero reflecting rough market conditions and fuzzy set theory are combined with jumps, representing sudden market changes to capture more realistic asset price movements. The gap between complex stochastic equations and solvable differential equations is bridged using tools like Feynman–Kac approach and Girsanov transformation. Simulations illustrating plausible scenarios ranging from pessimistic to optimistic demonstrate how this model can behave in practice, highlighting potential advantages over classical models like the Merton jump diffusion and Black–Scholes. Overall, the proposed model represents an advancement in mathematical finance by integrating fractional stochastic processes with fuzzy set theory, thus revealing new perspectives on derivative pricing and risk–free valuation in uncertain environments.

In financial modelling, traditional approaches often rely on deterministic frameworks that inadequately capture the inherent complexities of market dynamics. For example, option pricing models typically use crisp parameters, yet empirical evidence shows these variables often exhibit fuzziness and uncertainty. To address these limitations, this thesis proposes a novel unified framework that integrates fractional Brownian motion (fBM) and

fuzzy processes to model financial systems characterised by both randomness and fuzziness. By defining a joint product measure space for H $\in$ (0,1) where H is the Hurst parameter that characterises the degree of self–similarity and persistence in fractional Brownian motion, this work provides a comprehensive framework to account for long–range dependence, self–similarity, and individual risk preferences in market behaviour. Building on the Merton jump–diffusion model, we extend the framework to include fuzzy fractional Brownian motion, fuzzy Poisson processes, and fuzzy volatility. These innovations enable the modelling of hybrid systems under uncertain variables, offering practitioners a flexible and robust tool for understanding and forecasting financial market dynamics and proposing a novel methodology for its application in forecasting. This thesis examines the fractal characteristics of volatility time series. It provides empirical evidence that supports the proposition of self–similarity across different temporal scales and validates the fractal nature of volatility through Hurst exponent estimation. This finding motivates the development of the novel theoretical framework presented herein. The analysis highlights how fractal measures can capture structured patterns amidst apparent chaos, offering insights into intermittent bursts of volatility and periods of persistence.

Further, the thesis distinguishes between expected and unexpected volatility, linking endogenous shocks (e.g., economic changes) to episodic uncertainties and exogenous shocks (e.g., geopolitical events) to aleatoric uncertainties. By incorporating reinforcement learning (RL) and generative adversarial networks (GANs), the proposed framework manages volatility across diverse shock events, maintaining memory of older data points while continuously adapting to new information. Comparisons between traditional econometric models and modern machine learning approaches are used to benchmark current forecasting capabilities, highlighting the limitations that motivate our new theoretical approach. The research culminates in a novel hybrid framework that lays the groundwork for a new class of more advanced forecasting systems.

This research significantly advances the field by bridging theoretical advancements in fuzzy stochastic processes and fractal analysis with a clear roadmap for practical applications in financial markets. The findings underscore the importance of incorporating uncertainty and complexity in financial modelling, providing a robust foundation for future studies in stochastic hybrid systems.

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Publications

The following publications have arisen from this thesis:

[[P1]] G. Urumov and P. Chountas, “Clustering stock price volatility using intuitionistic fuzzy sets,” in *Proceedings of the International Conference on Intuitionistic Fuzzy Sets*, Sofia, 2022.

[[P2]] G. Urumov, P. Chountas, and T. Chaussalet, “Fuzzy fractional brownian motion: Review and extension,” *Algorithms*, vol. 17, no. 289, 2024.

[[P3]] G. Urumov, P. Chountas, and T. Chaussalet, “Fuzzy amplitudes and kernels in fractional brownian motion: Theoretical foundations,” *Symmetry*, vol. 17, p. 550, 2025.

[[P4]] G. Urumov and P. Chountas, “Essay on volatility clusters and time series prediction,” in *IEEE International Conference*, Warsaw, 2022.

Glossary

In order not to confuse the readers, in this section interchangeable use of fractal and fractional is explained in this manuscript and a glossary of financial terms is provided.

- **Fractals**: are intricate geometric shapes that exhibit self–similarity, meaning they look similar at any level of magnification. Fractals are characterized by their fractional dimensions, which are typically non–integer and indicate how completely the fractal fills space as you zoom in. Common examples of fractals include (i) the Mandelbrot Set – a complex and infinitely detailed boundary shape that exhibits self–similarity (ii) the Sierpinski Triangle – a recursively defined triangle that has the same appearance at any scale (iii) the Koch Snowflake – a snowflake–shaped curve that has an infinitely long perimeter but a finite area.
- **Fractional calculus**: is a generalisation of traditional calculus that extends the concepts of differentiation and integration to non–integer (fractional) orders. While traditional calculus deals with integer–order derivatives and integrals, fractional calculus allows for operations like half–derivatives, quarter–integrals, etc. Key concepts include:
  - **Fractional Derivatives**: These are derivatives of arbitrary order, providing a tool for describing memory and hereditary properties in various materials and processes.
  - **Fractional Integrals**: These are integrals of arbitrary order, often used in solving differential equations with fractional orders.
- **At–the–Money (ATM)**: An option is considered at–the–money if the current price of the underlying asset is equal to the option's strike price.
- **Out–of–the–Money (OTM)**: An option is out–of–the–money if it has no intrinsic value. For a call option, this means the underlying asset's price is below the strike price. For a put option, it means the

underlying asset's price is above the strike price.

- **In–the–Money (ITM)**: An option is in–the–money if it has intrinsic value. For a call option, this means the underlying asset's price is above the strike price. For a put option, it means the underlying asset's price is below the strike price.
- **Volatility**: A measure of the amount by which an asset price is expected to fluctuate over a given period. Higher volatility typically increases the premium of options.
- **Historical Volatility**: The measure of how much the price of an asset has varied in the past over a certain period. It is calculated by analysing past price movements, typically using statistical methods like standard deviation.
- **Statistical Volatility**: Often synonymous with historical volatility, it refers to the quantifiable measure of the dispersion of returns for a given security or market index, derived from historical data using statistical methods.
- **Implied Volatility (IV)**: The market's forecast of a likely movement in the underlying asset's price. It is derived from the price of an option and is a key factor in options pricing.
- **Observed Volatility (IV)**: The actual level of volatility that has been witnessed in the market over a specific period. It refers to the real, measurable fluctuations in asset prices that have already occurred.
- **Invented Volatility (IV)**: A theoretical concept suggesting that volatility is a construct or measure created by market participants and models, rather than a naturally occurring phenomenon. It reflects the idea that volatility, to some extent, can be influenced by the perception and actions of traders and analysts similar to quantum measurement.
- **Strike Price (Exercise Price)**: The price at which the underlying asset can be bought (in the case of a call option) or sold (in the case of a put option) when the option is exercised.

- **Premium**: The price paid by the buyer to the seller to acquire an option contract. It is composed of intrinsic value and time value.
- **Intrinsic Value**: The difference between the underlying asset's current price and the option's strike price. For call options, it is the amount the asset price exceeds the strike price. For put options, it is the amount the strike price exceeds the asset price.
- **Time Value**: The portion of the option premium that exceeds the intrinsic value, representing the potential for the option to gain value before expiration. It diminishes as the option approaches its expiration date.
- **Expiration Date**: The date on which an option contract becomes void and the right to exercise it no longer exists.
- **Underlying Asset**: The financial asset upon which an option or futures contract is based, such as stocks, bonds, commodities, or indices.
- **Call Option**: A financial contract giving the option buyer the right, but not the obligation, to buy a specified quantity of the underlying asset at a specified price within a fixed period.
- **Put Option**: A financial contract giving the option buyer the right, but not the obligation, to sell a specified quantity of the underlying asset at a specified price within a fixed period.
- **Exercise (or Strike) Price**: The fixed price at which the option holder can buy (call) or sell (put) the underlying asset.
- **Delta**: A ratio that compares the change in the price of an option to the corresponding change in the price of the underlying asset. It ranges from 0 to 1 for call options and from –1 to 0 for put options.
- **Gamma**: Measures the rate of change in delta with respect to changes in the underlying asset's price. It indicates the convexity of an option's value in relation to the underlying asset.
- **Theta**: The rate of decline in the value of an option due to the passage

of time. It represents the time decay of an option's price.

- **Vega**: Measures the sensitivity of an option's price to changes in the volatility of the underlying asset.
- **Rho**: Measures the sensitivity of an option's price to changes in interest rates.
- **Hedge**: An investment made to reduce the risk of adverse price movements in an asset, typically by taking an offsetting position in a related security, such as an options contract.
- **Black–Scholes Model**: A mathematical model used to calculate the theoretical price of European call and put options, based on factors such as current price, strike price, time to expiration, risk–free rate, and volatility.
- **American Option**: An option that can be exercised at any time before its expiration date.
- **European Option**: An option that can only be exercised on its expiration date.
- **Binary Option**: A type of option where the payoff is either a fixed amount or nothing at all, depending on whether a certain condition is met.
- **Greeks**: The collective term for the various measures of the sensitivity of an option's price to its underlying parameters, including delta, gamma, theta, vega, and rho.

CHAPTER 1 - INTRODUCTION AND RESEARCH OVERVIEW

1.1 INTRODUCTION

To understand the financial markets, it is critical to study the volatility of the market. Policymakers such as Central Bankers in particular, pay very close attention to the stresses in the system and to tightness or looseness of financial conditions. In fact, we know from recent publications, that both European Central Bank (ECB) and Federal Reserve (FED) in the US have used volatility in their models as a factor of how loose or tight the financial conditions really are. Hence, volatility modelling has direct impact on mortgage rates, inflation, and unemployment. This thesis will help the reader in understanding the stock volatility and possible ways of modelling and predicting it. Volatility is a mathematical measure of the dispersion of returns of a time series data. Ceteris paribus, the higher the volatility, the riskier the security. Volatility is often but not always measured as either the standard deviation or variance between returns from that same security or market index. It can also be measured as a range equal to maximum minus minimum or average true range which is a moving average of range.

In financial markets, volatility is often associated with big swings in either direction. For example, when the stock market rises and falls more than one percent over a sustained period, it is called a “volatile“ market. An asset’s volatility is a key factor when pricing options contracts. ‘Vega’ or the sensitivity of the options’ prices with respect to volatility is frequently the largest component (or partial derivative in math parlance) of the options' price.

Volatility often refers to the amount of uncertainty or risk related to the size of changes in a security's value. A higher volatility means that a security's value can potentially be spread out over a larger range of values. This means that the price of the security can change dramatically over a short time in either direction. A lower volatility means that a security's value does not fluctuate dramatically and tends to be steadier.

In the financial world described by Bachelier–Samuelson, where returns follow a Gaussian bell–shaped distribution, all returns are standardised using a measure called standard deviation. In this Gaussian world, it becomes straightforward to quantify the probability of observing a particular return magnitude. The following Table 1.1 presents the return magnitudes. The first column gives the list of returns (X) from 1 to 10. The second column gives the probability that the absolute value of the return is found larger than X times the standard deviation. The third column translates this probability into the number of periods (days) one would need to wait to witness such a return amplitude. The last column translates this waiting time to calendar time using convention that one month is equivalent to approximately 20 trading days and one year has 250 trading days. According to the table, a daily return exceeding 3% in amplitude would typically be observed approximately once every 1.5 years. A daily return exceeding 4% would typically occur only once every 63 years, while a return exceeding 5% would never have been observed in our available historical data.

Table 1.1: Source: [1] Probability and expected waiting time for an $X\%$ move.

X (%)	Probability	One in N events	Calendar waiting time
1	0.307	3	3 days
2	0.045	22	1 month

X (%)	Probability	One in N events	Calendar waiting time
3	0.0027	370	1.5 year
4	6.3 x 10^{-5}	15,787	63 years
5	5.7 x 10^{-7}	1.7 x 10^{6}	7 millennia
6	2.0 x 10^{-9}	5.1 x 10^{8}	2 million years
7	2.6 x 10^{-12}	3.9 x 10^{11}	1562 million years
8	1.2 x 10^{-15}	8.0 x 10^{14}	3 trillion years
9	2.3 x 10^{-19}	4.4 x 10^{18}	17,721 trillion years
10	1.5 x 10^{-23}	6.6 x 10^{22}	260 million years

Market volatility can also be measured by the VIX or Volatility Index. The VIX was created by the Chicago Board Options Exchange as a measure of the 30–day expected volatility of the U.S. stock market derived from real–time S&P 500 call and put options prices. It is effectively an instrument for bets and hedges by investors and traders on the future direction of the markets and individual securities' volatility. A high reading on the VIX of above 20 implies a riskier market. A reading above 30 implies a high level of fear among investors. Whereas a reading below 15 indicates a calm orderly market with low expected volatility. A reading between 15 and 20 represents normal market conditions. For example, a VIX of 16 would imply 1% daily volatility. This is because as will be shown in later section, volatility is proportionate to the square root of time and 16 is the approximate square root of the 256 trading days in a year. So, if we divide VIX by 16, we get an approximate expected daily move in S&P 500 [2].

Figure 1.1 below illustrates volatility surface of SPY[1] ETF. Even though VIX is calculated using a combination of SPX options, the two are nearly identical in terms of price movements, so plotting a SPY–based volatility

[1]SPY ETF is an exchanged traded fund that tracks S&P 500 Index.

surface can serve as a close heuristic approximation to that of the SPX (and thus indirectly the VIX). VIX is a broad measure of market volatility and is derived from a variety of option contracts. The plot has 3 dimensions which are strike price (K), days to expiration (T) and implied Volatility (IV). This volatility surface is far from the flat, uniform implied volatility profile one would expect under the simplifying assumptions of Geometric Brownian Motion.

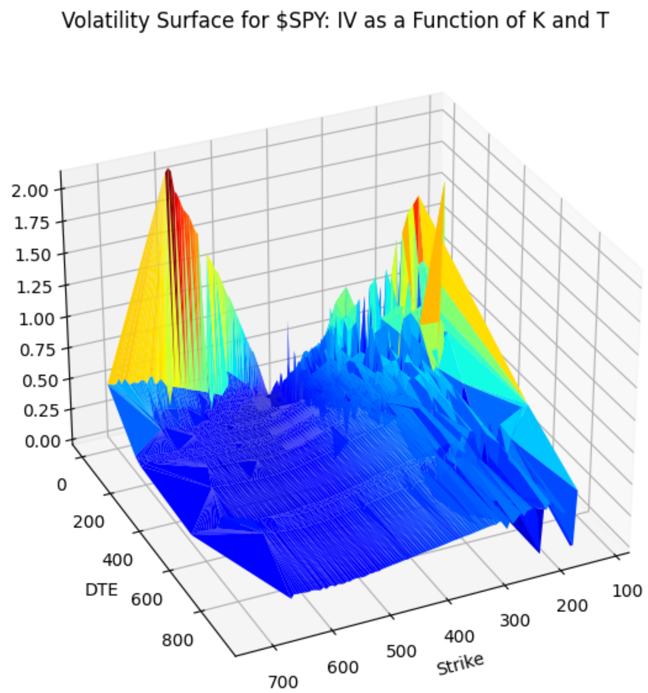

Figure 1.1: 3D Volatility Surface

As the mathematics of volatility modelling evolved researchers questioned the assumption of Geometric Brownian Motion (GBM). GBM assumes that the process has no memory i.e., each step is a Markovian process, so the noise has zero autocorrelation. If that assumption is true, then according to Itô calculus the standard deviation of price changes scales with the square root of time. However, if the volatility increments are positively and/or negatively correlated, then volatility process has memory and time–series behaviour cannot be modelled by GBM.

Thus, the researchers turned to another type of Brownian motion —- fractional Brownian motion (fBm). Fractality of the object means that it does not change its form at different scales (scale can mean distance, time). These are self– similar objects, where we can zoom in and out and note that the shapes will be approximately the same. In the example of volatility time series, if we look at intraday or weekly patterns, we will roughly see the same shape.

It was not until Gatheral et al., [3] seminal paper that the idea of roughness in volatility modelling took off. Their work introduced the concept of rough volatility, where volatility evolves as a fractional process with a Hurst exponent less than $\frac{1}{2}$, challenging the conventional assumption of smoothness implied by GBM. In their paper, the authors showed the first applications in both pricing and volatility forecasting, as well as established a rigorous theory as to why volatility exhibits roughness. This development occurred several decades after the introduction of fractional Brownian motion by Mandelbrot and Van Ness [4]. According to the new theory that volatility is rough, the roughness lies in the market microstructure and order flow distribution and behaviour. The wide proliferation of high frequency trading simply exacerbates these effects. The theory behind this is that buy orders generate more buy orders, and vice versa. However, the cascade of sell orders often has more impact on the order book and through feedback loop on the underlying spot market. This is what economists typically refer to as boom–and–bust cycles or reflexivity between processes. There are other more nuanced volatility dampening effects such as order splitting, but this is not the scope of this paper.

Also referred to as statistical volatility, historical volatility (HV) is a backward–looking measure that quantifies the extent of past price fluctuations over predetermined time intervals. In contrast to implied volatility, which incorporates expectations of future uncertainty, HV relies solely on historical data and thus provides no direct insight into future market conditions. Because of this retrospective focus, HV is less commonly used as a predictive tool.

Nonetheless, changes in HV remain informative: an increase often signals that the security's price has recently undergone more pronounced movements——suggesting a possible shift in market dynamics——while a decrease implies that some uncertainty has been resolved, allowing prices to stabilize and potentially revert to their long–term averages.

While fractional and rough volatility models attempt to capture the complex nature of volatility dynamics, practitioners often rely on market–derived measures such as the VIX and VVIX. As stated earlier, CBOE Volatility Index (VIX) is a real–time measure of the market's expectations of the near–term changes in the S&P 500 index. The index is derived from the prices of at the money and out of the money SPX index options with near–term expiration dates. The calculation derives a 30–day forward projection of volatility. In financial markets, volatility is a measure of how fast prices change and measures the degree of fear or greed amongst the market participants.

In 2012, CBOE introduced a new index called VVIX or volatility of volatility. The VVIX index is a volatility of volatility measure as it represents the expected volatility of the 30–day forward price of the CBOE Volatility Index (the VIX). It is this expected volatility that drives the price of the VIX options. VVIX is calculated from the price of a portfolio of liquid at–the–money and out–of–the money VIX options.

Having introduced both VIX and VVIX, we now turn to the range of modelling techniques that this study will employ. These methods were chosen to capture the multifaceted nature of volatility, incorporating memory, non–normality, and scale–invariance. We will use various algorithms to model VIX and VVIX: (i) Naïve Models (ii) GARCH family, (ii) Monte Carlo, (iii) LSTM, (iv) SVM, (v) Random Forest Regressor and more modern approaches encompassing (vi) NBEATS (vii) MSM, (viii) MMAR, (ix) RL (x) GAN and (xi) Mixed Fuzzy Fractional Brownian motion.

As one of the parameters that influences the movement in stock prices, volatility is a crucial input to numerous financial market operations, and it becomes an important tool to assess the risk of ruin for portfolio managers, investors, and other interested parties [5]. The modelling of volatility is a complex task because it cannot be observed directly. Indeed, it can only be measured by looking at the extent of the movement of the option prices and deriving it mathematically from that.

The mathematical estimators are complicated also by volatility clusters, fat tails, non–normality of the distribution, and structural breaks in the distribution of the returns. These features cannot be captured by simple classical models such as the autoregressive moving average (ARMA) process.

The next chapters will present an overview of the main estimators of the volatility with particular attention to the heteroskedastic models.

1.2 RESEARCH QUESTION AND POSTULATED HYPOTHESIS

This research investigates three primary research questions and their corresponding hypotheses.

i. **Research Question 1:** To what extent do volatility time series and their first derivatives (VVIX) exhibit fractal properties, and can these characteristics be accurately modelled using mixed fuzzy fractional Brownian motion (mfFBm)?

   Hypothesis: The central hypothesis is that both the volatility time series and its first derivative (the "volatility of volatility" or VVIX) exhibit fractal properties. Fractality implies self–similarity across different time scales, meaning that these time series cannot be accurately modelled using classical diffusion laws, such as Fick's laws, which assume memoryless and random–walk behaviours. In fractal processes, the Hurst exponent (H) plays a crucial role. If $H > \frac{1}{2}$, the time series displays long–range dependence (persistence), so high volatility today suggests it will likely remain high in the near future. Conversely, if $H < \frac{1}{2}$, the series is anti–persistent (rough), and increases in volatility tend to be followed by decreases, reflecting negative autocorrelation. The time series is more erratic and less smooth. For example, a market where volatility changes rapidly and frequently, with no clear trend over time. Empirical evidence often shows that H $\neq \frac{1}{2}$, indicating that simple random–walk assumptions do not hold. To capture these fractal characteristics, this thesis develops the necessary theoretical framework required to forecast volatility by first forecasting its derivative (VVIX). This derivative quantifies the fractal dimension of the time series, providing insight into its degree of self–similarity. The higher the fractal dimension, the more self–similar the time series is. Building on the foundational work of Mandelbrot, Lux, Liu, and Calvet, we extended a novel mathematical process——mixed *fuzzy* fractional Brownian motion (mfFBm) [P1] [P2] [P3] [P4] that incorporates both fuzziness and jump processes into fractional Brownian motion.

ii. **Research Question 2:** Does the fractal nature of volatility imply significant time–scale invariance, and can a theoretical framework be developed to capture these repeating statistical patterns?

   Hypothesis: The second hypothesis proposes that, due to the observed fractal nature of volatility, a degree of time–scale invariance should exist, meaning that statistical patterns repeat across different scales. The empirical findings in this thesis regarding the shifting Hurst exponent and the success of fractal based modelling in the literature support this proposition. This motivates the development of a scale–invariant theoretical framework as a core goal of this research.

iii. **Research Question 3:** How does the proposed fractal–based theoretical framework compare in forecasting performance against traditional econometric and deep learning models?

   Hypothesis: Finally, we hypothesize that the novel theoretical approach will outperform conventional techniques. We compare our results to more traditional econometric and deep learning models as discussed in [P1] [P4] and [P3]. This comparison of conventional techniques establishes a performance benchmark, which serves to justify the need for the novel theoretical approach developed in this thesis.

1.3 CONTRIBUTION TO KNOWLEDGE

The aims of this research are threefold. First, state–of–the–art methods related to fuzzy fractional Brownian motion are reviewed to establish the theoretical soundness of these approaches. Second, extension and refinement of several recent theoretical results are presented, providing novel proofs and extensions [P2] [P3]. Third, empirical evidence demonstrating the practical performance of various modelling techniques is presented. In undertaking this work, a narrative review methodology is employed to synthesise existing research and identify areas where the proposed framework can contribute new perspectives. This approach enables a comprehensive overview of the literature, as well as expert interpretation and synthesis, both of which are essential for advancing theoretical frameworks and introducing novel proofs in this domain.

This thesis develops a unified framework that integrates fractional Brownian motion with fuzzy processes, resulting in a product measure space encompassing both the intrinsic randomness of fractional Brownian motion and the fuzziness of associated parameters [P2] [P3]. Although existing literature provides numerous insights into forecasting the VIX time series directly, our research develops the theoretical foundation for a novel approach where VIX could be forecasted through its first derivative, VVIX, using fractional calculus techniques developed herein and combine them with fuzzy sets and jump processes. Since the VIX represents implied volatility rather than a direct price series, it behaves differently from standard financial time series. We hypothesize that by focusing on VVIX and employing multifractal models——such as the Multifractal Model of Asset Returns (MMAR) and its Markov–switching multifractal (MSM) variant enhanced with fuzzy variables——we can improve forecasting accuracy. This approach contributes uniquely to the existing body of knowledge on VIX forecasting by providing a new theoretical framework and a proposed methodology for its application, as these techniques have rarely been explored in this context.

The scarcity of research and literature on fractional mathematical methods in finance, particularly regarding volatility modelling, indicates that this area remains significantly under–researched. This work thus fills a critical gap, laying the groundwork for further exploration and development of fuzzy fractional approaches in the financial domain.

1.4 DOCUMENT STRUCTURE

The remainder of this thesis is structured as follows. Chapter 2 serves as the literature review chapter, providing a detailed theoretical overview of the proposed models. It begins by discussing the fractal nature of financial data and multifractal models (e.g., MMAR), and then progresses through various volatility forecasting techniques, including GARCH, LSTM, Monte Carlo simulations, Support Vector Machines, Random Forest Regressor, NBEATS, Reinforcement Learning, GANs, and Fuzzy Neural Networks.

Chapter 3 focuses on data and methodology. It introduces smoothing and transforming techniques, tests for stationarity using the Augmented Dickey–Fuller test, and outlines the methodology for generating multifractal simulations to better understand complex market behaviour. In addition to reviewing these methods, Part 3 also includes a dedicated section presenting the results of these models, detailing the performance and clustering techniques employed (K–Means and Fuzzy C–Means) and exploring deep neural network applications.

Chapter 4 presents the evaluation metrics used to assess the forecasting models, while Chapter 5 delves into the mixed fuzzy fractional Brownian motion (mfFBm) framework with jumps. This includes theoretical preliminaries, handling cases where $H > \frac{1}{2}$ and $H < \frac{1}{2}$, introducing fuzziness and jump processes, and providing rigorous proofs and applications.

Chapter 6 presents conclusion and future work, summarises the key research findings, discusses their implications for volatility modelling, acknowledges the study's limitations, and suggests potential avenues for future research. Finally, the thesis concludes with appendices containing code samples (e.g., for the Augmented Dickey–Fuller test and multifractal simulations), table A3 mapping the results and models to Hurst parameter regimes and a comprehensive list of references.

CHAPTER 2 – LITERATURE REVIEW

2.1 INTRODUCTION

An accurate model should be subjected to rigorous testing and adhere to the principle of parsimony, which holds that models with fewer parameters are generally preferable [6].

Measuring and forecasting volatility has attracted considerable attention from both academia and market practitioners ever since the Chicago Board Options Exchange (CBOE) introduced the VIX in 1993. In 2003, the methodology for calculating the VIX was revised in collaboration with Goldman Sachs, resulting in a widely accepted measure of expected volatility that has since been extensively used by scholars, traders, and risk managers. When volatility is interpreted as uncertainty, it becomes a key input in portfolio investment decisions, especially for pricing derivative instruments such as options. In 1996, the Basel Accord began requiring the modelling and forecasting of volatility as a mandatory input to risk management for financial institutions worldwide.

Ruipeng Liu and Rangan Gupta [7] studied the role of investor uncertainty as measured by the conditional volatility of investor sentiment in forecasting the volatility of six major stock markets using a bivariate Markov–switching multifractal (MSM) model. Their results showed that incorporating information on investor uncertainty led to smaller forecasting errors compared to univariate MSM models that did not include this metric. This superior performance underscores the importance of accounting for long memory and structural breaks when forecasting stock market volatility. Moreover, since financial market volatility is influenced by investor uncertainty, policymakers should recognize that such uncertainty can prolong negative economic impacts through feedback loops affecting real macroeconomic variables, as well as delaying investment and consumption decisions.

C–Rene Dominique and Luis Eduardo Solis Rivera [8] studied short–term and long–term dependence of the S&P500 index using a mixed fractional Brownian motion (mfBm) model. The mfBm model is a continuous–time stochastic process and is generalisation of the fractional Brownian motion (fBm) model that allows for both short–term and long–term dependence. The process is characterised by two Hurst parameters, H_S and H_L. H_S measures the short–term dependence, and H_L measures the long–term dependence. By capturing both volatility clustering and heteroscedasticity, mfBm better represents market microstructure noise in the short run and macroeconomic factors such as interest rates and inflation in the long run. The authors found that long–term dependence in the S&P 500 index is stronger during periods of economic expansion than during recessions.

In 2012, the CBOE introduced the Volatility of Volatility Index (VVIX), often referred to as "vol of vols." VVIX is calculated in a manner similar to VIX, except that it uses the VIX time series instead of S&P 500 options. VVIX measures the expected volatility of the VIX itself. This chapter reviews academic studies focusing on various volatility forecasting models including GARCH, Monte Carlo simulations, Support Vector Machines, LSTM, Random Forest, and other state–of–the–art techniques. While much of the existing literature focuses on forecasting VIX, there is a relative scarcity of research on forecasting VVIX using modern approaches such as fractal methods, deep reinforcement learning, or GANs. Some studies have explored Bayesian methods and Markov chain Monte Carlo (MCMC) techniques for forecasting VVIX, but overall, this remains an under–researched area.

2.1.1 FRACTAL NATURE OF THE DATA

The central thesis of this paper is that volatility and its first derivative time series data are fractal. In other words, if we examine monthly, weekly, daily, or tick–level data, we would not be able to distinguish one timescale from another unless we already know the time frame. While there are cut–off points that theoretically allow one to differentiate between scales, these points are often so minuscule (e.g., fractions of a nanosecond) or so vast (e.g., multiple decades) that we choose to disregard them in practical applications.

Mathematicians have long recognised discrepancies between standard Brownian motion models and real financial data. Firstly, the data exhibit temporal dependence, with periods of large changes alternating with periods of small changes. Secondly, the tails of the return distribution are far fatter than those predicted by a Gaussian distribution. Although the data may become less peaked and have thinner tails as the sampling interval extends to decades, they never truly become Gaussian. Empirical studies indicate that multifractal processes never converge to a Gaussian distribution. Consequently, researchers have explored alternative distributions and approaches such as the Student's t–distribution, Poisson jumps, and nonparametric representations to better capture these characteristics.

GARCH(p, q) models, introduced by Bollerslev, address volatility clustering and help mitigate the issue of fat tails. In these models, a higher α coefficient tends to produce sharper changes in volatility, while β corresponds to more gradually evolving volatility dependencies. Since $(\alpha + \beta) < 1$ typically holds, there is a trade–off between immediate large changes and sustained long cycles in volatility. This limitation means that GARCH cannot simultaneously capture abrupt shifts and prolonged volatility patterns.

Multifractality was introduced by Mandelbrot in 1972 and was first applied in physics to describe distributions of energy and matter such as minerals, turbulent dissipation, stellar matter. Mandelbrot extended multifractality to measure the heterogeneity in the financial time series and extended it to stochastic processes. Most diffusions are characterised by increments that grow locally at the rate $(\Delta t)^{1/2}$ throughout their sample path. The exception is FBM which has growth rate $(\Delta t)^{H}$, where H is invariant with t. Multifractals, on the other hand, have a multiplicity of local growth rates for increments (where $h = 1/2$, this becomes normal Brownian motion).

Mandelbrot's seminal 1997 paper introduced the Multifractal Model of Asset Returns (MMAR), viewing asset prices as multiscaling processes with both long memory and heavy tails. In the MMAR, volatility of volatility is introduced via a random trading time, generated as the cumulative distribution function of a random measure. Under this framework, the distribution of returns and volatility remains invariant across time scales, and the compounding effect of random trading time allows direct modelling of variance without influencing the direction or correlation of increments. In other words, the introduction of random trading time T(t) in the Multifractal Model of Asset Returns (MMAR) allows direct modelling of time–varying variance. This is achieved by scaling the variance through T(t) without altering the underlying direction (sign) or correlation structure of the price increments, which remain determined by the properties of the driving process (e.g., Brownian or fractional Brownian motion).

In their seminal paper, Thomas Lux and Ruipeng Liu [9] extended Generalised Method Moment (GMM) approach of Lux (2008) to multivariate settings and applied it to discrete and continuous distributions of the volatility components. Their results showed that the performance of the GMM estimator remained robust when moving from univariate to bivariate or trivariate models. In subsequent work, Ruipeng Liu et al [10] utilised a bivariate Markov–switching multifractal (MSM) model to examine the interplay between exchange rates and stock market volatility in various developed (Canada, France, Germany, Italy, Japan, UK) and emerging (BRICS) economies. The MSM model captured key statistical anomalies in volatility dynamics. The authors compared GARCH models with MSM and found that GARCH offered superior volatility forecast for short horizons whereas multifractal models offer significant improvements for longer horizons across most markets. Bivariate model outperformed univariate model across developed markets but had mixed performance in emerging markets.

2.1.1.1 FRACTALS

A fractal is defined as a curve or geometric shape, each part of which has the same statistical properties as the whole. Fractals are used in modelling structures such as eroded coastlines and snowflakes in which similar patterns recur at progressively smaller scales, and in describing partly random or chaotic phenomena such as crystal growth, fluid turbulence, and galaxy formation. Fractals are created by repeating a simple process in a never–ending feedback loop. A fractal dimension of 1D time series embedded in a 2D plot measures the roughness of a surface and has the following simple relationship with hurst exponent in 2D space: $d_H = 2 - H$. It is defined as the Hausdorff dimension, which is the dimension of a fractal set, is inversely related to the Hurst exponent (H) which is a measure of the long–term dependence of a time series. This relationship can be explained in the following way. A time series with a high Hausdorff dimension is said to be "rough", meaning that it has a lot of small–scale variations. A time series with a low Hausdorff dimension is said to be smooth, meaning that it has few small–scale variations. Hurst exponent is defined as the exponent of the power law that describes the decay of the autocorrelation function of the time series. In other words, The Hausdorff dimension, d_H, describes how a fractal curve fills space, offering insights into its roughness and detail across scales. For instance, when d_H *=1,* the curve is smooth and resembles a straight line. Conversely, if d_H *=2,* the curve becomes so jagged that it effectively fills an entire two–dimensional plane, like an irregular coastline. The relationship $d_H = 2 - H$ establishes a connection between the smoothness of the curve and its fractal dimension. As the Hurst exponent H increases, the curve becomes progressively smoother, and the fractal dimension d_H decreases, indicating fewer small–scale variations.

In time series analysis, the Hurst exponent H governs the behaviour of a process over time. If $H>0.5$, the series is persistent, meaning trends tend to continue in the same direction, creating smoother, long–range dependent paths. At $H=0.5$, the series behaves like a random walk, exhibiting no memory and resembling standard Brownian motion with d_H =*1.5.* For *H<0.5*, the series is anti–persistent, where trends tend to reverse frequently, resulting in rougher, more erratic paths with d_H *>1.5.* Thus, the relationship $d_H = 2 - H$ maps the persistence or anti–persistence of time series to their geometric roughness.

The physical interpretation of this relationship differs between fractals and time series. In fractals, the Hausdorff dimension quantifies the spatial roughness of a curve, reflecting how its complexity changes as one zooms in to finer scales. In time series, the Hurst exponent measures the persistence or memory of increments over time, linking temporal behaviour to spatial roughness. Despite these different contexts, the mathematical relationship bridges the two domains, providing a geometric interpretation of time series roughness while offering a quantitative measure of the self–similar nature of fractals and their temporal equivalents.

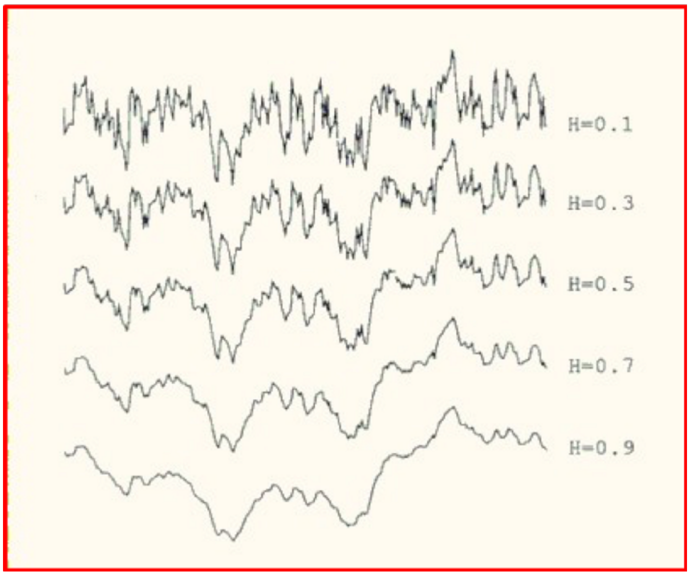

Figure 2.1: Source: [11] Hurst exponents and roughness.

Figure 2.1 [11] illustrates how the Hurst exponent H influences the roughness of a time series. As H increases, the curves become smoother, indicating fewer small–scale variations. This corresponds to smaller fractal dimensions d_H, where the relationship $d_H = 2 - H$ holds. Fractals exhibit statistical self–similarity, meaning that subsets of the process resemble the full set when viewed at different scales. A Hurst exponent measures the diffusion speed of the series. The problem with simulating even a high Hurst exponent is that on its own, it still produces a Gaussian distributed random walk and with the assumption of Levy stable distributions that imply an infinite variance, which can exhibit randomness, is that the key feature of the data is lost i.e. the sporadic nature of volatility which tends to go wild for some time and show multi sigma moves for several days in a row, and then abruptly go back to low periods of volatility. To replicate market panics, it is simply not enough to generate a leptokurtic distribution that generates high kurtosis occasionally, it also must cluster these days together.

2.1.1.2 LONG RANGE DEPENDENCE

Processes that have long range dependence occur when past events are nontrivially correlated with future events, even if they are far apart and exhibit a high persistence degree. This persistence arises from a slow, power–law decay in the autocorrelation function rather than the exponential decay seen in short–memory processes. One example by Granger, Joyeux and Hosking [12] is given by following a fractionally differenced time series:

$$(1-L)^d S_t = \varepsilon_t, \qquad d \notin \mathbb{Z}. \tag{2.1.1}$$

where L is the lag operator shifting S_t to $S_{t\text{-}1}$, the exponent d is non–integer that determines the degree of fractional differencing, and ϵ is a white noise error term. Fractional differencing allows for long–memory behaviour, where correlations decay slowly, following a power–law function, as opposed to traditional short–memory processes.

2.1.1.3 HURST EXPONENT AS A MEASURE OF DIFFUSION

To understand the nature of the price series it is helpful to analyse its speed of diffusion. Diffusion measures the spread of the series from a location where its concentration is higher than in most other places. It can be measured by examining how the variance of displacements scales with the time lag τ:

$$\mathrm{var}(\tau) = \left\langle |x_{t+\tau} - x_t|^2 \right\rangle . \tag{2.1.2}$$

where τ is the time interval between measurements, x_t represents the log–transformed series $x = log\ S(t)$, and $\langle \cdot \rangle$ denotes an average over time. Log transformations are often applied to stabilise variance and reduce heteroskedasticity in the series.

The mean squared displacement (MSD) is a measure of the deviation of the position of the particle with respect to a reference position over time. It is the most common measure of the spatial extent of random motion and is calculated as the average of the squared displacements of the particle over a given time interval. MSD behaves differently depending on the type of diffusion. In the case of normal diffusion, the MSD scales linearly with time, MSD$\sim\tau$, which is typical of a random walk with no memory or persistence. For subdiffusion, the MSD grows slower than linearly, scaling as MSD $\sim \tau^H$, where $H < 0.5$. This indicates anti–persistence, meaning the series exhibits rough behaviour with frequent trend reversals. In contrast, superdiffusion occurs when the MSD grows faster than linearly, following MSD$\sim \tau^H$, where $H > 0.5$. Superdiffusion reflects persistent behaviour, where trends tend to continue in the same direction, resulting in a smoother and more predictable trajectory.

For example, Figure 2.2 illustrates the MSD for Brownian motion will increase linearly with time, while the MSD for a system with anti–persistence and frequent trend reversals will increase more slowly than linearly [13].

Volatility of asset return time series strongly depends on the frequency one chooses to measure it. Measurement at one–minute intervals differ significantly from daily variance. Empirically, it has been established that if returns follow a Geometric Brownian Motion (GBM), the variance increases linearly with the lag τ, and the returns are normally distributed. On the other hand, if there are even small deviations from a pure random walk as is often the case,

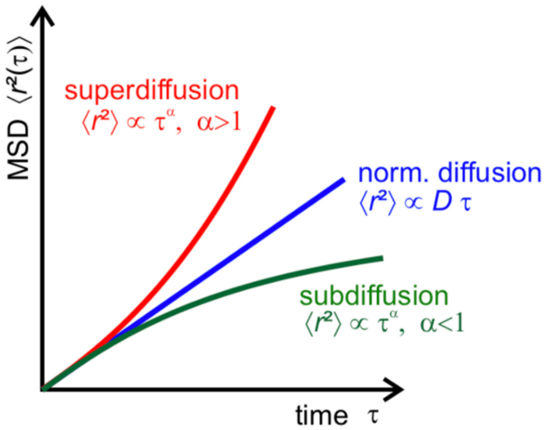

Figure 2.2: Source: [13] Relationship between Mean Squared Displacement and Time.

the variance for lag τ is not proportionate to τ anymore but instead follows a power–law relationship:

$$\text{var}(\tau) \propto \tau^{2H}. \tag{2.1.3}$$

where H is the Hurst exponent. For trending assets $H > \frac{1}{2}$ indicating positive persistence and for mean reverting assets $H< \frac{1}{2}$, indicating negative persistence. The Hurst exponent measures the level of persistence of a time series. The level of persistence is a measure of how likely it is to continue in its current trend. A time series with high persistence is more likely to continue in its current trend, while a time series with low persistence is more likely to reverse its trend. If $H = \frac{1}{2}$ then the time series has no persistence and follows a Geometric Brownian Motion which is defined by the following process $X(t) = X(0) * \ exp\left(\mu t + \sigma W(t)\right)$, where *W(t)* is a standard Brownian motion, μ is the drift term, and σ is the volatility. GBM has a few properties that makes them easy to use such as the distribution of the underlying is lognormal, the process is a martingale, so it is memoryless, and the underlying is always positive. The last property is not necessary if we use Arithmetic instead of Geometric Brownian motion because the underlying processes are quite often negative. Both Arithmetic and Geometric Brownian Motions are quite popular because they give easy closed form solutions, but they are a very simplified model of reality and in our view unsuitable for making real life decisions under uncertainty. The real world is far more complex than GBM, and there are many other filtrations that affect the underlying processes. Examples of each case of persistent, antipersistent and GBM are illustrated in below diagrams.

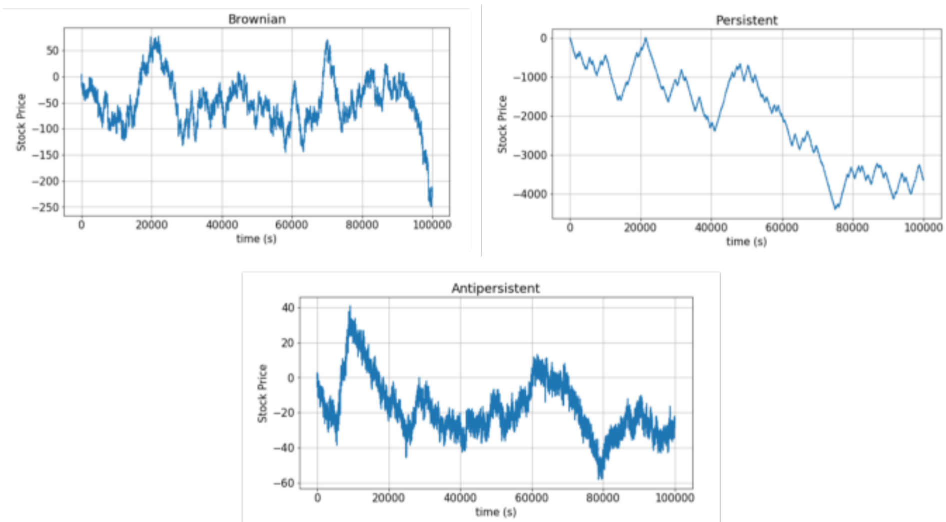

Figure 2.3: Source: [14] Different Levels of Persistence.

2.1.1.4 AUTOCORRELATION

The autocorrelation function measures the linear relationship between a time series and its lagged values and is defined as follows:

$$\rho(\tau) = \frac{1}{\sigma^2}\,\mathbb{E}\Big[\Big(S_t - \overline{S}\Big)\Big(S_{t+\tau} - \overline{S}\Big)\Big]\,. \tag{2.1.4}$$

where σ^2 is the variance of the series, E is the expectation operator, and τ is the lag.

Processes with autocorrelations that decay slowly are termed long memory processes, meaning past events nontrivially influence future events even over large lags. Such processes have some memory of past events which have diminishing influence on future events.

Long memory processes are characterised by autocorrelation function with power–law decay:

$$\rho(\tau \to \infty) \propto \tau^{-\alpha}. \tag{2.1.5}$$

where α *>0.* The relation between a and H is given by $a = 2(1\text{–}H)$ and as $H \to 1$, the decay becomes slower since a approaches zero indicating persistent behaviour. These processes where H is greater than $\frac{1}{2}$ but less than 1 are long memory processes and are often referred to as Fractional Brownian motion (FBM). The value of H strongly depends on the choice of lags τ.

2.1.2 MULTIFRACTAL MODEL OF ASSET RETURNS (MMAR)

Unlike earlier versions which focused on L–stable distributions, MMAR is a generalisation of the L–stable distribution model and the fractional Brownian motion model. It combines long tails and finite variance of the L–stable distribution model with the long dependence of the fractional Brownian motion model over discrete sampling intervals. Secondly, the MMAR contains long dependence which is the characteristic feature of Fractional Brownian Motion (FBM), which was formally introduced by Mandelbrot and van Ness in 1968 [4]. Finally, the last feature of MMAR is the concept of trading time, introduced by Mandelbrot and Taylor in 1967 [15]. This time concept is the explicit modelling of the relationship between clock time which we can observe and measure, and unobserved natural timescale of the returns process. They argued that the time at which trades occur is not evenly spaced but rather is clustered around certain times. This clustering of trades leads to a distortion of clock time. The MMAR takes into account the concept of trading time by modelling the return process as a function of trading time. This allows the model to capture the clustering of trades and distortion of clock time. All three features are contained in MMAR which accounts for empirical observations of financial time series with long tails relative to normal distribution and long memory in the absolute value of returns and volatility.

Stochastic fluctuations in volatility are measured by a random trading time.

Multifractals are used in MMAR model by generating a multiplicative cascade. A multiplicative cascade is a process in which a set of intervals is split into smaller intervals, and the sizes of the smaller intervals are determined by a random multiplicative process. For instance, at each stage of the cascade, intervals can be split in $b > 2$ intervals of equal size. Subintervals, indexed from left to right by β $(0 \leq \beta \leq b-1)$ receive fractions of the total mass equal to $m_0,\ m_{b-1}.$ The term "mass" in the context of MMAR model mass is not the same as probability mass. In the MMAR model, mass refers to a conceptual measure used to describe how the total quantity (e.g., volatility intensity or some scaling factor) is distributed across progressively smaller time intervals. This can be thought of as the total number of particles, or the amount of energy in the system (probability mass refers to the probability that a particular outcome will occur e.g., the probability mass of rolling 6 on a die is 1/6).

In MMAR, the mass of each interval is multiplied by a random factor. The random factors are chosen so that the total mass of the system is preserved. By the conservation of mass, the total mass of the system must remain constant over time, so these fractions add up to 1: $\sum m(\beta) = 1.$ For example, the line at step 0 starts with a mass of 10,000. At each step, the line is divided into two sections, one of which will have its mass multiplied by 0.6, and the other by the left over 0.4 At the next step, these two sections are divided into two further sections, for a total of 4, and the same multiplication is applied to it. This means that the probability mass of each interval is not affected by the multiplicative cascade process. In other words, the probability mass of each interval is independent of the mass of the interval. This is an important distinction because it means that the MMAR model can be used to model systems where the probability mass of each outcome is not constant.

By applying multiplicative cascade process at finer and finer time scales, we end up with a pattern where volatility is distributed unevenly. Some very short intervals might receive disproportionately high "mass" (intense volatility), while neighbouring intervals receive less. Over time, this uneven distribution across scales creates a multifractal structure——one that closely resembles the clustering of high and low volatility periods observed in real financial markets.

One can also introduce randomness to this process when generating these cascades by assigning a random multiplicative factor to each interval. The multiplicative cascade process divides the line smaller and smaller intervals, and then multiplies the mass of each interval by a random factor. The random factors are chosen so that the total mass of the line is preserved. This means that the total mass of the system is always the same, even though the mass of individual intervals may change. A simple analogy to understand this concept of mass in MMAR is a bag of marbles. The total mass of marbles in the bag is constant. We can divide the marbles into smaller and smaller groups, but the total mass of the marbles will always be the same. This is similar to the way the total mass of the system is preserved in the MMAR model. Figure 2.8 illustrates an example of a basic deterministic multiplicative cascade.

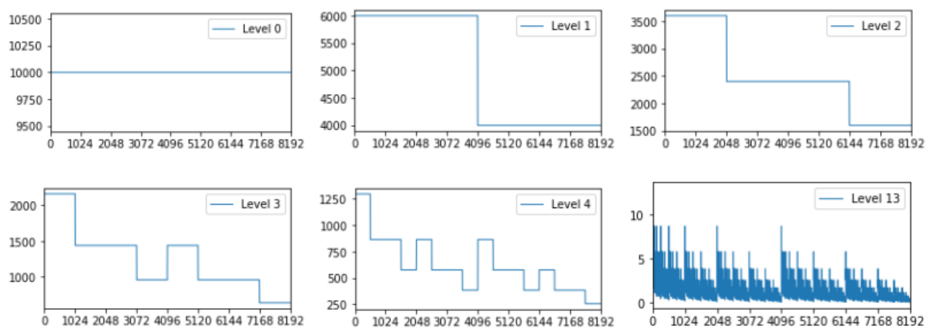

Figure 2.4: Source: [14] Multiplicative cascade.

Volatility clustering is the tendency for periods of high volatility to follow other and vice versa. The MMAR model, due to its multifractal construction and integration of trading time and long–range dependence, has been shown to be an effective tool in modelling this phenomenon. By capturing both long–term dependence and the hierarchical distribution of volatility across different time scales, MMAR aligns closely with the empirical behaviour of asset returns.

2.1.2.1 TRADING TIME AND COMPOUND PROCESSES

According to MMAR, the price change of an asset is considered a function of both the passage of trading time and random movements known as Brownian motion. Price is a function of trading time which is a function of clock time. So, the two functions of time and Brownian motion work jointly to produce a model different from its two composition parts. By combining these two components, MMAR aims to provide a more comprehensive representation of asset returns, that accounts for both the temporal structure of trading and the inherent randomness in market dynamics. Mandelbrot presented rigorously the concept in his 1997 paper “A Multifractal Model of Asset Returns” [16]. Following Mandelbrot’s notation, MMAR is defined as:

$$X(t) = \ln P(t) - \ln P(0). \tag{2.1.6}$$

where *P(t)* denotes a point P at time t and *P(0)* is the point at time zero. If $H = 0.5$, then these price movements are mutually independent, isotropic, and the price process is Brownian Motion.

Assumption 1

$$X(t) \equiv B_H\Big[\theta(t)\Big]. \tag{2.1.7}$$

X(t) is a compound process where *BH(t)* is a fractional Brownian Motion and $\theta(t)$ is a stochastic trading time or the time deformation process. When plotted, FBM exhibits self–affinity, which means the plot looks the same at any magnification. So, it is fractal.

Assumption 2

Mandelbrot stated that “Trading time is the key concept facilitating the application of multifractals to the financial markets”. He observed that “The salient of trading time is explicit modelling of the relationship between unobserved natural time–scale of the returns process, and clock time, which is what we observe.” Heuristically, this means that not every day “feels” the same in the financial markets. Some days "feel" faster than others and on other days trading is slow and boring as if there is very little happening. Volume of trading would be high on fast days and low on slow days. This concept of multifractal "trading time" is the important contribution that Mandelbrot introduced into quantitative finance.

Assumption 3

BH(t) and Θ*(t)* are independent.

2.1.2.2 PROPERTIES OF THE MMAR

MMAR is a model for simulating growth over time. It consists of two components– a multifractal cascade and a fractional Brownian motion. MMAR attempts to capture three important properties of financial time series: i) non–Gaussian distributions ii) heteroskedasticity in price changes ($H \neq 0.5$) iii) high and low volatility clustering together (concept of trading time). MMAR is estimated using 4 parameters —- H, α, σ^2, and λ. These must be estimated empirically before we can construct a market simulation. Here are the high–level steps to estimate the parameters of the MMAR model:

(i) Estimate the Hurst exponent (H) of the data using a variety of methods, such as the R/S analysis or the detrended fluctuation analysis.

(ii) Estimate the scaling exponent (α) of the data using a variety of methods, such as the box–counting method or the diffusion approximation.

(iii) Estimate the variance (σ^2) of the data using a variety of methods, such as the sample variance or the maximum likelihood estimator.

(iv) Estimate the lag parameter (λ) of the data using a variety of methods, such as the least squares method or the maximum likelihood estimator.

2.2 GENERALISED AUTOREGRESSIVE CONDITIONAL HETEROSCEDASTICITY (GARCH) MODEL

Generalized Auto–regressive Conditional Heteroscedasticity (GARCH) [17] refined Engle's [18] ARCH model by inserting the AR process in the heteroscedasticity from the variance into Generalized Auto Regressive Conditional Heteroscedasticity. Chang et al. [19] modelled the volatility of foreign exchange rate RM/Sterling using the Stationary GARCH–in–Mean model and concluded that the volatility of the exchange rate is constant. GARCH–in–Mean outperformed other GARCH models in out–of–sample and one–step–ahead forecasting. A. Thavaneswaran et al. [20] introduced non–centred nth order possibilistic moments of fuzzy numbers and extended the results to centred moments of fuzzy numbers. They combine GARCH model of option pricing and volatility forecasting with Fuzzy Weighted Coefficient Autoregressive (FWCA) time series model, in which the random coefficient is treated as a fuzzy number and derive the weighted possibilistic moments of the random coefficient. The simplest GARCH (p, q) model is the GARCH(1, 1) model, which can be written as:

$$y_t = \mu_t + u_t. \tag{2.2.1}$$

$$where\ \mu_t : the\ expected\ value\ of\ y_t\ and\ u_t : the\ error\ term\ or\ innovation$$

$$u_t = \sigma_t \varepsilon_t, \tag{2.2.2}$$

$$\sigma_t^2 = w + a_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \tag{2.2.3}$$

$where\, \varepsilon_t \sim i.i.d(0,1);\; a_1 u_{t-1}^2 \;\&\; \beta_1 \sigma_{t-1}^2\; are\; squared\; error\; and\; variance\; from\; t-1;\; \sigma_t^2\; is\; the\; conditional\; variance\; of\;\; u_t$

The GARCH (1, 1) model is a model of conditional variance that says that the conditional variance of u at time t depends not only on the squared error term in the previous period but also on its conditional variance in the previous period. The model assumes that the error term is white noise, which means it is i.i.d. with mean 0 and variance $\sigma^2 = 1$. This model can be generalized to a GARCH (p, q) model in which there are p lagged terms of the squared error term and q terms of the lagged conditional variances. GARCH is an advancement on ARCH models.

2.2.1 ***VOLATILITY ESTIMATION MODEL***

To measure the volatility of ARCH/GARCH, there are some steps to select the best model. The first step which should be conducted is examining the significance of ARCH coefficient or residual coefficient (α) and GARCH coefficient or variance coefficient (β). The ARCH coefficient measures the impact of past squared residuals on the current variance, while the GARCH coefficient measures the impact of past variances on the current variance.

A coefficient is significant if the probability of z–statistic value is less than the probability of critical value. The critical value is a threshold that we use to determine whether a coefficient is significant. The critical value is determined by the significance level which is typically set at 0.05. If the ARCH coefficient is significant, it means that past squared residuals have a significant impact on the current variance. This suggests that the ARCH model is a good fit for the data. If the GARCH coefficient is significant, it means that the past variances have a significant impact on the current variance. This suggests that the GARCH model is a good fit for the data. Once we have determined that the ARCH and GARCH coefficients are significant, we can then select the best model by comparing the AIC and BIC values. These are statistical measures that are used to compare the fit of different models. The model with the lowest AIC or BIC value is generally considered to be the best fit for the data.

2.3 VOLATILITY FORECASTING USING LSTM

Long Short–Term Memory Network is an advanced RNN, a sequential network, that allows information to persist. It can handle the vanishing gradient problem faced by RNN. A recurrent neural network is also known as RNN is normally used for persistent memory.

The RNNs remember the previous information and use it for processing the current input. The shortcoming of RNN is, they cannot remember long term dependencies due to vanishing gradient. LSTMs are explicitly designed to avoid long–term dependency problems.

S.K. Mitra [21] combined Neural Networks with traditional Black Scholes Formula to improve accuracy of volatility estimation and option pricing. A. Bucci ([22] compared the predictive performance of feed–forward and recurrent neural networks (RNNs) focusing on LSTM network and nonlinear autoregressive model with exogenous input (NARX) network with traditional econometric methods. Bucci showed that RNNs were able to outperform all the traditional econometric approaches. B. Yang et al. [23] introduced a likelihood–based loss function to train the deep learning models and test all the models by the likelihood of the test sample. They use deep neural network (DNN) and LSTM model to forecast the volatility of stock index. The results show that DNNs with likelihood–based loss function can forecast volatility more precisely than the econometric model and the deep learning models with distance loss function. LSTM was a better one amongst the two DNNs. A. Thavaneswaran et al. [24] combined randomness and fuzziness, and introduced a novel data–driven feed forward (Linear–>RELU–>Linear) neural network model for forecasting conditional variance. The paper demonstrated the superiority of proposed model option pricing over other volatility models. I. Livieris et al. [25] proposed a new framework to enhance deep learning time–series models. The proposed framework focused on transforming the noisy, non–stationary time–series data into de–noised, stationary time–series data suitable for efficiently training and fitting a deep learning model. A. Site et al. [26] and J.Sullivan [27] used Gated Recurrent Unit (GRU) and Long Short–Term Memory (LSTM) based models to forecast stock market time series. With LSTM, the closing prices of time series were predicted more accurately than with the other models. The LSTM model was shown to be more successful than the GRU model with similar error metrics. M. Fabbri et al. ([28] experimented with a long consecutive period of 18 years of historical DJIA series from 2000 to 2017 to train and test Dow Jones Industrial Average using RNNs. In 8 years of test set from 2009 to 2017 the model outperformed the traditional buy–hold strategy. T.Shiratori et al. [29] proposed a structured regularization method for completing two phases: computing base forecast and reconciling those forecasts based on their inherent hierarchical structure. They also developed a backpropagation algorithm specialised for applying their method to artificial neural networks for time series prediction.

Patel et al. [30] used and compared accuracy of artificial neural networks (ANN), Support Vector Machines (SVM), Random Forest (RF), and Naïve Bayes (NB) classifier algorithms. The best result was obtained using RF algorithm.

2.4 VOLATILITY FORECASTING USING MONTE CARLO SIMULATION

Monte Carlo methods, or Monte Carlo experiments, are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. The underlying concept is to use randomness to solve problems that might be deterministic in principle. They are often used in physical and mathematical problems and are most useful when it is difficult or impossible to use other approaches. Monte Carlo methods are mainly used in three problem classes– optimization, numerical integration, and generating draws from a probability distribution [31].

X. Zhang et al. [32] studied the continues–time dynamics of the VIX index with stochastic volatility and jump diffusion processes in VIX and volatility. A linear relationship model was derived between stochastic volatility factor and VVIX index. The studies suggested relationship in the jumps of VIX and VVIX. With VVIX as a proxy for the stochastic volatility, Monte Carlo Markov Chain (MCMC) method was used to estimate the dynamics of VIX. The authors show that the jumps in VIX and the VVIX are statistically significant.

In principle, Monte Carlo methods can be used to solve any problem having a probabilistic interpretation [33]. By the law of large numbers, integrals described by the expected value of some random variable can be approximated by taking the sample mean of independent samples of the variable. When the probability distribution of the variable is parametrized, mathematicians often use a Markov chain Monte Carlo (MCMC) sampler. The central idea is to design a judicious Markov chain model with a prescribed stationary probability distribution [34]

Monte Carlo methods vary, but tend to follow a particular pattern [35]:

- Define a domain of possible inputs
- Generate inputs randomly from a probability distribution over the domain
- Perform a deterministic computation on the inputs
- Aggregate the results

Uses of Monte Carlo methods require large amounts of random numbers, and it was their use that spurred the development of pseudorandom number generators, which were far quicker to use than the tables of random numbers that had been previously used for statistical sampling.

For example, the height of a person is a random number so it can be anything really which is how randomness works. Thus, when you sample a population, by randomly picking up elements of that population and measuring their height to approximate the average height, since each person from your sample is likely to have a different height then each measure is a random number. Mathematically, a sum of random numbers, is another random number. So, the result of the sum is a random number, and the numbers making up the sum are random numbers too.

The height of a population is a random variable because height among people making up this population varies randomly. We denote random variables with upper case letters such as the letter X and the elements making up that population are denoted with a lower capital letter, generally x as well. For example, if we write x_2, this would denote the height of the second person in the population. All these x's can also be seen as the possible outcomes of the random variable X. If we call the population height as the random variable X then we can express the concept of approximating the average adult population height from a sample with the following pseudo–mathematical formula [36]:

$$\overline{x} = \frac{1}{N} \sum_{n=1}^{N} x_n. \tag{2.4.1}$$

Which you can read as, the approximation of the average value of the random variable X, is equal to the sum of the height of N adults randomly chosen from that population i.e., the samples, divided by the sample size N. This is called a Monte Carlo approximation. It consists of approximating some property of a very large number of random variables, by averaging the value of that property for N of these random variables chosen randomly among all the others. You can also say that Monte Carlo approximation, is in fact a method for approximation using samples. As mentioned before the height of the adult population of a given country can be seen as a random variable X. However, note that its average value, which you get by averaging all the height for example of each person making up the population, where each one of these numbers is also a random number, is unique. In order to avoid the confusion with the sample size, which is usually denoted with the letter N, we will use M to denote the size of the entire population:

$$\text{Average}(X) = \frac{1}{M}\left(x_1 + x_2 + x_3 + \cdots + x_M\right). \tag{2.4.2}$$

where the x_1, x_2,...x_m corresponds to the height of each person making up the entire population as suggested before.

2.5 VOLATILITY FORECASTING USING SUPPORT VECTOR MACHINES

Support Vector Machine (SVM) is a supervised machine learning algorithm that can be used for both classification and regression tasks. However, it is mostly used in classification problems. R. Rosillo et al [37] analysed the influence of VIX using SVM to forecast the weekly change in the S&P 500 Index. Their study showed that SVMs with VIX produced better results when a down move in the market is expected to happen. That is because the volatility jumps up in a bearish market movement. SVMs managed to produce decent results and allowed for reduction in the Maximum Drawdown.

In the SVM algorithm, we plot each data item as a point in n–dimensional space (where n is a number of features you have) with the value of each feature being the value of a particular coordinate. Then, we perform classification by finding the hyper–plane that differentiates the two classes very well. This boundary is a hyperplane that splits space into two parameter's half spaces.

Finding such a hyperplane, see Figure 2.13, is an optimisation problem that involves finding a function that has the most ε deviation from the obtained targets y_i for all the training data, and at the same time is as flat as possible [39]. The SVM classifier is a frontier that best segregates the two classes in the hyper–plane.

If we take a linear function:

$$f(x) = \langle w, x\rangle + b, \quad \text{with } w \in X,\ b \in \mathbb{R}. \tag{2.5.1}$$

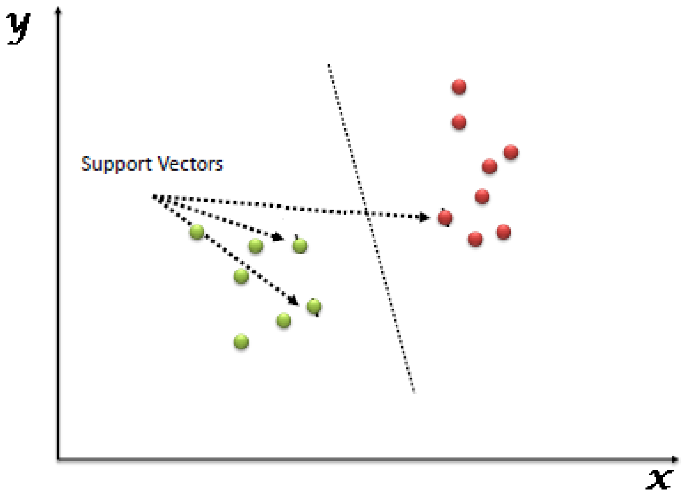

Figure 2.5: Source: [38] Hyperplane with support vectors.

where ⟨ , ⟩ is the dot product in X. Flatness in the case of f(x) means that we seek a small w. One way to ensure this is the case is to minimise the norm, i.e. $//w//^2 = \{w,\ w\}$. In other words, we are looking for as small 'w' as possible and we can convert this into an optimisation problem:

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2}\|w\|^2 \\ \text{s.t.} \quad & y_i - \langle w, x_i \rangle - b \leq \varepsilon, \\ & \langle w, x_i \rangle + b - y_i \leq \varepsilon, \qquad i = 1, \ldots, M. \end{aligned} \tag{2.5.2}$$

The convex optimisation problem above implies that there is a function f that approaches all pairs of $x_i,\ y_i$ under the accepted constraint ε.

Given a training set of instance–label pairs (x_i, y_i) $i = 1, \ldots, n$ where $x \in R^n$ and $y \in \{1, -1\}^T$, the SVM require the solution of the following convex problem:

$$
\begin{aligned}
\min_{w,b,\xi} \quad & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \\
\text{s.t.} \quad & y_i\Big(\langle w, x_i\rangle + b\Big) \geq 1 - \xi_i, \qquad i = 1, \ldots, n, \\
& \xi_i \geq 0, \qquad i = 1, \ldots, n.
\end{aligned} \tag{2.5.3}
$$

where ξ_i are the slack variables introduced by the method which measures the degree of misclassification of the data X_i. If X_i is correctly classified, then $\xi_i = 0$, otherwise $\xi_i >0$. The larger the ξ_i is, the more the SVM is penalized for misclassifying X_i. w is the normal vector orthogonal to the hyperplane; b is the offset of the hyperplane from the origin along the normal vector w and $C > 0$ is the penalty/cost parameter of the error term. The parameter C controls the trade–off between minimizing the error term and maximizing the margin. A larger value of C means that the SVM will be more penalized for misclassifications, and a smaller value of C means that the SVM will be more tolerant of misclassifications. Different values of parameter C are tested to achieve the best results in the forecasting problem. This formulation is for the soft margin SVM. The cost function includes a term that penalizes misclassifications, while still trying to find a hyperplane that separates the two classes as well as possible.

The soft margin SVM is a more flexible model than the hard margin SVM which requires all the data points to be correctly classified. The soft margin SVM is more robust to noise and outliers, can tolerate some misclassifications and is a more flexible model than a hard margin SVM.

As Figure 2.14 below shows, only the points outside the drawn area have impact to the cost. The points inside the drawn area are correctly classified, and the points outside are misclassified. Deviations are penalised linearly, which means the cost of misclassifying a point increases linearly as the point moves further away from the decision boundary.

Using the Lagrange multiplier to solve the convex optimisation problem above we get the regression function:

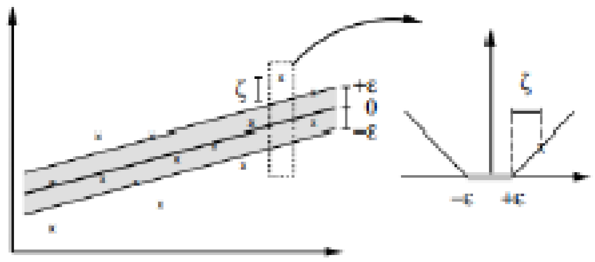

Figure 2.6: Source: [38] Decision Boundary of Soft Margin SVM.

$$y = f(x) = \sum_{i=1}^{M} (a_i - a_i^*) \ \langle x, x^i \rangle + b. \tag{2.5.4}$$

where $a_{i,}$ a^{*} >0 are the Lagrange multipliers that correspond to the support vectors.

If the dataset is not linear, SVR will not transform the data to a high–dimensional feature space, but instead it uses the kernel trick decision function, and the regression function becomes the following:

$$y = f(x) = \sum_{i=1}^{M} (a_i - a_i^*) \ k(x, x^i) + b. \tag{2.5.5}$$

where $k(x,\ x^i)$ is the kernel function and it can be any function that is a dot —- product transformed to a high dimensional space.

The linear and radial kernel functions are applied where:

- Linear means the following:

$$k(x, x^i) = x^{\top} x^i. \tag{2.5.6}$$

where $k(x,\ x^i)$ is the kernel function and it can be any function that is a dot —- product transformed to a high dimensional space.

- Radial Basis Function (RBF) means the following:

Radial basis function (RBF) kernel in SVM is a function used within Support Vector Machines (and other kernel methods) to measure similarity between two data points x and x'.

$$k(x, x^i) = \exp\left(-\frac{\|x - x^i\|^2}{2\sigma^2}\right). \tag{2.5.7}$$

where σ is the hyper–parameter which corresponds to the width of the Gaussian kernel centred at x^i. This kernel is used by the SVM algorithm to handle nonlinear classification or regression boundaries without explicitly mapping the data into a higher–dimensional feature space (i.e., the kernel trick).

SVR often enjoys better performance than simple MLP Neural Networks. The reasons for this being:

- Fewer hyperparameters. In general, SVM requires less grid searching to obtain a reasonably accurate model. SVMs with RBF kernels generally perform well. Unlike gradient problems with Neural Networks and local optima problem, SVM guarantees global optimum.
- Overall, SVMs are good because the training algorithm is efficient and it has a regularisation parameter, which forces you to think about regularisation and overfitting.
- In a classic Neural Network, the number of parameters is enormously high. For example, for vectors of the length 100 that must be classified into two classes, one hidden layer of the same size as an input layer will lead to more than 100,000 free parameters. One can imagine how badly the model can overfit and think how many training points the model will need to prevent that and how much time it will take to train it then.
- Usually, one must be a real expert to choose the topology at a glance. That means that to get good results, data scientist should perform lots of experiments. That is why it's easier to use SVM and far harder to

achieve similar results with NN.

- Usually, NN results can vary unless random seeds and hyperparameters are controlled.

An RBF network is a type of artificial neural network commonly used for function–approximation tasks. It typically consists of three layers: an input layer, a hidden layer of radial basis function units, and an output layer [40].

In the hidden layer, a Gaussian function is often used as the transfer (activation) function. This allows the network to act as a universal approximator with a relatively fast learning rate. RBF model training is considered complete when it meets a specified error threshold (e.g., 0.01) or when a fixed number of training iterations (e.g., 500) has passed. For instance, one might choose an RBF network with 10 nodes in its hidden layer.

Because of the Gaussian activation, the RBF network can converge more quickly than some Multi–Layer Perceptrons (MLPs). Once training is finished, both MLPs and RBF networks can then be tested under new conditions. The agreement between model predictions and observed data is compared, and the final model is selected based on whichever yields the least error.

2.6 VOLATILITY FORECASTING USING RANDOM FOREST REGRESSOR

A random forest regressor is a meta–estimator that fits several regression trees on sub–samples of the dataset and uses averaging to improve predictive accuracy and control over–fitting. The sub–sample size is controlled with the *max_samples* parameter if *bootstrap = True (default),* otherwise the whole dataset is used to build each tree. These sub–samples are created through a process called bootstrap sampling, where each sub–sample is generated by randomly selecting instances from the original dataset with replacement.

Random forests or random decision forests are an ensemble learning method for classification, regression and other tasks that operates by constructing a multitude of decision trees at training time. For classification tasks, the output of the random forest is the one chosen by most trees. For regression tasks, the mean or median prediction of the individual trees is returned. Random decision forests correct for decision trees' habit of overfitting to their training set. Random forests generally perform better than decision trees, but their accuracy is lower than gradient–enhanced trees. However, data characteristics can affect their performance.

Taylor [41] proposed a Quantile Regression (QR) to forecast financial time series volatility by estimating the conditional probability distribution of multi–period asset returns. D. Pradeepkumar [42] presented a novel Particle Swam Optimization (PSO) trained Quantile Regression Neural Network (PSOQRNN) to forecast volatility from financial time series. The authors compared the effectiveness of PSOQRNN with that of the traditional volatility time series forecasting models i.e. GARCH, ANNs and MLP, General Regression Neural Network (GRNN), Group Method of Data Handling (GMDH), Random Forest (RF) and two Quantile Regression (QR) —- based hybrids including Quantile Regression Neural Network (QRNN) and Quantile Regression Random Forest (QRRF). Their results indicate that the proposed PSOQRNN outperformed the other models in terms of Mean Squared Error (MSE) in most of the cases.

Random Forest Regression is a supervised learning algorithm that uses ensemble learning method for regression. Ensemble learning method is a technique that combines predictions from multiple machine learning algorithms to make a more accurate prediction than a single model - see Figure 2.15 below. Random Forest combines multiple decision trees into one model.

N_estimators is the parameter that controls the number of trees in the forest. Increasing the number of trees typically improves accuracy but at the cost of increased computation time. Since Random Forest is an ensemble method comprising of creating multiple decision trees, this parameter is used to control the number of trees to be used in the process.

In this thesis, the number of trees used is n_estimator. As we know,

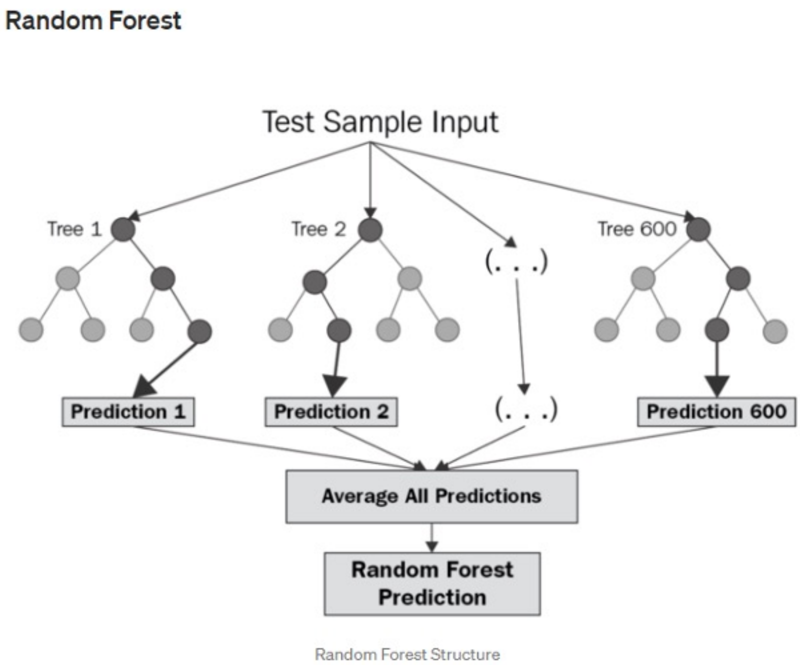

Figure 2.7: Source: [43] RF.

n_estimator is one of the key parameters in Random Forest Regressor. The input taken here is Volatility column of the dataset.

2.7 FORECASTING FINANCIAL TIME SERIES VOLATILITY USING ARIMA FRAMEWORK

A. Banerjee [44] used ARIMA (AutoRegressive Integrated Moving Average) ARIMA model to forecast VVIX. ARIMA is a powerful statistical method used to forecast time series data. It models the dependency between an observation and a number of lagged observations (autoregressive component, p), the differences required to make the series stationary (integrated component, d), and the dependency between an observation and a lagged forecast error (moving average component, q). Based on literature from Park [45], they considered the VVIX as an indicator of 'fat tail' risk i.e., the risk of extreme events. They found ARIMA (1,0, 4) was the fittest model to forecast future VVIX values.

This means the model includes one autoregressive term (p=1), no differencing (d=0), and four moving average terms (q=4). The moving average component (q=4) plays a crucial role in adjusting predictions based on recent forecast errors. In financial time series like VVIX, large deviations from predictions can occur due to unexpected market news or geopolitical events. By using the last four errors, the model smooths out these fluctuations, creating a more stable forecast that aligns with observed behaviour. Furthermore, the absence of differencing (d=0) indicates that VVIX data is inherently stationary, fluctuating around a mean level without long–term trends or seasonality, which is typical for volatility indices.

The choice of ARIMA (1, 0, 4) reflects an optimisation process that involved minimising information criteria AIC and BIC to achieve the best balance between model fit and complexity. Higher–order models with additional parameters might overfit the data, while simpler configurations could fail to capture VVIX's nuanced short–term dynamics. Banerjee's model thus provides an effective yet computationally manageable approach to forecasting.

In validating the model, the study assessed its reliability using residual diagnostics and metrics such as Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). These measures confirmed the model's ability to accurately predict VVIX values while maintaining the statistical assumptions of white noise residuals. By forecasting VVIX, the ARIMA (1, 0, 4) model enables financial market participants to anticipate extreme volatility periods, supporting more informed risk management strategies. While ARIMA provides a robust baseline for forecasting, it also highlights potential for hybrid approaches to account for nonlinearities in financial time series data.

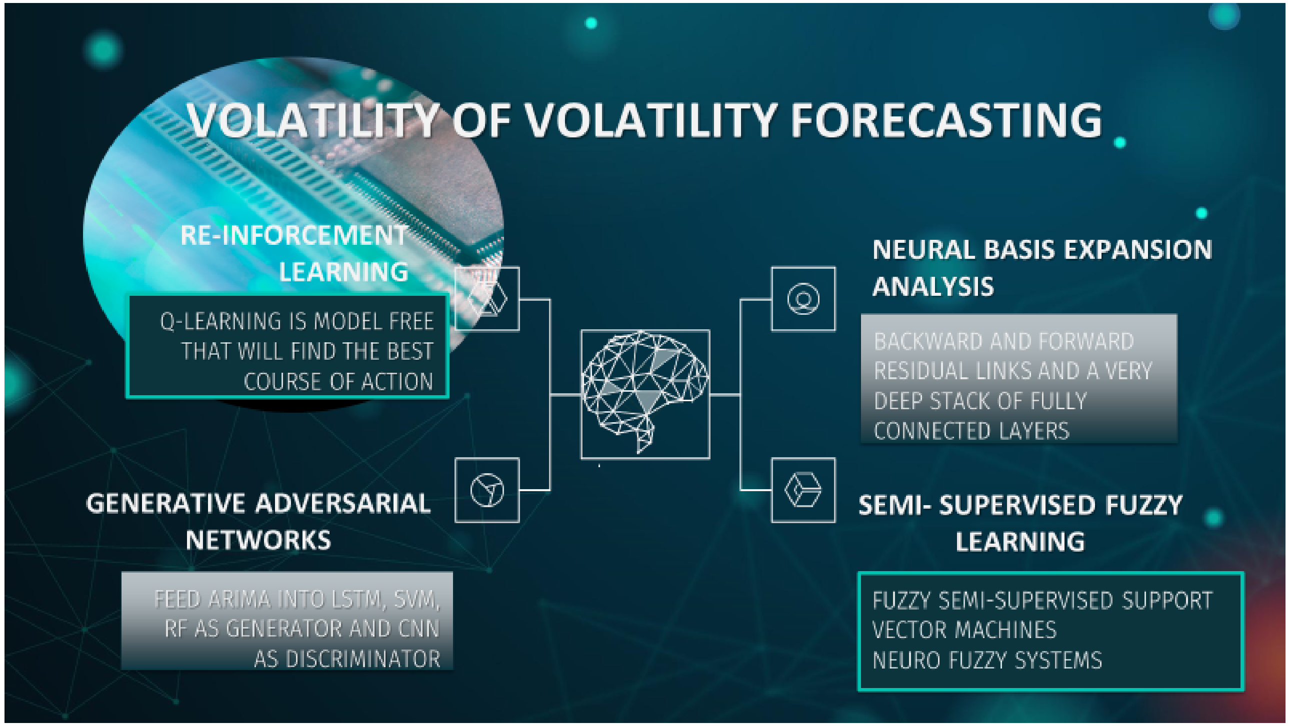

Figure 2.8: ADVANCED MACHINE LEARNING MODELS

2.8 *NEURAL BASIS EXPANSION (NBEATS)*

Oreshkin et al., proposed a novel architecture for univariate time series forecasting. They applied it to a variety of time series forecasting problems using non–overlapping competition datasets: M4, M3 and Tourism. The results significantly outperformed traditional econometric forecasting techniques [46].

N–BEATS is a novel deep learning architecture designed for interpretable univariate time series forecasting. The focus is on solving the univariate times series point forecasting problem using deep learning architecture based on backward and forward residual links and a very deep stack of fully connected layers. Although N–BEATS was initially designed for univariate forecasting, extensions have been used for multivariate time series forecasting by combining outputs from multiple univariate models. The backcast link takes the historical data as input and produces a representation of the past. In other words, it removes the portion of the historical data that the block can approximate, leaving the residual error for subsequent blocks to model. The forecast link takes the representation of the past and produces a partial forecast for future time steps. The architecture has several desirable properties, being interpretable, applicable without modification to a wide array of target domains, and fast to train.

The model consists of a sequence of stacks, each of which combine multiple blocks. The blocks connect feedforward networks via forecast and backcast links. A block "removes the portion of the signal ... it can approximate well" [46]. Then the block sets its focus on the residual error, which the preceding blocks could not disentangle. This residual error becomes the input for the subsequent blocks. Each block generates a partial forecast, with its focus set on the local characteristics of the time series. The stack aggregates partial predictions on the blocks it consists of, and then passes the results on to the next stack. The purpose of the stack is to identify non–local patterns along the complete time axis by "looking back." Each stack in N–BEATS aggregates partial forecasts from its constituent blocks. This hierarchical arrangement allows the model to progressively capture both local characteristics (within blocks) and global patterns (across stacks) along the time axis. Finally, the partial forecasts are aggregated into an overall forecast at the model level.

N–BEATS takes as its hyperparameters:

i. The size of the input and output layers (constants INLEN and N_FC) must be sufficient to assign a node for each feature in the source data. Each feature should have a corresponding node in the input and output layers. The length of the input segment should not be less than the order of seasonality, otherwise the learning process will have more difficulty combining the segments. For efficient memory usage, set them to a power of 2. For univariate time series, the size of the input layer corresponds to the length of the historical window, while the output layer represents the number of future time steps to forecast.

ii. The number of blocks in a stack (BLOCKS).

- N–beats consists of a sequence of stacks, each containing multiple blocks.
- The number of blocks in a stack (BLOCKS) is a hyperparameter that determines the depth and complexity of the model.

iii. The width of each fully connected layer in each block of a stack: its node count (LWIDTH).

- The width of each fully connected layer in each block(LWIDTH) refers to the node count of these layers.
- The width determines the capacity of the network to learn and represent complex patterns in the time series data.

The batch size determines the number of cases the model will process before updating its matrix weights. To effectively align it with the memory structure of our system, we set it to a power of 2. Very large batch sizes can skew gradients down only in one direction, and the model can get stuck at a sub–optimal minimum. Smaller batch sizes will cause the gradient descent to bounce around in different directions and can result in lower accuracy, but they also tend to prevent the model from overfitting. The most frequent recommendation is to choose an initial batch size of 32. Since our dataset has a frequency of 24 hours per day, we set the batch size to the following binary limit that can handle 24–time steps: 32. The epochs tell the model the number of cycles the exercise has to perform in each epoch, the model processes the entire training set, performing one forward pass and one back pass. Allowing for some oversimplification, the product of these hyperparameters defines the tensor size of the model. Large parameter values can cause it to reach the system's memory limit and cause exponentially longer processing times. While small parameter values may not be sufficient to reflect complex patterns in the source data. Probabilistic forecasting can be achieved in N–BEATS by incorporating quantile regression. This technique allows the model to estimate not only a central forecast but also uncertainty intervals, such as 10%/90% quantiles. This is an option we can exercise in all deep forecasting models. The loss function of a neural network can be constructed using *quantile loss function.* Then quantile regression will not only compute a central forecast value at each time step, a point estimate, but will draw uncertainty bands about it. Pairs of quantiles like 1%/99% or 10%/90% express the range over which the forecast value can vary, above or below the central value.

2.9 ***REINFORCEMENT LEARNING (RL)***

Reinforcement learning (RL) is a framework where an agent interacts with its environment by taking actions, transitioning between states, and receiving rewards. The agent aims to maximize cumulative rewards over time by learning an optimal policy, often modelled as a Markov Decision Process (MDP). A reinforcement learning agent can perceive and interpret its environment, take actions, and learn through trial and error. Reinforcement learning is one of the three fundamental models of machine learning, along with supervised and unsupervised learning. Model–free RL algorithms, such as Q–learning, do not require explicit knowledge of the environment's dynamics (transition probabilities and reward functions). This makes them suitable for financial markets, where the environment is often unknown and non–stationary. Obviously, if we knew the model of financial markets, we would just follow that model. This is what makes forecasting financial data so difficult. We plan to implement Q–learning in our approach. Q–learning updates the Q–value for each state–action pair using the Bellman equation, which incorporates the immediate reward and the estimated value of the next state. This iterative process enables the agent to learn an optimal policy by balancing exploration and exploitation. In Q–learning we will learn the value of taking an action from a given state. Q–value is the expected return after taking such action. We plan to utilise 'Rainbow,' an advanced deep reinforcement learning algorithm that combines several enhancements to the Deep Q–Network (DQN), including prioritised experience replay, double Q–learning, duelling network architectures, and distributional Q–learning. Reinforcement learning differs from supervised learning in that it does not need to present labelled input/output pairs and does not need to be explicitly corrected for suboptimal actions. Instead, the focus is on finding a balance between exploration of uncharted territory and exploitation of current knowledge [47]. Partially supervised RL algorithms can combine the advantages of supervised and RL algorithms [48].

An environment in reinforcement learning is often modelled as a Markov Decision Process (MDP) because many reinforcement learning algorithms in this context use dynamic programming techniques. It is defined by states, actions, transition probabilities, and reward functions. The Markov property ensures that the future state depends only on the current state and action, simplifying the decision–making process. MDPs form the theoretical foundation of many RL algorithms, enabling the agent to learn policies that maximise cumulative rewards. The main difference between classical dynamic programming methods and reinforcement learning algorithms is that the latter do not assume knowledge of an exact mathematical model of the MDP, and they target large MDPs where exact methods become infeasible.

A Markov chain is a stochastic model, whereby the probabilities connecting a sequence of partially random events is examined [49]. Markov series describe relationships between events in a sequence, which is necessary for models used in financial analysis where underlying trends in data must be observed.

In the reinforcement model, the agent is the Markovian decision maker and is responsible for trying to make the optimal decision based on his understanding of the data. This Agent's objective becomes a continuous process, where the Agent acts upon its environment then evaluates the outcome of their action, then takes this outcome into consideration in their next action. See Figure 2.17 below.

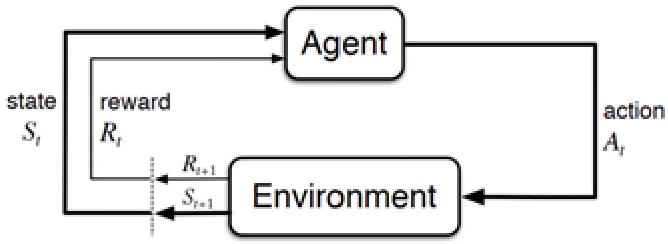

Figure 2.9: Source: [50] *Reinforcement Learning.*

M. Taghian [51] used a novel Deep Reinforcement Learning (DRL) method to extract asset–specific trading rules from vast amounts of historical data to increase the return and reduce risk of their portfolio. DRL is implemented to learn the new trading rules for each asset. The authors proposed a novel DRL model with various feature extraction techniques. The proposed model outperformed the other state–of–the–art models in learning single asset–specific trading rules. W. Bao et al. [52] combined transforming the data using wavelet transforms (WT), stacked autoencoders (SAEs) and long–short term memory (LSTM) for stock market forecasting. First, the time series were decomposed by WT to eliminate the noise. Then, SAEs were applied to generate a deep high–level feature for predicting the stock price and lastly, the high–level de–noised features are fed into LSTM to forecast the next day's closing price. Results showed that the proposed model outperformed other similar models in both predicting accuracy and profitability performance. M. Taghian et al. [53] employed a novel encoder–decoder framework combined with DRL to learn single instrument trading strategies from a long sequence of raw data of the instrument. The model consisted of an encoder which is a neural structure responsible for learning informative features from the input sequence and a decoder which is DRL model responsible for learning profitable strategies based on the features extracted by the encoder. The parameters of the encoder and the decoder structures were learned jointly which enabled the encoder to extract features specifically tailored to the task of the decoder DRL. Jae Won Lee [54] presented a method of calculating state values different from conventional indicators in that they represent directly the expected cumulative reward, the price rate of change, and not the auxiliary information for making buying and selling decisions. While stock price prediction is not strictly Markovian due to its dependence on external and long–term factors, it can be approximated as a Markov process to enable reinforcement learning models such as TD(0) and TD(λ) to operate effectively. TD(λ) generalises TD(0) by introducing a parameter λ that weights immediate and future rewards, enabling the model to better handle delayed returns——a critical feature for financial forecasting. We plan to extend this method to other algorithms such as TD(λ) where our definition of reward will be different from Lee's paper. Nuno Horta et al [55] described a new system for short–term speculation in the foreign exchange market based on RL. Neural Networks with three hidden layers of ReLU neurons are trained as RL agents under Q–Learning algorithm by an algorithm that consistently induces stable learning that generalizes to out–of–sample data. This framework includes new state and reward signals, and a method for more efficient use of available historical tick data that provides training quality and testing accuracy. Financial markets present unique challenges for reinforcement learning compared to deterministic environments like Atari or Go. These challenges include non–stationary data, where underlying market dynamics change over time; partial observability, where not all relevant variables are available to the agent; and high levels of noise in the signals, which can obscure meaningful patterns. It was shown by the authors that learning in the training dataset is stable and it is apparent from several validation learning curves that the Q–network is capable of finding relationships in financial data that translate to out–of–sample decision making.

2.10 *A GENERATIVE ADVERSARIAL NETWORK (GAN)*

A generative adversarial network (GAN) is a class of machine learning frameworks designed by Ian Goodfellow and his colleagues in June 2014 [56]. Two neural networks compete in a zero–sum game, where one agent's gain is another agent's loss.

With a training set, this technique learns to generate new data with the same statistics as the training set. For example, an image–trained GAN can produce new images that appear realistic at least to a human observer, with many realistic features. Though originally proposed as a form of generative model for unsupervised learning, GANs have also proved useful for semi–supervised learning, fully supervised learning, and reinforcement learning.

The core idea of a GAN is based on the "indirect" training through the discriminator, another neural network that can tell how much an input is "realistic", which itself is also being updated dynamically. This basically means that the generator is not trained to minimize the distance to a particular frame, but rather to fool the discriminator. This enables the model to learn in an unsupervised manner. The generator indirectly minimizes a divergence measure between the real and generated data distributions, such as Jensen–Shannon divergence or Wasserstein distance.

GANs are like mimicry in evolutionary biology, with an evolutionary arms race between both networks. To learn the generator's distribution p_g over data x, we define a prior on input noise variables $p_z(z)$ then represent a mapping to data space as $G(z; \Theta_g)$, where G is a differentiable function represented by a multiplayer perceptron with parameters Θ_g. We also define a second multilayer perceptron D $(x; \Theta_d)$ that outputs a single scalar. D(x) represents probability that x came from the data rather than p_g. We train Discriminator (D) to maximise the probability of assigning the correct label to both training examples and samples from G. Simultaneously we train G to minimise log(1 —- D(G(z))):

So, in other words, D and G play the following two–player minimax game with value function V(G, D) [56]:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\Big[\log D(x)\Big] + \mathbb{E}_{z\sim p_z(z)}\Big[\log\Big(1 - D(G(z))\Big)\Big]. \tag{2.10.1}$$

Kang Zhang et al [57] proposed a novel architecture of GAN with Multi–Layer Perceptron (MLP) as the discriminator and LSTM as the generator for forecasting the closing stock prices. The generator is built by LSTMN to mine the data distribution of stocks from given data and generate data in the same distributions, whereas the discriminator designed MLP to discriminate between real and fake data. Their results showed that the novel GAN structure gave promising results compared with other models in machine learning and deep learning. HungChun Lin et al [58] proposed stock price prediction model using GAN with Gated Recurrent Units (GRU) used as a generator that inputs historical stock price and generates future stock price and Convolutional Neural Network (CNN) as a discriminator to discriminate between real and fake distributions. Different to one step ahead forecasts only, the authors used deep learning to make multi–step ahead predictions more accurately. The authors also found that Wasserstein GAN performed better during unexpected events like COVID whereas conventional GAN performed better during normal times. They also showed that RNN–based GANs can become unstable due to the sequential nature of RNNs, which complicates gradient updates during adversarial training and requires careful tuning of hyperparameters. The key is tune parameters in each layer to make the whole model more accurate. Rainbow, an RL–based framework, has been explored for hyperparameter optimisation in GANs, though its practical implementation in adversarial networks remains experimental. Martin Erdman et al [59] used adversarial network with the Wasserstein distance to generate simulated detector data. The authors investigated two variants of GANs for detector simulations. In both cases, the transfer of probability distributions from one data set to another using GAN training worked well using the Wasserstein distance in the loss function. WGAN replaces the Jensen–Shannon divergence with the Wasserstein distance, providing a more stable loss function that alleviates vanishing gradient issues and improves training stability. Instead of training a deep network with simulations that differ in detail from data, simulations can be adapted to match data prior to network training. With this method, authors showed that results are better compared to training with the originally simulated traces.

2.11 *FUZZY NEURAL NETWORK (FNNS)*

"Fuzzy set is a class of objects with a continuum of grades of membership. Such a set is characterized by a membership (characteristic) function which assigns to each object a grade of membership ranging from zero to one. Concepts of inclusion, union, intersection, complement, relation, convexity, etc., are extended to such sets, and various properties of these notions in the context of fuzzy sets are established" [60]. Fuzzy neural networks (FNNs) for pattern classification often use the backpropagation or C–cluster type learning algorithms to learn the parameters of the fuzzy rules and membership functions from the training data. However, such kinds of learning algorithms usually cannot minimize training and testing errors simultaneously and thus cannot reach a good classification performance in the testing phase. To overcome this drawback, a support–vector–based fuzzy neural network classification (SVFNNC) is proposed. The SVFNNC leverages the robust classification capabilities of SVMs and the ability of FNNs to model uncertainty through fuzzy logic. The learning algorithm consists of two learning steps. In step 1, the fuzzy rules and membership functions are generated using clustering principles, enabling the system to model data uncertainty. In step 2, an adaptive fuzzy kernel function enhances SVM performance by fine–tuning FNN parameters for classification tasks. Deng et al [61] developed and tested a system combining deep Neural Network with Direct Reinforcement learning using fuzzy learning to summarise the market trend. The system was able to select features from raw data and learn effectively using DL mechanism. Fuzzy logic is particularly effective in financial markets due to its ability to handle noisy and uncertain data, capturing subtle patterns that may not be apparent with traditional machine learning techniques.

2.12 CONCLUSION

Despite decades of research and renewed attention following major market dislocations (e.g., the 2008 global financial crisis and the COVID-19 shock), the volatility forecasting literature remains characterised by *plurality rather than consensus*: performance is strongly horizon- and regime-dependent, and simple benchmarks can remain surprisingly competitive relative to more elaborate specifications. This motivates the thesis to move beyond purely conditional-variance or purely data-driven approaches by explicitly targeting the *fractal/long-memory structure* of volatility and integrating an *uncertainty-aware* representation via fuzzy modelling within a fractional Brownian framework.

This chapter has reviewed a broad spectrum of volatility and volatility–of–volatility modelling approaches, ranging from classical econometric specifications to multifractal processes and modern machine learning. Across these strands, a consistent empirical message emerges: volatility is not well represented by memoryless Gaussian diffusion. Instead, volatility dynamics exhibit *clustering*, *heavy tails*, *regime dependence*, and *scale effects* that are naturally described through long–memory and (multi)fractal viewpoints. In this context, models such as mixed/fractional Brownian frameworks and multifractal constructions (e.g., MMAR/MSM) are motivated by their ability to encode persistence (or anti–persistence), time deformation (trading time), and heterogeneous activity across scales. By contrast, standard GARCH–type models capture conditional heteroskedasticity effectively at short horizons but struggle to simultaneously represent abrupt shifts and prolonged volatility cycles, while purely data–driven machine learning models often achieve strong predictive performance yet may under–represent long–range dependence and scale invariance unless these properties are explicitly engineered into the feature set and experimental design.

A second theme is that *volatility of volatility* is central to modern volatility markets, but remains comparatively under–developed in the forecasting literature. While VIX has been extensively studied, the evidence base for VVIX forecasting is thinner and less systematic, particularly regarding approaches that integrate fractal/multifractal structure with state–of–the–art deep learning, reinforcement learning, or generative modelling. Moreover, even when advanced models are proposed, there is limited consensus on how to (i) robustly handle structural breaks and non–stationarity, (ii) ensure out–of–sample stability under changing market regimes, and (iii) quantify uncertainty in a way that is interpretable and decision–relevant for risk management.

These gaps motivate the direction of the thesis. First, rather than treating VVIX as a secondary quantity, the thesis elevates VVIX as the *primary* object for understanding and forecasting the dynamics that drive implied volatility. This is consistent with the chapter's core framing that volatility and its first derivative have fractal signatures, and that a scale–aware methodology is therefore required. Second, the thesis builds on the multifractal and fractional literature by developing a modelling pipeline that explicitly targets *time–scale invariance*: fractal measures (e.g., Hurst exponent and related roughness diagnostics) are used not only as descriptive statistics but as structural inputs to model selection and forecasting. Third, the thesis introduces an uncertainty–aware extension through fuzzy modelling, allowing parameters and latent state descriptions to represent imprecision in a controlled mathematical form. In particular, the broader thesis framework aligns with recent developments that unify fractional drivers, fuzziness, and jump risk within a coherent stochastic setting (including risk–neutral considerations), thereby providing a principled bridge between fractal dynamics and uncertainty quantification. This is especially relevant because the literature indicates that fully formal treatments of fuzziness in the rough (non–semimartingale) regime remain limited, and careful construction is required to preserve mathematical consistency of inference and simulation across regimes. Finally, the thesis responds to the commonly noted need for stronger empirical substantiation of complex fractional/fuzzy frameworks by adopting a comparative, benchmark–driven evaluation strategy against established econometric baselines and modern machine learning models, with explicit attention to robustness and practical viability.

Chapter 3 now operationalises these objectives by detailing the datasets, pre–processing, stationarity and scaling diagnostics, and the full experimental design used to evaluate classical, multifractal, deep learning, and uncertainty–aware modelling approaches for VIX and VVIX.

CHAPTER 3 – DATA AND METHODOLOGY

This chapter details the data sources, pre–processing protocols, and the complete experimental design used to conduct the empirical analysis in this thesis. A transparent and replicable methodology is essential for validating the findings and grounding the theoretical work in a robust empirical foundation. We outline the process for daily and high–frequency data acquisition and cleaning, the statistical tests for data stationarity, the specific design for training and evaluating the machine learning models, and the methodology for multifractal simulations.

3.1 *DATA ACQUISITION AND DESCRIPTION*

To investigate the research questions, this study utilises two distinct datasets sourced from Kaggle and the Bloomberg Terminal. Each dataset was subjected to a preliminary inspection to assess its size, variable types, and data quality.

3.1.1 Data Description and Quality Assessment

The primary source for long-term historical volatility data is the "VIX Daily" dataset obtained from Kaggle (Yahoo finance). This dataset provides a daily time series of the CBOE Volatility Index (VIX) [P1], [P4]. Kaggle is a platform hosting datasets and competitions related to machine learning and data science. The primary dataset was collected from the following link - https://www.kaggle.com/jonathanbesomi/cboe--volatility--index--vix.

- Dimensions and Scope: The dataset covers the period from January 2, 1990, to April 20, 2022, containing approximately 8,143 observations.
- Variables: The raw file (`VIX_daily.v1.csv`) consists of two primary columns:
  - Date: Recorded in DD/MM/YYYY format.
  - Last Price: The daily closing value of the VIX index.
- Quality Assessment:
  - Completeness: The dataset is highly structured with no obvious formatting errors in the timestamp index.
  - Stationarity: Preliminary visual inspection confirms the data exhibits mean–reverting behaviour typical of volatility indices, though it contains extreme outliers corresponding to market crises (e.g., 2008, 2020) which act as "shocks" in the time series.

To capture high-frequency dynamics and the "volatility of volatility," a second dataset was extracted directly from the Bloomberg Professional Terminal. This dataset includes both daily and 30-minute interval data for the VIX and the CBOE VIX of VIX Index (VVIX).

- Dimensions and Scope:
  - Daily Data: Covers VIX and VVIX closing prices. The VVIX daily series begins on January 3, 2007 resulting in a shorter historical range compared to the VIX. The raw VVIX daily file contains 3,997 observations.
  - Intraday Data: Includes 30–minute snapshots covering the recent period from August 2021 to April 2022. The intraday VIX dataset contains 4,702 observations, while the corresponding VVIX dataset contains 2,542 observations.
- Variables:

  - Date: Timestamp including date and time (e.g., `2022-04-20 09:15:00`).
  - Last Price: The index value at the specific interval.
  - #Ticks: The number of ticks recorded in the interval (measure of activity).
  - SMAVG (15): A 15–period Simple Moving Average included in the raw export.

- Quality Issues and Pre–processing:
  - Formatting Artifacts: The raw Bloomberg exports (`VVIX AND VIX DATA.xlsx`) contain leading empty columns (e.g., „`Date,Last Price`) and metadata headers that required removal before analysis.
  - Missing Values: Specific instances of missing data were identified. For example, the `VVIX DAILY.csv` file shows a single null value for Last Price on 2022-04-20 at the very end of the series.
  - Timestamp Alignment: The 30–minute intervals required alignment to ensure that VIX and VVIX observations corresponded to the exact same timestamps, necessitating the removal of unmatched rows during the data cleaning phase.

During the initial inspection, a small subset of data points was identified containing a recorded price or volume of zero. For the VIX and VVIX indices, a value of zero is theoretically inconsistent with the underlying pricing models (e.g., variance swaps), as it implies a complete absence of market volatility. Consequently, these values were classified as data feed anomalies or reporting errors rather than genuine market signals.

Justification for Deletion vs. Imputation- the decision was made to remove these artifacts rather than attempt imputation (such as forward-filling or mean substitution) for two primary reasons:

1. Preservation of Ground Truth: Since the VIX is the target variable for

prediction, imputing missing values would effectively train the models on synthetic data, potentially introducing bias.

2. Sample Integrity: The instances of zero or missing values were extremely rare, constituting approximately 0.025% of the total dataset. Given this negligible proportion, deletion preserves the statistical integrity of the time series without reducing statistical power.

Analysis was conducted using Python programming via Google Colaboratory.

To ensure stationarity and work with a more statistically stable series, we focused on the daily log returns of the VIX rather than the raw VIX index [P4]. Returns were calculated using the formula:

$$\text{Return}(t) = \ln\left(\frac{\text{VIX}(t)}{\text{VIX}(t-1)}\right). \tag{3.1.1}$$

Then calculate volatility of VIX by, first, calculating its standard deviation and then multiplying by square root of 252 trading days in a year. This transformation assumes that returns are independently and identically distributed (i.i.d.), even though clustering and jumps are known characteristics of volatility. The reason for this transformation and data manipulation is to convert the standard deviation of percentage changes in VIX for a sample of days into the standard deviation of percentage changes annualised. Hence, we take daily returns as log of (VIX2/VIX1), then we calculate standard deviation of this log return and multiply by the square root of 252 trading days in a year.

Theoretical Derivation based on Hull and CBOE

The VIX formula is derived from the pricing of a variance swap. As shown by Hull [62], the fair delivery price for a variance swap can be replicated by a static portfolio of out–of–the–money (OTM) calls and puts weighted by the inverse squared strike price ($1/K^2$).

The theoretical value of the annualized variance is given by the replication of the “Log Contract”:

$$E[\bar{\sigma}^2] = \frac{2}{T}\left(rT - \left(\frac{S_0}{S_*}e^{rT} - 1\right) - \ln\frac{S_0}{S_*}\right) \tag{3.1.2}$$

By discretizing this integral across available option strikes, we obtain the standard operational formula used by the CBOE:

$$\sigma^2 = \frac{2}{T}\sum_{i=1}^{N}\frac{\Delta K_i}{K_i^2}\, e^{RT}\, Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2. \tag{3.1.3}$$

[1]

VIX = σ * 100

T = Time to expiration

F = Forward index level derived from index option prices

Ko = First strike (the strike price is the price at which an option holder can exercise the option) below the forward index level, F

K_i = Strike price of i_{th} out of the money option; a call if $K_i > K_0$ and a put if $K_i < K_0$; both put and call if $K_i = K_0$ (put and call options are two types of derivatives that give the holder the right to sell or buy the underlying asset at the strike price on or before the expiration date in case of American option or on expiration date if the option is European).

$$\Delta K_i = \frac{K_{i+1} - K_{i-1}}{2}$$

R = Risk–free interest rate to expiration

Q(Ki) = The midpoint of the bid–ask spread for each option with strike Ki.

Proof:

[1] K. Demeterfi, E. Derman, M. Kamal and J. Zou, *More Than You Ever Wanted to Know About Volatility Swaps (But Less Than Can Be Said)*, Goldman Sachs Quantitative Strategies Research Notes, March 1999.

If return of the day i is r_i, r_i is independently and identically distributed across i time periods and the standard deviation of r_i, is σ, then variance of n day returns, $r_1 + r_2 + \ldots r_n$, equals expected value of $\{(r_1 + r_2 + \ldots r_n)$ —- $E[r_1 + r_2 + \ldots r_n]\}^2$. The expression can be expanded to n square terms of the form $(r_i$ —- $E[r_i])^2$ each of which has expected value of σ^2 and n(n–1)/2 cross terms of the form $(r_i$ —- $E[r_i])($ $(r_j$ —- $E[r_j])$, each of which has expected value of 0 because of independence assumption. So variance of $r_1 + r_2 + \ldots r_n$ equals $n\sigma^2$ and standard deviation equals $\sigma\sqrt{n}$. With n=252 trading days, the standard deviation equals to $\sigma\sqrt{252}$.

This transformation helps to normalise the data and is a standard procedure in the financial time–series analysis. All models, unless otherwise specified, were trained on this transformed series.

3.2 *STATIONARITY TEST OF THE DATA*

As detailed in the literature review, stationarity is essential for time series analysis. We formally tested for the presence of a unit root in the daily log–return series of the VIX using the Augmented Dickey–Fuller (ADF) test. Suppose a random variable S(t) is stationary. This behaviour can be more formally described by the following stochastic differential equation (SDE).

$$\mathrm{d}S_t = \theta\Big(\mu - S_t\Big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{3.2.1}$$

where, W(t) is a Wiener process (Brownian motion) at time t, the rate of reversion θ to mean, the equilibrium of mean value of the process μ and its volatility σ. According to this SDE, the variation of price at *t+1* is proportional to the difference between the price at time t and the mean. So, the price change is more likely to be positive or negative if the price is smaller or greater than the mean.

3.2.1 *HYPOTHESIS TEST AND STATIONARITY TEST*

A set of data is stated to be stationary if the average value, variance, and autocorrelation of the series are constant. In other words, the statistical properties do not change over time. The stationarity test is conducted to ensure that the return data have been stationary. The stationarity test of return data is conducted by using ADF (Augmented Dickey Fuller) Test. ADF test aims to find out whether the return data still contain the unit root or not. A unit root is a value that causes the series to trend over time. If the return data does not contain the unit root, it means that the data is stationary. Then it can be used in the further estimation processes. However, if the ADF test results show that the return data still contains the unit root (not stationary) then the differencing process of the data should be conducted until its condition becomes stationary [25]. Differencing is a process of subtracting the previous value from the current value. This process removes the trend from the series and makes it stationary.

The ADF test is conducted using the hypothesis test as follows:

Ho : $\gamma = 1$*, there are unit roots and non–stationary data*

H1: stationary data

If the ADF test statistic is smaller than the critical value or possessing the probability which is smaller than 5%, this means that the data does not contain the unit roots anymore, so it is stationary. If the data is not stationary yet, then it is necessary to conduct the differencing process of the data until the ADF test value is smaller than its critical value.

3.2.2 *STATIONARITY CONCEPTS*

As cited before, modelling financial time series is a complex task, due to some properties of financial data. Stock prices have the following stylised facts [63]:

i. The price series are usually non–stationary and close to a random walk.

A random walk assumes that price changes are random, and they wander unpredictably.

ii. This model makes impossible to predict future patterns. Therefore, stationarity is a crucial element in time series analysis.

There are two different definitions of stationarity. Firstly, "a process (Xt) is said to be strictly stationary if the vectors $(X1, . . . , Xk)$' and $(X1+h, . . . , Xk+h)$' have the same joint distribution, for any $k \in \mathbb{N}$ and any $h \in \mathbb{Z}$" [63].

In other words, a process is strictly stationary if its unconditional joint probability distribution does not change when shifted in time, and so do parameters, such as mean and variance. Another weaker definition of stationarity is known as second–order stationarity, and it requires only the first moment and the autocovariance not to vary with the time. The second moment, or the variance, should only be finite.

In statistics and econometrics, an augmented Dickey—-Fuller test (ADF) tests the null hypothesis that a unit root is present in a time series sample. The alternative hypothesis is different depending on which version of the test is used but is usually stationarity or trend–stationarity. It is an augmented version of the Dickey—-Fuller test for a larger and more complicated set of time series models that is normally used.

The augmented Dickey—-Fuller (ADF) statistic [64] used in the test, is a negative number. The more negative it is, the stronger the rejection of the hypothesis that there is a unit root at some level of confidence.

Testing procedure:

The testing procedure for the ADF test is the same as for the Dickey—-Fuller test but it is applied to the following model:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \delta_1 \Delta Y_{t-1} + \cdots + \delta_{p-1} \Delta Y_{t-p+1} + \varepsilon_t. \qquad (3.2.2)$$

where α is a constant, β is the coefficient on a time trend and ρ is the lag order of the autoregressive process. Imposing the constraints $\alpha = 0$ and $\beta = 0$ corresponds to modelling a random walk and using the constraint $\beta = 0$ corresponds to modelling a random walk with a drift. Consequently, there are three main versions of the test, analogous to the ones discussed on Dickey—-Fuller test.

By including lags of the order p, the ADF formulation allows for higher–order autoregressive processes. This means that the lag length p must be determined when applying the test. One possible approach is to test down from high orders and examine the t–values on coefficients. An alternative approach is to examine information criteria such as the Akaike information criterion, Bayesian information criterion or the Hannan—-Quinn information criterion.

3.2.2.1 APPLICATION AND RESULTS

The ADF test was performed using the adfuller() function from Python statsmodels library. Appendix A1 provides the full code and outputs for reference. The Augmented Dickey–Fuller (ADF) test yielded a test statistic of -6.71, which is significantly lower than the 1% critical value of -3.43. Furthermore, the p-value of 3.66×10^{-9} provides overwhelming evidence to reject the null hypothesis of a unit root. Consequently, the time series is confirmed to be stationary, satisfying the prerequisite conditions for the proposed forecasting models.

3.3 *EXPERIMENTAL DESIGN FOR MACHINE LEARNING MODELS*

3.3.1 *LSTM ARCHITECTURE*

We give a brief overview of LSTM architecture here. This section is related to section 3.5.3 where we present results from various deep neural network models. LSTM is a particular type of deep neural network developed by Hochreiter and Schmidhuber [65]. By inheriting the benefits from the recurrent neural network (RNN), LSTM has shown great power to model sequential data, like time series. Compared to vanilla RNN, LSTM uses gates to control information flows through the sequence so that it can be capable of learning long–term dependencies and solving the vanishing gradient problem.

At a high–level LSTM is like an RNN cell. Here is the internal functioning of the LSTM network. The LSTM consists of three parts, as shown in the Figure below and each part performs an individual function [66].

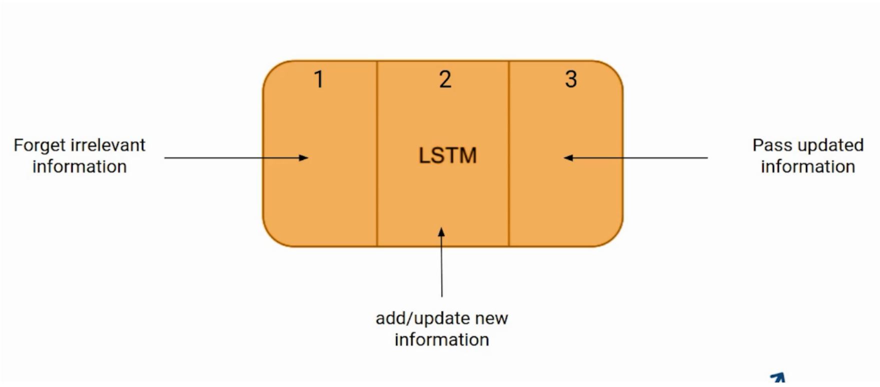

Figure 3.1: Source: [66] LSTM cell.

The first part chooses whether the information coming from the previous timestamp is to be remembered or is irrelevant and can be forgotten. In the second part, the cell tries to learn new information from the input to this cell. In the third part, the cell passes the updated information from the current timestamp to the next timestamp [67] .

These three parts of an LSTM cell are known as gates. The first part is called Forget gate, the second part is known as the Input gate and the last one is the Output gate.

The Figure below shows the typical architecture of LSTM. Like vanilla RNN, inputs are imported into the model at each time step, and outputs can be given at each time step. LSTM uses memory block to take place of neuron in RNN. A memory block is composed of a memory cell, an input gate, a forget gate, and an output gate. We can use vector formulas to describe LSTM, as follows [23]:

- **Input gate:**
$$i_t = \sigma(W_{ix}x_t + W_{ih}h_{t-1} + b_i)\,. \tag{3.3.1}$$
- **Forget gate:**
$$f_t = \sigma(W_{fx}x_t + W_{fh}h_{t-1} + b_f)\,. \tag{3.3.2}$$
- **Output gate:**
$$o_t = \sigma(W_{ox}x_t + W_{oh}h_{t-1} + b_o)\,. \tag{3.3.3}$$
- **Candidate cell state:**
$$\widetilde{c}_t = \tanh(W_{cx}x_t + W_{ch}h_{t-1} + b_c)\,. \tag{3.3.4}$$
- **Cell state:**
$$c_t = f_t \odot c_{t-1} + i_t \odot \widetilde{c}_t. \tag{3.3.5}$$
- **Hidden state:**
$$h_t = o_t \odot \tanh(c_t). \tag{3.3.6}$$

where:

- X_t is the input at time t
- H_{t-1} is the hidden state at time t-1
- C_t is the cell state at time t
- i_t is the input gate at time t
- f_t is the forget gate at time t
- o_t is the output gate at time t
- $\widetilde{c}_t$ is the candidate cell state at time t that is used to store information in an LSTM cell
- W are the weight matrices
- b are the bias vectors

- σ is the activation function with sigmoid function being popular
- tanh is the hyperbolic tangent function
- ∇ denotes pointwise multiplication

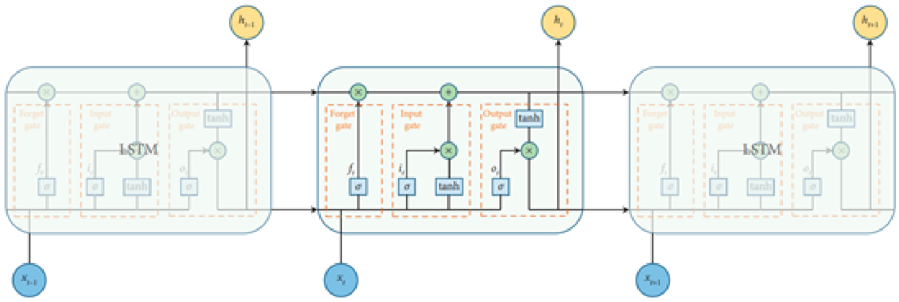

Figure 3.2: Source: [23] LSTM architecture.

We use a likelihood–based loss function when training our DNN and LSTM models. The likelihood function is defined as the probability of observing the actual return series given the model's predictions and maximises the likelihood of observing the actual returns given the predicted volatilities. If $\mathrm{r_t}$ is the return series that we observe from time 1 to time T and σ_t is the volatility that our models forecast at the same time step, we can calculate the sample likelihood under the assumption of normal distribution.

$$L(\{r_t\}_{t=1}^{T} \mid \{\sigma_t\}_{t=1}^{T}) = \prod_{t=1}^{T} \frac{1}{\sqrt{2\pi}\,\sigma_t} \exp\left(-\frac{r_t^2}{2\sigma_t^2}\right). \tag{3.3.7}$$

Since the likelihood functions involves multiplying many probabilities together, it is often more convenient to work with the logarithm of the likelihood. Taking the logarithm transforms the product into a sum simplifying the calculations. So, after taking logs of both sides we have the log–likelihood functions [23]:

$$\log L(\{r_t\}_{t=1}^{T} \mid \{\sigma_t\}_{t=1}^{T}) = \sum_{t=1}^{T} \left(-\frac{1}{2}\log(2\pi) - \log\sigma_t - \frac{r_t^2}{2\sigma_t^2}\right). \tag{3.3.8}$$

The log–likelihood function should be maximized through optimization, while the deep learning models are always trained by minimizing their loss function. So, we need to use the negative log likelihood to be the loss function in deep learning models. To further reduce the computational cost of deep learning models, we use a simplified function as the loss function of our DNN and LSTM models:

$$\text{Loss} = \sum_{t=1}^{T} \left(2 \log \sigma_t + \frac{r_t^2}{\sigma_t^2} \right). \tag{3.3.9}$$

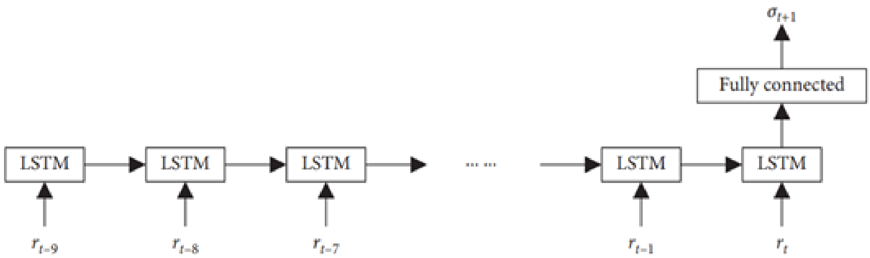

Figure 3.3: Source: [23] LSTM with fully connected layer.

LSTM model combines LSTM with a fully connected layer in the last time step. Its architecture is shown in the Figure above.

3.3.2 *GRADIENT VANISHING IN LSTM*

Same as in RNNs, LSTM outputs a prediction vector h(k). The knowledge encoded in the state vector c(t) captures the long–term dependencies and the relations in the sequential data. The gradient descent is computed to update the network parameters over T time steps. The vanishing gradient problem occurs when the gradients in a neural network diminish exponentially as they are backpropagated through layers and time steps.

The error term gradient is given by the same partial derivative as in RNNs and the gradient of the error for time step k has the same form given by variation of Taylor rule.

Backpropagation through time (BPTT) in RNN use gradients that are calculated recursively from the final time step(T) to the initial time step (t=1).

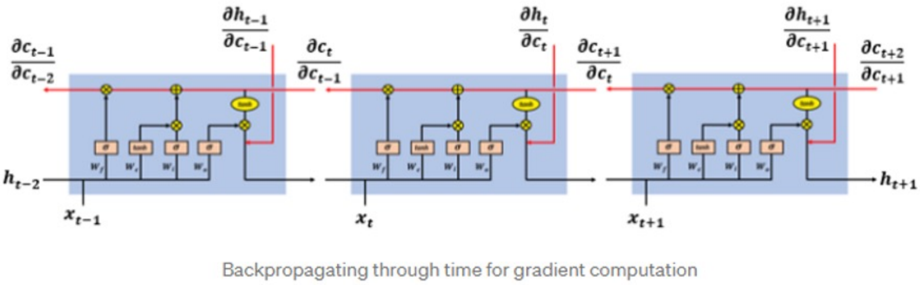

Figure 3.4: Source: [23] Gradient vanishing in LSTM.

The gradient at each step is obtained by multiplying the gradient from the next time step with the derivative of the activation function.

$$\delta_t = \delta_{t+1} \frac{\partial f(z_{t+1})}{\partial z_{t+1}}. \tag{3.3.10}$$

$\mathrm{\delta_t}$ represents the gradient at time step t, and $\mathrm{f(z_t)}$ is the activation function applied to the weighted sum $\mathrm{z_t}$ at time step t. When calculating the gradients through multiple time steps, the gradients are multiplied at each time step due to the chain rule of differentiation. This leads to an exponential decay of gradients as they are backpropagated to earlier time steps. The following product causes the gradients to vanish:

$$\Delta W = \prod_{t=1}^{T} \frac{\partial f(z_t)}{\partial z_t}. \tag{3.3.11}$$

ΔW represents the accumulated gradient term that is multiplied at each time step.

Unlike traditional RNNs, LSTMs utilise a memory cell (c) and various gates (input, forget and output gates) to control the flow of information and prevent the vanishing gradient problem. The additive property of the cell state gradients enables the network to update the parameter in such a way that the different sub gradients do not necessarily agree and behave in a similar manner making it less likely that the functions converge to zero. In other words, LSTM allows for better balancing and manipulation of gradient values during backpropagation. This property along with the presence of the forget gate, makes it less likely for the gradients to vanish. The forget gate in an LSTM allows the network to control the gradients at each time step and decide which information should be retained or forgotten. It helps prevent the vanishing gradient problem by enabling the model to update the parameters in a way that preserves important information.

3.3.3 *EXPERIMENTAL METHODOLOGY*

To ensure a robust and fair comparison of the machine learning models discussed in Chapter 2, a standardised experimental design was implemented.

Train–Test Split Protocol: Given the computational intensity of the deep learning architectures (specifically the GAN and N–BEATS ensembles), a chronological hold–out validation strategy was employed. The daily VIX time series (1990–2022) was split chronologically to prevent lookahead bias:

- Training Set (70%–80%): The initial majority of the dataset was used to train the models and capture long–term historical volatility regimes.
- Test Set (20%–30%): The final segment of the data was held out purely for out–of–sample evaluation.

Feature Engineering: The input features (X) were engineered differently depending on the model architecture's specific requirements:

i. N–BEATS Input: For the N–BEATS model, a univariate lookback window approach was used. The model was fed a sequence of lagged log–returns (e.g., a 7–day window) to forecast the subsequent step $(t + 1)$.

ii. GAN/CNN Input: For the GAN and CNN models, a novel image–based encoding was applied. Sequential segments of 100 days were reshaped into 10×10 grids and subsequently padded to 32×32 matrices. This transformation allowed the use of 2D Convolutional layers to detect spatial patterns within the time–series data.

Hyperparameter Selection: Optimal hyperparameters were selected through a combination of grid search and literature–derived baselines:

- LSTM: The LSTM model architecture consisted of 2 LSTM layers with 50 hidden units each, followed by a Dropout layer (rate 0.2) to mitigate overfitting. The model was optimized using Adam ($lr = 0.001$) by minimizing the Mean Absolute Error (MAE).
- N–BEATS: To establish a rigorous baseline, the N–BEATS architecture adopted the optimal configuration recommended in the original literature [46]. The generic architecture utilised a stack depth of 30, with 4 layers per block and a layer width of 512 units, trained over 5,000 epochs to ensure convergence.
- GAN: The Generative Adversarial Network was designed with a specialized architecture to handle the image–encoded time series. The Generator utilised Dense layers upsampled via Conv2DTranspose, while the Discriminator employed a 2D Convolutional Neural Network (Conv2D) to distinguish between real and synthetic volatility patterns. Training utilised the Adam optimiser with a learning rate of 0.0002 and $\beta_1 = 0.5$.

Performance Evaluation: The 93.3% accuracy metric associated with the GAN model refers strictly to the internal discriminator's ability to classify real versus generated data samples during training. It is not a forecasting metric. The true predictive performance of the hybrid framework is evaluated via the MAE and MAPE scores of the downstream regressors (SVM and Random Forest) that utilise the GAN–generated synthetic features for final prediction.

3.4 *METHODOLOGY OF GENERATING MULTIFRACTAL SIMULATIONS*

We will simulate approximately 30 years of data which should capture at least 4 different economic cycles —- 1990s boom and bust, 2000s housing boom and bust, 2010s blockchain and cryptocurrency boom and bust and COVID. This should give us the best chance to measure realistic parameters.

3.4.1 Definition of a multifractal stochastic process:

A stochastic process *X(t)* is defined as multifractal if it has stationary increments, and satisfies:

$$\mathbb{E}\Big[|X(t)|^q\Big] = c(q)\, t^{\tau(q)+1}, \qquad \forall\, t \in T,\ q \in Q. \tag{3.4.1}$$

where T and Q are intervals on the real line and have positive lengths.

In this formulation, $c(q)$ represents a deterministic, moment–dependent prefactor that is independent of the time scale t. The function $\tau(q)$ is the multifractal scaling exponent, which characterizes the scaling behaviour of the process. The non–linearity of $\tau(q)$ with respect to q is the defining condition for multifractality; a linear $\tau(q)$ would imply a monofractal process such as fractional Brownian motion.

Now we choose some values of q (raw statistical moments) which will be used for our empirical estimations. [16] used 99 values of q between 0.01 and 30 to estimate $\tau(q)$. Next step is to find the partition function by splitting the data into N non–overlapping subintervals, whose length Δt is less than our total time T:

$$S_q(T, \Delta t) = \sum_{i=0}^{N-1} \Big| X\Big((i+1)\Delta t\Big) - X\Big(i\Delta t\Big)\Big|^q. \tag{3.4.2}$$

where $N = T/\Delta t$ is the total number of time increments for that increment size. The partition function is the sum of the absolute values of all increments of price change, raised to some given power q which is a raw statistical moment for every size of time increment Δt.

The time increments Δt are the independent variable that we manipulate (x–axis), and the partition function S_q *(t,* Δt*)* is the dependent variable. This function is computed several times over for different values of q. Then we estimate the scaling function τ(q) by running OLS linear regression on the different lines of the graph for different values of q [68].

$$\ln\Big(\mathbb{E}\Big[S_q(T,\Delta t)\Big]\Big) = \widehat{\tau}(q)\ \ln(\Delta t) + \widehat{c}(q)\ \ln(T). \tag{3.4.3}$$

We run OLS regression on the partition function curve for each q. The gradient of the straight line is the value of τ(q) for that value of q. Through all the different OLSs, we will have many different values for τ(q) and the intercept *c(q)* for different values of q.

Next, we will use non–linear regression to estimate non–linear curve τ(q) that goes through all the estimated values as closely as possible. Next, we find the Hurst exponent H. Using the estimated scaling function τ(q) and the following formula, we solve for H. This is derived in [16].

$$\tau_X\left(\frac{1}{H}\right) = 0. \tag{3.4.4}$$

$$\therefore \quad \widehat{H} = \frac{1}{\widehat{\tau}_X^{-1}(0)}. \tag{3.4.5}$$

We find hurst exponent taking the inverse of the estimated scaling function. Then we estimate the multifractal spectrum f(a) by performing a Legendre transform on τ(q). The Legendre transform is defined as a technique that is used to convert functions of one quantity such as position, pressure, or temperature into functions of the conjugate quantity i.e., momentum, volume, entropy, respectively. In Mechanics, there are two common approaches to describing motion —- the Lagrangian and the Hamiltonian. The Legendre transform is a technique that is used to convert a Lagrangian function *L (x, dx/dt, t)* which describes a particle's velocity to a Hamiltonian function *H (x, dL/(dx/dt), t)* which describes its momentum. The function is the following:

$$H(p) = \sup_{x \in \mathbb{R}} \Big\{ p\,x - f(x) \Big\}. \tag{3.4.6}$$

where $p = df(x)/dx$ and *sup* is the supremum function. The supremum of a subset S of a partially ordered set T is the least element in T that is greater than or equal to all elements of S if such an element exists. If the transform has a negative sign, then instead of supremum we use infimum. The scaling function should be concave, so we can define the multifractal spectrum of the price process as:

$$\widehat{f}(\alpha) = \inf_{q \in Q} \{ \, q\,\alpha - \widehat{\tau}(q) \, \} + D. \tag{3.4.7}$$

where inf refers to infimum. Infimum of a subset S of a partially ordered set T is the greatest element in T that is less than or equal to all elements of S if such an element exists. Multifractal analysis aims to describe the singularity structure of a dataset, which reflects the varying degrees of irregularity, concentration of values, and scaling properties across different scales. Singularities are points in the data where certain properties deviate significantly from the expected regular behaviour. In financial time series, these singularities could correspond to extreme events, crashes or volatile periods. The multifractal spectrum characterises the scaling exponents that quantify the local regularity or singularity at various points within the dataset. These scaling exponents correspond to the Hölder exponent (α), which describe the local scaling properties of the data. High α values show regularity, while low α values indicate singularity. Hölder exponent should not be confused with Hurst exponent or Hausdorff dimension. These are related concepts but not the same. Hölder exponents are used in multifractal analysis to characterise the local scaling properties of a dataset. They indicate how the regularity or singularity of a function varies at different points within the data. The Hölder exponent α quantifies the rate of decay or growth of the function's oscillations. In the context of multifractal analysis, α can vary across different points, allowing for the characterisation of complex and irregular datasets. The multifractal spectrum τ(q) (sometimes also known as mass exponent or scaling function) is related to the singularity spectrum *f(α)* through Legendre transform which is used to obtain τ(q) from *f(α)* and vice versa. The transformation ensures that τ(q) is a concave function of *q*, and *f(α)* is its convex conjugate. Concavity of the infimum operator is related to the multifractal scaling properties and reflects the existence of different scaling behaviours at various points within the dataset. Using the estimated multifractal spectrum *f(a)*, we can find the most probable Hölder exponent.

$$\widehat{\alpha}_0 = \widehat{f}^{-1}\Big(\max_{\alpha} \widehat{f}(\alpha)\Big). \tag{3.4.8}$$

α_0 is the value of the Hölder exponent α for which $f(\alpha_0) = 1$. This should also be the maximum value of the multifractal spectrum. If the data is multifractal, then it has different Hölder exponents at different timepoints and α_0 is the most commonly occurring exponent in the data. Using the estimated values for H and α_0, estimate the log–normal distribution parameters for the multiplicative cascade —- λ and σ^2. The MMAR involves generating a multiplicative cascade, whose masses are drawn from a continuous probability distribution. The most common distribution used for these cascades is the lognormal distribution with a mean λ and a variance σ^2. [16] showed that these two parameters are determined using H and α_0:

$$\widehat{\lambda} = \frac{\widehat{\alpha}_0}{\widehat{H}}. \tag{3.4.9}$$

$$\widehat{\sigma^2} = \frac{2\big(\widehat{\lambda} - 1\big)}{\ln(b)}. \tag{3.4.10}$$

We will use these two estimators to generate randomly distributed Multipliers M_i to construct multifractal cascades for the MMAR simulation. First, we choose the number of b–adic intervals into which we will divide our time series at every step *k*. So, at every step *k*, there should be b^k intervals in total. For example, if we choose the number of intervals as 2, then we have 1 interval at step zero; 2 intervals at step one; 4 at step two and so on. Then choose some masses m_0 and m_1. The height of the graph for that interval is the mass of the interval will be multiplied by the allocated mass m_i. This would give us a simple cascade but if we were to continue this process, we would get a graph with a series of bars that look like scaled versions of each other. We need to introduce randomness into the system so instead of having static mass allocations, give them some probabilities of occurrence. Our mass allocations have been mutually exclusive so if one side had m_0 then the other side had m_1 and $m_0 + m_1 = 1$. Since the mass is conserved as in laws of physics, such multiplicative cascade is called conservative. The mass of each interval, which can be interpreted as the probability mass, is redistributed such that the total always sums to 1. However, if the mass allocations are independent of each other, this would mean that their sum does not have to add up to 1 each time but the expected sum of the distribution should be one. In other words, the total weight of all masses on average should add to one. In fact, we don't even need to have discrete probabilities or multipliers.

We can draw them from a continuous probability distribution. We can replace our discrete multipliers with some random multipliers draw from a distribution of our choice. The expected sum of the multipliers should equal to one. This second type of multiplicative cascade is called a canonical cascade, and it is the most used type in simulating MMAR model and the one we will be using as well. The mass of each interval in canonical cascade can vary independently, and while the sum might not always be 1 in every specific instance, the expected sum over a large number of instances or iterations is 1. This maintains the overall conservation of mass (or probability mass) in a statistical sense. We will also use binomial cascade meaning we will split each interval into two i.e. b = 2.

From [16], we get that:

$$V = -\log_b M \sim \mathcal{N}\left(\widehat{\lambda},\, \widehat{\sigma}^2\right). \tag{3.4.11}$$

$$\mathbb{E}[M] = \mathbb{E}\left[b^{-V}\right] = \mathbb{E}[\exp(-V \ln b)] = \exp\left(-\widehat{\lambda} \ln b + \frac{1}{2}\widehat{\sigma}^2 (\ln b)^2\right) = \frac{1}{b}. \tag{3.4.12}$$

where $V \sim N\ (\lambda,\ \sigma^2)$ and as we established before in 3.5.9:

$$\widehat{\lambda} = \frac{\widehat{\alpha_0}}{\widehat{H}}$$

$$\widehat{\sigma^2} = \frac{2(\widehat{\lambda} - 1)}{\ln[b]}$$

The trading time cumulative distribution function (CDF) $\theta(t)$ is the main component of MMAR and requires both high kurtosis and volatility clustering to generate our simulated returns. To compute CDF we will compute the cumulative sum of the cascade from time 0 to the end of time T for each value of t.

$$\theta(t) = F_\mu(t) = \sum_{i=0}^{t} \mu(i), \qquad t \in \mathbb{Z}_{\geq 0}. \tag{3.4.13}$$

Fractional Brownian Motion is the other component of MMAR model. This is the component that allows us to simulate random walks with Hurst component other than 0.5.

$\mathrm{B_H}(\mathrm{t})$ is a stochastic process characterized by the following equation:

$$\begin{aligned} B_H(t) = \frac{1}{\Gamma\left(H + \frac{1}{2}\right)} \Bigg[\int_{-\infty}^{0} \left((t-s)^{H-\frac{1}{2}} - (-s)^{H-\frac{1}{2}} \right) \mathrm{d}B(s) \\ + \int_{0}^{t} (t-s)^{H-\frac{1}{2}} \, \mathrm{d}B(s) \Bigg], \qquad t > 0. \end{aligned} \tag{3.4.14}$$

where:

$\Gamma(z)$ represents the Euler Gamma function $\int_{-\infty}^{0} [x^{z-1} e^{-x}]\, dx$;

H represents the Hurst exponent, $0 < H < 1$;

$B(s)$ represents a typical stochastic Brownian motion with $H = 0.5$;

t represents time.

These equations are used to simulate a random process with H $\neq$ 0.5. Imagine that we have a white noise process, which we can think of as a river of constantly fluctuating white noise. We want to integrate this white noise process to obtain a process with a Hurst exponent other than 0.5. We can do this by building a dam across the river. The dam will smooth out the white noise process, and the Hurst exponent of the resulting process will depend on the height of the dam. The Wiener integral is the mathematical equivalent of building a dam across a river of white noise. The Hurst exponent of the resulting process will depend on the parameters of the Wiener integral [69]. Simulating fBm requires setting the length of the fBm process. [16] has not said anything of their fBm settings. We would manually adjust the simulated data until the median standard deviation of returns was approximately equal to that of real data.

The last step is to merge the trading time CDF with the fBm for simulation of overall returns X(t). Wengert [70] offered MMAR code for users of MATLAB software which we adjusted for Python when constructing multiplicative cascade.

The methodology for generating multifractal simulations can be summarised as follows. Estimate the Hurst exponent and the multifractal spectrum $f(a)$. The latter describes the distribution of the fluctuations of the time series. So, we find the four empirical parameters H, α, λ, σ^2 by calculating the partition function $S_q(T, \Delta t)$, the scaling function τ(q), and the multiplicative spectrum $f(a)$. Then we generate a lognormal multiplicative cascade. This is a random process that is used to generate multifractal data. It is created by repeatedly dividing a time series into smaller and smaller intervals and then assigning random multipliers to each interval. The multipliers are drawn from a lognormal distribution, which means they have a long right tail. This ensures that the cascade has a high degree of variability. Next, we convert the lognormal multiplicative cascade into a cumulative distribution function (CDF). This function maps each possible value of a random variable. CDF can be computed using the following simplified mapping: $CDF(x) = \exp(-|(\ln(x) - \lambda)|^2 / (2\sigma^2))$. In other words, we simulate two random processes which are used as the two components of the MMAR i) trading time function $\Theta(t)$ made by converting a lognormal multiplicative cascade into a cumulative distribution and ii) fBm which is like a random walk but the changes do not have to be independent. fBm is a random process that is used to model long term trends in a time series. It is created by adding together a series of independent random walks, each of which has a Hurst exponent of H. Then we plug trading time into fBm so we can interpret the compound process as simulation of price growth over time $X_t = B_H[\Theta(t)]$. In other words, we combine the CDF of the lognormal multiplicative cascade with the fBm process. The result is a time series that has both long term trends and short–term fluctuations.

The pseudo–code in Appendix A2 provides a simplified implementation of this process. Achieving a complete implementation is an area of ongoing work and constitutes a significant part of our planned future developments.

3.4.2 LONG–RANGE DEPENDENCE AND THE RESCALED RANGE

Mandelbrot used the Rescaled Range (R/S) test statistic to test for long range dependence. The R/S statistic is the range of partial sums of deviations of a series from its mean divided by the standard deviation. Mandelbrot showed that R/S statistic led to superior results when compared to analysis of autocorrelations and variance ratios. The R/S statistic is obtained using the following method. Consider the time series of length n:

$$\{x_j\}_{j=1}^{n} = \{x_1, x_2, \ldots, x_n\}. \tag{3.4.15}$$

The partial sum of k deviation from the mean is given by:

$$\Pi_k \equiv \sum_{j=1}^{k} (x_j - \overline{x}_n) = (x_1 - \overline{x}_n) + (x_2 - \overline{x}_n) + \cdots + (x_k - \overline{x}_n). \qquad (3.4.16)$$

The R/S statistic is proportional to the difference between the maximum and the minimum of such sums where $k \in [1, \text{n}]$:

$$\frac{R}{S}(n) = \frac{1}{\sigma(n)} \left[\max_{1\leq k\leq n} \Pi_k - \min_{1\leq k\leq n} \Pi_k \right]. \qquad (3.4.17)$$

The denominator $\sigma(n)$ is the maximum likelihood standard deviation estimator. The rescaled range and the number n of observations have the following relation:

$$\frac{R}{S} \propto n^H. \qquad (3.4.18)$$

where H is the Hurst exponent. This relationship was observed empirically by Hurst in his analysis of river flows. He found that the range of cumulative deviations (rescaled by standard deviation) exhibited self–similar behaviour across time scales. This observation is now generalised to time series with long memory and fractal properties. After taking logarithms of both sides in the equation above, we obtain the relation between the rescaled range and the number of observations n:

$$\frac{R}{S} \approx A + Bn^H \;\Rightarrow\; \log\left(\frac{R}{S}\right) \approx C + H \log n. \qquad (3.4.19)$$

So, one can estimate the value of H from the log–log plot as the slope of the line. This is a standard procedure for identifying long–term dependencies in time series.

In the next diagrams, we illustrate how the Hurst coefficient changes with different assets, time frames and lags. The *'lag1'* and *'lag2'* variables define the range of time lags we considered for the analysis. For each time lag in the range, we calculated '*tau*', which represents the standard deviation of the differenced time series. Specifically, '*tau*' was obtained by computing the standard deviation of the differences between two portions of the time series data at each lag. This provided insight into how variability changed as the time lag increased and was essential for estimating the Hurst exponent, which characterises the long–term memory of the time series. Next, we performed a linear regression analysis on the logarithms of *'lags*' and '*tau*' using the '*polyfit*' function. The result, stored in 'm' corresponds to the slope of the log–log plot. Finally, we used the slope from the linear regression to calculate the Hurst exponent H, which quantifies the persistence or mean–reverting nature of the time series.

Figures 3.1–3.2 show long term time series of S&P exhibits strong persistent behaviour (H=0.64) and short–term S&P 500 is strongly mean reverting (H = 0.25). Stock prices rarely display mean–reversion behaviour long–term as asset prices tend to follow random walk and trend up with inflation over long time horizons. In contrast, Figures 3.3–3.4 show that volatility is more mean reverting long term and has more persistent behaviour but still mean reverting in the short–term. It is quite intuitive as volatility tends to be within one standard deviation long term but can become unbounded in the short term. We can see it through bursts of volatility during crisis and subsequent subsiding as it returns to its average corridor of one standard deviation.

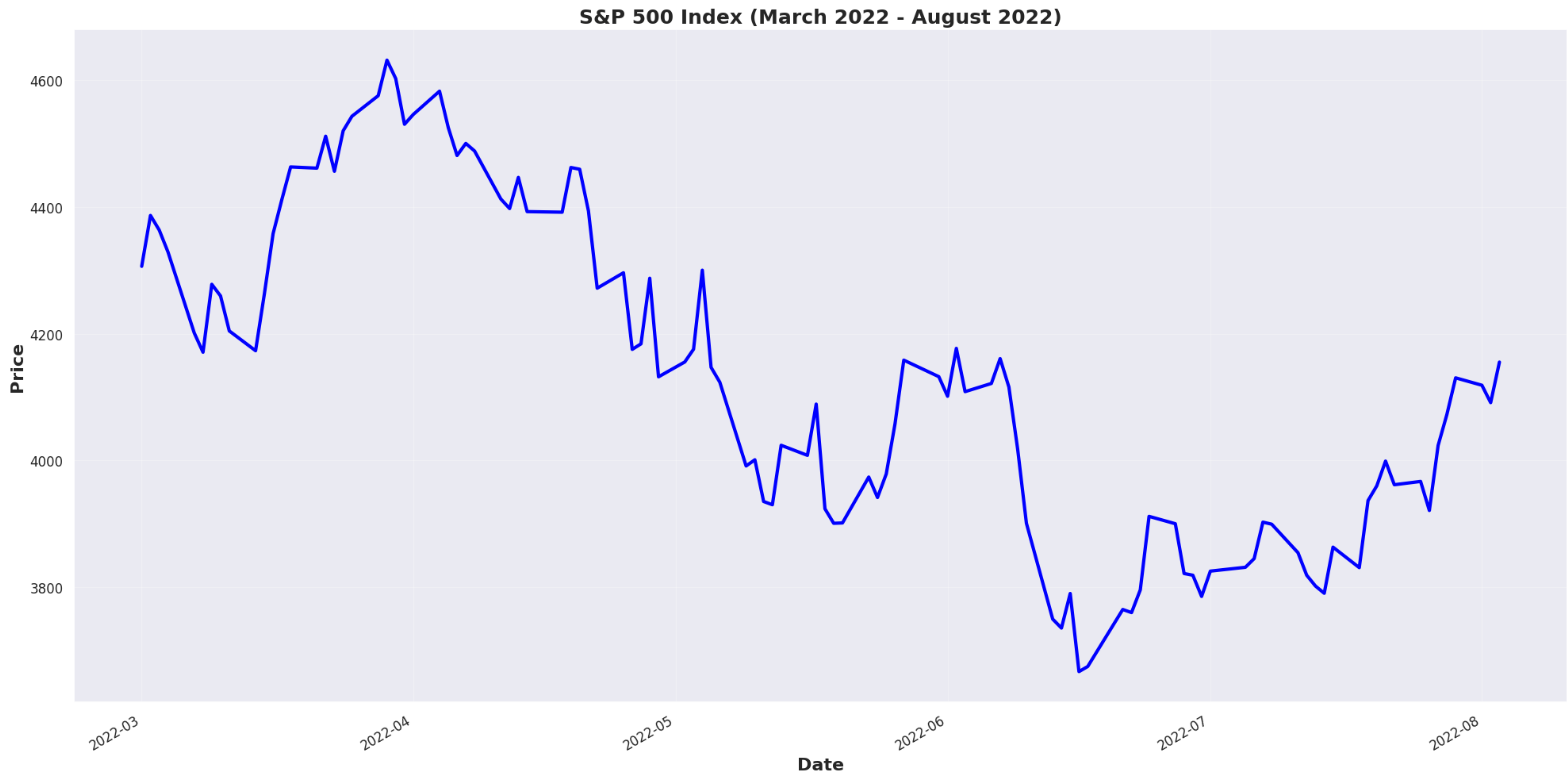

Figure 3.5: Time series: S&P500 Hurst = 0.399, Lag 1, Lag 2 = 2, 20 Time: 3/1/2022 — 4/8/2022.

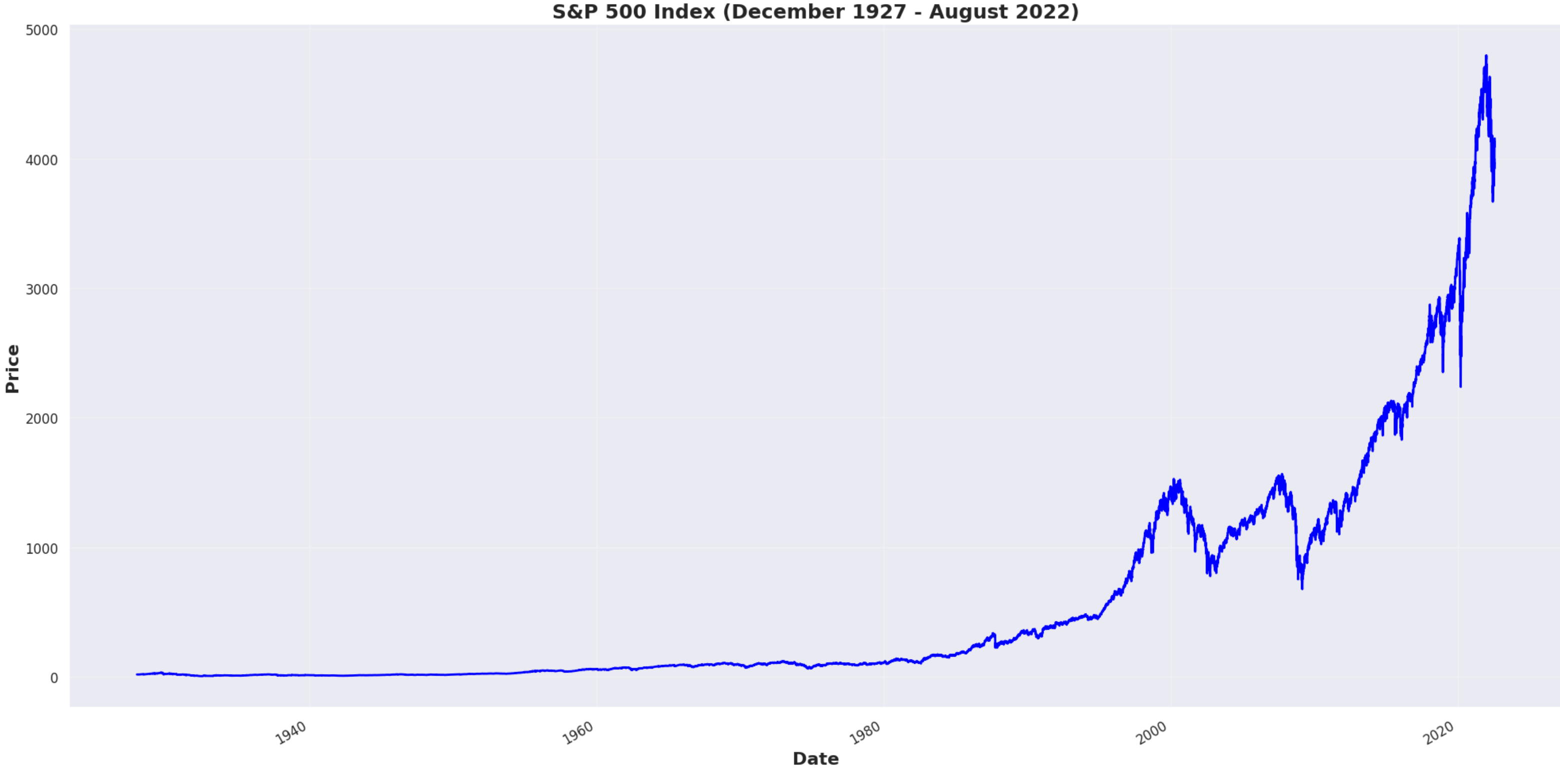

Figure 3.6: Time series: S&P 500 Hurst = 0.637, Lag 1, Lag 2 = 300, 400, Time: 30/12/1927 —- 4/8/2022.

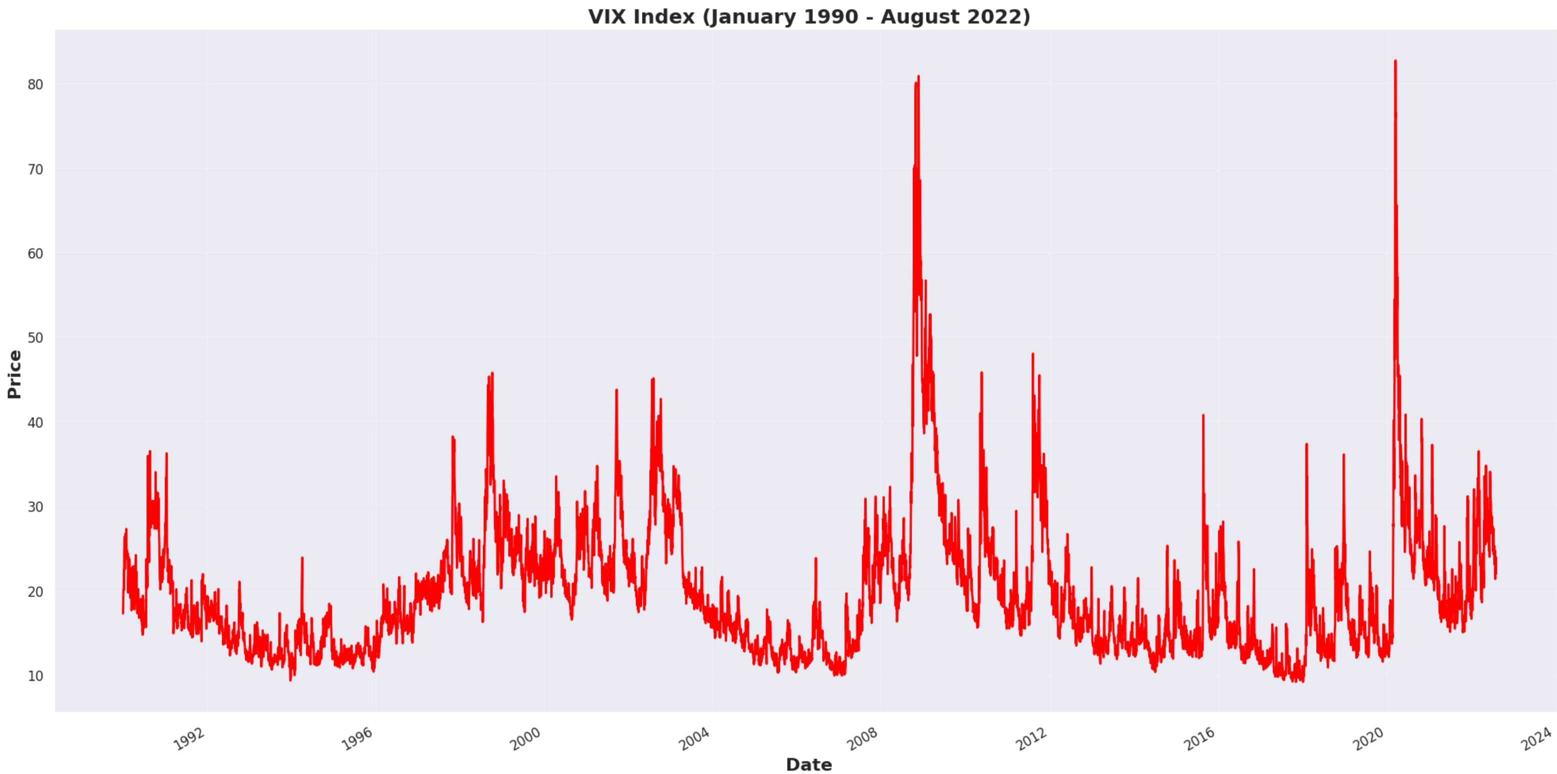

Figure 3.7: Time Series: VIX Hurst = 0.249, Lag 1, Lag 2 = 300, 400, Time: 02/01/1990 —- 4/8/2022.

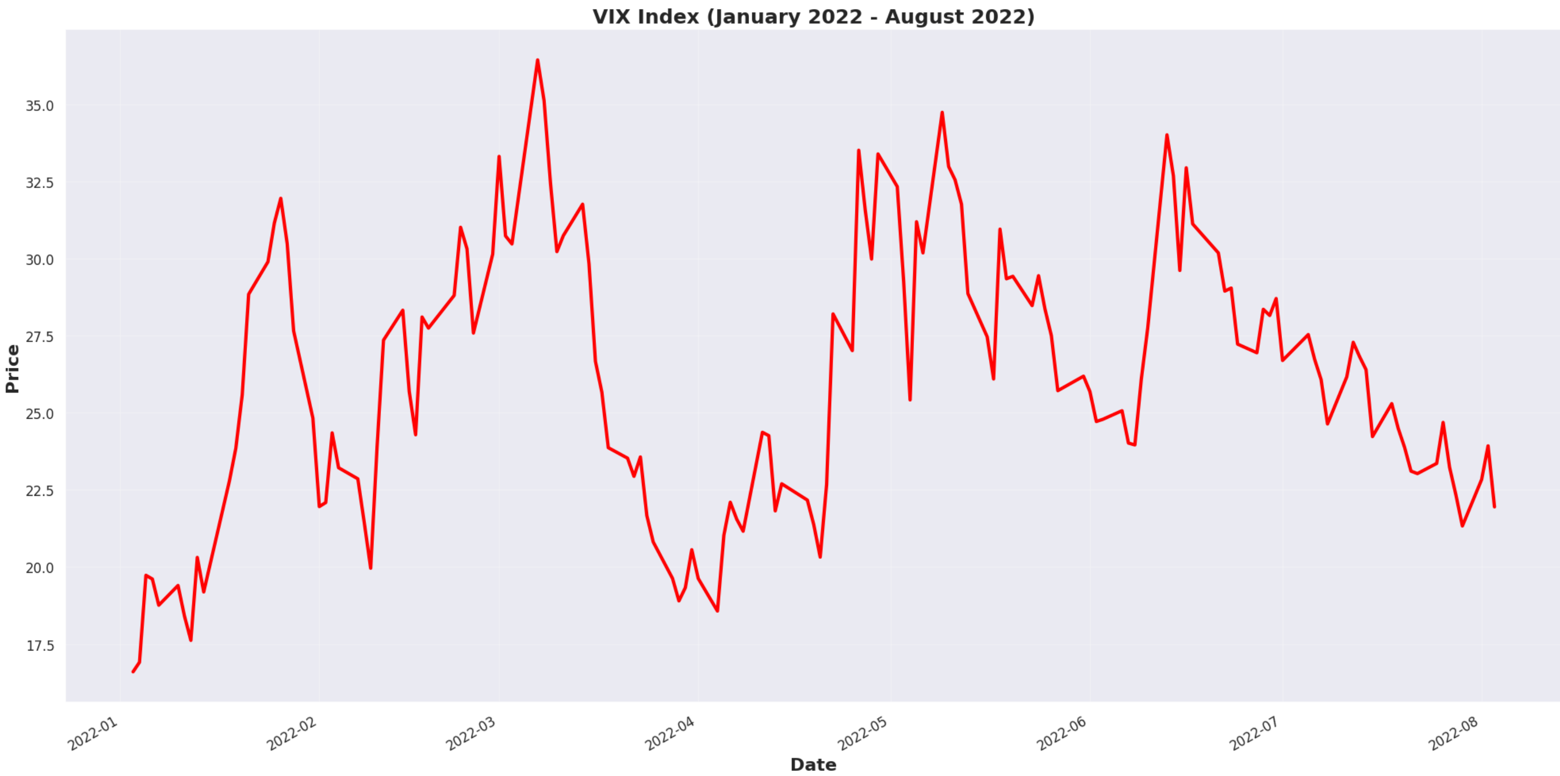

Figure 3.8: Time Series: VIX Hurst = 0.398, Lag 1, Lag 2 = 2, 20, Time: 03/01/2022 —- 4/8/2022.

3.5 *EMPIRICAL ANALYSIS AND MODEL BENCHMARKING*

In this section we present the empirical groundwork for the thesis. The analysis herein, based on our published work, serves two primary objectives. First, we explore the underlying structure of the VIX time series using clustering methodologies to identify distinct volatility regimes. Second, we establish a robust performance benchmark by implementing and evaluating a suite of forecasting models, from classical methods to advanced machine learning techniques. The limitations and characteristics identified in this empirical analysis provide the essential motivation for the novel theoretical framework developed in Chapter 5. For full model results and discussions please see 'Notes on Intuitionistic Fuzzy Sets' [P1] and IEEE paper [P4].

3.5.1 *K–MEANS AND FUZZY C–MEANS CLUSTERING*

Determining the optimal number of clusters (K) is crucial for the effectiveness of the K–Means algorithm. [P1] employed the 'Elbow' method and 'Silhouette' analysis to find the appropriate value for K. The Elbow Method plots the sum of squared distances from each point to its assigned centroid for different values of K. The optimal K is chosen at the "elbow" point, where the rate of decrease sharply slows down. Silhouette Analysis measures how similar a point is to its own cluster compared to other clusters, with higher silhouette values indicating better–defined clusters. In this study, data was normalized using the StandardScaler technique, which transforms the data to have a mean of 0 and a standard deviation of 1. Normalization is a preprocessing step used to standardize the range of independent variables or features of data. This step is essential to ensure that the algorithm treats all features equally, preventing features with larger ranges from dominating the clustering process. Figure 3.5 shows as the number of clusters increase, the error is minimised more and more. Visually, we are looking for an elbow of this curve.

It appears in the region of 2—-4 clusters.

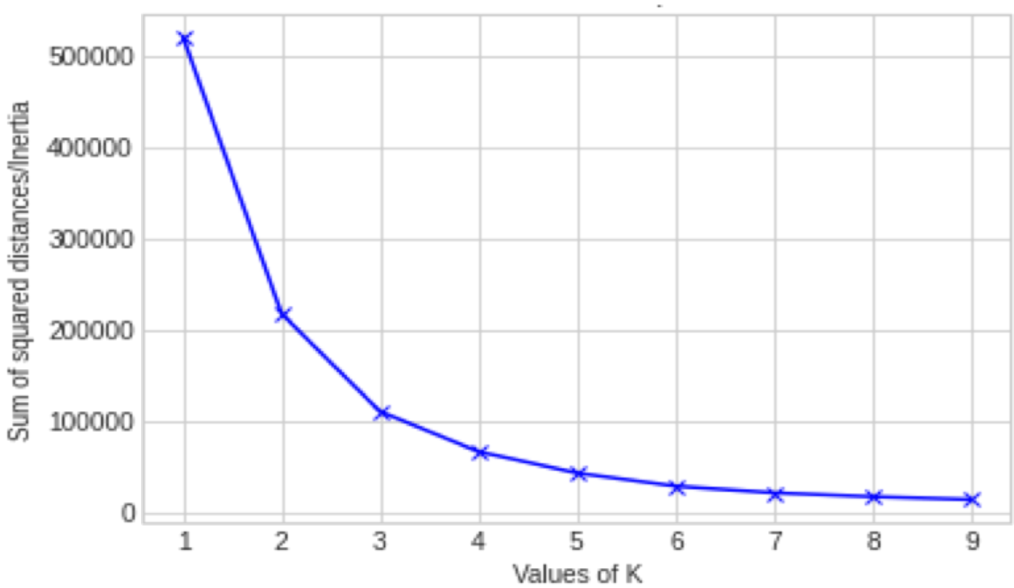

Figure 3.9: Source: [P1] *K–Means Clustering.*

The Silhouette method is often compared to the "Elbow" method, which involves plotting the within–cluster sum of squares (WCSS) against the number of clusters and identifying the point where the rate of decrease sharply slows. Both methods are used to determine the optimal number of clusters, but they approach the problem from different angles.

To apply the Silhouette method, we trained K–means clustering for each value of k (the number of clusters). For each clustering configuration, we computed the silhouette coefficient for every data point and then averaged these values to obtain the overall silhouette score for that k.

The Silhouette score for the five–cluster model was found to be 0.554. This score indicates a reasonably good clustering configuration, suggesting that the five–cluster model provides a meaningful division of the data. The Silhouette score, being above 0.5, implies that the majority of data points are well–matched to their assigned clusters and only a few points are on the boundary between clusters. While a score above 0.5 indicates reasonable clustering, the proximity of the score to 0.5 suggests potential overlap between clusters or suboptimal feature selection. Figure 3.6 shows the silhouette score on y–axis and the number of clusters on the x–axis. Prior to plotting the clusters, we find the centroids of the clusters: [26.81939959], [19.69309811], [13.48929072], [62.16890411], [38.19894207].

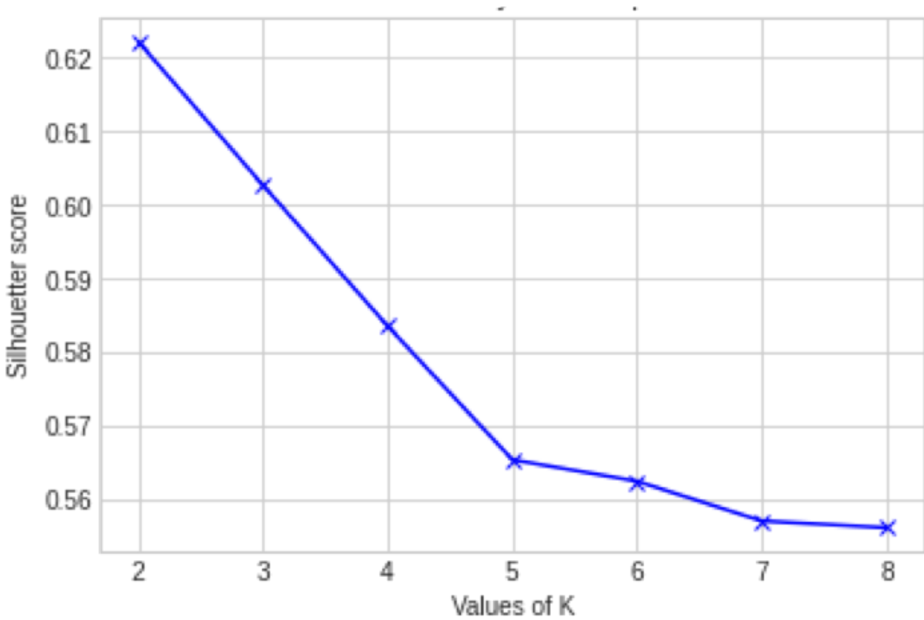

Figure 3.10: Source: [P1] *Silhouette method.*

Next, we plot the graph with 2 and 5 clusters. The top graph in Figure 3.7 shows two clusters identified by horizontal lines and the bottom graph with 5 clusters identified also by the horizontal lines. The top graph simply splits the data into periods of low and high volatility. The graph shows when the volatility is low it tends to stay low and when it is high it tends to remain high. The bottom graph has 5 centroids which are represented as horizontal lines. The graph splits the data into periods of very low volatility, low volatility, medium volatility, high volatility, and extremely high volatility.

To understand the dynamics of volatility transitions, we calculated the transition matrix for the clusters. The transition matrix showed how many times the Volatility Index (VIX) moved from one cluster to another over the observed period. This matrix is crucial for capturing the temporal behaviour of volatility and understanding the stability and transitions between different volatility states.

The expectation was that the transition matrix would be symmetric if there were no significant jumps in VIX. However, the matrix was not symmetric, indicating the presence of jumps and non–uniform transitions between volatility states. This asymmetry suggests that certain states of volatility are more likely to transition into others, rather than remaining stable or reverting, highlighting the dynamic nature of financial markets.

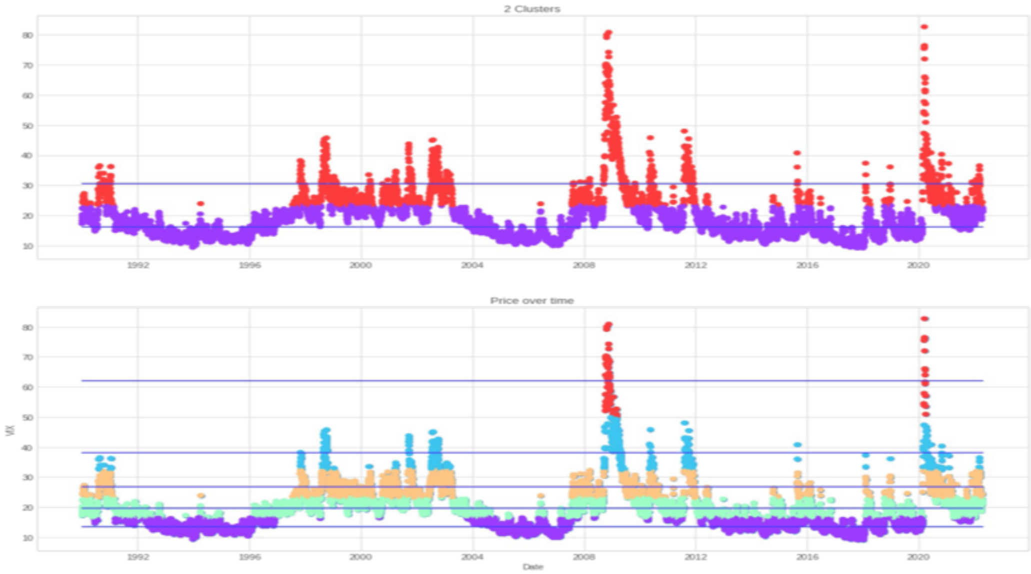

Figure 3.11: K-means clustering of VIX levels comparing different cluster configurations (1990–2022). Top panel: Binary classification into low volatility (purple) and high volatility (red) regimes. Bottom panel: Five-cluster solution distinguishing very low, low, medium, high, and extreme volatility states. Horizontal lines indicate cluster boundaries. Asymmetric transitions between clusters reveal the presence of volatility jumps. Source: [P1].

Fuzzy C–Means (FCM) clustering is an advanced clustering algorithm that extends the traditional K–means clustering by allowing data points to belong to multiple clusters with varying degrees of membership. This approach is particularly advantageous in scenarios where the boundaries between clusters are not well–defined, enabling a more flexible and nuanced understanding of the data structure. In hard clustering, data are divided into distinct clusters, where each data point can only belong to one cluster. In fuzzy clustering, data points can belong to multiple clusters. Fuzzy C–means clustering's ability to assign fractional membership is particularly valuable in financial volatility analysis, where transitions between states (e.g., medium to high volatility) may exhibit gradual shifts rather than discrete jumps. For example, an apple can be red or green (hard clustering), but an apple can also be red and green (fuzzy clustering). Here, the apple can be red to a certain degree as well as green to a certain degree. Instead of the apple belonging to green [green = 1] and not red [red = 0], the apple can belong to green [green = 0.5] and red [red = 0.5]. These values are normalized between 0 and 1; however, they do not represent probabilities, so the two values do not need to add up to 1. Membership grades are assigned to each of the data points. These membership grades indicate a degree to which data points belong to each cluster. Thus, points on the edge of a cluster, with lower membership grades, may be in the cluster to a lesser degree than points in the centre of cluster. Figure 3.8 illustrates Fuzzy C–Means Clusters 0 — 4.

In figure 3.9 we plot the best Cluster 5 with 5 centroids, which are represented as horizontal lines. The graph splits the data into periods of very low volatility, low volatility, medium volatility, high volatility, and extremely high volatility according to FCM algorithm.

The Silhouette Score is 0.555 which in this case is not very different to K–Means and is probably due to random state initialisation. It suggests that the clusters are somewhat well–separated but may not be perfectly distinct. While it's not as high as it could be, it still indicates a reasonable level of cluster quality. The proximity of the score to 0.5 suggests that there might be some overlap or inconsistency in the clusters.

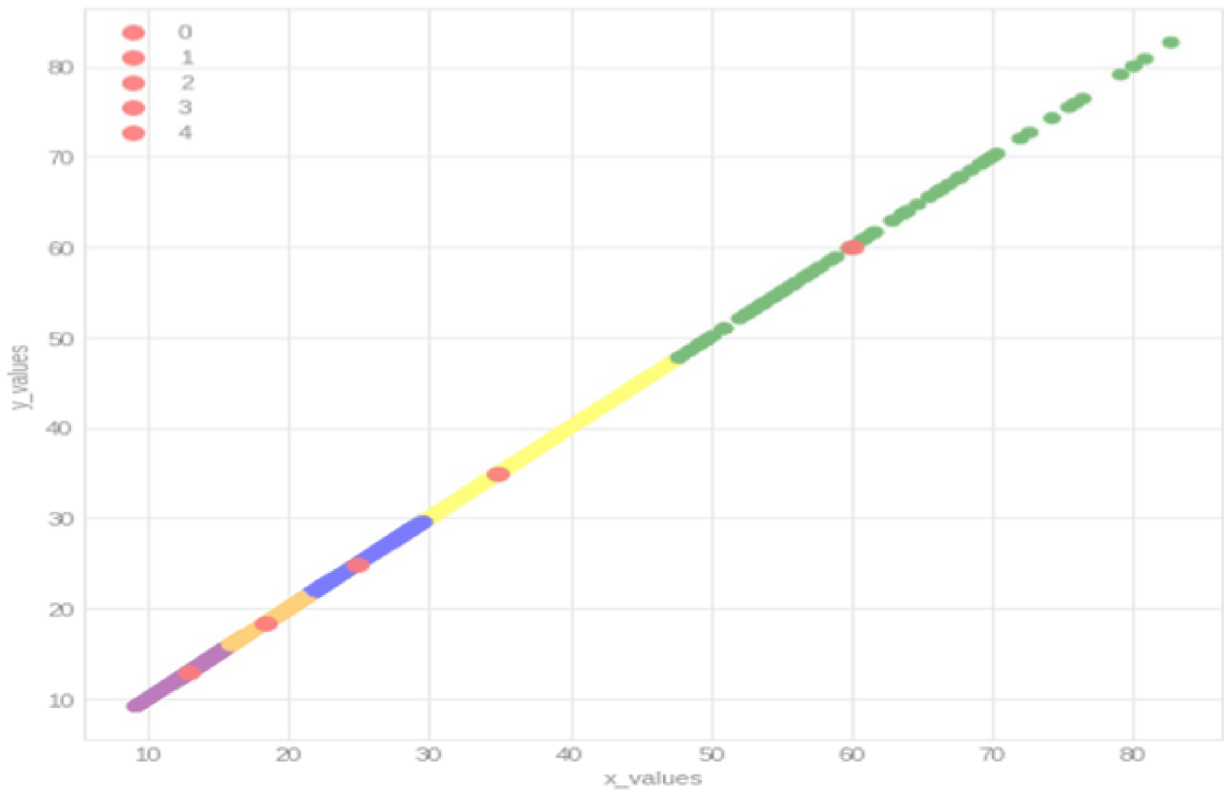

Figure 3.12: Fuzzy C-Means (FCM) cluster centroids for VIX volatility regimes. The five clusters (0–4) represent increasing volatility levels from very low (10–15) to extreme (80). Unlike hard clustering, FCM assigns fractional membership degrees, allowing data points near cluster boundaries to belong partially to multiple regimes. Source: [P1].

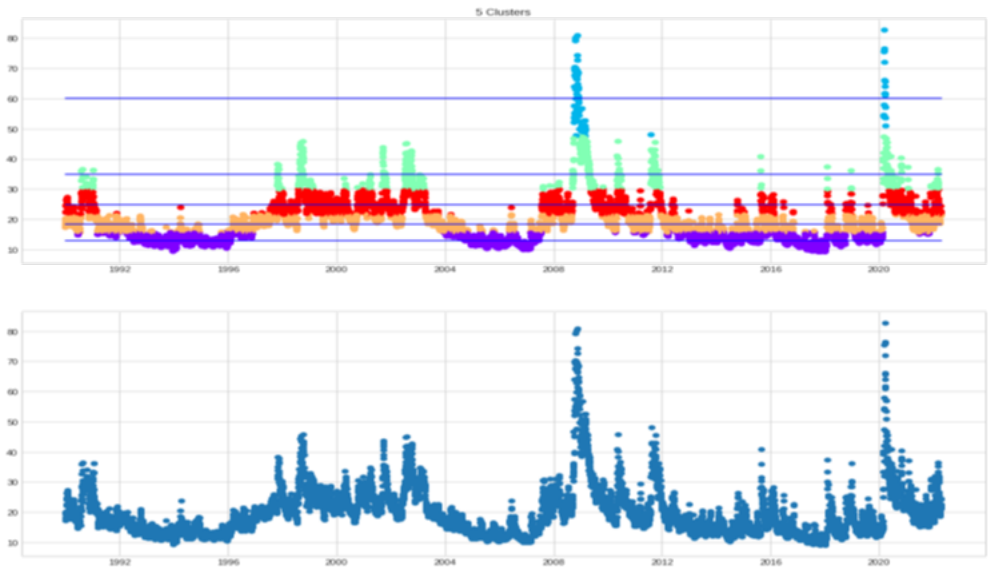

Figure 3.13: K-means clustering of VIX levels into five volatility regimes (1990–2022). Top panel: VIX values color-coded by cluster assignment, with horizontal lines indicating cluster boundaries. Bottom panel: Raw VIX time series for reference. Notable spikes correspond to the 2008 financial crisis and COVID-19 pandemic. Source: [P1].

3.5.2 *DEEP NEURAL NETWORK MODELS*

While clustering methods provide valuable insights into volatility regimes, deep neural networks offer complementary tools for time–series forecasting by capturing temporal dependencies and nonlinear patterns. To establish a performance baseline, a range of forecasting models were implemented and tested on the daily VIX log–return series.

3.5.2.1 Methodology

In the previous sections we already presented the theoretical framework of the models discussed here. A walk–forward validation approach was used. The models ranged from a Naïve forecast and ARIMA to more complex architectures like LSTM, NBEATS, and a hybrid GAN–SVM/RF framework. In the hybrid approach, an ARIMA model was first used to capture linear trends. A Generative Adversarial Network (GAN) was then trained, with an LSTM–based Generator and a CNN–based Discriminator, to learn the non–linear data distribution and generate synthetic time–series features. These generated features were then used as inputs for Support Vector Machine (SVM) and Random Forest (RF) models to make the final forecast. The 93.3% accuracy metric refers specifically to the internal training accuracy of the GAN's discriminator in distinguishing real from generated data. It is not a forecasting performance metric.

In this section we outline results from [P4]. As was already mentioned, we ran our data from January 1990 to April 2022 for a total of 1200 epochs and it gave us the accuracy rate of 93.3%.

In this paper, we used normalised log scale return VIX (Figure 3.10) as our train data and normalised log scale return VVIX as our validation data. While standard validation practice involves using a hold–out set of the same time series, a different approach was intentionally employed here to test a deeper aspect of the research hypothesis. The central premise of this thesis is that VVIX, as the "volatility of volatility", is not merely a separate index but is intrinsically linked to the process that generates VIX dynamics. VVIX

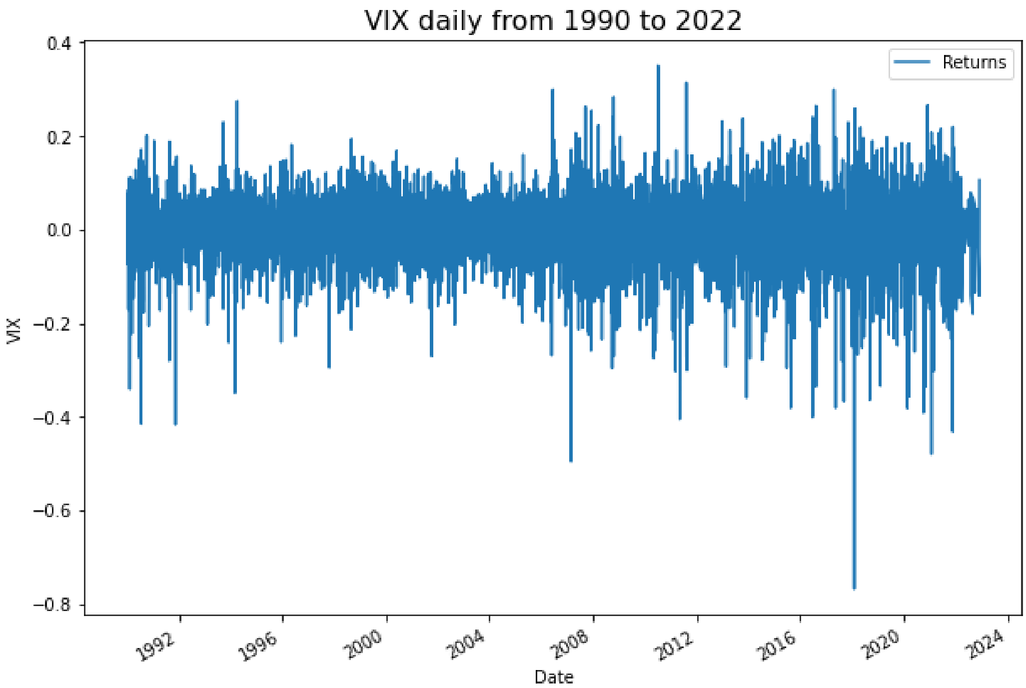

Figure 3.14: Daily log returns of the VIX index (1990–2022). The series exhibits volatility clustering, with periods of high volatility (e.g., 2008, 2020) followed by gradual mean reversion. The stationary nature of log returns is confirmed by ADF testing. Source: [P4].

can be considered a market–based proxy for the latent stochastic volatility process driving the VIX (Figure 3.11).

Therefore, using VVIX as the validation set was not intended to measure direct forecasting accuracy in the traditional sense. Instead, it was designed as a stringent test of model generalisation and robustness. The objective was to determine if a model trained exclusively on VIX data could learn the underlying structural dynamics of volatility deeply enough to generate outputs that correlate with, or statistically similar to, the behaviour of its own driving process (VVIX).

A successful validation in this context means the model's outputs on the validation set share statistical properties with the actual VVIX data and would provide strong evidence that the model has captured more than just superficial autocorrelation in the VIX series. It would suggest that the model has learned a fundamental aspect of the volatility generation process itself, thereby strengthening the case for using VVIX–related dynamics in future, more advanced forecasting models. See the graph below please.

We built ARIMA on scaled data and predict the fitted values on validation data. We also calculated the future trend growth with ARIMA. We then made a data frame of the fitted values and split it into train and test sets for our GAN model. ARIMA was used to capture long–term trends and reduce noise in the data, providing a stable baseline for our GAN–based feature generation. The combination leverages ARIMA's strengths in linear modelling and GANs' ability to model complex, nonlinear relationships. We added CNN conv1 layer as our discriminator and choose accuracy as our metric. Then in generator we used two layers of CNN and LSTM together and again we chose accuracy as our metrics. For training the model, we compiled discriminator and generator together and used the learning rate of 0.001 and beta of 0.5. We ran the model for a total of 1200 epochs and it gave us the internal training accuracy of the GAN's discriminator of 93.3%. We then used the fitted (predicted) values of GAN as features/inputs in SVM and RF. We defined the RF model and used train features for training and fitting and test features for predicting. The results for MAPE and MAE are quite encouraging. For SVM, we used

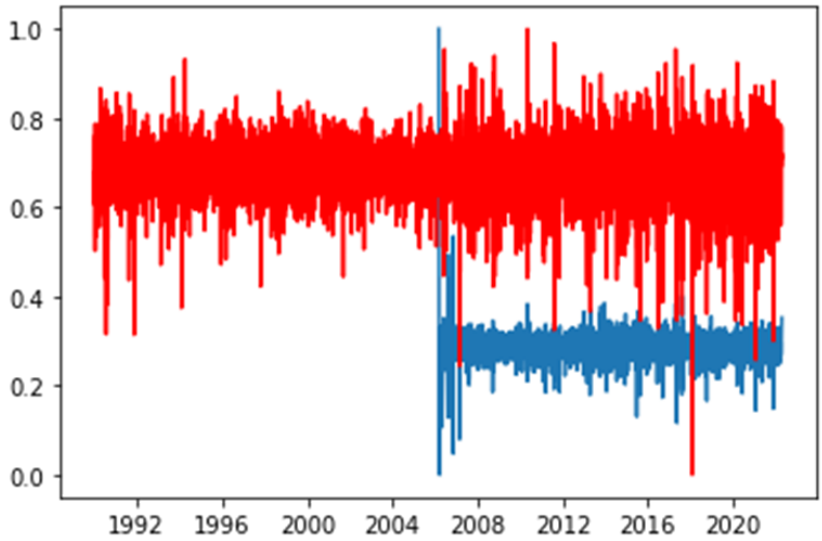

Figure 3.15: Comparison of normalised log returns: VIX (red) and VVIX (blue) from 1990–2022. VVIX data begins in 2006. Both series are scaled to [0,1] range to facilitate comparison. The co-movement suggests VVIX (volatility of volatility) captures similar market dynamics as VIX. Source: [P4].

grid search and fit the model to get a score of 98.5% on fitting.

Methodology

This study proposes a hybrid forecasting framework that combines a classical time series model (ARIMA), a deep generative model (GAN), and traditional machine learning regressors (SVM, RF) to predict financial volatility. The workflow consists of three main stages.

Stage 1: Data Pre–processing and Trend Extraction with ARIMA

First, the historical time series data was scaled to the range [–1, 1]. An ARIMA model was then fitted to this scaled data. The primary purpose of the ARIMA model was to capture the long–term linear trends and autoregressive patterns (i.e., the “memory“) in the series. The fitted values from the ARIMA model serve as a smoothed, detrended baseline, which is then used as the input for the subsequent generative model. This approach leverages ARIMA’s strength in linear modelling to provide a more stable foundation for the GAN.

Stage 2: Synthetic Feature Generation with a GAN

A Generative Adversarial Network (GAN) was developed to learn the complex, non–linear dynamics of the ARIMA–fitted time series and generate synthetic data with the same statistical properties. The generated data is then used as an input feature for the final prediction models.

- Generator Architecture: The generator is an LSTM–based recurrent neural network. It takes a random noise vector from a latent space of size 100 as input. This input is processed through two sequential LSTM layers, which are designed to learn and generate temporal patterns. The final layer is a Dense layer that outputs a sequence of the desired length, matching the input time series.

- Discriminator Architecture: The discriminator is a Convolutional Neural Network (CNN) designed for 1D sequence classification. It consists of a Conv1D layer followed by LeakyReLU activation and Dense layers. Its task is to distinguish between the real ARIMA–fitted sequences and the synthetic sequences produced by the generator.

- Training: The GAN was trained for 1200 epochs. The generator and discriminator were compiled together using an Adam optimizer with a learning rate of 0.001 and a beta_1 of 0.5. The training objective was to achieve high accuracy in the discriminator, indicating that the generator was producing high–quality, realistic time series data. The final internal training accuracy of the discriminator reached 93.3%.

Stage 3: Final Prediction with SVM and Random Forest

The trained GAN generator was used to produce a sequence of predicted values. These generated values, which encapsulate the non–linear patterns learned by the GAN, were used as input features for two final regression models: a Support Vector Machine (SVM) and a Random Forest (RF).

- Random Forest (RF): The RF model was trained on a training set of the GAN–generated features and evaluated on a test set.

- Support Vector Machine (SVM): A grid search was performed to optimize the hyperparameters for the SVM model, which was then trained and fitted on the same data, achieving a high fitting score.

The performance of both models was evaluated using Mean Absolute Percentage Error Mean Absolute Error (MAE).

Comprehensive Architectural Summary

1. Data Representation and Preprocessing

Each 100-point time-series segment is first reshaped into a 10×10 grid, padded to 32×32 (added zeros around it to become 32x32), and treated as a single-channel "image" so that convolutional layers can process it. Both training and test sets are cast into (N, 32, 32, 1) tensors before model training.

2. Generative Adversarial Network (GAN)

Discriminator

- Input: (32, 32, 1) images. (height, width, number of channels —- 1 greyscale, 3 for RGB).
- Architecture: Four Conv2D blocks with 128 filters, 3×3 kernels, stride 2, each followed by LeakyReLU ($\alpha = 0.2$) and Dropout (0.25); flattened to a Dense(1, sigmoid) output.
- Optimizer: Adam (lr $= 2\times10^{-4}$, $\beta_1 = 0.5$).
- Loss: Binary cross-entropy

Generator (Conv2DTransposed variant)

- Input: latent vector z of dimension 100.
- Architecture: Dense(128·4·4) → LeakyReLU → Reshape(4,4,128); three Conv2DTranspose(128, 4×4, stride 2, padding "same") blocks each with LeakyReLU; final Conv2D(1, 3×3, tanh) producing a (32,32,1) sample.
- Optimizer: Adam (lr $= 2\times10^{-4}$, $\beta_1 = 0.5$).
- Loss: Binary cross-entropy

Alternative Generator (1-D Conv + LSTM)

An additional prototype replaces Conv2DTranspose with a Time Distributed Conv1D stack, followed by two Bidirectional LSTM(100) layers and dense blocks; output is again reshaped to (32,32,1)

GAN Assembly and Training

- The GAN freezes the discriminator during generator updates and trains with Adam ($\mathrm{lr} = 10^{-4}$, $\beta_1 = 0.5$)
- Real sample generation: random index sampling from the training set with label 1
- Fake sample generation: generator output with label 0
- Training loop: for each step, update discriminator on real and fake batches, then update generator via GAN loss; invoked with latent_dim=100, 20 epochs, batch size 64

3. Downstream Forecasting (Conceptual)

The GAN's synthetic samples are intended as features for a hybrid CNN—-LSTM forecaster. The alternative generator (Conv1D + Bidirectional LSTM) illustrates how convolutional layers extract local patterns, while LSTMs model temporal dependencies before projecting back to the 32×32 grid.

Component	Key Settings
Latent dimension	100
Batch size	64
Epochs	20
Discriminator optimizer	Adam (2×10^{-4}, β_1=0.5)
Generator optimizer	Adam (2×10^{-4}, β_1=0.5)
GAN optimizer	Adam (1×10^{-4}, β_1=0.5)
Dropout	0.25 in discriminator blocks
Activation	LeakyReLU (α=0.2); tanh output

- Data Representation (10x10 grid, 32x32 image): A pre–processing trick.

The 1D time series (100 points long) is reshaped into a 2D 10x10 grid, which is then “padded“ (zeros added around it) to become a 32x32 grid. This is done so that 2D CNNs, which are powerful pattern detectors, can be used.

- Tensor: The primary data structure used in deep learning. It’s a multi–dimensional array. A tensor with shape (N, 32, 32, 1) represents:
  - N: Batch size (number of samples).
  - 32: Height of the “image.“
  - 32: Width of the “image.“
  - 1: Number of channels (1 for grayscale, 3 for a color RGB image).
- Conv2D: A 2D Convolutional Layer. It slides a small window (kernel) over a 2D input (like the 32x32 image) to detect spatial patterns.
- Filters (e.g., 128 filters): The number of different pattern–detectors in a convolutional layer. More filters allow it to learn more complex patterns.
- Kernel (e.g., 3x3): The size of the “window“ that scans the input. A 3x3 kernel looks at a 3x3 pixel area at a time.
- Stride (e.g., stride 2): The number of pixels the kernel jumps as it scans the input. A stride of 2 (instead of 1) moves 2 pixels at a time, which effectively downsamples the image (makes it smaller).
- Dropout (e.g., 0.25): A regularization technique to prevent overfitting (where the model just memorizes the training data). During training, it randomly “drops“ (sets to zero) a fraction (e.g., 25%) of neurons, forcing the network to learn more robust patterns.
- Flattened: A layer that converts a multi–dimensional tensor (like the output of a CNN) into a single 1D vector, typically to feed it into a Dense layer.
- Conv2DTranspose (or Deconvolution): The “opposite“ of a Conv2D

layer. It's used to upscale a small input into a larger output. The generator uses this to go from its small latent vector (reshaped to 4x4x128) up to the final 32x32x1 "image."

- Padding ("same"): A setting for a convolutional layer. It adds pixels (usually zeros) around the border of the input so that the output feature map has the same height and width as the input (when stride=1).

- Conv1D: A 1D Convolutional Layer. Similar to Conv2D, but it scans a 1D sequence. It's excellent for finding local patterns in time series or text.

- TimeDistributed: A special wrapper layer. It applies another layer (like a Conv1D) to every individual time–step of an input sequence. This allows a CNN to process each part of a sequence before it's fed to an RNN (like an LSTM).

- Batch Size (e.g., 64): The number of data samples processed in one iteration of training before the model's weights are updated.

We decompose volatility dynamics into linear and nonlinear components. An ARIMA model, fitted in a rolling, leakage–free manner to normalised VIX log–returns, provides a denoised baseline and/or residuals. On top, we train a GAN with an LSTM–based generator and a lightweight Conv1D discriminator to produce time–series features that are statistically indistinguishable from ARIMA–processed sequences. The downstream forecasters (SVM and Random Forest) consume these generated features to predict the next–step return. Training uses Adam with $\beta1$=0.5 and a small learning rate; in practice we prefer TTUR (lrD_DD>lrG_GG) and WGAN–GP for stability, together with early stopping monitored by distributional and temporal similarity metrics (e.g., MMD, ACF/PSD alignment) rather than discriminator accuracy. We evaluate forecast skill on VIX via MAE, RMSE, and sMAPE/MASE with Diebold—-Mariano tests against Naïve, ARIMA, residual LSTM, and N–BEATS baselines. Separately, we validate representation robustness on VVIX (volatility–of–volatility) as a domain–shift test, reporting correlation, spectral and Hurst/DFA similarity, and TSTR performance, to evidence that the learned representation captures structural drivers beyond mere autocorrelation.

Figure 3.12 represents the architecture of a deep neural network with several dense (fully connected) layers and Leaky ReLU activation functions that we used in our modelling. This deep neural network architecture is composed of multiple layers that work together to process and predict data effectively. It begins with an input layer, labelled input_1, which accepts input data of shape (None, 100). Here, “None“ indicates that the network can handle batches of any size, while each sample in the batch has a feature vector of size 100. This input layer serves as the foundation for subsequent transformations and computations throughout the network.

The architecture incorporates several fully connected (dense) layers that progressively refine and process the input data. The first dense layer maps the 100–dimensional input to a 256–dimensional space, outputting (None, 256). A second dense layer retains this dimensionality, allowing deeper feature representation. Following these, the third dense layer expands the feature space to 512 dimensions, and the fourth layer continues processing within this 512–dimensional space. The final dense layer aggregates all the features to produce a single scalar output per input sample, which we use for predicting continuous values.

To enhance the learning capabilities of the network, each dense layer is followed by a Leaky ReLU activation function. Unlike the standard ReLU, which can cause “dying neurons“ due to zero gradients for negative inputs, Leaky ReLU allows a small gradient for negative values, ensuring better gradient flow throughout the network and maintaining learning efficiency.

A reshape layer is included near the output, modifying the dense layer’s output into a specific shape of (None, 32, 32, 1). We used this to structure the data for further processing for convolutional operations. The final output layer is sequential and produces a single scalar value per input sample.

This architecture is highly flexible and versatile. The fully connected layers enabled it to handle a variety of input data types, while the use of Leaky

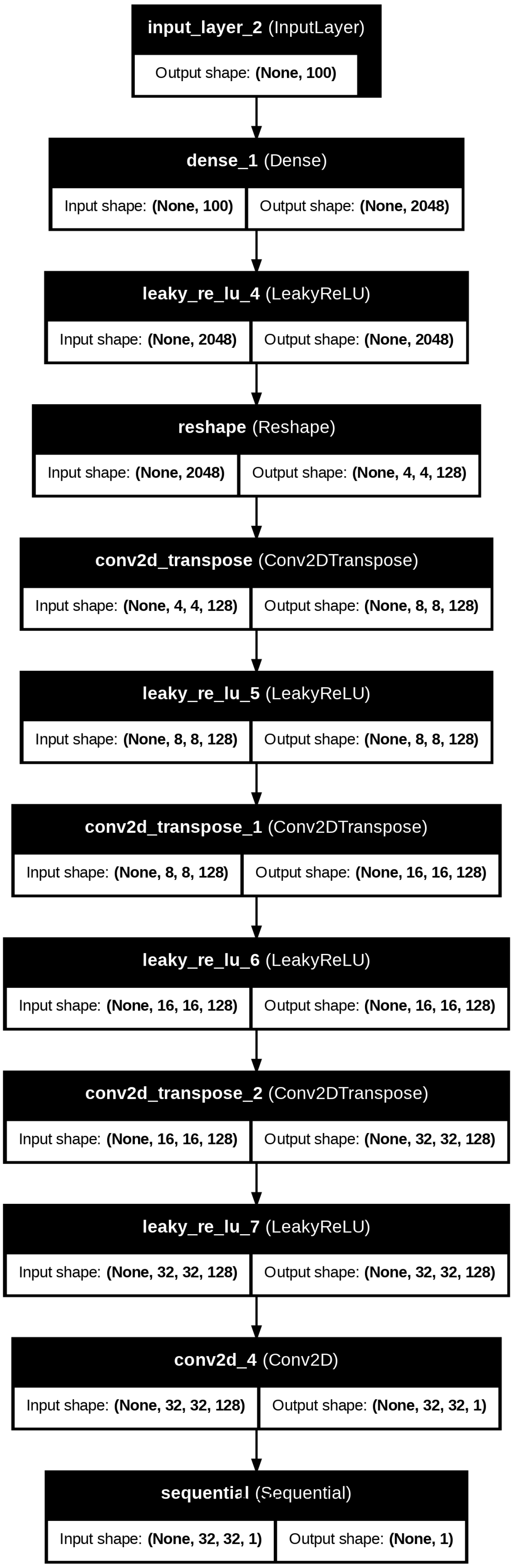

Figure 3.16: Source: [P4] DNN Architecture.

ReLU ensured stable training and prevents the vanishing gradient problem. Its depth allows it to capture complex hierarchical features, and the reshape layer adds adaptability for downstream tasks

3.5.2.2 Results

RF MAPE: 0.007 and MAE: 0.005. SVM MAPE: 0.006 and MAE: 0.004.

The MAPE and MAE values of 0.006 and 0.004 for SVM indicate excellent predictive accuracy, outperforming baseline models like ARIMA and even certain deep learning approaches. While NBEATS slightly outperforms Conv1D and LSTM in terms of MAE and MSE, its architecture may be less effective for high–frequency financial data where local patterns dominate, highlighting the need for task–specific tuning.

The performance of all models was evaluated using MAE, MSE, RMSE and MASE metrics. Table 3.1 below summarizes the key findings.

Table 3.2: Performance metrics

Model	**mae**	**mse**	**rmse**	**mase**
naïve_model	0.0865	0.0153	0.0865	0.999
model_2_LSTM	0.0412	0.00325	0.0570	0.672
model_3_N_BEATs	0.0411	0.00325	0.0570	0.671
model_5_ensemble	0.0413	0.00327	0.0572	0.673
model_6_turkey	0.0418	0.00342	0.0424	0.681

The results show that deep learning models like NBEATS (MAE: 0.0411) and LSTM (MAE: 0.0412) significantly outperform the Naïve forecast (MAE: 0.0865). Furthermore, the hybrid GAN–SVM approach yielded a highly competitive MAE of 0.004 and MAPE of 0.006, demonstrating the power of using GANs for sophisticated feature generation. The NBEATS model slightly outperforming LSTM with an MAE of 0.0412. This marginal difference may reflect the ability of NBEATS to generalise over long time horizons, whereas LSTM is better suited for capturing localised patterns in high–frequency

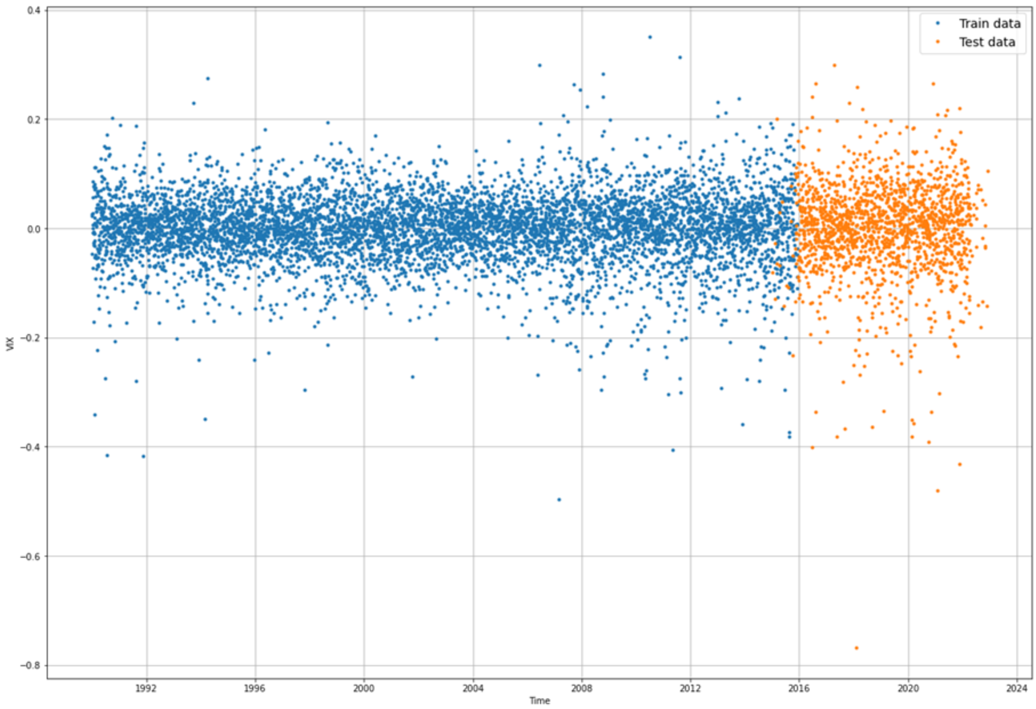

Figure 3.17: Train and test split (source: [P4]).

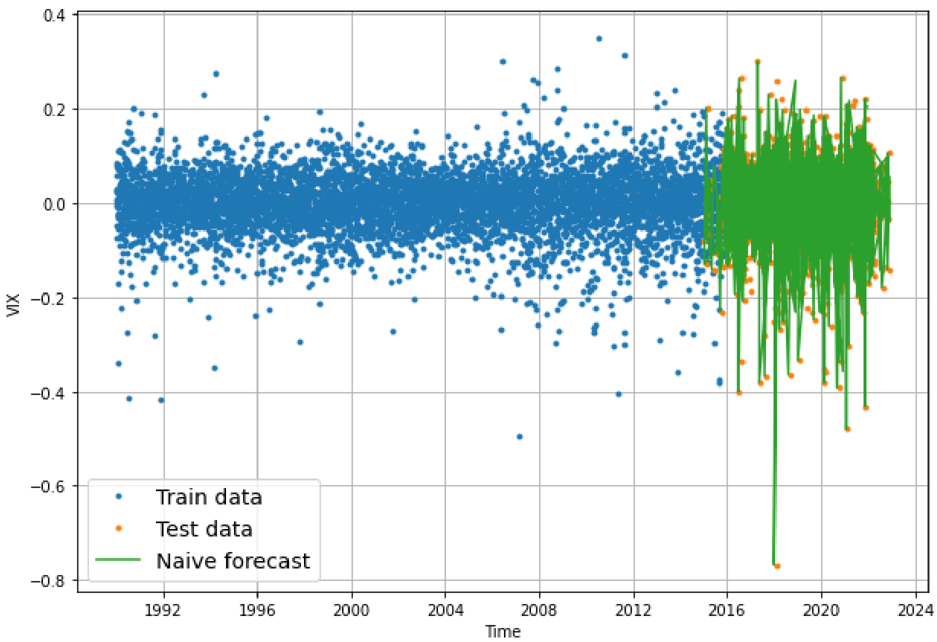

Figure 3.18: *Naïve forecast* (source: [P4]).

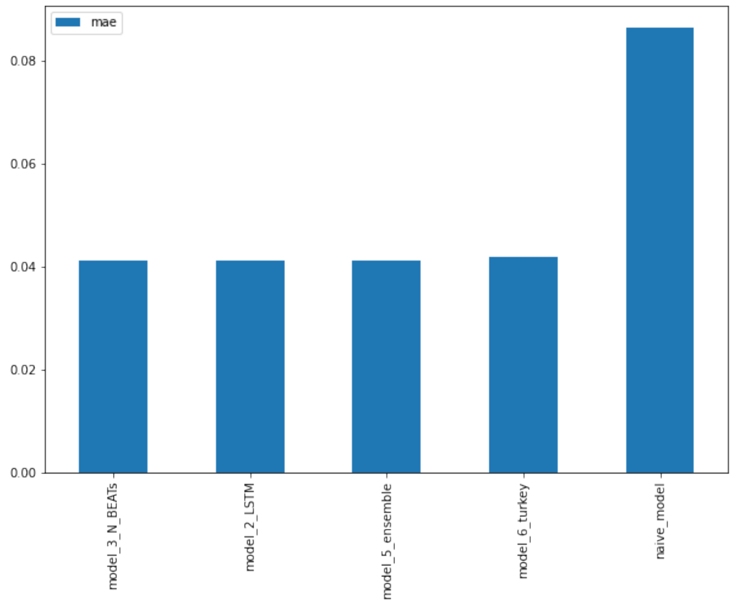

Figure 3.19: Model performance (source: [P4]).

data. However, both models demonstrate strong performance in identifying volatility trends, as evidenced by their MSE values.

The empirical investigations presented in this chapter have served two critical purposes in the overarching argument of this thesis. First, by implementing and evaluating a wide array of models from classical econometric methods like GARCH and ARIMA to advanced machine learning techniques including LSTM, NBEATS, and GAN–SVM hybrids, we have established a robust performance benchmark for existing volatility forecasting methodologies. The results, detailed in [P1], [P4] and summarised herein, demonstrate that while modern machine learning models can achieve strong predictive accuracy, they function primarily as sophisticated pattern–recognition systems. Their architectural design does not inherently reflect the deep structural properties of the data–generating process.

Second, and more fundamentally, this chapter provides the empirical grounding for the central hypothesis of this thesis: that financial volatility is a fractal phenomenon. The analysis of volatility clustering using K–Means and Fuzzy C–Means revealed distinct, persistent regimes that are not well–captured by models assuming simple random walks. More formally, the calculation of the Hurst exponent for the VIX time series consistently yielded values of $H \neq 0.5$, confirming the presence of long–range dependence and a "rough" character that defies classical diffusion–based assumptions.

These findings, taken together, create a clear and compelling justification for the theoretical work undertaken in the subsequent chapters. While the models benchmarked in this chapter are powerful, their limitations highlight a clear gap in the literature. To truly advance our understanding and ability to model volatility, it is not sufficient to apply ever–more–complex "black box" algorithms to the data. It is necessary to develop a new theoretical framework, which is presented in Chapter 5, from first principles that explicitly embeds the observed properties of fractality, memory, and uncertainty into its core mathematical structure.

Therefore, the empirical results of this chapter serve as the essential motivation for the development of the mixed fuzzy fractional Brownian motion framework, which is the primary subject of Chapter 5.

3.5.3 Conclusion

This chapter has established the experimental and methodological foundations used throughout the empirical part of the thesis. We first motivated the use of recurrent deep learning for sequential financial data by outlining the Long Short–Term Memory (LSTM) architecture, its gating mechanism (input, forget, output), and the associated state dynamics in (3.3.1)–(3.3.6). Building on this, we specified a likelihood–based training objective for volatility forecasting, deriving the Gaussian log–likelihood in (3.3.8) and adopting the corresponding negative log–likelihood loss in (3.3.9) to align model optimisation with probabilistic fit to observed returns.

We then discussed the vanishing–gradient phenomenon in recurrent training, highlighting how backpropagation through time (BPTT) produces multiplicative gradient chains (3.3.11) that can decay exponentially. The LSTM memory cell and gate structure were presented as the key architectural intervention that stabilises gradient flow via additive state updates and selective information retention, thereby improving learnability of long–horizon dependencies that are characteristic of volatility dynamics.

To ensure fair and leakage–free model comparison, the chapter introduced a standardised walk–forward validation design with rolling train–validation–test windows. Feature engineering was formalised via lagged returns, moving averages, and rolling volatility, while hyperparameter selection used grid search optimisation within each fold. This experimental protocol was applied consistently across classical baselines and modern architectures (LSTM, N–BEATS, GAN hybrids), enabling performance metrics to be aggregated across folds and interpreted as out–of–sample forecasting skill rather than artefacts of a single split.

A second major contribution of the chapter was the methodology for generating multifractal simulations under the Multifractal Model of Asset Returns (MMAR). We defined multifractality via scaling of moments (3.4.1), constructed partition functions (3.4.2), estimated the scaling function $\tau(q)$ via log–linear regression (3.4.3), and obtained the Hurst exponent through the root condition (3.4.4)–(3.4.6). The multifractal spectrum was then recovered through a Legendre transform (3.4.7), allowing estimation of the most probable Hölder exponent (3.4.8) and the cascade parameters λ and σ^2 in (3.4.9)–(3.4.11). These parameters support simulation of a lognormal multiplicative cascade and its cumulative trading time $\theta(t)$ (3.4.13), which is subsequently time–changed through fractional Brownian motion (3.4.14) to produce the composite MMAR process $X_t = B_H\big[\Theta(t)\big]$. The chapter also reviewed long–range dependence testing via the rescaled range statistic (3.4.17)–(3.4.19), empirically illustrating the contrast between persistent behaviour in long–horizon equity indices and stronger mean reversion in volatility, consistent with volatility clustering and regime shifts.

Finally, the chapter connected these methodological elements to empirical benchmarking. Volatility regimes were identified via K–Means and Fuzzy C–Means clustering, with silhouette–based model selection and transition–matrix asymmetries providing evidence of non–uniform regime switching and jump–

like behaviour. Forecasting benchmarks spanning naive methods, ARIMA, deep networks, and hybrid GAN–SVM/RF pipelines were then summarised, clarifying that discriminator accuracy reflects adversarial training dynamics rather than forecast skill, and emphasising that model evaluation must rely on out–of–sample error metrics. Collectively, these results motivate the thesis perspective that strong predictive performance alone is insufficient: volatility exhibits memory, fractality, and regime structure that warrant explicit modelling assumptions rather than purely black–box pattern recognition.

In summary, this chapter delivers (i) a reproducible experimental design for robust comparison of machine learning forecasters, and (ii) a principled multifractal simulation pipeline capable of generating long–memory, clustered–volatility synthetic data. These foundations provide the empirical motivation and computational apparatus required for the subsequent theoretical development, where the thesis advances a modelling framework that embeds fractality, memory, and uncertainty directly into the underlying stochastic structure.

CHAPTER 4 – EVALUATION METRICS

4.1 INTRODUCTION

Evaluation metrics are fundamental tools for assessing the performance of predictive models. This chapter focuses on three widely used metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). These metrics quantify prediction accuracy by comparing predicted values against observed data. While this chapter does not present novel evaluation metrics, it provides a detailed exploration of these standard measures, highlighting their strengths, limitations, and applications.

4.2 ROOT MEAN SQUARED ERROR

Root mean squared error (RMSE) [71] or root mean square deviation is one of the most used metrics to assess the quality of a forecast. It shows how far the predictions deviate from the actual values measured using the Euclidean distance. RMSE is commonly used in supervised learning applications because the RMSE uses and requires real measurements at each predicted data point.

RMSE is the square root of the mean of the square of all the errors. The use of RMSE is very common and is considered an excellent general–purpose measure of error for numerical predictions.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\Big(S_i - O_i\Big)^2}. \tag{4.2.1}$$

where Oi are the observations, Si predicted values of a variable, and n the number of observations available for analysis. RMSE is a good measure of accuracy, but only for comparing the prediction errors of different models or model configurations for a particular variable and not between variables, as it is scale dependent.

4.2.2 is a restatement of 4.2.1 but uses different notations for predicted and observed values. It emphasises the normalisation step by n, which ensures that RMSE reflects the average error magnitude rather than a cumulative measure. This normalisation is critical for datasets with varying sizes, as it provides a consistent scale for comparison. To calculate RMSE, compute the difference between prediction and actual for each data point, calculate the norm of residual for each data point, compute the mean of residuals and take the square root of this average.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\Big(S_i - O_i\Big)^2}. \tag{4.2.2}$$

where n is the number of observations, y(i) is the observed value, and $\hat{y}$(i) is its predicted value.

4.2.3 introduces the use of a norm (||·||) for computing residuals, connecting RMSE to broader mathematical contexts such as vector norms. The use of norms highlights RMSE's interpretation as a Euclidean distance between predicted and observed values in $\mathbb{R}^n$. This interpretation is especially relevant for geometric or spatial data where distances are meaningful.

$$\mathrm{RMSE} = \frac{1}{\sqrt{n}}\left\|y - \widehat{y}\right\|_2. \tag{4.2.3}$$

where N is the number of data points, *y(i)* is the i–th measurement, and *$\hat{y}$(i)* is its corresponding prediction.

By squaring errors and calculating a mean, the RMSE can be strongly influenced by some predictions much worse than others. If this is undesirable, using the absolute value of residuals and/or calculating median can provide a better idea of how a model performs for most predictions, without further influence from the anomalously poor predictions.

In machine learning, RMSE is a single number that is extremely useful for evaluating a model's performance, whether during training, cross–validation, or post–deployment monitoring. It is a proper scoring rule that is intuitive to understand and compatible with some of the most common statistical assumptions.

From a mathematical perspective and ignoring the division by n under the square root, the first thing we can notice is a resemblance to the formula for the Euclidean distance between two vectors in $\mathbb{R}^n$:

$$\operatorname{dist}(x, y) = \sqrt{\sum_{i=1}^{n} \left(x_i - y_i\right)^2}. \tag{4.2.4}$$

This tells us heuristically that RMSE can be thought of as some kind of (normalized) distance between the vector of predicted values and the vector of observed values.

If we keep the number of observations fixed as n, all it does is rescale the Euclidean distance by a factor of $\sqrt{(1/n)}$. If we assume that our observed values are determined by adding random "errors" to each of the predicted values, we obtain as follows:

$$y_i = \hat{y}_i + \epsilon_i \text{ for } i = 1, \ldots, n$$
$$\epsilon_1, \ldots, \epsilon_n \text{ independent, identically distributed errors}$$

These errors are random variables and might have Gaussian distribution with mean μ and standard deviation σ, but any other distribution with a square–integrable probability density function would also work. We want to think of $\hat{y}_i$ as an underlying physical quantity, such as the exact distance from A to B at a particular point in time. Our observed quantity y_i would then be the distance from A to the B as we measure it, with some errors coming from mis–calibration of our measure tool and measurement noise from external interference.

The mean μ of the distribution of our errors will correspond to the continuous bias that comes from poor calibration, while the standard deviation σ would correspond to the amount of measurement noise. Now imagine that we know the mean μ of the distribution for our errors exactly and would like to estimate the standard deviation σ. We can see that:

$$\begin{aligned}\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}\left(\widehat{y}_i - y_i\right)^2\right] &= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2\right] \\ &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\varepsilon_i^2\right] \\ &= \mathbb{E}\left[\varepsilon^2\right] \\ &= \mathrm{Var}(\varepsilon) + \left(\mathbb{E}[\varepsilon]\right)^2 \\ &= \sigma^2 + \mu^2.\end{aligned} \tag{4.2.5}$$

Here *E[. . .]* is the expectation, and *Var(. . .)* is the variance. We can replace the mean of the expectations $E[\epsilon_i^2]$ with the $E[\epsilon^2]$ where ϵ is a variable with the same distribution as each of the ϵ_i, because the errors ϵ_i are identically distributed, and thus their squares all have the same expectation.

We assumed that we already knew μ exactly. In other words, the persistent bias in our instruments is a known bias, rather than an unknown bias. So, we can also correct this bias from the start by subtracting μ from all our raw observations. That is, we can also assume that our errors have been distributed with mean $\mu = 0$. Plugging this into the equation above and taking the square root of both sides then yields:

$$\sqrt{\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}\left(\widehat{y}_i - y_i\right)^2\right]} = \sqrt{\sigma^2 + 0^2} = \sigma. \tag{4.2.6}$$

If we removed the expectation *E[. . .]* from inside the square root, it is exactly our formula for RMSE from before if the errors are unbiased ($\mu = 0$). The Central Limit Theorem tells us that as n increases, the variance of the quantity $\Sigma_i\ (\hat{y}_i - y_i)^2\ /\ n = \Sigma_i\ (\varepsilon_i)^2\ /\ n$ should converge to zero. In fact, a more robust form of the central limit theorem tells us its variance should converge to 0 asymptotically like $1/n$. This tells us that $\Sigma_i\ (\hat{y}_i - y_i)^2\ /\ n$ is a good estimator for $E[\Sigma_i\ (\hat{y}_i - y_i)^2\ /\ n] = \sigma^2$. But then the RMSE is a good estimator of the standard deviation σ of our error distribution. We should also now have an explanation for division by n under the square root of the RMSE. It allows us to estimate the standard deviation σ of the error for a typical single observation rather than some kind of "total error". By dividing by n, we keep this measure of error consistent as we move from a small collection of observations to a larger collection, and it just becomes more accurate as we increase the number of observations. To paraphrase it another way, RMSE is a good way to answer the question: "*How far off should we expect our model to be on its next prediction?*"

RMSE is a good measure to use if we want to estimate the standard deviation σ of a typical observed value from our model's prediction, assuming that our observed data can be decomposed in accordance with Doob's theorem:

Observed value= predicted value + predictably distributed random noise with mean zero

There is a limitation here which is that Doob's theorem requires that the stochastic process be a semimartingale but the fBm processes we are modelling are neither a semimartingale nor a weak martingale. It is a Gaussian process with stationary increments, but its sample paths are not of finite variation and its increments are not orthogonal to the past. Recently a concept of fractional semimartingale has gained traction but there is no consensus on the definition and the concept is still under development. The random noise here could be anything that our model does not capture (e.g., unknown variables that might influence the observed values). If the noise is small, as estimated by the RMSE, it generally means that our model predicts our observed data well, and if the RMSE is large, it usually means that our model does not consider important features located in our data. In machine learning and this paper, RMSE has a double purpose to serve as a heuristic for training models and to evaluate trained models for usefulness and accuracy. This raises an important question: What does it mean for RMSE to be 'small'? Firstly, "small" will depend on our choice of unit and the application we want. For training models, it doesn't matter which unit we use, because all we care about during training is having a heuristic to help us reduce the error at each iteration. We are only interested in the relative size of the error from one step to the next, not the absolute size of the error.

But in evaluating trained models in data science for usefulness and accuracy, we do care about units, because we aren't just trying to see if we're doing better than last time: we want to know if our model can help us solve a practical problem. The nuance here is that judging whether the RMSE is small enough will depend on how precise we need our model for our given application. There is never going to be a mathematical formula for this because it depends on things like human intentions i.e., purpose of the model, and risk aversion such as how much harm would be caused if this model made an inaccurate prediction. In other words, our choice of machine learning model (e.g., SVM, LSTM, naïve Bayes) and RMSE for predicting cancer or heart attack will be very different to the models and RMSE used for predicting stock market changes. Equally, the social and ethical considerations of the magnitude of errors resulting from self–driving *Uber* cars or facial recognition software are quite different to the social impact coming from errors in *Youtube* algorithm or routine *satnav* algorithms.

Besides the units, there is also another consideration: 'small' also needs to be measured relative to the type of model being used, the number of data points, and the history of training the model went through before you evaluated it for accuracy. At first this may sound counter–intuitive, but not when you consider the problem of over–fitting.

There is a risk of overfitting whenever the number of parameters in your model is large relative to the number of data points you have. For example, if we are trying to predict one real quantity y as a function of another real quantity x, and our observations are (x_i, y_i) with $x_1 < x_2 < x_3 \ldots < x(n)$, a general interpolation theorem tells us there is some polynomial $f(x)$ of degree at most $n+1$ with $f(x_i) = y_i$ for $i = 1, \ldots , n$. This means that if we choose our model to be a polynomial of order *n + 1*, by adjusting the parameters of the model (the polynomial coefficients) we should be able to bring the RMSE down to 0. This is true regardless of what our y values are. In this case the RMSE tells us nothing about the accuracy of our underlying model. We have been assured that we can modify the parameters to get RMSE=0 as measured on our existing data points, whether there is a relationship between two real quantities or not.

But it's not only when the number of parameters exceeds the number of data points that we might run into problems. Even if we don't have an unreasonable number of parameters, it is still possible that general mathematical principles along with moderate assumptions about our data may guarantee us with a high probability that by adjusting the parameters of the model, we can bring the RMSE below a certain threshold. If we are in such a situation, the fact that the RMSE was below this threshold might not say anything significantly about the predictive power of our model. Statistically speaking, the question we are going to be asking is not whether RMSE of our trained model is small but rather, what is the probability that the RMSE of our trained model on these observations is this small by a random chance.

4.3 MEAN ABSOLUTE ERROR (MAE)

Mean Absolute Error (MAE) [72] measures the average magnitude of errors between paired observations for the same variable. It is commonly used to evaluate prediction accuracy in time series analysis, regression tasks, and other contexts where interpretability is important. Examples include comparing predicted values to observed values, measuring changes between baseline and follow–up, or assessing differences between two measurement techniques.

MAE is calculated as:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i| = \frac{1}{n}\sum_{i=1}^{n}|e_i|\,. \tag{4.3.1}$$

It is an arithmetic average of the absolute error $/e_i/=/y_i$ —- $x_i/$, where y_i is the prediction and x_i is the true value. Other formulations may include relative frequencies as weight factors. One of MAE's key advantages is its robustness to outliers compared to metrics that square errors, such as Root Mean Squared Error (RMSE). However, MAE is scale–dependent, meaning it cannot be used to compare error magnitudes across datasets with different units or scales. This limitation should be considered when selecting MAE as an evaluation metric.

4.4 MEAN ABSOLUTE PERCENTAGE ERROR

Mean absolute percentage error (MAPE) also known as regression loss is added to all the models along with RMSE and MAE. MAPE's best value is 0.0 and inaccurate predictions can lead to arbitrarily large MAPE values, especially if some y_true values are very close to zero. This function returns undefined value of infinity when y_true is zero because division by zero is not defined. Here is an actual formula.

$$\text{MAPE} = \frac{100\%}{n} \sum_{t=1}^{n} \left| \frac{A_t - F_t}{A_t} \right|. \tag{4.4.1}$$

where A_t is the actual value and F_t is the forecast value. Their difference is divided by the actual value A_t. The absolute value in this ratio is added for each prediction time point and divided by the number of adjusted points n.

4.5 *CONCLUSION*

This chapter has presented the evaluation criteria used throughout the empirical studies of the thesis, focusing on three standard point–forecast error measures: RMSE, MAE, and MAPE. We defined each metric formally and clarified the distinct loss geometries they induce. RMSE, as an ℓ_2 measure, penalises large deviations quadratically and is therefore highly sensitive to outliers; it can be interpreted as a (normalised) Euclidean distance between the vectors of predictions and observations and, under unbiased square–integrable error assumptions, relates to the typical dispersion of the residuals. MAE, as an ℓ_1 measure, provides a more robust summary of typical error magnitude by reducing the influence of extreme residuals, at the cost of being less sensitive to rare but economically important tail events. MAPE offers a scale–free percentage interpretation that can be convenient for cross–series comparison, but it is unstable when realised values are close to zero and is undefined at zero, which limits its suitability for return series or other data with near–zero denominators.

Beyond definitions, the chapter has emphasised that the meaning of a "small" error is application–dependent and that metric choice should be consistent with the decision problem (e.g., tail risk sensitivity versus typical accuracy) and with the data scale. The discussion also highlighted that low in–sample errors can be misleading in high–capacity models due to overfitting, reinforcing the need for out–of–sample evaluation and appropriate validation protocols when comparing forecasting approaches.

Finally, we noted an important modelling nuance that motivates later chapters: many common interpretations of squared–error criteria are rooted in classical semimartingale diffusion settings and independence assumptions, whereas the volatility processes considered in this thesis may exhibit long–range dependence and roughness (e.g., fractional Brownian components). While RMSE/MAE/MAPE remain useful as practical, model–agnostic scoring rules for benchmark comparisons, their limitations underscore the importance of pairing them with careful experimental design and with diagnostics that reflect dependence structure and scaling behaviour. Taken together, the metrics formalised here provide a transparent and consistent basis for evaluating the forecasting models developed and benchmarked in subsequent chapters.

Accordingly, RMSE, MAE and, where numerically appropriate, MAPE, are used as the primary scoring rules for the comparative experiments in later chapters, supplemented by dependence- and scale-aware diagnostics when assessing models designed to capture long-memory or rough dynamics.

CHAPTER 5 – MIXED FUZZY FRACTIONAL BROWNIAN MOTION WITH JUMPS

5.1 *FUZZY FRACTIONAL BROWNIAN MOTION*

In the field of stochastic process modelling, the fractional Brownian motion (fBm) is often recommended as a replacement for Brownian motion as the driving process, particularly when studying phenomena characterised by long–range dependence. The fBm, with a Hurst exponent (H) ranging between 0 and 1, is a type of Gaussian process that possesses several advantageous properties, including long–range dependence, self–similarity, and stationarity of increments. Notably, when the Hurst exponent is different from 1/2, the fBm ceases to be a semi–martingale. This chapter is somewhat reliant on an earlier work of Doob–Meyer decomposition theorem which provides a representation of a submartingale as the sum of a martingale and an increasing predictable process. This theorem lays foundation for the decomposition of submartingales into their components. If we let *X(t)* be a submartingale with respect to a filtration (a sequence of sigma algebras), then there exists a unique, increasing, predictable process *A(t)* such that the martingale process *M(t)* can be defined as *M(t) = X(t) —- A(t).* In other words, according to Doob–Meyer the submartingale can be decomposed into the sum of a martingale *M(t)* and an increasing predictable process *A(t).* The proof for a continuous case is very complicated and can be found in [73]. However, there is a contradiction here as it was said that the fBM is not even a semi–martingale.

In [P2] we proposed to review and extend proofs in mixed fuzzy stochastic Brownian motion as novel equations suitable for modelling hybrid systems. These equations provide a flexible framework that can effectively capture the dynamics of complex systems characterized by both stochastic and fuzzy components. By combining the concepts of fuzzy logic and stochastic processes, we aim to address the modelling challenges associated with hybrid systems. Applications of these processes can be made in the telecommunication industry, where measurements of variables suffer from latency and financial industry, where participants are operating under uncertainty.

To illustrate the applicability of our approach, we utilise the Liouville form of the fractional Brownian motion. The Liouville form represents a particular representation of the fBm that allows for an enhanced analysis and interpretation of its properties. By leveraging this form, we can gain further insights into the behaviour of the mixed fuzzy stochastic Brownian motion, thereby facilitating their use in a wide range of modelling scenarios.

In summary, this paper proposes the utilisation of the mixed fuzzy fBm as a replacement for Brownian motion in stochastic process modelling, specifically focusing on its advantageous properties such as long–range dependence, self–similarity, and stationarity of increments. Furthermore, we introduce proofs and extensions for some novel equations, namely the mixed fuzzy stochastic Brownian motion with jumps, to address the modelling needs of hybrid systems.

5.1.1 PRELIMINARIES

In this section, the fundamental mathematical definitions and notations that underpin the theoretical framework of this chapter are established. The stochastic basis on which the asset price and volatility processes are constructed is defined as follows. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space equipped with a natural filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions of right–continuity and completeness. This filtration represents the information set available to market participants at any time t, encompassing both the continuous market evolution and discontinuous jump events.

Central to the modelling approach is the concept of Fractional Brownian Motion (fBm), which serves as the primary stochastic driver for the underlying asset dynamics. A standard fractional Brownian motion $B^H = \{B_t^H, t \geq 0\}$ with Hurst parameter $H \in (0,1)$ is defined as a continuous Gaussian process with zero mean, $B_0^H = 0$, and the covariance function $\mathbb{E}[B_t^H B_s^H] = \frac{1}{2}(t^{2H} + s^{2H} - |t-s|^{2H})$. This process generalises standard Brownian motion (obtained when $H = \frac{1}{2}$), providing the necessary mathematical flexibility to capture the self–similarity and long–range dependence ($H > \frac{1}{2}$) or anti–persistence ($H < \frac{1}{2}$) observed in the volatility time series analysed in the preceding chapters.

Building upon these foundational concepts, the specific stochastic process utilised in this framework is defined below.

Definition (Mixed Multifractal Fuzzy Fractional Brownian Motion – mmffBm)

The Mixed Multifractal Fuzzy Fractional Brownian Motion (mmffBm) is defined as a generalized stochastic process operating within a fuzzy probability space. It is constructed as a superposition of a standard Brownian motion $W(t)$ and an independent fractional Brownian motion $B^H(t)$, governed by fuzzy coefficients.

The continuous process $M^H(t, \tilde{\xi})$ is defined by:

$$M^H(t, \tilde{\xi}) = \tilde{a}W(t) + \tilde{b}B^H(t), \quad t \geq 0, \tag{5.1.1}$$

where:

- $W(t)$ is a standard Brownian motion (Hurst parameter $H = \frac{1}{2}$).
- $B^H(t)$ is an independent fractional Brownian motion with Hurst parameter $H \in (0,1)$ and $H \neq \frac{1}{2}$.
- $\tilde{a}$ and $\tilde{b}$ represent fuzzy numbers characterizing the weights of the standard and fractional components, respectively.

This mixed structure allows the process to exhibit properties of both semi-martingales (via $W(t)$) and long–memory processes (via $B^H(t)$). Furthermore, the combination of distinct Hurst exponents (0.5 and H) results in a process with time–varying local regularity, satisfying the condition for multifractality. This framework is specifically designed to derive results for regimes where $H > \frac{1}{2}$ (persistence) and $H < \frac{1}{2}$ (anti–persistence) under conditions of fuzzy uncertainty.

5.1.2 FRACTIONAL BROWNIAN MOTION

The fBm BH is a zero mean Gaussian process with the following covariance function:

$$C_{B^H}(t,s) = \mathbb{E}[B_H(t)\, B_H(s)] = \frac{1}{2}\left(t^{2H} + s^{2H} - |t-s|^{2H}\right), \qquad s,t \geq 0. \tag{5.1.2}$$

This was derived in Kolmogorov [74] and further studied in Mandelbrot and van Ness [4], where a stochastic integral representation was derived in terms of the Brownian motion —- see equation 3.5.14. $\int_0^t [(t-s)^{H-\frac{1}{2}}]dB(s))$ is called the Liouville form of a fBm and it holds many properties of the fBm except that it has non–stationary increments. The classical Itô theory cannot handle a stochastic integral in terms of fBm because B^{H} is not a semi–martingale if $H \neq \frac{1}{2}$. Two approaches have been used to evaluate stochastic integrals with respect to fBm. If *H> 1/2*, the Riemann–Stieltjes stochastic integral can be defined using Young's integral [75].

Suppose we have a financial time series representing the price of a stock over a certain period of time, denoted by *S(t).* We want to calculate the Riemann–Stieltjes stochastic integral of the stock price with respect to a cumulative trading volume, denoted by *V(t),* which represents the total number of shares traded up to time *t.*

1. Partition: Choose a partition of the time interval into subintervals. For simplicity, consider a uniform partition with n subintervals.

2. Piecewise Linear Approximation: Approximate $V(t)$ on each subinterval using a piecewise linear function. This can be achieved by interpolating the cumulative trading volume between the available data points.

3. Young's Integral Calculation: Calculate the Riemann–Stieltjes integral using Young's integral approach.

   (a) For each subinterval, compute the integral of $S(t)$ with respect to the piecewise linear approximation of $V(t)$.

   (b) This operation is performed by multiplying the stock price at each subinterval by the change in the approximated cumulative trading volume.

4. Refinement and Convergence: Repeat the above steps with increasingly finer partitions.

   (a) Refine the piecewise linear approximation of $V(t)$ at each step.

   (b) Recalculate the Riemann–Stieltjes stochastic integral using Young's integral.

   (c) By repeating this process, the integral will converge to a well–defined value, representing the integral of the stock price with respect to the cumulative trading volume over the given time period.

This methodology provides insights into the relationship between trading activity and stock price movements. In practice, to define $\int S(t)\, dV(t)$ via a Riemann—-Stieltjes/Young framework, one partitions *[0,T]* into $0 = t_0 < t_1 < \cdots < t_n = T$, uses piecewise approximations of V, and sums $S(t_i)[V(t_{i+1}) - V(t_i)]$. One refines this partition to ensure convergence. For $H > \frac{1}{2}$, fBm has sample paths of sufficient Hölder regularity to admit such an integral.

The general obstruction to using the classical Riemann–Stieltjes or Lebesgue–Stieltjes integral with B^H is fBm paths are not of bounded variation (a.s.) for any $H \, \epsilon$ (0,1) and even Brownian motion does not have bounded variation. This makes the Young's integral more flexible and easier to use in stochastics.

The Riemann–Stieltjes integral cannot be used to define an integral of a function with respect to fBm, because the Stieltjes measure is *sometimes* not defined for stochastic processes with non–stationary increments like fBm. A Stieltjes measure is a generalisation of the Lebesgue measure to allow for integration over sets with irregular boundaries. A Stieltjes measure is defined on a measurable space $(X,\ A)$, where X is a set and A is a σ–algebra of subsets of X. A Stieltjes measure is a function μ: $A \rightarrow [0,\ \infty]$ that satisfies the following properties:

i. $\mu(\emptyset) = 0$.

ii. *Finite Additivity:* $\mu(A \cup B) = \mu(A) + \mu(B)$ for all disjoint sets $A, B \in \mathcal{A}$.

iii. If $A\_1 \subset A\_2 \subset \cdots$ is a sequence of sets in A such that $A\_n \uparrow A$, then $\mu(A) = lim\_n{\rightarrow}\infty\ \mu(A\_n)$.

The third property is called the continuity property. It ensures that the Stieltjes measure is not sensitive to small changes in the sets that are being measured. Stieltjes measures can be used to define integrals of functions with respect to sets with irregular boundaries. For example, the Stieltjes measure can be used to define the integral of a function with respect to a Cantor set. These are fractal sets that are constructed by repeatedly removing a middle third of a line segment. Cantor sets have no Lebesgue measure, but they do have a Stieltjes measure. This means that we can define the integral of a function with respect to a Cantor set. Here is an example of how to use a Stieltjes measure to define the integral of a function with respect to a set with an irregular boundary:

Let $X = [0,\ 1]$ and let A be the σ–algebra of all Borel subsets of X. Let $g(t)$ be the Cantor function. The Cantor function is a continuous function that is defined on the interval [0, 1] and has the following properties:

i. $g(0) = 0$ and $g(1) = 1$.

ii. $g(t)$ is constant on each interval of the form $[a,\ b]$, where a and b are points of the Cantor set.

iii. The value of $g(t)$ on each interval of the form $[a,\ b]$ is equal to the average of the values of *g(a)* and *g(b).*

The Cantor function is a pathological function, which means it has some unusual features. The Cantor function g: [0,1]→ [0,1] (sometimes called the Devil's staircase) is the classic example of a continuous, non–decreasing function that is constant on the intervals removed in constructing the Cantor set and increases from *g(0)=0* to *g(1)=1.* It is a set of points that are sparse but not empty. Despite the Cantor set having Lebesgue measure zero, the function g induces a nontrivial Stieltjes measure μ_g defined by:

$$\mu_g((a, b]) = g(b) - g(a),$$

$$\mu(A) \;=\; g(b) \;-\; g(a),$$

where A is a Borel subset of X and *[a, b]* is the smallest interval of the form *[a, b]* that contains A which itself must be an interval. We can now use the Stieltjes measure to define the integral of a function *f(t)* with respect to the Cantor function g:

$$\int_0^1 f(t)\,\mathrm{d}g(t) = \int_{[0,1]} f(t)\,\mathrm{d}\mu_g(t),$$

where: *f(t)* is the function that we want to integrate; *g(t)* is the Cantor function or the cdf corresponding to the pdf ; μ is the Stieltjes measure induced by g.

The measure μ_g is singular with respect to Lebesgue measure: it is supported on the Cantor set (which has Lebesgue measure 0) and assigns it total mass 1. This integral is defined for all functions *f(t)* that are measurable with respect to the Lebesgue measure. The Cantor set has Lebesgue measure 0, but we can still define the Stieltjes measure for the Cantor set with respect to g. This is because the Stieltjes measure is a more general measure than the Lebesgue measure. This illustrates that Stieltjes measures can assign positive measure to sets (like the Cantor set) that have Lebesgue measure zero.

The Lebesgue measure is a measure that is defined on the Borel sets of the real line. The Borel sets of the real line are the sets that can be formed from the open sets of the real line using the operations of countable union, countable intersection, and complementation. The Stieltjes measure is a measure that is defined on the Borel sets of the real line. When we define the Stieltjes measure for the absolutely continuous function such as probability density function, or for the Cantor function, we are using the Borel σ*–algebra* (the same domain as Lebesgue measure) to define the sets on which the Stieltjes measure is defined on. However, we are not using the Lebesgue measure to define the values of the Stieltjes measure. The values of the Stieltjes measure are defined by the probability density function or the Cantor function. The Stieltjes measure for the Cantor function or for a pdf can be used to define the integral of a function with respect to that function.

The integral of a function with respect to the Cantor function can be used to calculate the area under the graph of the Cantor function. This does not equal the geometric area under the graph of g, which equals $\frac{1}{2}$. Moreover, while Young's integral provides a pathwise calculus suitable for rough integrators such as fBm with H $> \frac{1}{2}$, Lebesgue —- Stieltjes measures are measure–theoretic device built from distribution functions such as the Cantor function and, in the Cantor case, yield the Cantor (Lebesgue– Stieltjes) measure, which assigns total mass of 1 to the Cantor set.

In summary, both Young's integral and Stieltjes measures are valuable mathematical tools, but their applicability depends on the characteristics of the stochastic process being studied and the mathematical framework in which they are used. Young's integral is tailored to the needs of fBm and related processes, while Stieltjes measures are more general and can be applied to a wider range of scenarios.

If $H < \frac{1}{2}$, Malliavin calculus techniques are used to approximate $B_H(t)$ by a semi–martingale process [76]. The scope for this paper is to derive results for mixed fuzzy fractional Brownian motion (MFFBM) processes when $h > \frac{1}{2}$

and $h < \frac{1}{2}$. The theoretical framework we are relying on is from the Molchan–Golosov integral and adopted by [77]. It allows us to change the Hurst exponent using a specified transformation. Going back to the proposition that if H > $\frac{1}{2}$, then Riemann–Stieltjes integral can be defined using Young's integral [75].

$$B_{H,\varepsilon}(t) = \int_0^t (t-s+\varepsilon)^{H-\frac{1}{2}}\,\mathrm{d}B(s), \qquad \varepsilon > 0, \quad a = H - \tfrac{1}{2}. \tag{5.1.3}$$

$$B_{H,\varepsilon}(t) = a\int_0^t \varphi^{\varepsilon}(s)ds + \ \varepsilon^a B(t)$$

where $\varphi^{\varepsilon}(s) = \int_0^t (t-s+\varepsilon)^{a-1}]dB(s)$

Proof using integration by parts and chain rule technique:

$$\int u(x)\,v(x)\,\mathrm{d}x = u(x)\int v(x)\,\mathrm{d}x - \int \left(u'(x)\int v(x)\,\mathrm{d}x\right)\mathrm{d}x.$$

So,

$$\begin{aligned}
B_{H,\varepsilon}(t) &= \int_0^t (t-s+\varepsilon)^a\,\mathrm{d}B(s)\\
&= (t-s+\varepsilon)^a\int_0^t \mathrm{d}B(s) - \int_0^t a(t-s+\varepsilon)^{a-1}\left(\int_0^t \mathrm{d}B(s)\right)\,\mathrm{d}t\\
&= [(t-s+\varepsilon)^a B(t)]_0^t - \left[-a\int_0^t (t-s+\varepsilon)^{a-1}\left(\int_0^t \mathrm{d}B(s)\right)\,\mathrm{d}t\right]\\
&= (t-t+\varepsilon)^a B(t) - (0-s+\epsilon)^a B(0) + a\iint_0^t (t-s+\varepsilon)^{a-1}\,\mathrm{d}B(s)\,\mathrm{d}s\\
&= \varepsilon^a B(t) + a\iint_0^t (t-s+\varepsilon)^{a-1}\,\mathrm{d}B(s)\,\mathrm{d}s\\
&= a\int_0^t \varphi^{\varepsilon}(s)\,\mathrm{d}s + \varepsilon^a B(t),
\end{aligned}$$

$$\text{where} \qquad \varphi^{\varepsilon}(s) = \int_0^t (t-s+\varepsilon)^{a-1}\,\mathrm{d}B(s).$$

The process $\mathrm{B}_{\mathrm{H},\epsilon}$ (t) converges to B_{H}(t) in $\mathrm{L}^2(\Omega)$ when ϵ tends to zero because it is square integrable [78]. For an alternative proof of this theorem please see Thao Theorem 2.1 [78].

5.1.3 FUZZY RANDOM VARIABLE, FUZZY STOCHASTIC PROCESS, AND INTEGRAL OVERVIEW

This section provides some necessary theoretical background for fuzzy processes, and we formally prove certain results. In section 5.1.6 we will introduce fuzzy Poisson process $\hat{N}$ and how it is applied to fuzzy mean and variance. This section deals generally with fuzzy variables without going into specific membership functions. The material mainly comes from the fuzzy set theory of Zadeh [60]. Here we will define several measures that are used in this dissertation.

- Lebesgue Measure: Widely used to measure lengths, areas, volumes, and higher–dimensional extents of sets. It is defined on the Borel σ–algebra of subsets of Euclidean space. For example, the Lebesgue measure of an interval $[a, b]$ in one dimension is simply the length of the interval, given by $b - a$.
- Counting Measure: A fundamental measure in combinatorics and discrete mathematics that assigns the number of elements in a set. It is defined on any set and is particularly useful for counting finite sets. For instance, the counting measure of a finite set A is equal to the cardinality (number of elements) of A.
- Probability Measure: In probability theory, probability measures assign probabilities to events in a sample space. A probability measure is defined on a σ–algebra of subsets of the sample space and satisfies the properties of non–negativity, assigning probability 0 to the empty set, and countable additivity. For example, the uniform distribution on the interval $[0, 1]$ is a probability measure that assigns equal probability to all sub–intervals of $[0, 1]$.

- Hausdorff Measure: In geometric measure theory, the Hausdorff measure is used to quantify the "size" or "extent" of fractal sets. It is defined on metric spaces and captures the concept of dimension. For instance, the Hausdorff measure of a fractal set like the Cantor set or the Sierpinski triangle measures the "length" or "area" of the set in a fractal sense.

Let *K(R)* be the family of all nonempty, compact, and convex subsets of *R* (field). The Hausdorff metric d_H is defined by the following:

$$d_H(A, B) = \max\left\{\sup_{a\in A}\inf_{b\in B}|a-b|,\ \sup_{b\in B}\inf_{a\in A}|a-b|\right\}. \tag{5.1.4}$$

A metric space is said to be separable if there exists a countable dense subset. If *A, B, C* $\subset$ *K(R)* then from Minkowski addition and translation invariance:

$$d_H(A \oplus C,\ B \oplus C) \leq d_H(A,\ B). \tag{5.1.5}$$

Let Ω*, A, P* be a probability space. The mapping *F*: Ω–> *K(R)* is called A–measurable if it satisfies $\{w \, \epsilon \, \Omega : F(w) \cap C \neq \varphi\} \, \epsilon \, A$, for every closed set $C \subset R$. This means that for any element *w* that belongs to space Ω such that *F(w)* is disjoint with *C*, it is not equal to zero for every closed set $C \subset R$. A closed set means that all limiting and interior points belong to a set. A multifunction $F \subset M$ is said to be L^p —- integrably bounded if the p^{th} power is integrable for $p \geq 1$, iff (if and only if) there exists $h \subset L^p(\Omega,$ $A,\ P,\ R_+)$ such that the norms metric (the norm ||.|| induces a metric on *F*, known as the metric induced by the norm or the norm metric) such that $//F// \leq h$ P–a.e (almost everywhere – event holds for almost all elements in a set, with the exception of a negligible subset of elements according to the probability measure *P*), $R_+ = [0, \infty)$ and $//F// = d_H(F, \{0\}) = sup\ /f/,\ f \,\epsilon$ *F*. This means that the norm in terms of metric is the greatest value of *F*. For example, imagine we have a function *F* that takes some inputs and gives us sets (groups of things) as outputs. Now, this function *F* is special in two ways:

i. It doesn't make sets that are too big or too wild.

ii. we can measure how big these sets are.

The first point means that if we look at the sets produced by this function, they're not too crazy. They don't stretch out forever or become enormous. The second point is about measuring. We can measure the size of these sets, like giving them a number that says how big they are. Now, this function F can't be too unpredictable. It has to follow certain rules. If we square the sizes of its sets and add them all up, that should be a reasonable number (integrable). This is where the "L^p" comes in. "p" is just a number, and when it's 1 or bigger, it means things are behaving well.

To make sure these sets aren't too big, there's another function (let's call it "h") that keeps an eye on them. It's like a supervisor. This supervisor function, "h," is also measured in a special way, and it makes sure our function's sets don't get too big. So, in simple terms, an L^p–integrably bounded multifunction is like a machine that produces sets. These sets are controlled and supervised to make sure they're not too big, and we can measure their sizes using numbers.

The membership function u on an interval closed set: *R –> [0,1]* is defined for a fuzzy set $u \subset R$, where *u(x)* is the degree of membership of x in the fuzzy set u. Let us denote by *F(R)* the fuzzy sets u: *R –> [0,1]* such that $[u]^a \subset K(R)$ for every $a \subset [0,1]$, where $[u]^a = \{x \epsilon R: u(x) \geq a\}$. Define domain d_∞*: F(R) x F(R) –> [0,∞)* codomain /space of images and *F(R)* where d_∞ is a complete metric space by $d_\infty(u, v) = sup\ d_H\ ([u]^a, [v]^a)$, where $a \subset [0,1]$. This means that we are defining a function d_∞ that takes two inputs from the set *F(R)* which represents the set of all real valued functions and returns a value in the range *[0, ∞)* —- the non–negative real numbers. We will now show that d_∞ is a metric in F(R) and *(F(R) d_∞)* is a complete metric space. To accomplish this, we are considering *F(R)* equipped with the metric d_∞ to establish if it is a complete metric space. A metric space consists of a set equipped with a metric, which is a function that measures the distance between elements in the set. In this case, *F(R)* is the set, and d_∞ is the metric we are defining. The metric $d_\infty(u,v)$ is defined as the supremum or the least upper bound of the Hausdorff distance between the equivalence classes $[u]^a$ and $[v]^a$, where a belongs to the interval $[0,1]$. Here $[u]^a$ and $[v]^a$ represent alpha–cuts of functions u and v, respectively, under the relation defined by the parameter a. The Hausdorff distance measures the "closeness" between two sets by considering the furthest distance from each set to the other. It is the maximum distance of a set to the nearest point in the other set. In this case, the sets $[u]^a$ and $[v]^a$ are being compared, and their Hausdorff distance is computed. The supremum is then taken over all values of a in the interval [0,1]. By defining the metric d_∞ this way, we establish that $F(R)$ is a complete metric space. Completeness means that every Cauchy sequence in $F(R)$ converges to a limit within $F(R)$. In other words, $F(R)$ with the metric d_∞ satisfies the property that any sequence of functions that gets arbitrarily close to each other eventually has a limit within the same set. So, we have defined the metric d_∞ on the set of real valued functions $F(R)$ and established $F(R)$ equipped with d_∞ as a complete metric space. The metric measures the Hausdorff distance between equivalence classes of functions under a parameter a in the interval [0,1]. For every $u, v, w, z \in F(R), \lambda \in R$, we have the following properties:

i. Additive property $d_\infty(u + w,\, v + w) \leq d_\infty(u,\, v)$.

ii. Distribution property $d_\infty(u + v,\, w + z) \leq d_\infty(u,\, w) + d_\infty(v,\, z)$.

iii. Transitive (triangular) property $d_\infty(u,v) \leq d_\infty(u,w) + d_\infty(w,v)$;

iv. Scalar multiplication property $d_\infty(\lambda u,\ \lambda v) = |\lambda|\ d_\infty(u,v)$;

v. Non–negativity: $d_\infty(u, v) \geq 0$ for all $u, v \in F(\mathbb{R})$.

vi. Identity of indiscernibles: $d_\infty(u, v) = 0$ if and only if $u = v$.

vii. Symmetry: $d_\infty(u, v) = d_\infty(v, u)$ for all $u, v \in F(\mathbb{R})$.

viii. Consistency: $d_\infty(u, v) = d_\infty(v, u) = 0$ implies $u = v$.

Expanding the work of [74], we will now define a fuzzy random variable as a function that assigns fuzzy sets to elements in probability space. We will introduce the concept of measurability with respect to different metrics such as d_∞ and d_s and describe the properties and between these metrics and notion of a fuzzy random variable. We will break it down step by step:

Let (Ω, A, P) be a probability space, where Ω represents the sample space, A is a σ–algebra of subsets of Ω, and P is a probability measure defined on A. Given a set Ω, a σ–algebra of subsets of Ω, denoted as Σ, is a collection of subsets of Ω that satisfies certain properties. It is a fundamental concept in measure theory and serves as a foundation for defining measurable sets and functions. To formally define a σ–algebra, we require it to satisfy the following properties:

- The empty set ($\emptyset$) is in Σ: $\emptyset \in \Sigma$.
- Σ is closed under complementation: if A is in Σ, then its complement, denoted as A^c, is also in Σ.
- Σ is closed under countable unions: if $A_1, A_2, A_3, \ldots$ are subsets of Ω and each A_i is in Σ, then the union of all the A_i, denoted as $\bigcup A_i$, is also in Σ.

In other words, a σ–algebra is a collection of subsets of Ω that includes the empty set, is closed under taking of complements, and is closed under countable unions.

The purpose of defining a σ–algebra is to identify a set of subsets of Ω that can be considered "measurable" in a certain sense. The elements of Σ are often referred to as measurable sets, and they form a structure that allows us to define measures (such as probability measures) on these sets.

By specifying a σ–algebra on Ω, we establish a framework for defining and working with measurable functions, integrating functions, and constructing measures. The σ–algebra provides a systematic way to determine which subsets of Ω are considered "measurable" within the given context, allowing us to analyse and reason about these subsets in a rigorous mathematical manner. In summary, a σ–algebra of subsets of Ω is a collection of subsets that satisfies specific properties, including containing the empty set, being closed under complementation, and being closed under countable unions. It provides a foundation for defining measurable sets and functions, allowing us to study measures and perform various mathematical operations on these sets.

A fuzzy random variable is defined as a function $X\colon \Omega \to F(R)$, where $F(R)$ represents the set of fuzzy sets over the real numbers. This means that X assigns each element ω in Ω a fuzzy set $X(\omega)$ in $F(R)$.

For every $\alpha \in [0, 1]$, the mapping $[X]^{\alpha}\colon \Omega \to K(R)$ (where $K(R)$ denotes the set of fuzzy sets over R) should be A–measurable. This implies that the pre–images of measurable sets in $K(R)$ under $[X]^{\alpha}$ should be measurable sets in A.

Consider a metric ρ on the set $F(R)$ and let B_{ρ} be the σ–algebra generated by the topology induced by ρ. A fuzzy random variable X can be seen as a measurable mapping between the measurable spaces (Ω, A) and $(F(R), B_{\rho})$, and we refer to X as being $A/B\rho$–measurable.

The metric $ds(u, v)$ is defined as follows: $d_s(u, v) := inf\ max\ \{sup|\lambda(t)$ —- $t|$, $sup\ d_H(X_u(t), X_u(\lambda(t)))\}$, where $\lambda \in \Phi$, $t \in [0,1]$ and $t \in [0,1]$, Φ denotes the set of strictly increasing continuous functions $\lambda\colon [0,1] \to [0,1]$ satisfying $\lambda(0) = 0$ and $\lambda(1) = 1$. X_u and X_v are càdlàg (right–continuous with left limits) representations of the fuzzy sets u and v in $F(R)$, respectively. d_H represents the Hausdorff distance between fuzzy sets.

The space $(F(R), d_{\infty})$ is complete (meaning all Cauchy sequences converge) and non–separable (does not contain a countable dense subset), while the space $(F(R), d_s)$ is a Polish metric space (a complete and separable metric space). The reason for the difference in these properties lies in their respective definitions and metrics. The metric d∞, defined as the supremum of the Hausdorff distances between α–cuts $[u]^a$ and $[v]^a$, is a robust way to measure the “closeness” of fuzzy sets. It can be shown that the space $(F(R), d_{\infty})$ is complete, meaning all Cauchy sequences converge. However, $d\infty$ doesn't lend itself to finding a countable dense subset within the space $(F(R), d_{\infty})$. This makes $(F(R), d_{\infty})$ non–separable because it lacks the property of separability, even though it is complete. On the other hand, the metric d_s is defined differently, and results in a metric space that is both complete and separable. This metric is designed in such a way that it allows for the existence of a countable subset within the space ($F(R)$, d_s). Because ($F(R)$, d_s) is complete (all Cauchy sequences converge) and separable (contains a countable dense subset), it mees the criteria to be classified as a Polish metric space.

For a mapping $X\colon \Omega \rightarrow F(R)$ on the probability space (Ω, A, P), X is a fuzzy random variable iff X is A/B_{ds}–measurable and if X is $A/B_{d\infty}$–measurable, then it is a fuzzy random variable, but the opposite is not necessarily true. For X to be considered a fuzzy random variable, it must meet the condition of being A/B_{ds}–measurable. This means that X can be measured using d_s metric. The d_s metric is a specific way to compare how close or similar two fuzzy sets are. If X is also $A/B_{d\infty}$–measurable, it indicates that X can be analysed using the $d\infty$ metric. These two metrics, d_s and d_∞, are used to assess the closeness of fuzzy sets, but they do so differently. d_s looks at the behaviour of the fuzzy sets through continuous λ function and their values at different points. In contrast, d_∞ considers the worst–case scenario by looking at the maximum difference between the fuzzy set at any point. It is important to note that while A/B_{ds}–measurable is inclusive, the opposite is not necessarily true. Just because X is A/B_{ds}–measurable doesn't automatically mean it is $A/B_{d\infty}$–measurable. Therefore, $A/B_{d\infty}$–measurable is a more specific and essential criterion for defining a complete fuzzy random variable. In summary, the given explanations define a fuzzy random variable as a function that assigns fuzzy sets to elements in probability space. It introduces the concept of measurability with respect to different metrics, such as d_s and d_∞, and describes the properties and relationships between these metrics and notion of a fuzzy random variable.

Next, we expand the work of [79] in a fuzzy random variable and fuzzy stochastic process. A fuzzy random variable $X\colon \Omega \rightarrow F(R)$ is said to be L^p –integrably bounded for $p \geq 1$ if the α–cut $[X]^\alpha$ belongs to the space L^p $(\Omega, A, P; K(R))$ for every $\alpha \in [0,1]$. Here, the random variables $X, Y \in L^p$ $(\Omega, A, P; K(R))$ are identical if $P(d_\infty(X, Y) = 0) = 1$. This means that for X and Y to be considered identical, the probability that their distance d_∞

is equal to zero should be 1. d_∞ measures the "distance" or dissimilarity between two fuzzy variables X and Y. When the distance d_∞ between X and Y is zero, it indicates that X and Y are indistinguishable from each other in terms of their fuzzy characteristics. For a fuzzy random variable X: $\Omega \to$ *F(R)* and $p \geq 1$, the following conditions are equivalent: a) $X \in L^p(\Omega, A, P; F(R))$. b) The equivalence class $[X]^0$ belongs to $L^p(\Omega, A, P; F(R))$. c) The norm of $[X]^0$ denoted as $||[X]^0||$ belongs to $L^p(\Omega, A, P; R+)$, where *R+* represents the non–negative real numbers. The statement then asserts that these three conditions (a, b, and c) are equivalent. In other words, if any one of these conditions is true, the other two conditions will also be true, and vice versa. This demonstrates the relationships between the integrability of a fuzzy random variable, its equivalence class, and the norm of that equivalence class in different spaces. The notation $I := [0, T]$ signifies the interval from 0 to T, which means it is a closed interval from 0 to T inclusive and (Ω, A, P) represents a complete probability space equipped with a filtration $\{F_t\}$ $t \in I$. A filtration is a sequence or family of sub–σ–algebras $\{F_t\}$ $t \in I$ defined on a probability space. In this case, the filtration is defined for each t in the interval $I = [0, T]$. A complete probability space refers to a probability space where every subset of a null set (a set with measure zero) is also measurable and has measure zero. A complete probability space means that not only is the sample space, the σ–algebra, and the probability measure defined, but it also has an additional property that every subset of a null set (a set with measure zero) is also measurable. In other words, if you have a set of outcomes that is so unlikely (e.g., its probability measure is zero), then any subset of that unlikely set is also considered measurable. Being measurable means that it's part of the σ–algebra, so you can assign probabilities to it. The requirement that these subsets of null sets are measurable ensures that there are no hidden or unmeasurable sets that could potentially lead to issues in probability theory. It essentially fills in any gaps or irregularities in the probability space, making it more robust.

The filtration is an increasing and right–continuous family of sub–σ–algebras of A, which contains all P–null sets. The filtration $\{F_t\}$ $t \in I$ is said to be increasing, which means that as t increases, the sub–σ–algebra A_t becomes larger or contains more sets. In other words, if $s \leq t$, then A_s is a subset of A_t. This property ensures that the information or knowledge available at time s is also available at any later time $t \geq s$. The filtration $\{F_t\}t \in I$ is also right–continuous, which means that for any t in the interval I, the sub–σ–algebra A_t is equal to the intersection of all sub–σ–algebras A_s with $s > t$. In other words, the information available at time t is already available at any slightly earlier time t - ε, where ε is a small positive value. The filtration $\{F_t\}t \in I$ is required to contain all P–null sets. A P–null set is a subset of the sample space Ω with measure zero. By including all P–null sets in each sub–σ–algebra A_t, the filtration captures all negligible information about the underlying probability space.

A fuzzy stochastic process X is said to be d_∞–continuous if almost all its trajectories, denoted as $X(\cdot,\ \omega)$, are d_∞–continuous functions. Here, the trajectories refer to the mappings $X(\cdot,\ \omega)$ that associate each time point t in the interval I with a fuzzy set $X(t,\ \omega)$ in $F(R)$. The d_∞–continuity of these trajectories means that they exhibit continuity with respect to the d_∞ metric, which measures the dissimilarity between fuzzy sets. A fuzzy stochastic process X is considered measurable if the α–cut $[X]^\alpha\colon I \times \Omega \rightarrow K(R)$ is a measurable multifunction for all α in the interval [0, 1]. Here, $[X]^\alpha$ represents the α–cut of X with respect to the parameter α, and $K(R)$ denotes the set of fuzzy sets over the real numbers. A measurable multifunction $[X]^\alpha\colon I \times \Omega \rightarrow K(R)$ maps a measurable set in the time domain I and the sample space Ω to a set of measurable fuzzy sets in the value domain R. For example, Let $X(t)$ be a fuzzy stochastic process that represents the price of a stock at time t. The parameter α can represent the level of confidence that we have in the price prediction. For example, $\alpha = 0.95$ could represent a 95% confidence level. We can define the equivalence class $[X(t)]^\alpha$ as follows:

$[X(t)]^\alpha = \{Y \in K(R)\ /\ Y(x) \geq \alpha \text{ for all } x \in R\}$

The parameter α represents the level of confidence in the price prediction. It quantifies the minimum membership value required for a fuzzy set to be considered as part of the equivalence class $[X(t)]^\alpha$. This equivalence class contains all fuzzy sets that are at least as precise as *X(t)* with respect to the confidence level α. The equivalence class *[X(t)]*$^{\alpha}$ is defined as the set of fuzzy sets Y in *K(R)* that have a membership value greater than or equal to α for all x in the real numbers (R). In other words, *[X(t)]*$^{\alpha}$ contains all fuzzy sets that are at least as precise as *X(t)* with respect to the confidence level α. We can then show that the multifunction *[X(t)]*$^{\alpha}$*:* $I \times \Omega \to$ *K(R)* is measurable. Because the Borel σ–field on the fuzzy set space is generated by the α–cut projections, this means that the fuzzy stochastic process *X(t)* is measurable. By stating that the multifunction *[X(t)]*$^{\alpha}$*:* $I \times \Omega \to K(R)$ is measurable, we are asserting that the mapping from the Cartesian product of the index set I and the sample space Ω to the set of fuzzy sets *K(R)* is a measurable multifunction. This means that the pre–images of measurable sets in *K(R)* under *[X(t)]*$^{\alpha}$ are measurable sets in $I \times \Omega$. The pre–images, in this context, refer to the subsets of the domain $I \times \Omega$ that get mapped to a specific set in the target space *K(R)* under the mapping *[X(t)]*$^{\alpha}$*:* $I \times \Omega \to$ *K(R)*.To put it simply, if we have a set A in *K(R)*, the pre–image of A under *[X(t)]*$^{\alpha}$ is the set of all points *(i, ω)* in the domain $I \times \Omega$ such that *[X(t)]*$^{\alpha}$*(i, ω)* belongs to A.

In other words, the pre–image of a set A in *K(R)* consists of all the elements *(i, ω) in* $I \times \Omega$ that get mapped to A under the mapping *[X(t)]*$^{\alpha}$. By saying that the pre–images of measurable sets in *K(R)* under *[X(t)]*$^{\alpha}$ are measurable sets in $I \times \Omega$, we mean that if we consider a measurable set A in *K(R)*, the pre–image of A under *[X(t)]α* is a measurable set in $I \times \Omega$. This implies that the behaviour of the fuzzy stochastic process *X(t)* with respect to such pre–images is well–behaved and can be characterised by measurable sets in the domain $I \times \Omega$.

For example, we have a fuzzy stochastic process *X(t)* that represents the temperature at a particular location over time. The parameter α represents the confidence level in our temperature predictions. The domain of the process is the Cartesian product of the time index set I and the sample space Ω. Let's assume $I = \{1, 2, 3\}$ represents three time points and $\Omega = \{A, B\}$ represents two possible outcomes (e.g., sunny, or cloudy days). The target space $K(R)$ represents the set of fuzzy sets over the real numbers. Equivalence class– for a specific time point t, let's say $X(t)$ generates a fuzzy set that represents the temperature with respect to the confidence level α. Measurable sets in $K(R)$, let's consider a measurable set A that represents a specific range of temperatures. Now, let's define the pre–image of A under $[X(t)]^{\alpha}$: the pre–image of A under $[X(t)]^{\alpha}$ is the set of all (i, ω) in $I \times \Omega$ such that $[X(t)]^{\alpha}(i, \omega)$ belongs to A. In other words, it consists of all the time points and sample outcomes for which the fuzzy set generated by $X(t)$ at that particular time and outcome falls within the range represented by A. For example, let's assume A represents temperatures between 20 and 30 degrees Celsius. If $[X(t)]^{\alpha}(2, B)$ represents a fuzzy set with membership values indicating temperatures around 25 degrees Celsius, then $(2, B)$ would be in the pre–image of A because the temperature falls within the range represented by A. In this case, the pre–image of the measurable set A under $[X(t)]^{\alpha}$ would be a subset of $I \times \Omega$ containing all the time points and sample outcomes where the fuzzy sets generated by $X(t)$ fall within the temperature range represented by A.

We propose that the condition that $[X]^{\alpha}$ be measurable for all α in the interval [0, 1] ensures that the fuzzy stochastic process X can be integrated with respect to time and the sample space.

$B(I)$ is the Borel σ–algebra, which consists of all subsets of the interval I. To determine measurability, $[X]^{\alpha}: I \times \Omega \to K(R)$ should be $B(I) \otimes A$–measurable. Here, $B(I) \otimes A$ represents the product σ–algebra, which is generated by the Borel σ–algebra $B(I)$ and the σ–algebra A. This requirement ensures that the equivalence class $[X]^{\alpha}$ is measurable with respect to the joint σ–algebra generated by $B(I)$ and A. In this context, $K(R)$ denotes the set of fuzzy sets over the real numbers. Fuzzy sets are sets that assign membership degrees or degrees of truth to elements, allowing for a more flexible representation of uncertainty or partial truth. $B(I)$ refers to the Borel σ–algebra, which is the σ–algebra generated by the open sets in the interval I. The Borel σ–algebra consists of all open sets of I that can be formed by taking unions, intersections, and complements of open sets. $B(I) \otimes A$ represents the product σ–algebra, which is the smallest σ–algebra generated by the sets of the form $B \times A$, where B is a Borel set from $B(I)$ and A is a set from the σ–algebra A. The requirement of $[X]^{\alpha}$ being $B(I) \otimes A$–measurable ensures that the equivalence class $[X]^{\alpha}$ is measurable with respect to the joint σ–algebra generated by $B(I)$ and A. This joint σ–algebra allows us to analyse and reason about the measurability properties of the fuzzy stochastic process X across both the time domain (Borel sets in I) and the probability space (sets in A). Suppose we have two sets: $B(I)$, which represents the Borel σ–algebra generated by the open sets in the interval I, and A, which represents a σ–algebra defined on a sample space Ω. We want to generate a σ–algebra that captures the joint measurability of sets from $B(I)$ and A. The product σ–algebra $B(I) \otimes A$ is defined as the smallest σ–algebra generated by sets of the form $B \times A$, where B is a Borel set from $B(I)$ and A is a set from the σ–algebra A. Let's consider a specific example to illustrate this concept. Assume that I is the interval [0, 1] and A is the power set of $\Omega = \{a, b\}$, which contains all possible subsets of Ω. $B(I)$: $B(I)$ is the Borel σ–algebra on I, generated by the open sets in I. It includes sets such as open intervals, closed intervals, half–open intervals, and countable unions or intersections of such intervals. A: A is the power set of Ω, which includes all subsets of Ω. In this case, $A = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$. To construct the product σ–algebra $B(I) \otimes A$, we consider all possible combinations of Borel sets from $B(I)$ and sets from A. For example, let's take $B = (0.2, 0.8)$ (an open interval in I) and $A = \{a\}$ (a singleton set in A). The set $B \times A$ is the Cartesian product of B and A, which results in the set $\{(x, a) \mid 0.2 < x < 0.8\}$. Similarly, we can consider other combinations of Borel sets from $B(I)$ and sets from A. The product σ–algebra $B(I) \otimes A$ is then defined as the smallest σ–algebra that includes all such sets $B \times A$ for all possible choices of B from $B(I)$ and A from A. In this example, the product σ–algebra $B(I) \otimes A$ would consist of sets such as $B \times A$, where B represents Borel sets from $B(I)$ and A represents sets from A. It captures the joint measurability of sets from $B(I)$ and A, allowing us to reason about sets that involve both the real number domain and the sample space. Formally, if we have a function $X{:}I\times \Omega \rightarrow \boldsymbol{R}$, being "measurable" means that for every Borel set $B \subseteq R$, the pre–image $X^{--1}(B)$ is in the product σ

–algebra $B(I) \otimes A$. In other words, we want to handle pairs (i, ω) where $i \in I$ and $\omega \in \Omega$. Imagine we want to analyse rainfall data for different cities over time. We have a set of cities, say {New York, London, Paris}, and we want to consider different time intervals for our analysis. Our set of cities is like the interval I in our previous discussion. The time intervals could represent events or outcomes (such as days with rainfall data) in our probability space, denoted by the σ–algebra A. Now, we want to understand the measurability of rainfall data across both the cities and time intervals. This is where the product σ–algebra $B(I) \otimes A$ comes into play. For each city, we can create Borel sets, which are sets formed from open intervals. In this context, these could represent different subsets of time periods for each city. For example, we might have the Borel set for New York's rainy days, the Borel set for London's rainy days, and so on. σ-Algebra A – this represents our probability space, which includes various outcomes or events in our analysis. In this case, each outcome or event is a different time interval (e.g., a specific date or range of dates). Now, the product σ–algebra $B(I) \otimes A$ is the smallest σ–algebra that contains all possible combinations of sets formed by pairing a Borel set from $B(I)$ with a set from σ–algebra A.

In our example, it means we're looking at measurable sets that are related to both specific cities (Borel sets) and specific time intervals (σ–algebra A). It allows us to answer questions like: "How does rainfall in New York relate to specific time intervals, and how does it compare to rainfall in London during those times¿' So, $B(I) \otimes A$ helps us explore the measurability of events that occur at the intersection of these two domains: cities and time intervals. Although cities are not literally an interval in $\boldsymbol{R}$, because strictly, an interval in $\boldsymbol{R}$ is an uncountably infinite set with Borel structure, whereas {New York, London, Paris} is a finite set, heuristically, the product σ–algebra allows us to consider pairs of events "(City in some city–set) $\times$ (Time in some time–set)".

A process X is said to be nonanticipating if, for every $\alpha \in [0, 1]$, the multifunction $[X]^{\alpha}$ is measurable with respect to the σ–algebra N. The σ–algebra N is defined as follows: $N := \{A \in B(I) \otimes A : A^t \in A$ for every t

$\in I\}$, where A^t represents the set of outcomes ω such that the pair (t, ω) belongs to the set A. In other words, N consists of subsets of $I \times \Omega$ for which the corresponding time t belongs to the set A.

Before proceeding further, we need to state a few theorems.

Definition 5.1

The Fubini's theorem [80], also known as the Fubini–Tonelli theorem, is a fundamental result in measure theory and integration theory. It provides a condition under which the iterated integrals of a measurable function over a product space can be computed by performing separate integrations. Formally, let (Ω_1, A_1, μ_1) and (Ω_2, A_2, μ_2) be two measure spaces, where Ω_1 and Ω_2 are sets, A_1 and A_2 are σ–algebras of subsets of Ω_1 and Ω_2, respectively, and μ_1 and μ_2 are measures defined on A_1 and A_2, respectively. Consider the product measure space (Ω, A, μ) defined as the product of the two measure spaces, where $\Omega = \Omega_1 \times \Omega_2$, $A = A_1 \otimes A_2$ (the product σ–algebra), and μ is the product measure. Fubini's theorem states that if $f\colon \Omega \rightarrow [0, +\infty]$ is a non–negative measurable function, then the iterated integrals $\int\Omega_1\int\Omega_2$ $f(\omega_1, \omega_2)\ d\mu_1(\omega_1)\ d\mu_2(\omega_2)$ and $\int\Omega_2\int\Omega_1\ f(\omega_1, \omega_2)\ d\mu_2(\omega_2)\ d\mu_1(\omega_1)$ are both well–defined and equal, provided at least one of them is finite. In other words, under appropriate conditions, the order of integration can be interchanged without affecting the result.

Definition 5.2

The Aumann integral, also known as the integral with respect to a fuzzy measure, is a generalisation of traditional integration theory that applies to fuzzy sets to extend the concept of integration to accommodate uncertainty and partial truth [81].

Traditional integration measures the "area under the curve" or the accumulation of quantity over a given range. With fuzzy sets, where membership degrees represent degrees of truth or uncertainty, the concept of area or accumulation needs to be generalised. The Aumann integral achieves this by integrating a fuzzy set with respect to a measure, which assigns a degree of "importance" or "size" to subsets of a set, similar to how a traditional measure assigns a value to subsets. Formally, let X be a fuzzy set defined on a set Ω and let μ be a ~~fuzzy~~ measure defined on the power set of Ω. The Aumann integral of X with respect to μ, denoted as $\int X \, d\mu$, is a generalised notion of integration that takes into account the degrees of truth or membership of X. The Aumann integral is typically defined in terms of a level–wise integral. For each α in the interval $[0, 1]$, the Aumann integral $\int X \, d\mu$ is equal to the integral of the α–cut of X (i.e., the subset of elements with a membership degree of at least α) with respect to the α–cut of the measure μ. The Aumann integral allows for the accumulation and aggregation of fuzzy information by integrating each confidence band (level set) and reassembling the resulting bands into a fuzzy set.

Let's illustrate the difference between classical product measure and an Aumann integral of a fuzzy function under measure. In standard case, let's consider a product space where X is the set of real numbers $\mathbb{R}$, and L is the lattice of closed intervals $[a, b]$ on the real line, ordered by inclusion. Suppose we have a function $f\colon \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$ defined as $f(x, [a, b]) = x^2$. Here, x represents a point in the real line, and $[a, b]$ represents a closed interval. To compute an integral of f, we divide the lattice of closed intervals into levels based on the length of the intervals, such as intervals of length 1, intervals of length 1/2, intervals of length 1/4, and so on. We then integrate the function $f(x, [a, b]) = x^2$ over each level, considering the measure on the real line. Finally, we combine these integrals according to lattice operations to obtain the final integral. The standard integral of the function $f(x,[a,b]) = x^2$ over the product space X x L, where X is the set of real numbers and L is the lattice of the closed intervals on the real line is given by:

$$\int_{X \times L} f\left(x, [a, b]\right) d\mu(x)\, d\lambda\left([a, b]\right) = \int_{-\infty}^{\infty} \int_{0}^{1} x^2 \, d\lambda\left([a, b]\right)\, d\mu(x)$$

where μ is the Lebesgue measure on the real line and λ is the counting measure on the lattice L. Since the counting measure gives a weight of 1 to each interval, the integral over L is simply the sum of the values of $f(x,[a,b])$

over all intervals $[a,b] \epsilon L$. Therefore, the integral becomes:

$$\int_{X\times L} f\left(x,[a,b]\right) d\mu(x)\, d\lambda\left([a,b]\right) = \int_{-\infty}^{\infty} \sum_{a\leq b} x^2 d\mu(x)$$

The sum over all intervals [a,b] can be replaced by an integral over the real line, since the intervals are contiguous and non–overlapping as we are summing over disjoint collection of intervals.

$$\int_{X\times L} f\left(x,[a,b]\right) d\mu(x)\, d\lambda\left([a,b]\right) = \int_{-\infty}^{\infty} \int_{-\infty}^{x} x^2 dy dx$$

Evaluating the inner integral, we get:

$$\int_{X\times L} f\left(x,[a,b]\right) d\mu(x)\, d\lambda\left([a,b]\right) = \int_{-\infty}^{\infty} \frac{x^2}{3} dx$$

Evaluating the outer integral, we get:

$$\int_{X\times L} f\left(x,[a,b]\right) d\mu(x)\, d\lambda\left([a,b]\right) = \infty$$

Therefore, the standard integral of the function $f(x, [a,b]) = x^2$ over the product space $X\times L$ is infinite.

We construct a scenario showing a measure and a fuzzy function, then compute its Aumann integral level by level. This highlights how fuzzy integration can differ from standard measure integration.

Let $\Omega = \{\omega 1,\omega 2,\omega 3\}$. We define a measure μ on the power set of Ω with the following properties:

i. $\mu(\emptyset)=0$ and $\mu(\Omega)=1$ (normalization).

ii. Monotonicity: if $A\subseteq B\subseteq\Omega$, then $\mu(A)\leq\mu(B)$.

iii. Fuzzy property: μ need not be additive. For simplicity, we will just define it directly on subsets: $\mu(\{\omega 1\})=0.2$, $\mu(\{\omega 2\})=0.5$, $\mu(\{\omega 3\})=0.4$,

$\mu(\{\omega 1,\omega 2\})=0.6$, $\mu(\{\omega 1,\omega 3\})=0.5$, $\mu(\{\omega 2,\omega 3\})=0.8$, $\mu(\Omega)=1$. We ensure monotonicity by checking each subset is $\leq$ its superset in measure.

Define a fuzzy function X: $\Omega \to \widetilde{F}(R)$, i.e. each w maps to a fuzzy set $X(w) \in R$. Let,

Let $X(\omega_1)$ be a fuzzy set in $\mathbb{R}$, concentrated mostly around 1 and 2, with membership function

$$\mu_{X(\omega_1)}(x) = \begin{cases} 1, & x = 1, \\ 0.5, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Let $X(\omega_2)$ be a fuzzy set in $\mathbb{R}$, concentrated around 2 and 3, with membership function

$$\mu_{X(\omega_2)}(x) = \begin{cases} 0.8, & x = 2, \\ 0.4, & x = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Let $X(\omega_3)$ be a fuzzy set in $\mathbb{R}$, concentrated around 5, with membership function

$$\mu_{X(\omega_3)}(x) = \begin{cases} 1, & x = 5, \\ 0, & \text{otherwise.} \end{cases}$$

For $\alpha \in (0,1]$, define the α-cut by

$$X(\omega_i)_\alpha = \left\{ x \in \mathbb{R} \;\middle|\; \mu_{X(\omega_i)}(x) \geq \alpha \right\}.$$

Examples:

$$X(\omega_1)_{0.5} = \{1,2\}, \qquad X(\omega_1)_{0.9} = \{1\}.$$

$$X(\omega_2)_{0.4} = \{2,3\}, \qquad X(\omega_2)_{0.8} = \{2\}.$$

$$X(\omega_3)_\alpha = \{5\} \quad \text{for all } \alpha \in (0,1],$$

since $\mu_{X(\omega_3)}(5) = 1$. Next, we combine the α–cuts via the fuzzy measure μ:

$$S_\alpha = \bigcup_{\omega\in\Omega} \{\, \omega \mid X(\omega)_\alpha \neq \varnothing \,\}\,.$$

Weight these subsets by the fuzzy measure μ. In some definitions, it might form a "set of integrals of the α–cuts or a single numeric value by summation/integration.

$$\mathrm{Aum}(X) = \bigcup_{\alpha\in(0,1]} \alpha \cdot \left(\int_\Omega X(\omega)_\alpha \, \mathrm{d}\mu(\omega)\right),$$

Compute partial contributions from each $\omega \in \Omega$ with coefficient *$\mu(\{\omega\})$:*

$$Aum(X)(\alpha) = \sum_{\omega\epsilon\Omega} X(\omega)_\alpha \ \bullet \mu\left(\{\omega\}\right),$$

If we interpret the integral as a Minkowski–like sum. So, for the sets that appear at $\alpha = 0.5$, if we form a set–valued integral, the union over ω in Ω of α=0.5:

$$\bigcup_{w\in\Omega} X(\omega)_{0.5} = \{1, 2, 5\},$$

where 3 is excluded because it only had membership 0.4 in ω_2.

Let $\Omega = \{\omega_1, \omega_2, \omega_3\}$. Use a σ–additive probability measure P on Ω with singleton weights normalized from (0.2, 0.5, 0.4):

- $p_1 = P(\{\omega_1\}) = 2/11$,
- $p_2 = P(\{\omega_2\}) = 5/11$,
- $p_3 = P(\{\omega_3\}) = 4/11$.

Fuzzy–valued map and α–cuts. Define $X : \Omega \to \mathbb{F}(\mathbb{R})$ by its membership values:

- $X(\omega_1)$: $\mu(1) = 1$, $\mu(2) = 0.5$, 0 elsewhere.
- $X(\omega_2)$: $\mu(2) = 0.8$, $\mu(3) = 0.4$, 0 elsewhere.
- $X(\omega_3)$: $\mu(5) = 1$, 0 elsewhere.

Hence, for representative levels:

- $\alpha = 0.9$: $[X]_{0.9}(\omega_1) = \{1\}$, $[X]_{0.9}(\omega_2) = \emptyset$, $[X]_{0.9}(\omega_3) = \{5\}$.
- $\alpha = 0.8$: $[X]_{0.8}(\omega_1) = \{1\}$, $[X]_{0.8}(\omega_2) = \{2\}$, $[X]_{0.8}(\omega_3) = \{5\}$.
- $\alpha = 0.5$: $[X]_{0.5}(\omega_1) = \{1, 2\}$, $[X]_{0.5}(\omega_2) = \{2\}$, $[X]_{0.5}(\omega_3) = \{5\}$.
- $\alpha = 0.4$: $[X]_{0.4}(\omega_1) = \{1, 2\}$, $[X]_{0.4}(\omega_2) = \{2, 3\}$, $[X]_{0.4}(\omega_3) = \{5\}$.

Aumann α–cut formula (finite Ω). For each α:

$$\left[\int_\Omega X \,\mathrm{d}P\right]_\alpha = \sum_{i=1}^{3} p_i\, x_i, \qquad x_i \in [X(\omega_i)]_\alpha = p_1[X(\omega_1)]_\alpha + p_2[X(\omega_2)]_\alpha + p_3[X(\omega_3)]_\alpha.$$

Computed α–cuts of the Aumann integral.

1. $\alpha = 0.8$: single choice at each $\omega_i \Rightarrow$ singleton α–cut
   $[\int X\, dP]_{0.8} = \{ (2/11)\cdot 1 + (5/11)\cdot 2 + (4/11)\cdot 5 \} = \{ 32/11 \} \approx \{ 2.9091 \}$.
2. $\alpha = 0.5$: two choices at ω_1, fixed at ω_2, $\omega_3 \Rightarrow$ two–point α–cut
   $[\int X\, dP]_{0.5} = \{ 32/11, 34/11 \} \approx \{ 2.9091, 3.0909 \}$.
3. $\alpha = 0.4$: two choices at ω_1, two at ω_2, fixed at $\omega_3 \Rightarrow$ four–point α–cut
   $[\int X\, dP]_{0.4} = \{ 32/11, 34/11, 37/11, 39/11 \} \approx \{ 2.9091, 3.0909, 3.3636, 3.5455 \}$.
4. $\alpha = 0.9$: some $[X]_{0.9}(\omega_i)$ is empty (ω_2), so no selections exist $\Rightarrow$ empty α–cut $[\int X\, dP]_{0.9} = \emptyset$.

At each confidence level α, the Aumann integral aggregates X by taking all measurable selections (here, all pointwise choices from the α–cuts) and averaging them with the probabilities p_i. The resulting α–cut is the weighted Minkowski sum of the input α–cuts.

Following the works of [79], we will now use Fubini's theorem and Aumann integral to build the fuzzy integral and discuss its properties. A fuzzy stochastic process X is called L^p –integrably bounded ($p \geq 1$) if there exists a real–valued stochastic process $h \in L^p(I \times \Omega,\ N;\ R+)$, where I is the interval and Ω is the probability space, such that the fuzzy set $||[X(t,\ \omega)]^0||$ is bounded by $h(t,\ \omega)$ for almost all $(t,\ \omega)$ in $I \times \Omega$. The notation $||[X(t,\ \omega)]^0||$ represents the magnitude or absolute value of the equivalence class $[X(t,\ \omega)]^0$. $L^p(I \times \Omega,\ N;\ F(R))$ denotes the set of nonanticipating and L^p–integrably bounded fuzzy stochastic processes. Nonanticipating means that the process X does not rely on future information, but only on the current and past information. By employing Aumann–Fubini theorem which is the extension of Fubini theorem, the fuzzy integral of a fuzzy stochastic process X, denoted as $\int_0^T X(s,\ \omega)\ ds$ is defined. Here, T is a fixed time point, and the integral is taken over the interval from 0 to T. The integral is defined for ω in the set Ω minus N_X, where N_X is a subset of Ω belonging to the σ–algebra A with $P(N_X) = 0$, meaning it has measure zero. The fuzzy integral $\int_0^T X(s,\ \omega) ds$ can be defined level–wise. For each α in the interval [0, 1] and for each ω in the set Ω minus N_X, the Aumann integral $\int_0^T [X(s,\ \omega)]^a\ ds$ belongs to the set $K(R)$, which represents fuzzy sets over the real numbers. Therefore, the fuzzy random variable $\int_0^T X(s,\ \omega)\ ds$ belongs to the set $F(R)$ for every ω in the set $\Omega \setminus N_X$.

Now, we define the fuzzy stochastic Lebesgue–Aumann integral of $X \epsilon L^1(I\ x\ \Omega,\ N;\ F(R))$.

$$L_x(t,\omega) = \begin{cases} \displaystyle\int_0^T \mathbf{1}_{[0,t]}(s)\, X(s,\omega)\, \mathrm{d}s, & \text{for every } \omega \in \Omega \setminus N_x, \\ \langle 0 \rangle, & \text{for every } \omega \in N_x. \end{cases} \tag{5.1.5}$$

We will now discuss properties of this integral L_X. First, we introduce some more terminology. The material comes from Kolmogorov [82] and Zadeh [60]. The $d\infty$ metric, also known as the supremum or uniform metric, is a mathematical measure of dissimilarity or distance between fuzzy sets. It quantifies the maximum discrepancy between the membership degrees of corresponding elements in two fuzzy sets. Formally, let A and B be two fuzzy sets defined on a common universe of discourse. The $d\infty$ metric between A and B, denoted as $d\infty(A, B)$, is defined as the supremum or maximum of the absolute differences in membership degrees:

$d\infty(A, B) = sup |\mu A(x) - \mu B(x)|$, where $\mu A(x)$ and $\mu B(x)$ represent the membership degrees of the element x in the fuzzy sets A and B, respectively. The supremum is taken over all elements x in the universe of discourse. The $d\infty$ metric provides a measure of the largest difference in membership degrees between corresponding elements of two fuzzy sets. It captures the maximum discrepancy or dissimilarity between the two sets, indicating how much they differ in terms of their membership degrees. Now we will define L^1 norm. L^1 refers to a mathematical space or norm in the context of functions and integrals.

Specifically, L^1 represents a function space of integrable functions with a finite integral norm. Formally, L^1 is a function space defined over a given measure space (Ω, A, μ). It consists of all measurable functions $f\colon \Omega \to \mathbb{R}$ (or $f\colon \Omega \to \mathbb{C}$) such that the integral of the absolute value of f, denoted as $\int|f|\, d\mu$, is finite: $L1(\Omega, A, \mu) = \{f\colon \Omega \to \mathbb{R}\ (or\ \mathbb{C})\ /\ \int|f|\, d\mu < \infty\}$.

In simpler terms, L^1 consists of functions whose absolute value is integrable over the measure space, meaning that the area under the curve of the absolute value function is finite. The norm associated with the L^1 space, denoted as $\|f\|_1$, is defined as the integral of the absolute value of f: $\|f\|_1 = \int|f|\, d\mu$. The L^1 norm measures the size or magnitude of the function f in terms of its integral, capturing how spread out or concentrated the function is. L^1 represents a function space of integrable functions, where the absolute value of the function is integrable. The associated norm, $\|f\|_1$, measures the size of the function based on its integral. In the context of stochastic processes, the L^1 space refers to a function space that consists of processes that are integrable with respect to a given measure.

Formally, let (Ω, A, P) be a probability space and let X be a stochastic process defined on that space. The L^1 space, denoted as $L^1(\Omega, A, P)$, consists of all stochastic processes X that satisfy the condition of integrability. This means that the expected value or integral of the absolute value of X is finite: $L^1(\Omega,\ A,\ P) = \{X\colon \Omega \rightarrow \mathbb{R}\ (or\ \mathbb{C})\ /\ E[/X/] < \infty\}$.

Here, $E[\cdot]$ denotes the expected value or the integral with respect to the probability measure P. The absolute value $/X/$ ensures that both positive and negative values contribute to the integrability condition. In simpler terms, a stochastic process X belongs to the L^1 space if the expected value of its absolute value is finite. This indicates that the process is well–behaved in terms of its average behaviour and does not exhibit extreme fluctuations or infinite values. Now we define the L^P space. The L^P space, also known as a p–norm space, is a function space that consists of functions with finite p–norms.

Formally, the L^P space, denoted as $L^p(\Omega,\ A,\ \mu)$, is defined over a given measure space $(\Omega,\ A,\ \mu)$. It consists of all measurable functions $f\colon \Omega \rightarrow \mathbb{R}$ (or $\mathbb{C}$) such that the p–th power of the absolute value of f, denoted as $/f/^p$, is integrable with respect to the measure μ:

$L^p(\Omega,\ A,\ \mu) = \{f\colon \Omega \rightarrow \mathbb{R}\ (or\ \mathbb{C})\ /\ \int /f/^p\ d\mu < \infty\}$.

Here, p is a positive real number greater than or equal to 1. The p–norm of a function f, denoted as $||f||^p$, is defined as the p–th root of the integral of $/f/^p$: $||f||^p = (\int /f/^p\ d\mu)^{(1/p)}$.

The L^P space is a function space that captures functions whose p–th power is integrable, indicating that the function is well–behaved in terms of its magnitude and spread. The norm associated with the L^P space, the p–norm, measures the size or magnitude of the function f with respect to its integral, emphasizing the contribution of higher values due to the exponent p. The L^P spaces have various properties, including completeness, which means that every Cauchy sequence of functions converges to a limit within the space. Different values of p lead to different L^P spaces. For instance, when $p = 1$, we have the L^1 space, and when $p = 2$, we have the L^2 space. In summary, the L^P space consists of functions with finite p–norms, where the p–th power of the absolute value is integrable. It provides a framework for studying well–behaved functions in terms of their magnitudes and spread, with the p–norm quantifying their sizes. In the context of stochastic processes, the L^P space refers to a function space that consists of processes that are p–integrable with respect to a given measure. Formally, let $(\Omega,\ A,\ P)$ be a probability space and let X be a stochastic process defined on that space. The L^P space, denoted as $L^p(I \times \Omega,\ A \times N\ ;\ F(R))$, consists of all stochastic processes X that satisfy the condition of p–integrability. This means that the p–th power of the absolute value of X, denoted as $|X|^p$, is integrable with respect to the product measure $A \times N$: $L^p(I \times \Omega,\ A \times N\ ;\ F(R)) = \{X\colon I \times \Omega \to F(R)\ |\ \int\int |X(t,\ \omega)|^p\ dP(t,\ \omega) < \infty\}$. Here, p is a positive real number greater than or equal to 1. The integral is taken over the product measure P, which is the joint measure defined on the σ–algebra generated by the Cartesian product of the intervals I and the probability space $(\Omega,\ A,\ P)$. The p–integrability condition ensures that the p–th power of the absolute value of the process X is well–behaved and has a finite integral.

Having defined integrability conditions, we can now expand the work of [79] by exploring the Lebesgue–Aumann integral. If $X \in L^p(I \times \Omega,\ N\ ;\ F(R))$ for $p \geq 1$, then $L_X(\cdot,\ \cdot) \in L^P(I \times \Omega,\ N\ ;\ F(R))$: this property states that if the fuzzy stochastic process X belongs to the L^P space, which consists of processes that are p–integrable, then the fuzzy stochastic Lebesgue–Aumann integral L_x, defined over the same probability space, also belongs to the L^P space. In other words, the integrability of X carries over to the integrability of L_x.

Let $X \in L^1(I \times \Omega,\ N\ ;\ F(R))$, then $\{L_X(t)\}_{t \in I}$ is d_∞–continuous. This property states that if the fuzzy stochastic process X belongs to the L^1 space, which consists of processes that are integrable, then the trajectories of the fuzzy stochastic Lebesgue–Aumann integral $\{L_x(t)\}_{t \in I}$ are d_∞–continuous. In other words, the integral $L_x(t)$ is a d_∞–continuous function of time t for each fixed outcome ω. This implies that the fuzzy integral exhibits continuity with respect to the d_∞ metric, capturing the dependence and smoothness of the integral over time. We will now define an inequality that provides a bound on the maximum difference between fuzzy sets $L_{t,x}(u)$ and $L_{t,\ y}(u)$ over the interval [0, t] in terms of the cumulative difference between fuzzy sets $X(s)$

and *Y(s)* over the sample interval. The inequality captures the relationship between the supremum and integral of the d_∞ metric almost everywhere (a.e.). This inequality states that the supremum of the d_∞ metric between $L_{t,x}(u)$ and $L_{t,\,y}(u)$ for u in in the interval *[0, t]* is bounded by the scaled integral of the d_∞ metric between *X(s)* and *Y(s)* for s in the interval *[0, t]* with the scaling factor depending on the value of p.

$$\sup_{u\in[0,t]} d_\infty^p(L_{t,x}(u),\, L_{t,y}(u)) \leq t^{p-1}\int_0^t d_\infty^p(X(s),\, Y(s))\,\mathrm{d}s, \qquad \text{a.e.} \qquad (5.1.6)$$

The L.H.S. represents the supremum (maximum) of the d_∞ metric between $L_{t,x}(u)$ and $L_{t,y}(u)$, where t is a fixed value and u ranges from 0 to t. Here, $d^p{}_\infty(a,\ b)$ denotes the d_∞ metric between two fuzzy sets or elements a and b. $d^p{}_\infty(L_{t,x}(u),\ L_{t,y}(u))$ is the d_∞ metric between the fuzzy sets $L_{t,x}(u)$ and $L_{t,y}(u)$ at each u. It measures the maximum difference in membership degrees between corresponding elements of the fuzzy sets at a given u. The R.H.S. represents an integral involving the d_∞ metric between the fuzzy sets *X(s)* and *Y(s)*, where s ranges from *0* to t. The integral is scaled by $t^{p—-1}$, where p is a positive real number. It is the integral of the d_∞ metric between *X(s)* and *Y(s)* at each s. It measures the cumulative difference in membership degrees between corresponding elements of the fuzzy sets over the interval *[0, t]*. $t^{p—-1}$ scales the integral by a factor that depends on the value of p. The value of p determines how the integral contributes to the overall inequality. The inequality holds almost everywhere (a.e), meaning that it holds for all values of t except possibly for a set of measure zero.

We will now define the fuzzy stochastic Itô integral by using fuzzy random variable $\int_0^T X(s)dW(s)$, where W is a Wiener process. Let $X \in L2(I \times \Omega,\ N;\ R)$; then, $\{\left\langle\int_0^T X(s)dW(s)\right\rangle\}$, where $t \in I$. This represents a fuzzy stochastic process defined over the interval *I*. For each fixed t, $\left\langle\int_0^T X(s)dW(s)\right\rangle$ is a fuzzy set. The process captures the evolution of these fuzzy sets as t varies. Here, *dW(s)* represents the differential of a Wiener process or Brownian motion. The next part indicates that the fuzzy stochastic process $\left\langle\int_0^T X(s)dW(s)\right\rangle$ belongs to the L^2 space over the product space $I \times \Omega$. The fuzzy sets generated by the integral have finite second moments or variances and take values in the space of fuzzy sets over the real numbers, denoted as *F(R)*. The last proposition states that if X belongs to a family of square integrables then the next integration will become fuzzy stochastic differential equation and then that integration also belongs to the family of square integrables. Formally, let $X \in L^2(I \times \Omega,\ N\ ;\ R)$, then X belongs to the L^2 space, which consists of processes that have finite second moments or variances. The process X takes real–valued values and $\left\langle \int_0^T X(s)dW(s) \right\rangle,\ t \in I$ is d$_\infty$–continuous. In other words, as t varies within the interval *I*, the fuzzy sets $\left\langle \int_0^T X(s)dW(s) \right\rangle$ exhibit continuity with respect to the d_∞ metric. As discussed earlier, the d$_\infty$ metric, also known as the supremum or uniform metric, measures the maximum difference in membership degrees between corresponding elements of fuzzy sets. Thus, the statement asserts that the fuzzy sets $\left\langle \int_0^T X(s)dW(s) \right\rangle$, obtained by integrating X with respect to the Wiener process, show continuity with respect to the d_∞ metric as t varies. This continuity property is significant because it indicates that the fuzzy stochastic process $\left\langle \int_0^T X(s)dW(s) \right\rangle,\ t \in I$ has a smooth and well–behaved evolution with respect to the d_∞ metric. It suggests that the variations in the fuzzy sets, induced by the integral, are gradual and predictable as t changes within the interval *I*. In summary, the given statement states that if the stochastic process X belongs to the L^2 space, then the fuzzy stochastic process $\left\langle \int_0^T X(s)dW(s) \right\rangle,\ t \in I$, obtained by integrating X with respect to a Wiener process, exhibits continuity with respect to the d^∞ metric. This property implies a smooth and predictable evolution of the fuzzy sets as t varies.

5.1.4 FUZZY STOCHASTIC DIFFERENTIAL EQUATION WHEN H > $\frac{1}{2}$

We will now focus on the special case of $H > \frac{1}{2}$ and apply Liouville form fBm [78] [79]. This is a type of fractional Gaussian process that is defined by a covariance function that exhibits a power–law scaling. Specifically, the covariance function is given by $C(t,\ s) = |t\ \text{-}\ s|^{(2H\ -\ 2)}$, where $0 < H < 1$ is the Hurst exponent. Now, in this case, $H>\frac{1}{2}$ for the process to have finite moments of order greater than or equal to two. If H $< \frac{1}{2}$, the quadratic variation is singular (infinite). When $H> \frac{1}{2}$, the Liouville fBM does not have mean reversion. When $H< \frac{1}{2}$, the fBm process does not possess the self–similarity and stationarity properties required for the Liouville form. A fractional Brownian motion with $H < \frac{1}{2}$ is often referred to as "rough" or "pathologically irregular." It exhibits strong long–range dependence, which means that distant points in the process are highly correlated. In the Liouville form, fBm is expressed as a stochastic integral involving a deterministic function and a standard Brownian motion. It has the following form [78] :

$$X(t) = X_0 + \int_0^t f\Big(s, X(s)\Big)\,\mathrm{d}s + \left\langle \int_0^t g\Big(s, X(s)\Big)\,\mathrm{d}B_H(s) \right\rangle, \qquad X_0 = X(0). \tag{5.1.7}$$

where the FSDE is given by equation 5.1.7, *X(t)* is the solution of the equation at time *t*, X_0 is the initial condition of *X* at time *0*, *f(s, X(s))* is a function that describes the drift of *X*, and *g(s, X(s))* is a function that describes the diffusion of *X*, and B_H is the Liouville fBm that drives the equation. The integral term in equation represents the stochastic integral of *g(s, X(s))* with respect to $B_H(s)$, which is a type of integral that is defined in terms of the Itô integral and the covariance structure of the fBm. The angle brackets around the integral indicate that it is taken in the sense of the Stratonovich integral. The corresponding approximation equation to 5.1.7 is given by $X^\varepsilon(t)$, where $X^\varepsilon(t)$ is an approximation to *X(t)* that is obtained by discretising the integral term using a numerical scheme. The integral term in the approximation equation is approximated using a discrete version of the stochastic integral, which is defined in terms of the Riemann sum and the covariance structure of the discretised fBm, denoted by $BH^\varepsilon(s)$ [79].

$$X^\varepsilon(t) = X_0 + \int_0^t f\Big(s, X^\varepsilon(s)\Big)\,\mathrm{d}s + \left\langle \int_0^t g\Big(s, X^\varepsilon(s)\Big)\,\mathrm{d}B_H^\varepsilon(s) \right\rangle. \tag{5.1.8}$$

This section discusses a class of stochastic differential equations that are driven by a Liouville fBm, which is a type of fractional Gaussian process with a power–law covariance structure. Building on [79], we will now make a couple of assumptions to prove that 5.1.8 has a strong unique solution. The first assumption(A1) is that for every *u, v* $\in$ *F(R)* and every *t* $\in$ I, there exists *L* $>$ *0* such that *max{*d_∞*(f(t, w,u), f(t, w, v)), |g(t, w, u) —- g(t, w, v)|}* $\leq$ *L*$d_{\infty(}$*u,v).* This assumption helps to estimate upper bound using constant *L*. The second assumption (A2) is the Cauchy–Schwarz inequality and suppose that there is C>0 such that *max{*d_∞*(f(t, w,u*$\langle 0\rangle$*), |g(t, w, u)| }* $\leq$ *C(1+*$d_{\infty(}$*u,*$\langle 0\rangle$*).* The last assumption (A3) is that the mapping f: *I x* Ω *x F(R) –> F(R)* and *g: I x* Ω *x F(R) –> F(R)* are measurable with respect (w.r.t) to Standard Brownian motion. Formally, $N \otimes B_{ds}/B_{ds}$ - measurable and $N \otimes B_{ds}/B(R)$ —- measurable, respectively. The product sigma–algebra is a way to combine two sigma–algebras (sets of events) from different sources. In this work, *N* represents the sigma–algebra associated with the underlying probability space, which captures the randomness or uncertainty inherent in the system. B_{ds}/B_{ds} denotes the sigma–algebra generated by the Brownian motion process B_{ds}. It contains all the events or information that can be described solely in terms of the Brownian motion process. The symbol $\otimes$ represents the product operation between sigma–algebras, which creates a larger sigma–algebra that combines the information from both sources. The product sigma–algebra $N \otimes$ *Bds/Bds* captures the combined information from the underlying probability space and the Brownian motion process. It allows us to consider events or random variables that depend on both the inherent randomness of the system (represented by *N*) and the randomness introduced by the Brownian motion process (represented by *Bds/Bds).*

This construction is important in the theory of stochastic processes and stochastic differential equations because it provides a way to define and work with stochastic integrals, which are integrals with respect to Brownian motion or other stochastic processes. The measurability condition (A3) in the context provided ensures that the drift and diffusion functions are well–defined and can be integrated with respect to the Brownian motion process, taking into account both the underlying probability space and the Brownian motion itself.

Next, we will consider the bound on the distance between two stochastic integrals involving Wiener process W(s). Let *X(s)* and *Y(s)* be two stochastic processes and let $d^2{}_{\infty}$ *(·, ·)* denote the *L2* distance between two processes. Then by the Burkholder and Gundy inequality [83]:

$$\begin{gathered}\mathbb{E}\left[\sup_{u\in[0,t]} d_{\infty}^{2}\Big(\Big\langle\int_0^u X(s)\,\mathrm{d}W(s)\Big\rangle,\Big\langle\int_0^u Y(s)\,\mathrm{d}W(s)\Big\rangle\Big)\right] \leq \\ 4\,\mathbb{E}\int_0^t d_{\infty}^{2}(\langle X(s)\rangle,\,\langle Y(s)\rangle)\;\mathrm{d}s.\end{gathered} \tag{5.1.9}$$

Here, $\langle\cdot\rangle$ denotes the quadratic variation of a stochastic process, which is a measure of the total variation of the process. The integral $\int_0^u X(s)dW(s)$represents the stochastic integral of *X(s)* with respect to the Wiener process up to time u. The inequality states that the distance between the stochastic integrals of *X(s)* and *Y(s)* with respect to *W(s)* is bounded by a multiple of the integral of their L_2 distance over time. The constant 4 in the inequality is a universal constant that arises from the properties of the Wiener process. 5.1.9 is a special case of the Burkholder–Davis–Gundy (BDG) inequality which we will explore further down the line and on which we will rely in other proofs. We will be using 5.1.9 and the assumptions above to show that 5.1.8 has a strong unique solution.

We need to establish two points: 1. Existence of a strong solution. There exists a process *X(t)* that solves the FSDE a.s. for all $t \geq 0$. 2. Uniqueness of the strong solution. The solution *X(t)* is unique a.s. Using equation 5.1.2 and integration by parts technique, we can re–write 5.1.8:

$$
\begin{aligned}
X^{\varepsilon}(t) = X_0 & \\
& + \int_0^t f\Big(s, X^{\varepsilon}(s)\Big)\,\mathrm{d}s \\
& + \Big\langle\, a\int_0^t \bigg(\int_0^t (t-s+\varepsilon)^{H-\frac{1}{2}}\,\mathrm{d}B(r)\bigg) g\Big(s, X^{\varepsilon}(s)\Big)\,\mathrm{d}s \\
& \qquad + \int_0^t \varepsilon^a\, g\Big(s, X^{\varepsilon}(s)\Big)\,\mathrm{d}W(s)\,\Big\rangle \qquad (5.1.10) \\
= X_0 + \int_0^t f\Big(s, X^{\varepsilon}(s)\Big)\,\mathrm{d}s & + \Big\langle\, a\int_0^t \varphi^{\varepsilon}(s)\, g\Big(s, X^{\varepsilon}(s)\Big)\,\mathrm{d}s \\
& \qquad + \int_0^t \varepsilon^a\, g\Big(s, X^{\varepsilon}(s)\Big)\,\mathrm{d}W(s)\,\Big\rangle.
\end{aligned}
$$

where

$$\varphi^{\varepsilon}(s) = \textstyle\int_0^t (t-s+\varepsilon)^{a-1}]dB(s),\ \varepsilon > 0,\ \ a = H - 1/2.$$

The proof begins by introducing the Picard iterations [80], denoted as $X_n^{\varepsilon}(t)$, which are obtained by iteratively applying the equation $X_n^{\varepsilon}(t) = X_0 + \int_0^t f\left(s,\ X_{n-1}^{\varepsilon}(s)\right) ds + \left\langle \int_0^t (a\varphi^{\varepsilon}(s) g\left(s,\ X_{n-1}^{\varepsilon}(s)\right) ds + \int_0^t \varepsilon^a g(s,\ X_{n-1}^{\varepsilon}(s) dW(s) \right\rangle$, almost everywhere (a.e.). These iterations aim to find an approximation for the solution of the equation. To approximate *X(t)*, the Picard iteration scheme defines a sequence of iterates $X_n^{\varepsilon}(t)$ that should converge to *X(t)* as n goes to infinity. Starting with $X_0^{\varepsilon}(t) = X_0$, we calculate $X_1^{\varepsilon}(t)$ by plugging X_0 into the right–hand side. Then $X_2^{\varepsilon}(t)$ is calculate using $X_1^{\varepsilon}(t)$ and so on. The key idea is that as n increases, the sequence $X_n^{\varepsilon}(t)$ should get closer and closer to the true solution *X(t)*, provided some condition on *f*, *g* and the initial data is satisfied to ensure convergence. The parameters ε and a are introduced to control the convergence behaviour, while φ is some approximating kernel. The Picard iterations provide a recursive way to get successively better approximations to the original equation's solution by iteratively updating an integral equation approximation. Define the sequence $j_n(t) = E\sup_{u\in[0,t]} d^2_\infty(X^\varepsilon_n(u), X^\varepsilon_{n-1}(u))$. The sequence measures the distance between two consecutive iterations. When we take *n=1*, and by the assumption that there is *C>0* such that $max\{d_\infty(f(t,\ w,u\langle 0\rangle),\ |g(t,\ w,\ u)|\ \} \leq C(1+d_{\infty}(u,\langle 0\rangle)$ and from triangular inequality, we know that:

$$\begin{aligned}
\mathbb{E}\Big|X^\varepsilon(t)\Big|^2 &\leq \mathbb{E}\left|X_0 + \int_0^t f\Big(s, X^\varepsilon(s)\Big)\,\mathrm{d}s + \int_0^t g\Big(s, X^\varepsilon(s)\Big)\,\mathrm{d}B(s)\right|^2 \\
&\leq 3\,\mathbb{E}|X_0|^2 + 3\,\mathbb{E}\left|\int_0^t f\Big(s, X^\varepsilon(s)\Big)\,\mathrm{d}s\right|^2 + 3\,\mathbb{E}\left|\int_0^t g\Big(s, X^\varepsilon(s)\Big)\,\mathrm{d}B(s)\right|^2.
\end{aligned}$$

Therefore, we have a constant *3* in:

$$\begin{aligned}
J_1(t) := \mathbb{E}\Bigg[\sup_{u\in[0,t]} d^2_\infty\Bigg(&\int_0^u f\Big(s, X^\varepsilon_0(s)\Big)\,\mathrm{d}s \\
&+ \Big\langle\int_0^u a\,\varphi^\varepsilon(s)\,g\Big(s, X^\varepsilon_0(s)\Big)\,\mathrm{d}s + \int_0^u \varepsilon^a\,g\Big(s, X^\varepsilon_0(s)\Big)\,\mathrm{d}W(s)\Big\rangle,\ \langle 0\rangle\Bigg)\Bigg] \\
\leq{}& 3\,\mathbb{E}\Bigg[\sup_{u\in[0,t]} d^2_\infty\Bigg(\int_0^u f\Big(s, X^\varepsilon_0(s)\Big)\,\mathrm{d}s,\ \langle 0\rangle\Bigg)\Bigg] \\
&+ 3\,\mathbb{E}\Bigg[\sup_{u\in[0,t]} d^2_\infty\Bigg(\Big\langle\int_0^u a\,\varphi^\varepsilon(s)\,g\Big(s, X^\varepsilon_0(s)\Big)\,\mathrm{d}s\Big\rangle,\ \langle 0\rangle\Bigg)\Bigg] \\
&+ 3\,\mathbb{E}\Bigg[\sup_{u\in[0,t]} d^2_\infty\Bigg(\Big\langle\int_0^u \varepsilon^a\,g\Big(s, X^\varepsilon_0(s)\Big)\,\mathrm{d}W(s)\Big\rangle,\ \langle 0\rangle\Bigg)\Bigg].
\end{aligned} \tag{5.1.11}$$

Using proposition 5.1.6:

$\sup_{u\in[0,t]} d^p_\infty\left(L_{t,\ x}(u),\ L_{t,y}(u)\right) \leq t^{p-1}\int_0^t d^p_\infty\left(X(s),\ Y(s)\right)ds$ a.e., proposition 5.1.9, the linearity of the expectations operator, and since our metric is d^p_∞, so we can pull out constants as square, we can restate that:

$$
\begin{aligned}
J_1(t) \le 3t\,\mathbb{E}\Big[\int_0^t d_\infty^2\Big(f\Big(s, X_0^\varepsilon(s)\Big), \langle 0\rangle\Big)\ \mathrm{d}s\Big] \\
+ 3a^2\,\mathbb{E}\Big[\sup_{u\in[0,t]} d_\infty^2\Big(\Big\langle\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big\rangle, \langle 0\rangle\Big)\Big] \\
+ 12\,\varepsilon^{2a}\,\mathbb{E}\Big[\int_0^t d_\infty^2\Big(g\Big(s, X_0^\varepsilon(s)\Big), \langle 0\rangle\Big)\ \mathrm{d}s\Big].
\end{aligned}
$$

Using assumption *max{d_∞(f(t, w,u⟨0⟩), |g(t, w, u)| } ≤C(1+$d_{\infty(}$u,⟨0⟩),* for C >0, and properties of the expectation operator, we substitute this into the following integral expression to obtain the following. Also, remember that $d_\infty^2\left(X_0^\varepsilon(s)\right), \langle 0\rangle$ measures the squared maximum distance between some fuzzy set X_0^ε at a specific time s and zero fuzzy set, where all membership values are zero and utilise the metric inner product. For a different version of this proof, see [79].

$$
\begin{aligned}
J_1(t) &\le 3\Big(T + 4\varepsilon^{2a}\Big)\Bigg[\mathbb{E}\Big(\int_0^t d_\infty^2\Big(f\Big(s, X_0^\varepsilon(s)\Big), \langle 0\rangle\Big)^2\ \mathrm{d}s\Big) \\
&\quad + \mathbb{E}\Big(\int_0^t d_\infty^2\Big(g\Big(s, X_0^\varepsilon(s)\Big), \langle 0\rangle\Big)^2\ \mathrm{d}s\Big) \\
&\quad + 3a^2\,\mathbb{E}\Bigg[\sup_{u\in[0,t]} d_\infty^2\Bigg(\Big\langle\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big\rangle^2\Bigg)\Bigg]\Bigg] \\
&\le 3\Big(T + 4\varepsilon^{2a}\Big)\Bigg[2\int_0^t \mathbb{E}\Big(C\Big(1 + d_\infty^2(X_0^\varepsilon(s), \langle 0\rangle)\Big)\Big)^2\ \mathrm{d}s \qquad (5.1.12) \\
&\quad + 3a^2\,\mathbb{E}\Bigg[\sup_{u\in[0,t]} d_\infty^2\Bigg(\Big\langle\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big\rangle^2\Bigg)\Bigg]\Bigg] \\
&= 6C^2\Big(T + 4\varepsilon^{2a}\Big)\Bigg[t\Big(1 + \mathbb{E}|X_0^\varepsilon|^2\Big) \\
&\quad + 3a^2\,\mathbb{E}\Bigg[\sup_{u\in[0,t]} d_\infty^2\Bigg(\Big\langle\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big\rangle^2\Bigg)\Bigg]\Bigg].
\end{aligned}
$$

This works only for $a = H$ —- $\frac{1}{2}$ *>0*, which was our assumption earlier and which is why it only works for $H > \frac{1}{2}$. For $H < \frac{1}{2}$, we use different integration methods. Next, we utilise the following results. The d_H^2 distance, or Hausdorff distance, measures the maximum discrepancy between two sets by considering the minimum distance from each point in one set to the other set. This means that the d_H^2 distance accounts for the smallest possible distance between the two sets. On the other hand, the d_∞^2distance, or supremum distance, measures the maximum difference between the values of two functions or sets. It calculates the maximum absolute difference between corresponding points of the two functions or sets. Since the d_H^2 distance considers the minimum distances, it takes into account the smallest possible difference between the two sets. Therefore, the d_H^2 distance will always be greater than or equal to the d_∞^2 distance, which measures the maximum difference. Using the above, and by the metric inner product:

$$\begin{aligned}
&\mathbb{E}\Bigg[\sup_{u\in[0,t]} d_\infty^2\Big(\Big\langle\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big\rangle\,,\,\langle 0\rangle\Big)\Bigg] \\
&\le \mathbb{E}\Bigg[\sup_{u\in[0,t]} d_H^2\Big(\Big\{\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big\}\,,\,\{0\}\Big)\Bigg] \\
&\le \mathbb{E}\Bigg[\sup_{u\in[0,t]} \Big(\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big)^2\Bigg].
\end{aligned} \tag{5.1.13}$$

We will now use the famous Hölder inequality [75]:

$$\int_a^b |f(x)g(x)|dx \le (\int_a^b |f(x)|^p dx)^{1/p}(\int_a^b |g(x)|^q dx)^{1/q}$$

and re–arrange terms r with s.

$$\begin{aligned}
&\mathbb{E}\Bigg[\sup_{u\in[0,t]} \Big(\int_0^u \varphi^\varepsilon(s)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big)^2\Bigg] \\
&= \mathbb{E}\Bigg[\sup_{u\in[0,t]} \Big(\int_0^u \Big(\int_0^s (s-r+\varepsilon)^{a-1}\,\mathrm{d}W(r)\Big)\, g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\Big)^2\Bigg].
\end{aligned}$$

Since the integration variables are independent, we can use stochastic Fubini's theorem [80] to rearrange the integration variables. This is the rule for swapping independent integration variables. If the function being integrated is continuous over a rectangular region $R = [a, b] \times [c, d]$, and the integral is of the form $\int\int R\ f(x, y)\ dxdy$, then the order of integration can be swapped, resulting in $\int\int R\ f(x, y)\ dydx$ or $\int\int R\ f(x, y)\ dxdy$. For example, consider the following double integral: $\int\int R\ exp(x^2 + y^2)dxdy$, where R is the region defined by $0 \leq x \leq 1$ and $0 \leq y \leq 2$. If we integrate with respect to x first, we have: $\int\int R\ exp(x^2 + y^2)dxdy = \int_0^2 \int_0^1 exp(x^2 + y^2)\ dxdy$. $\int$ If we integrate with respect to y first, we have: $\int\int R\ exp(x^2 + y^2)\ dxdy = \int_0^1 \int_0^2 exp(x^2 + y^2)\ dydx$. In this example, the integrand $exp(x^2 + y^2)$ is continuous over the rectangular region R, and therefore, we can swap the order of integration without affecting the result. It is important to note that not all integrals allow for swapping the order of integration. The conditions for applying Fubini's theorem include continuity of the integrand, boundedness of the region of integration, and absolute integrability meaning that the integral of the absolute value of the function over the region $\boldsymbol{R}$ is finite. If these conditions are not met, swapping the order of integration may lead to incorrect results.

We will now expand the work of [79]. First, we will use Fubini to re–arrange r with s in this integral. Then, we will apply Itô isometry or the Itô integral rule. It is a key tool in stochastic calculus and is used to calculate the expected value of stochastic integrals involving Brownian motion or other forms of stochastic processes. The identity essentially says that the square of the difference between two points in a Brownian motion path, raised to the power of $(\alpha - 1)$, can be expressed as the double stochastic integral of $(\alpha - 2)$ powers of the differences between the same points and the endpoints of the integration range. Intuitively, this identity can be understood as follows: the Brownian motion process is continuous and has infinite variation, meaning that the difference between two points in a path can take on any value in a certain range with positive probability. The Itô isometry [84] allows us to calculate the expected value of the square of these differences, which is a key step in deriving stochastic integrals and differential equations. In the rearrangement of terms given in the original question, the Itô isometry is used to express the term $(s - r + \varepsilon)^{a-1}$ as a double stochastic integral, which allows for the interchange of the order of integration and ultimately results in the desired form of the expression. $(W(t) - W(s))^{a-1} = \int_0^s (r - s + \varepsilon)^{a-2} dW(r) \int_0^t (u - t + \varepsilon)^{a-2} dW(u)$. Using this identity with t=r and s=s, we have, $(s - r + \varepsilon)^{a-1} = \int_0^r (u - r + \varepsilon)^{a-2} dW(u) \int_0^s (v - s + \varepsilon)^{a-2} dW(v)$.

Substituting this into the integral, we can now interchange the order of integration between s and r. There are three integrals in the intermediate step because the stochastic integral involving the Brownian motion process requires two integrals: one with respect to time and one with respect to the Brownian motion itself. There are two stochastic integrals involving Brownian motion: one over the range $[u,\ s]$ and one over the range $[r,\ t]$. This requires a total of three integrals: one over the range $[u,\ t]$ with respect to time, and two with respect to Brownian motion over the ranges $[u,\ s]$ and $[r,\ t]$, respectively. When the order of integration is interchanged between s and r, the resulting expression involves a double stochastic integral over the ranges $[u,\ s]$ and $[r,\ t]$, respectively. This requires two integrals with respect to Brownian motion and one integral with respect to time, which is consistent with the three integrals in the original expression.

The identity introduced with $t = s$ and $s = v$ allows for the simplification of the expression by expressing the term $(s - r + \varepsilon)^{a-1}$ as a double stochastic integral over the ranges $[s,\ u]$ and $[r,\ v]$, respectively. This ultimately leads to the desired form of the expression with only two integrals involving Brownian motion and one integral with respect to time.

$$\mathbb{E}\left[\sup_{u\in[0,t]}\left(\int_0^u\left(\int_0^s (s-r+\varepsilon)^{a-1}\,\mathrm{d}W(r)\right) g\Big(s, X_0^\varepsilon(s)\Big)\,\mathrm{d}s\right)^2\right]$$
$$= \mathbb{E}\left[\sup_{u\in[0,t]}\left(\int_0^u\left(\int_s^u g\Big(r, X_0^\varepsilon(r)\Big)\,\mathrm{d}r\right)(r-s+\varepsilon)^{a-1}\,\mathrm{d}W(s)\right)^2\right].$$

Using Itô isometry [84] i.e., that $(dW)^2 = dt$, and for the upper bound, simply multiply the expectation by 4 [80], we can further simplify the stochastic integral:

$$
\begin{aligned}
&\mathbb{E}\Bigg[\sup_{u\in[0,t]}\bigg(\int_0^u\bigg(\int_s^u g\Big(r,X_0^\varepsilon(r)\Big)\,\mathrm{d}r\bigg)(r-s+\varepsilon)^{a-1}\,\mathrm{d}W(s)\bigg)^2\Bigg]\\
&\le 4\,\mathbb{E}\Bigg[\int_0^t\bigg(\int_s^t g\Big(r,X_0^\varepsilon(r)\Big)(r-s+\varepsilon)^{a-1}\,\mathrm{d}r\bigg)^2\,\mathrm{d}s\Bigg]\\
&\le 4\,\mathbb{E}\int_0^t\bigg(\int_s^t g^2\Big(r,X_0^\varepsilon(r)\Big)(r-s+\varepsilon)^{a-1}\,\mathrm{d}r\bigg)\bigg(\int_s^t(r-s+\varepsilon)^{a-1}\,\mathrm{d}r\bigg)\,\mathrm{d}s\\
&\le 4\,\mathbb{E}\int_0^t\bigg(\int_s^t g^2\Big(r,X_0^\varepsilon(r)\Big)(r-s+\varepsilon)^{a-1}\,\mathrm{d}r\bigg)\left[\frac{(r-s+\varepsilon)^a}{a}\right]_{r=s}^{r=t}\,\mathrm{d}s\\
&\le 4\,\mathbb{E}\int_0^t\bigg(\int_s^t g^2\Big(r,X_0^\varepsilon(r)\Big)(r-s+\varepsilon)^{a-1}\,\mathrm{d}r\bigg)\left[\frac{(t-s+\varepsilon)^a}{a}-\frac{\varepsilon^a}{a}\right]^2\,\mathrm{d}s,\\
&\text{since } t,s,\varepsilon\ge 0,\quad \text{we have }\ \frac{(t-s+\varepsilon)^a}{a}-\frac{\varepsilon^a}{a}\le\frac{(t+\varepsilon)^a}{a}.
\end{aligned}
$$

following similar argument:

$$
\begin{aligned}
&4\,\mathbb{E}\int_0^t\bigg(g^2\Big(r,X_0^\varepsilon(r)\Big)\int_s^t(r-s+\varepsilon)^{a-1}\,\mathrm{d}r\bigg)\frac{(t+\varepsilon)^a}{a}\,\mathrm{d}s\\
&\le 4\,\frac{(t+\varepsilon)^a}{a}\,\mathbb{E}\int_0^t g^2\Big(r,X_0^\varepsilon(r)\Big)\bigg(\int_s^t(r-s+\varepsilon)^{a-1}\,\mathrm{d}s\bigg)\,\mathrm{d}r\\
&\le 4\,\frac{(t+\varepsilon)^a}{a}\,\mathbb{E}\int_0^t g^2\Big(r,X_0^\varepsilon(r)\Big)\frac{(t+\varepsilon)^a}{a}\,\mathrm{d}r\\
&\le \frac{4(t+\varepsilon)^{2a}}{a^2}\,\mathbb{E}\int_0^t g^2\Big(r,X_0^\varepsilon(r)\Big)\,\mathrm{d}r.
\end{aligned}
\tag{5.1.14}
$$

Because this is a linear integral, we can take the term $\frac{(t+\varepsilon)^a}{a}$ outside the integral. Now, using the Cauchy–Schwarz inequality [80],

$$
\max\Big\{d_\infty\Big(f(t,\omega,u),\langle 0\rangle\Big),\ |g(t,\omega,u)|\Big\}\le C\Big(1+d_\infty(u,\langle 0\rangle)\Big).
$$

$$
\mathbb{E}\Big[g^2\Big]\le 2C^2\left(1+\mathbb{E}\Big[d_\infty^2(X_0^\varepsilon,\langle 0\rangle)\Big]\right).
$$

we can evaluate the last expression:

$$\begin{aligned}\frac{4(t+\varepsilon)^{2a}}{a^2}\,\mathbb{E}\int_0^t g^2\Big(r, X_0^\varepsilon(r)\Big)\,\mathrm{d}r &\le 2C^2\,\frac{4(T+\varepsilon)^{2a}}{a^2}\Big(1+\mathbb{E}|X_0^\varepsilon|^2\Big)t\\ &\le C^2\,\frac{8(T+\varepsilon)^{2a}}{a^2}\Big(1+\mathbb{E}|X_0^\varepsilon|^2\Big)t.\end{aligned} \tag{5.1.15}$$

We will now substitute this result in J_1(t), simplify and take exponential out of the bracket.

$$\begin{aligned}J_1(t) &\le 6C^2\Big(T+4\varepsilon^{2a}\Big)\Big(1+\mathbb{E}|X_0^\varepsilon|^2\Big)t \;+\; 3a^2C^2\,\frac{8(T+\varepsilon)^{2a}}{a^2}\Big(1+\mathbb{E}|X_0^\varepsilon|^2\Big)t\\ &\le \Big(6C^2\Big(T+4\varepsilon^{2a}\Big)+24C^2(T+\varepsilon)^{2a}\Big)\Big(1+\mathbb{E}|X_0^\varepsilon|^2\Big)t\\ &\le 6C^2\Big(T+4\varepsilon^{2a}+4(T+\varepsilon)^{2a}\Big)\Big(1+\mathbb{E}|X_0^\varepsilon|^2\Big)t.\end{aligned}$$

We will explain what this all means and why we need it. The first term *$J_1(t)$* in the estimate is an upper bound on the error between the true solution *X(t)* and the Picard iterate *$X^\varepsilon(t)$* obtained at the first iteration. It represents the contribution to the error due to the approximation made at the current Picard iteration. The inequality for *$J_1(t)$* follows from the error estimate for the numerical method used to compute *$X^\varepsilon(t)$* at the current time step. This estimate indicates that the error in *$X^\varepsilon(t)$* at the current time step is proportional to the time step size t and to a constant factor that depends on the parameters of the numerical method and on the regularity of the solution *X(t)*. Similarly, we can obtain the same result for *$J_{n+1}(t)$* which, in the estimate, is an upper bound on the error between the numerical solution *$X^\varepsilon(t)$* obtained at the (n+1)th time step and the numerical solution *$X^\varepsilon(t)$* obtained at the nth time step. It represents the contribution to the error due to the propagation of the error from the current time step to the next time step. The inequality for the next time step, which we will prove now, is:

$$J_{n+1}(t) \le 3\,\Big(t+4\varepsilon^{2a}\;\;+4(T+\varepsilon)^{2a}\Big)L^2E\int_0^t d_\infty^2(X_n^\varepsilon(u), X_{n-1}^\varepsilon(u)ds$$

It follows from a stability estimate for the numerical method used to compute $X^{\varepsilon}(t)$ at the (n+1)th time step. This estimate indicates that the error in $X^{\varepsilon}(t)$ at the (n+1)th time step is proportional to the time step size t, to a constant factor that depends on the parameters of the numerical method and on the regularity of the solution $X(t)$, and to the norm of the difference between the numerical solutions for $X_n^{\varepsilon}(t)$ and $X_{n-1}^{\varepsilon}(t)$ obtained at the nth and (n–1)th time steps respectively. Heuristically, this makes sense because, as the iterations proceed, the Picard method is able to capture more accurately the behaviour of the true solution $X(t)$ at each time step. Second, let us consider the constant factor that depends on the parameters of the numerical method and on the regularity of the solution X(t). This constant factor represents the accuracy of the numerical method itself and is determined by the choice of the numerical scheme, the order of accuracy of the numerical method, and other parameters specific to the method. The regularity of the solution X(t) also plays a role in determining the accuracy of the numerical method, since more regular solutions can be approximated more accurately than less regular ones.

Finally, let's consider the norm of the difference between the numerical solutions $X_n^{\varepsilon}(t)$ and $X_{n-1}^{\varepsilon}(t)$ obtained at the nth and (n–1)th time steps, respectively. This term accounts for the propagation of the error from the previous time step to the current time step. If there is a large difference between the numerical solutions at the two–time steps, this indicates that the error has been propagated and has accumulated over time, leading to a larger error in the numerical solution at the current time step. The norm of this difference is used to quantify the magnitude of the error that has been propagated from the previous time step. Therefore, the estimate for the error in $X_n^{\varepsilon}(t)$ at the (n+1)th time step takes into account the time step size, the accuracy of the numerical method, and the propagation of the error from the previous time step. In summary, the term $J_{n+1}(t)$ accounts for the propagation of the error from the current time step to the next time step and provides an upper bound on the total error in the numerical solution $X^{\varepsilon}(t)$ at the (n+1)th time step. By summing up the contributions from all time steps, we obtain an estimate for the total error in the numerical solution $X^e(t)$ over the entire time interval [0, T].

This is our proof of the above. Let $X_n^\varepsilon(t)$ be the numerical solution obtained at n–th time step and $X_{n-1}^\varepsilon(t)$ be the numerical solution obtained at the (n–1)th time step. Then the error between these two solutions can be written as:

$\varepsilon_n(t) = X_n^\varepsilon(t) - X_{n-1}^\varepsilon(t)$.

Using the difference equation for the numerical method, we can write: $\varepsilon_{n+1}(t) = X^\varepsilon(t + \Delta t) - X_n^\varepsilon(t)$, where $X^\varepsilon(t + \Delta t)$ is the true solution at (n+1)th step and Δt is the time step size. Then, the error in the numerical solution $X^\varepsilon(t)$ at the (n+1)th time step can be written as:

$$\begin{aligned}\varepsilon_{n+1}(t) &= X^\varepsilon(t+\Delta t) - X_n^\varepsilon(t) - X_n^\varepsilon(t) + X_{n-1}^\varepsilon(t) + \Big(X_n^\varepsilon(t) - X_{n-1}^\varepsilon(t)\Big)\\ &= \Big(X^\varepsilon(t+\Delta t) - 2X_n^\varepsilon(t) + X_{n-1}^\varepsilon(t)\Big) + \Big(X_n^\varepsilon(t) - X_{n-1}^\varepsilon(t)\Big).\end{aligned}$$

Taking the norm ||.|| of both sides and using the triangle inequality property of norms in vector spaces, $||u + v|| \leq ||u|| + ||v|$, we obtain:

$$||\varepsilon_{n+1}(t)\,|| \leq\; ||X^\varepsilon(t+\Delta t) - \;2X_n^\varepsilon(t)|| + ||(X_n^\varepsilon(t) - \;X_{n-1}^\varepsilon(t)||$$

A function f(x) is said to be Lipschitz continuous if there exists a constant K such that:

$|f(x_1) - f(x_2)| \leq K|x_1 - x_2|$ for all x_1 and x_2 in the function's domain [80]. Intuitively, this means that the function's rate of change is bounded by a constant factor K, which is independent of the size of the interval over which the function is measured. Using Lipschitz continuity of the function f in the differential equation, we can write:

$$X^\varepsilon(t+\Delta t) - 2X_n^\varepsilon(t) \leq L^2 \Delta t \left| f\Big(t, X_n^\varepsilon(t), u\Big) - f\Big(t, X_{n-1}^\varepsilon(t), u\Big)\right|.$$

where L^2 is the Lipschitz constant. It is the smallest possible value of K that satisfies the Lipschitz condition $|f(x_1) - f(x_2)| \leq K|x_1$ —- $x_2|$. In other words, it is the smallest possible upper bound on the rate of change in the function over its entire domain and it provides a measure of how rapidly the function changes over its domain. We will use it in the following context. If the right–hand side of a differential equation is Lipschitz continuous with a constant K, then numerical methods indicate that a time step t, will converge to the true solution at a rate proportional to t/K. Using the assumption on the distance between the initial condition u_0 and the true solution $X(t)$, we can write:

$$\left||f\left(t,\ X_n^{\varepsilon}(t), u\right) - f\left(t,\ X_{n-1}^{\varepsilon}(t), u\right)\right|| \ \leq\ Ld_{\infty}(X_n^{\varepsilon}(t),\ X_{n-1}^{\varepsilon}(t))$$

Therefore, we can write the following and integrate both sides over time interval [t(n), t(n+1)]:

$$||\varepsilon_{n+1}(t)\left|\right| \leq\ Ld_{\infty}\left(X_n^{\varepsilon}(t),\ X_{n-1}^{\varepsilon}(t)\right) + |X^{\varepsilon}(t) -\ X_{n-1}^{\varepsilon}(t)\left|\right|$$

$$\int_0^t \left\|\varepsilon_{n+1}(s)\right\| \mathrm{d}s \leq L^2 \Delta t \left(\int_0^t d_{\infty}^2\left(X_n^{\varepsilon}(s), X_{n-1}^{\varepsilon}(s)\right) \mathrm{d}s\right)^{1/2} \left(\int_0^t \left\|X_n^{\varepsilon}(s) - X_{n-1}^{\varepsilon}(s)\right\|^2 \mathrm{d}s\right)^{1/2}.$$

Proof:

By BDG and Cauchy–Schwarz

Taking the expectation on both sides and using the fact that the expectation operator is linear, we obtain:

$$\mathbb{E}\int_0^t \left\|\varepsilon_{n+1}(s)\right\| \mathrm{d}s$$
$$\leq L^2 \Delta t\, \mathbb{E}\left[\left(\int_0^t d_{\infty}^2\left(X_n^{\varepsilon}(s), X_{n-1}^{\varepsilon}(s)\right) \mathrm{d}s\right)^{1/2}\right] \mathbb{E}\left[\left(\int_0^t \left\|X_n^{\varepsilon}(s) - X_{n-1}^{\varepsilon}(s)\right\|^2 \mathrm{d}s\right)^{1/2}\right].$$

Use the fact that the square of the norm of the difference between two solutions is bounded by three times the sum of the squares of their individual norms. This is a proof that combines the triangle inequality, a bounding arguement, and basic properties of the vector norms. It is not the tightest bound possible condition, and bounds with tighter constants exist in functional analysis. Nevertheless, for our purposes this result works.

So, we prove that: $||u - v||^2 \leq 3(||u||^2 + ||v||^2)$ using dot product and the norm. The dot product is $u \cdot v = ||u||\ ||v||\ cos(b)$, where b is the angle between u and v. Using this expression, we can write the square of the normal of the difference between u and v as: $||u - v||^2 = (u - v) \cdot (u - v) = ||u||^2 - 2(u \cdot v) + ||v||^2 = ||u||^2 + ||v||^2 - 2||u||\ ||v||\ cos(b)$, where we have used the distributive property of dot product. Note that the $cos(b)$ is bounded by –1 and 1 so: $-2||u||\ ||v|| \leq -2||u||\ ||v||\ cos(b) \leq 2||u||\ ||v||$. Adding this inequality to the expression $||u-v||^2$, we get: $||u - v||^2 + 2||u||\ ||v|| = ||u||^2 + ||v||^2 - 2(u \cdot v) + 2||u||\ ||v|| = ||u||^2 + ||v||^2 + 2||u||\ ||v|| - 2(u \cdot v)$ we obtain: $||u - v||^2 + 2||u||\ ||v|| \leq ||u||^2 + ||v||^2 + 2||u||\ ||v||$. Then take the square root we get: $||u - v|| \leq ||u|| + ||v||$ and use the fact that $||u-v||$ is non–negative, and that $(||u|| + ||v||)^2 \leq 3(||u||^2 + ||v||^2)$ for all real numbers $||u||$ and $||v||$, to get $||u - v||^2 \leq (||u|| + ||v||)^2 \leq 3(||u||^2 + ||v||^2)$. Now we continue with the original proof.

$$\begin{aligned} &\mathbb{E}\int_0^t \Big\|\varepsilon_{n+1}(s)\Big\|\,\mathrm{d}s \\ &\leq 3\Delta t \left(T_n + 4\varepsilon^{2a} + 4(T_n+\varepsilon)^{2a}\right)^{1/2} L^2 \left(\mathbb{E}\int_0^t d_\infty^2\Big(X_n^\varepsilon(s), X_{n-1}^\varepsilon(s)\Big)\,\mathrm{d}s\right)^{1/2}. \end{aligned} \tag{5.1.16}$$

Therefore, we have derived the expression for $\mathrm{j_{n+1}(t)}$ term in the error estimate:

$$
\begin{aligned}
j_{n+1}(t) &\leq 3\Big(T + 4\varepsilon^{2a} + 4(T+\varepsilon)^{2a}\Big)\, L^2\, \mathbb{E}\int_0^t d_\infty^2\Big(X_n^\varepsilon(s), X_{n-1}^\varepsilon(s)\Big)\, \mathrm{d}s \\
&\leq 3\Big(T + 4\varepsilon^{2a} + 4(T+\varepsilon)^{2a}\Big)\, L^2\, \int_0^t \mathbb{E}\Big[\sup_{u\in[0,s]} d_\infty^2\Big(X_n^\varepsilon(u), X_{n-1}^\varepsilon(u)\Big)\Big]\, \mathrm{d}s \\
&\leq 3\Big(T + 4\varepsilon^{2a} + 4(T+\varepsilon)^{2a}\Big)\, L^2\, \int_0^t j_n(s)\, \mathrm{d}s.
\end{aligned}
\tag{5.1.17}
$$

This inequality holds because the function $j_n(s)$ is defined as the supremum distance between values of the process X over the interval from n to $n+1$, as measured by the metric function d_∞^2. Therefore, integral of $j_n(s)$ over the interval t_0 to s gives an upper bound on the supremum distance between values of the process X over the same interval. The left–hand side of the inequality gives an upper bound on the function $j_{n+1}(t)$ while the right hand side gives a lower bound.

We will now apply Chebyshev inequality to the metric function [80]. The Chebyshev inequality is a probabilistic inequality that relates the probability of a random variable deviating from its mean to its variance. Specifically, the Chebyshev inequality states that for any random variable X with finite mean μ and finite variance σ^2, the probability that X deviates from its mean by more than k standard deviations is bounded by $1/k^2$. $P(|X–\mu|>k\sigma) \leq 1/k^2$. We will apply this inequality to the metric function $d_\infty^2\ [\Big(X_n^\varepsilon(u),\ X_{n-1}^\varepsilon(u)\Big)$ to obtain a probabilistic bound on the supremum distance between values of the process X over the interval n and $n+1$.

Proof. Define Y to be the random variable such that:

$$
Y = \quad \sup_{u\in[0,s]} d_\infty^2\ [\Big(X_n^\varepsilon(u),\ X_{n-1}^\varepsilon(u)\Big)\,.
$$

Then we have:

$\mu = E(Y) = \ E(\sup_{u\in[0,s]} d_\infty^2\ [\Big(X_n^\varepsilon(u),\ X_{n-1}^\varepsilon(u)\Big)$ and

$$\sigma^2 = Var(Y) = \ Var(\sup_{u\in[0,s]} d^2_\infty \ [\Big(X^\varepsilon_n(u), \ X^\varepsilon_{n-1}(u)\Big) .$$

Since the metric function is a distance measure, it is non–negative, and therefore Y is also non–negative. We can choose $k{=}2^n$ to obtain $P(Y{>}2^n\mu) \leq Var(Y)/(2^n\mu)^2$, and since Y is non–negative, we have $E(Y){=}\mu \geq 0$, therefore $(2^n\mu)^2 \geq 0$. This means that the denominator of the fraction is non–negative. We can multiply and invert the inequality to obtain:

$$\begin{aligned}
\mathbb{P}\Bigg(\sup_{u\in[0,s]} d^2_\infty\Big(X^\varepsilon_n(u), X^\varepsilon_{n-1}(u)\Big) \geq \frac{1}{2^n}\Bigg) &= \mathbb{P}(Y > 2^n\mu) \leq \frac{1}{4^n} \\
\leq (2^n)^2 \, &\frac{\mathrm{Var}\Big(\sup_{u\in[0,s]} d^2_\infty\Big(X^\varepsilon_n(u), X^\varepsilon_{n-1}(u)\Big)\Big)}{2^n \, \mathbb{E}\Big[\sup_{u\in[0,s]} d^2_\infty(X^\varepsilon_n(u), X^\varepsilon_{n-1}(u))\Big]^2} \\
&= 2^n \, \frac{\mathrm{Var}\Big(J_n(T)\Big)}{\mathbb{E}\Big(J_n(T)\Big)^2} \leq 2^n J_n(t).
\end{aligned} \tag{5.1.18}$$

We will now use Borel–Cantelli lemma [85]. Let $\{E^n\}$ be a sequence of events in the sample space Ω; then, if the sum of the probabilities of the events in the sequence is finite, then the probability of the intersection of all the events is zero i.e., $\sum_{n=1}^{\infty} \Pr(E_n) < \infty \ then \Pr(E^s) = 0 \ where \ E^s = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} E_m$. If the events $\{E^n\}$ are independent, then

$$\sum_{n=1}^{\infty} Pr = \infty \ then \Pr(E^s) = 1 \ where \ E^s = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} E_m.$$

Conversely, if the events are independent and the sum of their probabilities diverges, then the probability of the intersection of all the events is 1. To understand this, consider the first case where the sum probabilities is finite. We can define a new sequence of events $\{F_n\}$ as follows: $F_n = \bigcup_{m\geq n} E_n$. That is the union of all the events in the sequence E_n starting from index n. Intuitively, F_n represents the event that at least one of the events E_n occurs starting from index n. Since the sum of the probabilities of the events in the sequence is finite, we have: $\sum_{n=1}^{\infty} P(E_n) < \infty$. This implies that the sum of the probabilities of the events in the sequence is also finite since F_n is the union of a finite number of events E_n. Therefore, we have:

$$\sum_{n=1}^{\infty} \Pr\left(E_n\right) = \sum_{n=1}^{\infty} \left(\bigcup_{m\geq n} E_n\right) \leq \sum_{n=1}^{\infty} \sum_{m\geq n} P(E_n) \; = \; \sum_{n=1}^{\infty} P\left(E_n\right) + P\left(E_{n+1}\right) + \ldots)$$

The last step follows from the fact that the events E_n are non–negative and therefore the sum can be rearranged. Now, let us consider the event $E^s = \bigcap_{n=1}^{\infty} F_n$. Intuitively, E^s represents the event that infinitely many of the events E_n occur. We can write E^s as a countable intersection of the events $\{Fn\}$, i.e.,: $E^s = \bigcap_{n=1}^{\infty} F_n = \lim\ \sup F_n$ where *lim sup* is the limit supremum of the sequence $\{Fn\}$. In other words, E^s is the set of all outcomes that belong to infinitely many of the events F_n. Since the sum of the probabilities of the events in the sequence $\{Fn\}$ is finite, we can apply the first Borel–Cantelli lemma to conclude that: $Pr(lim\ sup\ F_n) = 0$. Therefore, we have shown that $Pr(E^s) = Pr(lim\ sup\ F_n) = 0$, which completes the proof of the first case. We don't need the second case here because $J_n(T)$ is convergent. Having shown the convergence, we will now apply the Borel–Cantelli lemma. Let $\{f_n(x)\}$ be a sequence of measurable functions on [0,1] with $/f_n(x)/ < \infty$ for a.e. $x \subset [0,1]$. Then, there exists a sequence $\{c_n\}$ of positive real numbers such that $f_n(x)/c_n$ $-> 0$ a.e. $x \subset$[0,1]. Suppose there is a sequence such that $m(E_n) \leq 2^{-n}$.

Then from the lemma, it follows that $P(/f(x)/> N/n) > 2^{-n}$. Going back to our proof, and using Borel–Cantelli lemma:

$P \sup \quad {}_{u\in[0,s]} d_{\infty}^2 \; [\left(X_n^{\varepsilon}(u),\ X_{n-1}^{\varepsilon}(u)\right) \geq \frac{1}{\sqrt{2}^n}\ infinitely\ often = 0.$ This means that the probability of the event that the supremum of the d^2 distance between $\left(X_n^{\varepsilon}(u),\ X_{n-1}^{\varepsilon}(u)\right)$ is greater than $\frac{1}{\sqrt{2}^n}$ for infinitely many values of n is zero. Following from this, we can say that:

$$\sup_{u\in[0,s]} d_{\infty}^2\left(X_n^{\varepsilon}(u),\, X_{n-1}^{\varepsilon}(u)\right) \leq \frac{1}{(\sqrt{2})^n}. \tag{5.1.19}$$

We will now formally prove this important result. The value $1/\sqrt{2}$ comes from the fact that we are interested in the probability of the event that the supremum of the d^2 distance between $\left(X_n^\varepsilon(u) \text{ and } X_{n-1}^\varepsilon(u)\right)$ is greater than $(1/\sqrt{2})^n$ for infinitely many values of n. The value $1/\sqrt{2}$ is chosen because it is a convenient threshold that ensures the convergence of the series $\sum_{N=1}^{\infty} J_n(t)$ in the statement of the problem. In particular, the converges of the series implies that the sequence $X_n^\varepsilon(u)$ converges almost surely in the d^2 metric, which is a stronger condition than the almost surely convergence in the Euclidean metric since the latter is a special case of a d^2 metric. The choice of the threshold $(1/\sqrt{2})^n$ is motivated by the fact that it ensures that the rate of growth of $E(\sup_{u\in[0,s]} d_\infty^2\ [\left(X_n^\varepsilon(u),\ X_{n-1}^\varepsilon(u)\right)$ is bounded by $\sqrt{2}$.

The sum of the variances of the random variables is equal to n, since they are independent standard normal random variables. Therefore, the sum of the variances of the random variables in the sequence $\{X_n^\varepsilon(u)\}$ up to time n is equal to: $Var \sum_{j=1}^{n} \varepsilon_{jn}(T)u_j = \sum_{j=1}^{n} u_j^2 Var\left[\varepsilon(T)\right] = n\sum_{j=1}^{n} u_j^2$.

We can use the fact that $\sum_{j=1}^{n} u_j^2$ is bounded by 1 for all n, since u belongs to the unit ball in $\mathrm{R^d}$. Therefore, we have $Var \sum_{j=1}^{n} \varepsilon_j^n(T)u_j \le n$. Using this result, we can re–write.

$$\mathbb{E}\left[\sup_{u\in[0,s]} d_\infty^2\left(X_n^\varepsilon(u), X_{n-1}^\varepsilon(u)\right)\right] \le \sqrt{2}\sum_{n=1}^{\infty} \mathrm{Var}\left(\sum_{j=1}^{n} \varepsilon_j^n(T)\, u_j\right)^{1/2} \le \sqrt{2}\sum_{n=1}^{\infty}\sqrt{n}.$$

where the last inequality follows from the fact that the sum of the variances of the random variables in the sequence $\{X_n^\varepsilon(u)\}$ up to time n is bounded by n.

Let $Y_n = \sup_{u\in[0,s]} d_\infty^2\ [\left(X_n^\varepsilon(u),\ X_{n-1}^\varepsilon(u)\right) \ \ge 0$. Set the threshold τ_n:=$1/\sqrt{2}$. By Chebyshev (Markov), $P\left(Y_n > \frac{1}{\sqrt{2^n}}\right) \le \frac{E\left[Y_n^2\right]}{\left(\frac{1}{\sqrt{2^n}}\right)^2} = 2^n E[Y_n^2]$.By the Borel–Cantelli lemma,

$$P\left(Y_n > \frac{1}{\sqrt{2^n}}\ infinitely\ often\ (i.o.\right) = 0.$$

Then, a.s. $Y_n \leq \frac{1}{\sqrt{2^n}}$ for all sufficiently large n, i.e. $Y_n \to 0\ a.s.$ at a geometric rate with the threshold $1/\sqrt{2}$. Now, recall that the sequence $\{X_n^\varepsilon(u)\} = X_{n-1}^\varepsilon(u) + \sum_{j=1}^n \varepsilon_j^n(t)$ represents independent standard normal variables. Therefore, the distance between $d^2(X_n^\varepsilon(u), X_{n-1}^\varepsilon(u) = \sum_{j=1}^n (\varepsilon_j^n(t) - - \varepsilon_j^{n-1}(t))^2 u_j^2$. Now we will use the fact that these are independent random variables with mean 0 and variance 1. Therefore $Var(\sum_{j=1}^n (\varepsilon_j^n(t) - \varepsilon_j^{n-1}(t))^2)$ is equal to the sum of their variances which is 2 and the covariance term is zero. Using this we can write that: $E[d^2(X_n^\varepsilon(u), X_{n-1}^\varepsilon(u))] = \sum_{j=1}^n (\varepsilon_j^n(t) - \varepsilon_j^{n-1}(t))^2 u_j^2 = 2\sum_{j=1}^n u_j^2$. With unit ball, $\sum_{j=1}^n u_j^2 \leq 1$, so taking the square root, we have:

$$E(\sup_{u\in[0,s]} d_\infty^2\ [\left(X_n^\varepsilon(u),\ X_{n-1}^\varepsilon(u)\right) \leq \sqrt{2}E[\sum_{n=1}^{\infty} u_j^2]^{1/2} \leq \sqrt{2}\sum_{j=1}^{n} E(|u_j|) \leq \sqrt{2}$$

Therefore, by the Borel–Cantelli lemma, we can conclude that the probability of the event that the supremum of the d^2 distance between $X_n^\varepsilon(u)\ and\ X_{n-1}^\varepsilon(u)$ is greater than $\frac{1}{\sqrt{2^n}}$ for infinitely many values of n is zero. So, we have proven that:

$$\sup_{u\in[0,s]} d_\infty^2\left(X_n^\varepsilon(u),\ X_{n-1}^\varepsilon(u)\right) \leq \frac{1}{(\sqrt{2})^n}. \tag{5.1.20}$$

We will now complete the proof of the existence of a strong and unique solution. We know that as n goes to infinity then $X_n^\varepsilon(t) \to$ *approaches to* $X^\varepsilon(t)$ so $d_\infty^2(X_n^\varepsilon(t), X^\varepsilon(t)) = 0$ as $d_\infty(x, x) = 0$ *d(a,a) = a - a = 0.* In other words, $X_n^\varepsilon(t)\ and\ X^\varepsilon(t)$ become the same. Let us take $X^\varepsilon(t)$ as some continuous fuzzy stochastic process. Then, by the above $d_\infty^2\ [\left(X_n^\varepsilon(t)^a,\ X_{n-1}^\varepsilon(u)^a\right) \to 0\ as\ n \to \infty$. Hence, as n goes to infinity:

$$\mathbb{E}\left[\sup_{t\in I}\left(d_\infty^2\left(X_n^\varepsilon(t), X^\varepsilon(t)\right) + d_\infty^2\left(X_n^\varepsilon(t),\ X_0^\varepsilon + \int_0^t f\left(s, X^\varepsilon(s)\right) \mathrm{d}s\right.\right.\right.$$
$$\left.\left.\left. + \left\langle \int_0^t g\left(s, X^\varepsilon(s)\right) \mathrm{d}B_H^\varepsilon(s)\right\rangle\right)\right)^2\right] - > 0.$$

and this shows the existence of a strong solution. Q.E.D.

$$\sup_{t\in I} d_\infty^2\left(X^\varepsilon(t),\ X_0^\varepsilon + \int_0^t f\left(s, X^\varepsilon(s)\right) \mathrm{d}s + \left\langle \int_0^t g\left(s, X^\varepsilon(s)\right) \mathrm{d}B_H^\varepsilon(s)\right\rangle\right) = 0. \tag{5.1.21}$$

Now, we will show the uniqueness of the solution. For this we need Gronwall's lemma and inequality [86]. Let *y(t), f(t), g(t)* be non–negative functions on a closed interval $[a,\ b]$, where $a< b$. If *f(t)* satisfies inequality: $f(t) \leq g(t) +\ \int_a^b h(s)f(s)ds$ for all t ⊂ [a,b] then it follows that:

$$f(t) \leq g(t) + \int_a^t h(s)\, f(s)\, e^{\int_s^t h(u)\,\mathrm{d}u}\, \mathrm{d}s, \qquad \text{for all } t \in [a, b].$$

A special case of Grownwall's lemma arises when *g(t)* is identically zero on the interval $[a,b]$ which then simplifies to: $f(t) \leq \int_a^b h(s)f(s)ds$ which provides an upper bound on *f(t)*. Grownall's inequality provides an upper bound on the function *f(t)* in terms of the constant A and the exponential of the integral of *g(s)* over the interval $[a,\ t]$. The inequality guarantees that the growth of *f(t)* is controlled by the exponential function. The general form of the inequality is as follows: let *f(t)* be a non–negative, continuous function on a closed interval $[a,b]$, where $a<b$. If *f(t)* satisfies the inequality: $f(t) \leq A +\ \int_a^b f(s)g(s)ds$, where A is a constant and g is a non–negative, continuous function, then it follows that: $f(t) \leq A \int_a^b e^{g(s)ds}$. Going back to our prove of uniqueness. Consider that:

$$J(t) = \mathbb{E}\left[\sup_{u\in[0,t]} d_\infty^\varepsilon(X^\varepsilon(u), Y^\varepsilon(u))\right].$$

where the estimation or upper bound is:

$$j(t) \leq 3\Big(T + 4\varepsilon^{2a} + 4(T+\varepsilon)^{2a}\Big)\, L^2\, \mathbb{E}\int_0^t d_\infty^2(X^\varepsilon(s), Y^\varepsilon(s))\; \mathrm{d}s$$
$$\leq 3\Big(T + 4\varepsilon^{2a} + 4(T+\varepsilon)^{2a}\Big)\, L^2\, \int_0^t j(s)\, \mathrm{d}s.$$

In the context of the *j(t)* function, we apply Gronwall's inequality with identical initial data, *j(0) = 0,* so *A=0* and $B = 3[(T + 4^{2a}\ +4(T+\varepsilon)^{2a}]\ \ L^2$. Using $f(t) \leq A\int_a^b e^{g(s)ds}$, we can state that,

$j(t) \leq 0 * e^{3[\left(T+4^{2a}\ \ +4(T+\varepsilon)^{2a}\right]L^2 dt} = 0$. Therefore, *j(t) = 0* for all t within the given interval. This bound shows that the function *j(t)* grows exponentially in time, with the rate of growth determined by the constant $3[(T + 4^{2a}\ \ + 4(T+\varepsilon)^{2a}]L^2$. However, the growth of *j(t)* is limited by its initial value *j(0).* If *j(t)=0* for all *t* in *I*, this means that the distance between the two solutions of the stochastic differential equation is zero for all times in the interval *I.* But since *j(t)* is non–negative, this implies that *j(t) =0* for all $t \geq t_0$ where t_0 is the endpoint of the interval *I.* Therefore, the two solutions of the equation coincide for all times $t \geq t_0$ which means they become arbitrarily close to each other as time goes to infinity. Proof of uniqueness is completed. Q.E.D.

$J(t) = E[\quad \sup_{u\in[0,t]} d_\infty^2\ \ (X^\varepsilon(u),\ Y^\varepsilon(u))\ ] = 0\ a.e.,$

We will next prove that $X^\varepsilon(t)$ converges to the solution *X(t)* in L^2 space as $\varepsilon\text{–}{>}0$ and $t \subset [0,\ T]$, which is fundamental if we want to apply Itô's lemma later on. However, first we need to prove that, for every $\varepsilon > 0$ and $0<a<\frac{1}{2}$,

$$\int_s^t ((r-s+\varepsilon)^{a-1} - \Big(r-s)^{a-1}\Big)\, dr \leq \frac{a+1}{a}\varepsilon^a$$

We will need this inequality to use in the former proof. Let x = r – s and T = t - s ≥ 0. Then,

$$\int_s^t ((r-s+\varepsilon)^{a-1} - \Big(r-s)^{a-1}\Big)\, dr =$$

$$= \int_0^T ((x+\varepsilon)^{a-1} - x^{a-1})dx$$

a > 0 so we can use the integral $\int_0^T x^{a-1}dx = \left[\frac{x^a}{a}\right]_0^T$:

$$\int_0^T (\left(x+\varepsilon\right)^{a-1} - x^{a-1}\right) dx = \frac{(T+\varepsilon)^a - \varepsilon^a}{a} - \frac{T^a}{a} = \frac{(T+\varepsilon)^a - T^a - \varepsilon^a}{a}$$

$(T+\varepsilon)^a \leq T^a + \varepsilon^a$. Hence, $(T+\varepsilon)^a - \ T^a - \varepsilon^a \ \leq 0$, so the integral is non–positive. Moreover, since $(T+\varepsilon)^a \geq \ T^a$, then $-\varepsilon^a \leq (T+\varepsilon)^a - \ T^a - \varepsilon^a \ \leq 0$, and therefore the tight estimate:

$$\mid \int_s^t ((r-s+\varepsilon)^{a-1} - \left(r-s\right)^{a-1}\right) dr \mid \leq \frac{\varepsilon^a}{a}$$

Since the left side is less than zero, then:

$$\int_s^t \left((r-s+\varepsilon)^{a-1} - (r-s)^{a-1}\right) \mathrm{d}r \leq \frac{a+1}{a}\,\varepsilon^a. \tag{5.1.22}$$

We will now use this result to prove the convergence of the solution and show how to apply it for the purposes of this thesis. We want to prove that $X^\varepsilon(t) = X_0 + \int_0^t f\left(s,\ X^\varepsilon(s)\right) ds + \left\langle \int_0^t g\left(s,\ X^\varepsilon(s)\right) dB_H^\varepsilon(s)\right\rangle)]$ converges to $X(t) = X_0 + \int_0^t f\left(s,\ X(s)\right) ds + \left\langle \int_0^t g\left(s,\ X(s)\right) dB_H(s)\right\rangle)]$. We will utilise equation 5.1.9. and Euler–Maruyama numerical method for that. This is a reminder of the equation 5.1.9.

$$\begin{aligned} \mathbb{E}\Bigg[\sup_{u\in[0,t]} d_\infty^2\Big(\Big\langle\int_0^u X(s)\,\mathrm{d}W(s)\Big\rangle, \Big\langle\int_0^u Y(s)\,\mathrm{d}W(s)\Big\rangle\Big)\Bigg] \\ \leq 4\,\mathbb{E}\int_0^t d_\infty^2(\langle X(s)\rangle, \langle Y(s)\rangle)\ \mathrm{d}s. \end{aligned}$$

Euler–Maruyama is a modification of the standard Euler method for solving ordinary differential equations (ODEs) to solve stochastic differential equations

(SDEs). The solution of an SDE is itself a random process and therefore cannot be computed exactly in closed form. The Euler–Maruyama approximates a solution of an SDE as a piecewise linear function. At each step, the approximation is updated by taking a small step in time and adding a random perturbation. Specifically, it takes the form: $dX(t) = f(t,X(t))dt + g(t,X(t))dBH(t)$. Choose a time step size Δt and define the time points $t_K = k\Delta t$ for $k = 0,1,2,\ldots,n$. Initialise the approximation by setting $X_0 = X(t_0)$. For $k = 1,2,\ldots,n$, compute the approximation at time t_K as: $X(t_K) = X(t_{K-1}) + f(t_{K-1}, X(t_{K-1}))\Delta t + g(t_{K-1},X(t_{K-1}))(B^H(t_K) - B^H(t_{K-1}))$

where $B^H(t_K)$ —- $B^H(t_{K-1})$ is the increment of the Brownian motion over the time interval $[t_{K-1}, t_K]$. Now, to obtain a bound on the distance between $X(t)$ and $X^\varepsilon(t)$, we use the triangle inequality and the fact that the distance between the two drift functions $f(X)$ and $f(X^\varepsilon)$ can be bounded by the integral of the distance between them over time. Specifically, we have:

$$\begin{aligned}\mathbb{E}\left[\sup_{u\in[0,t]} d^2_\infty(X(u),X^\varepsilon(u))\right] &\leq 2\,\mathbb{E}\left[\sup_{u\in[0,t]}\int_0^u d^2_\infty\Big(f\big(s,X(s)\big),\, f\big(s,X^\varepsilon(s)\big)\Big)\,\mathrm{d}s\right] \\ &\quad + 2\,\mathbb{E}\Big[\sup_{u\in[0,t]} d^2_\infty\Big(\Big\langle\int_0^u g\big(s,X(s)\big)\,\mathrm{d}B_H(s)\Big\rangle, \\ &\qquad \Big\langle\int_0^u g\big(s,X^\varepsilon(s)\big)\,\mathrm{d}B^\varepsilon_H(s)\Big\rangle\Big)\Big]. \end{aligned} \tag{5.1.23}$$

To obtain this result, we applied triangle inequality to the distance between X(u) and X$^\varepsilon$(u) to bound the first term.

$$\mathbb{E}\left[\sup_{u\in[0,t]} d^2_\infty(X(u),X^\varepsilon(u))\right] \leq \mathbb{E}\left[\sup_{u\in[0,t]} d^2_\infty(X(u),X^\varepsilon(u))\right] + \mathbb{E}\left[\sup_{u\in[0,t]} d^2_\infty(X^\varepsilon(u),X(u))\right].$$

Similarly, we can bound the second term and substitute these bounds we obtain 5.1.23. Then, we apply the Burkholder–Davis–Gundy (BDG) inequality in our proof. BDG is a tool used to bound the moments of stochastic integrals involving martingales. It provides for a way to control the growth of moments of stochastic integrals. In general form, the inequality states that:

$E[\sup_{0\leq s\leq t}|M_s|^p \leq C_p E[(M_t^*)^p]$, where $\mathrm{M_t}$ is a continuous square–integrable martingale, p>1 is a real positive number and $M_t^* = E[\sup_{0\leq s\leq t}|M_s|$ is the maximum value of the absolute value of the martingale up to time t. In other words, BGD inequality relates the supremum of the pth power of a continuous square–integrable martingale $\mathrm{M_t}$ to the pth moment of its maximum value M_t^*. To apply BGD, we first need to express the stochastic integrals in terms of the increments of the Brownian motion (so they are martingale). In particular, we use the 5.1.2 derivation of the fractional Brownian motion $B_{H,\varepsilon}(t)$:

$$B_{H,\varepsilon}(t) = a\int_0^t \varphi^\varepsilon(s)ds + \ \varepsilon^a B(t)$$

where $\varphi^\varepsilon(s) = \int_0^t (t-s+\varepsilon)^{a-1}]dB(s)$ and B(t) is a standard Brownian motion.

Using this representation, we can express the stochastic integral involving the difference between X and X^ε as:

$\int_0^u (g(s,\ X(s))(dB_H^\varepsilon(s) - dB_H(s)) = \int_0^u (g(s,\ X^\varepsilon(s))(dB_H^\varepsilon(s) - dB_H(s))\int_0^u (g(s,\ X(s)) - g(s,\ X^\varepsilon(s))dB_H^\varepsilon(s)$.

We can similarly express the other integrals involving X and X^ε in terms of the increments of the Brownian motion. Now that we have our representations, we can apply BDG inequality to bound the supremum of the stochastic integrals involving Brownian motion increments.

Recall that a special case of BDG that we used earlier:

$$\mathbb{E}\left[\sup_{u\in[0,t]} d_\infty^2\left(\left\langle\int_0^u X(s)\,\mathrm{d}W(s)\right\rangle, \left\langle\int_0^u Y(s)\,\mathrm{d}W(s)\right\rangle\right)\right] \leq 4\,\mathbb{E}\int_0^t d_\infty^2(\langle X(s)\rangle, \langle Y(s)\rangle)\ \mathrm{d}s.$$

Then,

$$
\begin{aligned}
&\mathrm{E}\left[\sup_{u\in[0,t]} d_\infty^2(X(u), X^\varepsilon(u))\right] \\
&\leq 2\mathrm{E}\sup_{u\in[0,t]}\int_0^u d_\infty^2(f(s,X(s)), f(s,X^\varepsilon(s)))ds \\
&+ 4\mathrm{E}\sup_{u\in[0,t]} d_\infty^2\left\langle\int_0^u g(s,X(s))dB_H(s)\right\rangle, \left\langle\int_0^u g(s,X(s))dB_H^\varepsilon(s)\right\rangle \\
&+ 4\mathrm{E}\sup_{u\in[0,t]} d_\infty^2\left\langle\int_0^u g(s,X(s))dB_H^\varepsilon(s)\right\rangle, \left\langle\int_0^u g(s,X^\varepsilon(s))dB_H^\varepsilon(s)\right\rangle \\
&= 2\mathrm{E}\sup_{u\in[0,t]}\int_0^u d_\infty^2(f(s,X(s)), f(s,X^\varepsilon(s)))ds \\
&+ 4\mathrm{E}\sup_{u\in[0,t]} d_\infty^2\left\langle\int_0^u g(s,X(s))dB_H(s)\right\rangle, \left\langle\int_0^u g(s,X(s))dB_H^\varepsilon(s)\right\rangle \\
&+ 4\mathrm{E}\sup_{u\in[0,t]} d_\infty^2\left\langle\int_0^u g(s,X(s))dB_H^\varepsilon(s)\right\rangle, \left\langle\int_0^u g(s,X^\varepsilon(s))dB_H^\varepsilon(s)\right\rangle \\
&= 2\mathrm{E}\sup_{u\in[0,t]}\int_0^u d_\infty^2(f(s,X(s)), f(s,X^\varepsilon(s)))ds \\
&+ 4\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u g(s,X(s))(dB_H(s) - dB_H^\varepsilon(s))\right|^2 \\
&+ 4\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u (g(s,X(s)) - g(s,X^\varepsilon(s)))dB_H^\varepsilon(s)\right|^2
\end{aligned}
$$

We will now substitute 5.1.2 $B_{H,\varepsilon}(t) = a\int_0^t \varphi^\varepsilon(s)ds + \varepsilon^a B(t)$, where $\varphi^\varepsilon(s) = \int_0^s \left[(s-r+\varepsilon)^{a-1} - (s-r)^{a-1}\right] dB(r)$, so

$$
\begin{aligned}
B_{H,\varepsilon}(t) &= a\int_0^t \left[(s-r+\varepsilon)^{a-1}\right] dB(s) \\
\text{and}\quad B(t) &= a\int_0^t \left[(s-r)^{a-1}\right] dB(s) \\
\text{where}\quad \varepsilon &> 0, \quad a = H - 1/2
\end{aligned}
$$

into above and use the (BDG) inequality in our proof. We need to replace the fBm with normal Brownian motion.

So,

$$\begin{aligned}
&\mathrm{E}\left[\sup_{u\in[0,t]} d_\infty^2\left(X(u), X^\varepsilon(u)\right)\right] \\
&\leq 2\mathrm{E}\sup_{u\in[0,t]}\int_0^u d_\infty^2\left(f(s,X(s)), f(s,X^\varepsilon(s))\right) ds \\
&+ 8e^{2a}\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u g\left(s, X^\varepsilon(s)\right) dB(s)\right|^2 \\
&+ 8a^2\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u g\left(s, X^\varepsilon(s)\right)\left(\int_0^s \left[(s-r+\varepsilon)^{a-1} - (s-r)^{a-1}\right] dB(r)\right) ds\right|^2 \\
&+ 8a^2\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u \left(g\left(s, X(s)\right) - g(s, X^\varepsilon(s))\right)\left(\int_0^s (s-r+\varepsilon)^{a-1} dB(r)\right) ds\right|^2 \\
&+ 8e^{2a}\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u \left(g(s,X(s)) - g\left(s, X^\varepsilon(s)\right)\right) dB(s)\right|^2
\end{aligned}$$

We now have:

$$\begin{aligned}
&\mathrm{E}\left[\sup_{u\in[0,t]} d_\infty^2(X(u), X^\varepsilon(u))\right] \leq 2\,\mathrm{E}\sup_{u\in[0,t]}\int_0^u d_\infty^2(f(s,X(s)),\ f(s,X^\varepsilon(s)))\ ds \\
&+ 8\varepsilon^{2a}\,\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u g(s,X^\varepsilon(s))\ dB(s)\right|^2 \\
&+ 8a^2\,\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u\int_s^u g(r,X^\varepsilon(r))\left[(r-s+\varepsilon)^{a-1} - (r-s)^{a-1}\right] dr\, dB(s)\right|^2 \\
&+ 8a^2\,\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u\int_s^u \Big(g(r,X^\varepsilon(r)) - g(s,X^\varepsilon(s))\Big)(r-s+\varepsilon)^{a-1}\, dr\, dB(s)\right|^2 \\
&+ 8\varepsilon^{2a}\,\mathrm{E}\sup_{u\in[0,t]}\left|\int_0^u \Big(g(s,X(s)) - g(s,X^\varepsilon(s))\Big)\, dB(s)\right|^2.
\end{aligned}$$

We will now apply the Doob's inequality for Brownian motion to the 2nd, 3rd, 4th and 5th terms which states that $E\ [\underset{u\in[0,t]}{sup}\ d_\infty^2\ |B_t|]^2 \leq 4E|B_t|^2$, Holder's inequality to the 3rd and 4th term f$\subset$L$^{\mathrm{P}}$, q$\subset$L$^{\mathrm{Q}}$, $\int fg \leq ||f||_P||g||_Q$ and Itô isometry to the 2nd, 3rd, 4th and last terms , which states that: $\mathrm{E}(\int_0^t \int f_t dW_t)^2) = E(\int_0^t (f_t)^2 dt)$.

$$
\begin{aligned}
&\mathrm{E}\Bigg[\sup_{u\in[0,t]} d_\infty^2(X(u),X^\varepsilon(u))\Bigg]\\
&\le 2\,\mathrm{E}\sup_{u\in[0,t]}\int_0^t d_\infty^2(f(s,X(s)),\ f(s,X^\varepsilon(s)))\ ds + 32\varepsilon^{2a}\,\mathrm{E}\int_0^u g^2(s,X^\varepsilon(s))\ ds\\
&+32a^2\,\mathrm{E}\int_0^t\Bigg(\int_s^t g^2(r,X^\varepsilon(r))\Big[(r-s+\varepsilon)^{a-1}-(r-s)^{a-1}\Big]\,dr\\
&\qquad\qquad\qquad\times\int_s^t\Big[(r-s+\varepsilon)^{a-1}-(r-s)^{a-1}\Big]\,dr\Bigg)\,ds\\
&+32a^2\,\mathrm{E}\int_0^t\Bigg(\int_s^t\Big(g(r,X^\varepsilon(r))-g(s,X^\varepsilon(s))\Big)^2(r-s+\varepsilon)^{a-1}\,dr\\
&\times\int_s^t(r-s+\varepsilon)^{a-1}\,dr\Bigg)\,ds+32\varepsilon^{2a}\,\mathrm{E}\int_0^u\Big(g(s,X(s))-g(s,X^\varepsilon(s))\Big)^2\,ds.
\end{aligned}
$$

Using 5.1.14, 5.1.15, 5.1.21 and assumptions A1–A3:

$$
\begin{aligned}
&\mathrm{E}\Bigg[\sup_{u\in[0,t]} d_\infty^2(X(u),X^\varepsilon(u))\Bigg]\\
&\le 2L^2\,\mathrm{E}\int_0^t d_\infty^2(X(s),X^\varepsilon(s))\ ds+64\varepsilon^{2a}C^2\int_0^t\Big(1+\mathrm{E}\|\,[X^\varepsilon(s)]^0\,\|^2\Big)\,ds\\
&\quad+64C^2(T+\varepsilon)^{2a}\frac{a+1}{a}\varepsilon^a\int_0^t\Big(1+\mathrm{E}\|\,[X^\varepsilon(s)]^0\,\|^2\Big)\,ds\\
&\quad+32L^2(T+\varepsilon)^{2a}\,\mathrm{E}\int_0^t d_\infty^2(X(s),X^\varepsilon(s))\ ds+32L^2\varepsilon^{2a}\,\mathrm{E}\int_0^t d_\infty^2(X(s),X^\varepsilon(s))\ ds\\
&\le\Big(2L^2+32L^2(T+\varepsilon)^{2a}+32L^2\varepsilon^{2a}\Big)\int_0^t\mathrm{E}\Bigg[\sup_{u\in[0,s]}d_\infty^2(X(u),X^\varepsilon(u))\Bigg]\,ds\\
&\qquad+\Big(64\varepsilon^{2a}C^2+64C^2(T+\varepsilon)^{2a}\tfrac{a+1}{a}\varepsilon^a\Big)\int_0^t\Big(1+\mathrm{E}\|\,[X^\varepsilon(s)]^0\,\|^2\Big)\,ds.
\end{aligned}
\tag{5.1.24}
$$

We will now show the convergence of the solution of a fuzzy stochastic differential equation driven by fBm. The first part of 5.1.24 provides a bound on the supremum of the distance between the solution *X(u)* and a perturbed version of the solution $X^{\varepsilon}(u)$ over the interval [0,t]. Here $\sup_{u\in[0,s]} d_{\infty}^{2}\ (X(u),\ X^{\varepsilon}(u))$ denotes the supremum distance between the two processes, L is a Lipschitz constant, T is the final time, ε is a small perturbation parameter, and a is the Hurst parameter of the fBm. The second part of 5.1.24 provides a bound on the second moment of the perturbed solution $X^{\varepsilon}(s)$. Here, C is a constant and $[X^{\varepsilon}(s)]^{0}$ is the initial value of the perturbed process. We will apply Gronwall's lemma to the inequality to show that the supremum of the distance between $X(u)$ *and* $X^{\varepsilon}(u)$ converges to zero as ε approaches zero, which implies the convergence of the solution $X(u)$ to a limit process as the perturbation parameter ε goes to zero. The equation provides an estimate for the convergence rate of the solution of the fuzzy stochastic differential equation driven by a fBm. To apply Gronwall's lemma to the inequality, we first define $Y(s) = \underset{u\in[0,t]}{E}\ [sup d_{\infty}^{2}\ (X(u),\ X^{\varepsilon}(u))\ ]$ and rewrite the inequality as:

$$\begin{aligned} Y(s) \leq &\left(2L^2 + 32L^2(T+\varepsilon)^{2a} + 32L^2\varepsilon^{2a}\right)\int_0^t Y(s)\,ds \\ &+ \left(64\varepsilon^{2a}C^2 + 64C^2(T+\varepsilon)^{2a}\tfrac{a+1}{a}\varepsilon^{a}\right)\int_0^t \left(1 + \mathrm{E}\|[X^{\varepsilon}(s)]^0\|^2\right) ds. \end{aligned}$$

We can now apply Gronwall's lemma, which states that if $f(t)$ and $g(t)$ are nonnegative continuous functions on [0,T] and if there exists a non-negative constant K such that $f(t)\ \leq\ K + \int_0^T g(s)f(s)\ \forall\ t\in[0,T]$ then $f(t) \leq Ke^{\int_0^T g(s)f(s)}\ \ \forall\ t\in[0,T]$. Applying this lemma we have:

$$\begin{aligned} Y(s) \leq\ &\left(2L^2 + 32L^2(T+\varepsilon)^{2a} + 32L^2\varepsilon^{2a}\right)\int_0^t Y(s)\,ds \\ &+ \left(64\varepsilon^{2a}C^2 + 64C^2(T+\varepsilon)^{2a}\frac{a+1}{a}\varepsilon^{a}\right)\int_0^t \left(1 + E\left\|\left[X^{\varepsilon}(s)\right]^0\right\|^2\right) ds. \end{aligned}$$

$$K = (64\varepsilon^{2a}C^2 + 64C^2(T+\varepsilon)^{2a}\frac{a+1}{a}\varepsilon^{a})\int_0^t (1 + E||[X^{\varepsilon}(s)]^0||^2)ds$$

$$g(s) = 2L^2 + 32L^2(T+\varepsilon)^{2a} + 32L^2\varepsilon^{2a}$$

Then, by lemma $f(t) \leq Ke^{\int_0^T g(s)f(s)} \quad \forall\, t \in [0,T]$, we obtain:

$$\begin{aligned}
&\mathrm{E}\left[\sup_{u\in[0,t]} d_\infty^2(X(u), X^\varepsilon(u))\right] \\
&\leq \left(64\varepsilon^{2a}C^2 + 64C^2(T+\varepsilon)^{2a}\frac{a+1}{a}\varepsilon^a\right)\int_0^t \left(1 + \mathrm{E}\left\| [X^\varepsilon(s)]^0 \right\|^2\right) ds \\
&\quad \times e^{2L^2+32L^2(T+\varepsilon)^{2a}+32L^2\varepsilon^{2a}}.
\end{aligned}$$

We can use the fact that the space of continuous functions equipped with the supremum norm is complete i.e., all Cauchy sequences converge, to conclude that the solution *X(u)* converges to a limit process as the perturbation parameter ε goes to zero. The prefactor $K = (64\varepsilon^{2a}C^2 + 64C^2(T+\varepsilon)^{2a}\frac{a+1}{a}\varepsilon^a)\int_0^t (1 + E||[X^\varepsilon(s)]^0||^2)ds$ goes to 0 because *a>0* and as ε approaches zero, $K_\varepsilon(t) = O(e^a) \to 0$.

This completes the proof that $X^\varepsilon(t)$ converges to the solution *X(t)* in L^2 space as $\varepsilon \to 0$ and $t \subset [0,\ T]$. Q.E.D.

5.1.5 Valuation of Derivatives under Mixed Fuzzy Fractional Brownian Motion

We will now describe the dynamics of a particle, considering the uncertainties involved. The underlying stochastic process is modelled by a fractional Brownian motion (fBm), known for its long–range dependence and self–similarity, which makes it suitable for financial modelling. Fractional Brownian motion is traditionally employed in financial modelling due to its ability to represent memory effects through the Hurst exponent with values significantly influencing the roughness or smoothness of volatility [78].

The model involves fuzzy processes, denoted by $x_\varepsilon(t)$, which are functions that map the time parameter t to a set of possible values in R with certain degrees of membership that quantify uncertainty or ambiguity inherent to market expectations. Unlike traditional models that typically incorporate either fBm or fuzzy logic individually, our approach combines both techniques to offer a more robust and realistic characterisation of market uncertainty.

We consider a stochastic differential equation (SDE) that includes a linear drift coefficient μ and a volatility coefficient σ, both initially assumed constant. To find the closed explicit fuzzy solution for $\mu \geq 0$, the system of equations is solved to obtain the fuzzy solutions $X^l_\varepsilon(t)$ and $X^u_\varepsilon(t)$ for the lower and upper bounds, respectively [79]. Similarly, the solution is derived for $\mu < 0$ by obtaining the fuzzy solutions $X^l_\varepsilon(t)$and $X^u_\varepsilon(t)$ for the lower and upper bounds, respectively.

We used existing results from [80] [87] to extend the unique solution to the fuzzy stochastic fractional differential equation (FSFDE) for $\mu \geq 0$. According to these results, which provided certain regularity conditions such as Lipschitz continuity with constant L of the function a, continuity of the function b, and σ which is locally Lipschitz continuous, then there exist constants K and p such that $//\sigma(x)// \leq K(1 + //x//^P)$ for all x. Then, for any initial condition $X(0) = x(0)$ with probability 1, there exists a unique solution to the FSFDE in the form:

$$dX(t) = \ [aX(t) + \ b]\, dt \ + \ \sigma\, (X(t))\, dW(t),$$

where *W(t)* is a standard d–dimensional Wiener process. Clarification: This existence–uniqueness template is the classical Yamada–Watanabe framework for Itô SDEs; in our fBm case with H $> \frac{1}{2}$ we work pathwise (Young/Stratonovich integral), and the analogous Lipschitz/linear–growth hypotheses ensure well–posedness in that calculus.

The key idea is to show that when $h > \frac{1}{2}$, the matrix–valued function a, the vector–valued function b, and the matrix–valued function σ satisfy certain regularity conditions that enable us to apply standard techniques from the theory of stochastic differential equations to prove the existence and uniqueness of a solution to the FSFDE. In particular, the Lipschitz continuity condition on 'a' ensures the local boundedness of the solution, while the local Lipschitz continuity condition on σ ensures that the solution is almost surely continuous. The continuity condition on 'b' ensures that the drift term is well–defined and continuous.

We start with the typical partial differential equation used in mathematical finance:

$$X(t) = X_0 + \int_0^t \mu X(s)ds + \int_0^t \sigma X(s)dB_H(s), \quad X_0 = X(0)$$

where *X(t)* is the value of the financial quantity at time *t*, *X(0)* is the initial value of X at time 0, μ and σ are constants representing the drift and volatility of *X*, respectively, and $B_H(s)$ is the fractional Brownian motion (fBm) process which has long–range dependence and self–similarity properties. We already covered why the fBm process is used in financial modelling. In summary, it can capture the long–term dependencies in financial data, which are not always captured by traditional Brownian motion processes. This makes it suitable for modelling financial quantities that exhibit long–term trends or memory effects. Unlike crisp partial differential equation, the equation involving uncertainties is modelled using fuzzy processes. In the case of linear coefficients, an explicit solution can be obtained. The fractional fuzzy stochastic differential equation (FFSDE) satisfies the assumptions of the existence–uniqueness framework (Lipschitz and linear growth) in the H $> \frac{1}{2}$ Young/Stratonovich setting. The Yamada–Watanabe–Kunita theorem provides conditions for the existence and uniqueness of solutions to certain types of SDEs [88]. [Note: Yamada–Watanabe is stated for Itô SDEs; we use its frame (Lipschitz/linear growth) while our actual integral with fBm and H $> \frac{1}{2}$ is treated via pathwise (Young/Stratonovich) calculus.] The equation we are deriving is a stochastic differential equation (SDE) describing a system with two state variables, $X_l^{\varepsilon a}(t)$ and $X_u^{\varepsilon a}(t)$. The SDE has a fractional Wiener–type process W, which represents the random noise in the system. The SDE can be divided into two cases depending on the value of the drift coefficient μ. When $\mu \geq 0$, H $> \frac{1}{2}$ and Stratonovich/Young Integral, the unique solution to the linear multiplicative SDE has closed form and can also also be obtained using Feynman–Kac Theorem after we pass to the ε–regularised Liouville form– $B_{H,\varepsilon}(t) = a\int_0^t \varphi^\varepsilon(s)ds + \varepsilon^a B(t)$ and lift the system to a Markov state $(X^\varepsilon, \varphi^\varepsilon)$ driven by the standard Brownian Motion B. For the lifted Markov system, classical Feynman–Kac applies to the transition functional $u^\varepsilon(t,x) = E[\Phi(X_t^\varepsilon)\,|X_0^\varepsilon = x]$, yielding a parabolic PDE whose solution reproduces the same exponential form for X^ε. Letting ε –> 0 then recovers the H $> \frac{1}{2}$ solution. We emphaise that plain fBm is not Markov when H $\neq \frac{1}{2}$, so the Feynman–Kac is used through this regularisation. The direct Stratonovich/Young derivation bypasses this step and reaches the same solution. The solution for $X_l^{\varepsilon a}(t)$ and $X_u^{\varepsilon a}(t)$ can be expressed in a matrix form. The solution involves exponentials of μt, α, and the Wiener process W.

When $\mu < 0$, the unique solution to the SDE can also be obtained using the same theorem. Again, the solution for $X_l^{\varepsilon a}(t)$ and $X_u^{\varepsilon a}(t)$ be expressed in a matrix form. The solution involves hyperbolic functions of μt (cosh and sinh), exponentials of α and the Wiener process W. In both cases, the solution involves an integral over the Wiener process W, which is a stochastic integral. The integral is evaluated using the Itô or Stratonovich sense (with the usual Itô–Stratonovich conversion when needed) , which is a stochastic calculus tool used to integrate stochastic processes with respect to the Wiener process. The integral is approximated using the Stratonovich integral, which is a different stochastic calculus tool to Itô and is based on symmetric Riemann sums. It gives a different result than the Itô integral. The result of the Stratonovich integral is denoted by the symbol $\langle\ \rangle$ in the equation we derive. The final result is an approximation of the solution to the SDE in terms of $X(0)$, $X_l^{\varepsilon 1}(0)$, $X_u^{\varepsilon 1}(0)$, α, σ, μ, and the Wiener process W. The solution is a fuzzy solution because the state is modelled as a fuzzy process (its values are fuzzy sets) and the outcome also depends on the realisation of the Wiener process. Now we will derive it.

FFSDE involves a stochastic process X that maps from $R+ \times \Omega$ to $F(R)$, where $F(R)$ is the space of fuzzy real numbers. $X_l^1(t)$ *and* $X_u^1(t)$ are functions that map from $R+ \times \Omega$ to R and define the lower and upper bounds of the fuzzy process $[X^1(t)] = X_l^1(t)$ *and* $X_u^1(t)$.

To obtain a closed explicit form of the solution for $\mu \geq 0$, a system of equations is derived. The solution to the system provides a unique solution, $X_l^{\varepsilon 1}(t)$ *and* $X_u^{\varepsilon 1}(t)$, which is expressed as an explicit formula involving the initial values $X_l^{\varepsilon 1}(0)$ *and* $X_u^{\varepsilon 1}(0)$. This formula involves the exponential function and the fBm process $\mathrm{B_H(s)}$. For every $\alpha \in [0, 1]$, a similar procedure is applied to obtain a new system of equations, which is similar to the previous system but involves the fBm process $dB_H^{\varepsilon}(s)$ instead of $\mathrm{dB_H(s)}$. The solution to this new system of equations also involves an explicit formula, $X_l^{\varepsilon a}(t)$ and $X_u^{\varepsilon a}(t)$ which is expressed as an integral involving the initial values and the fBm process $dB_H^{\varepsilon}(s)$.

To obtain an explicit solution, we first express *X(s)* in terms of its initial value X_0 and the fBm process $B_H(s)$:

$$X(s) = X_0 + \int_0^s \mu X(r)\, dr + \int_0^s \sigma X(r)\, dB_H(r).$$

and for $\mathrm{H} = \frac{1}{2}$, apply Itô's formula to the function $f(X(s) = \exp(aX(s))$ for some constant a:

$$df\Big(X(s)\Big) = a\ \exp\Big(aX(s)\Big)\, dX(s) + \frac{1}{2}\, a^2\ \exp\Big(aX(s)\Big)\, \Big(dX(s)\Big)^2.$$

Substitute for *dx*, we get:

$$df\Big(X(s)\Big) = ae^{aX(s)}\Big(\mu X(s)\, ds + \sigma X(s)\, dB_H(s)\Big) + \frac{1}{2}\, a^2\, e^{aX(s)}\, \sigma^2 X(s)^2\, ds.$$

Integrate both sides from 0 to t, we get:

$$e^{aX(t)} - e^{aX(0)} = \int_0^t \alpha e^{\alpha X(s)}\Big(\mu X(s)\, ds + \sigma X(s)\, dB_H(s)\Big) + \frac{1}{2}\alpha^2 \int_0^t e^{\alpha X(s)} \sigma^2 X^2(s)\, ds.$$

Solving for *X(t)* we obtain:

$$X_t = X_0 \exp(\Big(\mu - \frac{1}{2}\sigma^2\Big)\, t + \sigma W_t)$$

This is the solution to the crisp SDE with a linear coefficient. We are interested in the fuzzy SDE. Consider the FFSDE which satisfies the theorem we proved in [P2], that there is the existence of a strong and unique solution.

$$\sup_{t\in I}[d_\infty^2[(X^\varepsilon(t), X_0^\varepsilon + \int_0^t f\,(s,\ X^\varepsilon(s))\,ds + \Big\langle \int_0^t g\,(s,\ X^\varepsilon(s))\,dB_H^\varepsilon(s)\Big\rangle)] = 0$$

Then, the given equation is:

$$X(t) = \ X_0 + \int_0^t \mu X(s)ds + \Big\langle \int_0^t \frac{\sigma}{2}(X_l^1(s) + X_u^1(s))dB_H(s)\Big\rangle)$$

To obtain a closed form solution for $\mu \geq 0$, we obtain the following system of equations for lower and upper bounds.

$$\begin{cases} X_l^1(t) = \ X_l^1(0) + \int_0^t \mu X_l^1(s)ds + \Big\langle \int_0^t \frac{\sigma}{2}(X_l^1(s) + X_u^1(s))dB_H(s)\Big\rangle) \\ X_u^1(t) = \ X_u^1(0) + \int_0^t \mu X_u^1(s)ds + \Big\langle \int_0^t \frac{\sigma}{2}(X_l^1(s) + X_u^1(s))dB_H(s)\Big\rangle) \end{cases}$$

We add two equations and simplify to get.

$$\begin{aligned} X_l^1(t) + X_u^1(t) &= X_l^1(0) + X_u^1(0) + \int_0^t \mu\Big(X_l^1(s) + X_u^1(s)\Big)\,ds \\ &\quad + \Big\langle \int_0^t \sigma\Big(X_l^1(s) + X_u^1(s)\Big)\,dB_H(s)\Big\rangle. \end{aligned}$$

We will now transition from fBm to standard Brownian motion. To achieve this, we substitute a derivative of $B_{H,\varepsilon}(t)$ in 5.1.2 into above equation. Recall 5.1.2.

$$B_{H,\varepsilon}(t) = a\int_0^t \varphi^\varepsilon(s)\,ds + \varepsilon^a B(t).$$

where $\varphi^\varepsilon(s) = \int_0^t (t-s+\varepsilon)^{a-1}]dB(s)$

So, taking derivative $dB_{H,\varepsilon}(t) = \frac{d[a\int_0^t \varphi^\varepsilon(s)ds+\ \varepsilon^a B(t)]}{dt} = a\varphi^\varepsilon(t) + \varepsilon^a dB(t)$

Substitute these identities into expression for $X_l^1(t) + X_u^1(t)$ will convert this FFSDF into a Wiener process. $\langle\int()\rangle$ this symbol is Stratonovich integral, and this is why we needed the Stratonovich–Itô conversion so we could go back and fourth between Wiener and fractal processes. The Stratonovich–Itô conversion equation, which was mentioned earlier in the text, provides a way to relate SDEs driven by a Wiener process to SDEs driven by a fractional Brownian motion process or other fractional processes. This theorem allows for the conversion between the two types of processes, which is necessary in this case to convert the FFSDE involving a fractional Brownian motion into an SDE involving a standard Wiener process. By using the Stratonovich–Itô conversion theorem and the Stratonovich integral, it becomes possible to work with the more familiar Wiener process while still accounting for the fractal nature of the original process through the appropriate transformations and integrals.

$$\begin{aligned}
X_l^1(t) + X_u^1(t) &= X_l^1(0) + X_u^1(0) + \int_0^t \mu\Big(X_l^1(s) + X_u^1(s)\Big)\,ds \\
&\quad + \Big\langle \int_0^t \sigma\Big(X_l^1(s) + X_u^1(s)\Big)\Big(a\varphi^\varepsilon(t) + \varepsilon^a dB(t)\Big)\Big\rangle \\
&= X_l^1(0) + X_u^1(0) + \int_0^t \Big(\mu + \sigma a\varphi^\varepsilon(s)\Big)\Big(X_l^1(s) + X_u^1(s)\Big)\,ds \\
&\quad + \int_0^t \Big(X_l^1(s) + X_u^1(s)\Big)\sigma\varepsilon^a\,dB(t).
\end{aligned}$$

Now, we can use the solution that we used to solve the crisp stochastic differential equation and substitute back fBm expression from 5.1.2.

$$X_l^{\varepsilon 1}(t) + X_u^{\varepsilon 1}(t) = \Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)\exp\Big(\mu t + \sigma a \int_0^t \varphi^\varepsilon(s)\,ds - \frac{1}{2}\sigma^2\varepsilon^{2a}t + \sigma\varepsilon^a W(t)\Big)$$
$$= \Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)\exp\Big(\mu t + \sigma B_H^\varepsilon(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}t\Big). \tag{5.1.25}$$

Finally, we apply similar approach for every $a \in [0,\,1]$ to get the system of equations.

$$\begin{cases} X_l^{\varepsilon a}(t) = & X_l^{\varepsilon a}(0) + \int_0^t \mu X_l^{\varepsilon a}(s)ds + \Big\langle \int_0^t \frac{\sigma}{2}(X_l^{\varepsilon 1}(s) + X_u^{\varepsilon 1}(s))dB_H^\varepsilon(s)\Big\rangle) \\ X_u^{\varepsilon a}(t) = & X_u^{\varepsilon a}(0) + \int_0^t \mu X_u^{\varepsilon a}(s)ds + \Big\langle \int_0^t \frac{\sigma}{2}(X_l^{\varepsilon 1}(s) + X_u^{\varepsilon 1}(s))dB_H^\varepsilon(s)\Big\rangle) \end{cases}$$

For $\mu \geq 0$, we apply the solution derived above, namely 5.1.25 for upper and lower bounds:

$$\begin{aligned} X_l^{\varepsilon a}(t) &= X_l^{\varepsilon a}(0) + \int_0^t \mu X_l^{\varepsilon a}(s)\,ds \\ &\quad + \int_0^t \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)\exp\Big(\mu s + \sigma B_H^\varepsilon(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\ dB_H^\varepsilon(s), \\ X_u^{\varepsilon a}(t) &= X_u^{\varepsilon a}(0) + \int_0^t \mu X_u^{\varepsilon a}(s)\,ds \\ &\quad + \int_0^t \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)\exp\Big(\mu s + \sigma B_H^\varepsilon(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\ dB_H^\varepsilon(s). \end{aligned}$$

In terms of the Wiener process, we again get rid of fBm and replace it with normal Brownian motion. So, we will substitute a derivative of 5.1.2 into above equation same as we did before.

$$
\begin{aligned}
X_l^{\varepsilon a}(t) &= X_l^{\varepsilon a}(0) + \int_0^t \mu X_l^{\varepsilon a}(s)\, ds + \int_0^t \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\times \exp\Big(\mu s + \sigma a \int_0^s \varphi^{\varepsilon}(u)\, du + \sigma \varepsilon^a B(t) - \frac{1}{2}\sigma^2 \varepsilon^{2a} s\Big)\ ds\ \Big(a\varphi^{\varepsilon}(t) + \varepsilon^a dB(t)\Big) \\
&= X_l^{\varepsilon a}(0) + \int_0^t \mu X_l^{\varepsilon a}(s)\, ds + a\int_0^t \varphi^{\varepsilon}(t)\frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\times \exp\Big(\mu s + \sigma a \int_0^s \varphi^{\varepsilon}(u)\, du + \sigma \varepsilon^a B(t) - \frac{1}{2}\sigma^2 \varepsilon^{2a} s\Big)\ ds \\
&+ \varepsilon^a \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\times \int_0^t \exp\Big(\mu s + \sigma a \int_0^s \varphi^{\varepsilon}(u)\, du + \sigma \varepsilon^a B(t) - \frac{1}{2}\sigma^2 \varepsilon^{2a} s\Big)\ dB(t).
\end{aligned}
$$

Similarly:

$$
\begin{aligned}
X_u^{\varepsilon a}(t) &= X_u^{\varepsilon a}(0) + \int_0^t \mu X_u^{\varepsilon a}(s)\, ds + \int_0^t \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\times \exp\Big(\mu s + \sigma a \int_0^s \varphi^{\varepsilon}(u)\, du + \sigma \varepsilon^a B(t) - \frac{1}{2}\sigma^2 \varepsilon^{2a} s\Big)\ ds\ \Big(a\varphi^{\varepsilon}(t) + \varepsilon^a dB(t)\Big) \\
&= X_u^{\varepsilon a}(0) + \int_0^t \mu X_u^{\varepsilon a}(s)\, ds + a\int_0^t \varphi^{\varepsilon}(t)\frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\times \exp\Big(\mu s + \sigma a \int_0^s \varphi^{\varepsilon}(u)\, du + \sigma \varepsilon^a B(t) - \frac{1}{2}\sigma^2 \varepsilon^{2a} s\Big)\ ds \\
&+ \varepsilon^a \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\times \int_0^t \exp\Big(\mu s + \sigma a \int_0^s \varphi^{\varepsilon}(u)\, du + \sigma \varepsilon^a B(t) - \frac{1}{2}\sigma^2 \varepsilon^{2a} s\Big)\ dB(t).
\end{aligned}
$$

So, to summarise what we have done and give a high–level overview of the steps involved, this is what we did so far. To solve this SDE, we need to find the probability distribution of *X(t)* given its initial condition *X(0)*. This probability distribution is given by the rough Fokker–Planck equation [3], which is a partial differential equation that describes the time evolution of the probability density function of *X(t)*. However, in the case of our SDE, we can simplify the problem by noticing that $X_l^{\varepsilon a}(t)$ *and* $X_u^{\varepsilon a}(t)$ are linear combinations of X, which means that their probability distributions can be obtained from the probability distribution of X(t). We can then use Itô's lemma to transform the SDE into an equivalent SDE for $X_l^{\varepsilon a}(t)$ *and* $X_u^{\varepsilon a}(t)$ [80] [79]. Itô's lemma is a rule that allows us to find the SDE satisfied by a function of a stochastic process. The lemma states that if Y(t) is a function of X(t), then the SDE satisfied by *Y(t)* is given by: $dY(t) = (\partial Y/\partial t)\, dt + (\partial Y/\partial X)\, dX + \frac{1}{2}\, (\partial^2 Y/\partial X^2)\, (dX)^2$, where $(\partial Y/\partial t)$ is the partial derivative of Y with respect to time, $(\partial Y/\partial X)$ is the partial derivative of Y with respect to X, and $(\partial^2 Y/\partial X^2)$ is the second partial derivative of Y with respect to X. Using Itô's lemma, we can find the SDEs satisfied by $X_l^{\varepsilon a}(t)$ *and* $X_u^{\varepsilon a}(t)$. The SDEs have the same form as the original SDE, but with different drift and diffusion coefficients that depend on α. The final solution involves an integral over the Wiener process W, which is a stochastic integral. The integral is evaluated using the Itô integral or the Stratonovich integral, depending on the convention used. The solution is a fuzzy solution because it involves a fuzzy stochastic process, and the value of $X^e(t)$ is not deterministic but depends on the realisation of the Wiener process W. We will derive a generic solution for stochastic differential equation of the following form which can be applied to each specific case:

$$
\begin{aligned}
X(t) = X(0) &+ \int_0^t \mu X(s)\, ds \\
&+ a \int_0^t \frac{\sigma}{2} X(0)\, \exp\Big(\mu s + \sigma a \int_0^s \varphi^\varepsilon(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a} s\Big)\, ds \\
&+ \varepsilon^a \frac{\sigma}{2} X(0) \int_0^t \exp\Big(\mu s + \sigma a \int_0^s \varphi^\varepsilon(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a} s\Big)\, dB(t).
\end{aligned}
$$

Re–writing in terms of derivatives, we need to solve the stochastic differential equation of the form:

$$
\begin{aligned}
dX(t) = \mu X(t)\, dt &+ a \frac{\sigma}{2} X(0)\, \exp\Big(\mu t + \sigma a \int_0^t \varphi^\varepsilon(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a} s\Big) \\
&+ \varepsilon^a \frac{\sigma}{2} X(0)\, \exp\Big(\mu t + \sigma a \int_0^t \varphi^\varepsilon(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a} s\Big)\, dB(t).
\end{aligned}
$$

with initial condition *X(0),* we will use Itô's lemma to obtain the solution:

Let *Y(t, B(t))* = $e^{-\mu t}$*X(t).* Then, using Itô's lemma, product rule and definition of X to obtain the solution:

$$
\begin{aligned}
dY\Big(t, B(t)\Big) &= e^{-\mu t}\, dX(t) + e^{-\mu t}\frac{\sigma^2}{2}X(t)\, dt \\
&= e^{-\mu t}X(t)\, dt + e^{-\mu t}\, a\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^t \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^{a}B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dt \\
&\quad + e^{-\mu t}\,\Big(\varepsilon^{a}\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^t \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^{a}B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dB(t)\Big) \\
&= e^{-\mu t}\, a\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^t \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^{a}B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dt \\
&\quad + e^{-\mu t}\,\Big(\varepsilon^{a}\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^t \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^{a}B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dB(t)\Big)\,.
\end{aligned}
$$

Integrating both sides of this equation from 0 to t, we treat left hand side using fundamental theorem of calculus: *Y(t, B(t))* —- *Y(0, B(0)).* The first integral on the right–hand side can be simplified using change of variable. Since *Y(t, B(t))* = $e^{-\mu t}$*X(t),*

$$
\begin{aligned}
Y(t) &= e^{-\mu t}X(t) \\
&= X(0) + \int_0^t e^{-\mu t}\, a\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^t \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^{a}B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dt \\
&\quad + \int_0^t e^{-\mu t}\,\Big(\varepsilon^{a}\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^s \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^{a}B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dB(t)\Big)\,.
\end{aligned}
$$

Multiply by $e^{\mu t}$:

$$X^{\varepsilon}(t) = e^{\mu t}\Bigg[X(0) + \int_0^t e^{-\mu t}\, a\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^t \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dt$$

$$+ \int_0^t e^{-\mu t}\, \Big(\varepsilon^a\frac{\sigma}{2}X(0)\, \exp\Big(\mu t + \sigma a\int_0^s \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dB(t)\Big)\Bigg]$$

$$= e^{\mu t}\Bigg[X(0) + \frac{\sigma a}{2}X(0)\int_0^t \exp\Big(\sigma a\int_0^s \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^a B(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, ds$$

$$+ \frac{\sigma\varepsilon^a}{2}X(0)\int_0^t \exp\Big(\sigma a\int_0^s \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^a B(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dB(s)\Bigg].$$

The second integral on the right–hand side is a stochastic integral of the form:

$e^{\mu s}\left(\varepsilon^a\frac{\sigma}{2}(X(0))\int_0^s exp\left(\sigma a\int_0^s\varphi^{\varepsilon}(u)du + \ \sigma\varepsilon^a B(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\right)dB(s)\right)$

Substituting both sides back into the original equation, we get:

$$X(t) = e^{\mu t}X(0) + a\frac{\sigma}{2}X(0)\, e^{\mu t}\int_0^t \exp\Big(\sigma a\int_0^s \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, ds$$
$$+ e^{\mu t}\left(\varepsilon^a\frac{\sigma}{2}X(0)\int_0^t \exp\Big(\sigma a\int_0^s \varphi^{\varepsilon}(u)\, du + \sigma\varepsilon^a B(t) - \frac{1}{2}\sigma^2\varepsilon^{2a}s\Big)\, dB(s)\right).$$

So, the unique solutions are same as before.

$$
\begin{aligned}
X_l^{\varepsilon a}(t) &= e^{\mu t} X_l^{\varepsilon a}(0) + e^{\mu t} a \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\quad \times \int_t^T \phi^{\varepsilon}(s)\, e^{\sigma\alpha \int_t^T \phi^{\varepsilon}(u)\,du - \frac{1}{2}\sigma^2\varepsilon^{2a} t + \sigma\varepsilon^a W(t)}\, ds \\
&\quad + e^{\mu t}\varepsilon^a \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\quad \times \int_t^T e^{\sigma\alpha \int_t^T \phi^{\varepsilon}(u)\,du - \frac{1}{2}\sigma^2\varepsilon^{2a} t + \sigma\varepsilon^a W(t)}\, dW(s), \\
X_u^{\varepsilon a}(t) &= e^{\mu t} X_u^{\varepsilon a}(0) + e^{\mu t} a \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\quad \times \int_t^T \phi^{\varepsilon}(s)\, e^{\sigma\alpha \int_t^T \phi^{\varepsilon}(u)\,du - \frac{1}{2}\sigma^2\varepsilon^{2a} t + \sigma\varepsilon^a W(t)}\, ds \\
&\quad + e^{\mu t}\varepsilon^a \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big) \\
&\quad \times \int_t^T e^{\sigma\alpha \int_t^T \phi^{\varepsilon}(u)\,du - \frac{1}{2}\sigma^2\varepsilon^{2a} t + \sigma\varepsilon^a W(t)}\, dW(s).
\end{aligned}
\tag{5.1.26}
$$

Now, we will revert to fBm by making the same substitution as before. Recall that:

$B_{H,\varepsilon}(t) = a\int_0^t \varphi^{\varepsilon}(s)ds + \ \varepsilon^a B(t)$,

where $\varphi^{\varepsilon}(s) = \int_0^t (t-s+\varepsilon)^{a-1}]dB(s)$

So, taking derivative $dB_{H,\varepsilon}(t) = \frac{d[a\int_0^t \varphi^{\varepsilon}(s)ds + \ \varepsilon^a B(t)]}{dt} = a\varphi^{\varepsilon}(t) + \varepsilon^a dB(t)$, and in terms of fBm, we get:

$$
\begin{aligned}
X_l^{\varepsilon a}(t) &= e^{\mu t}\left[X_l^{\varepsilon a}(0) + \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)\int_0^t e^{\sigma B_H^{\varepsilon}(s) - \frac{1}{2}\sigma^2\varepsilon^{2a} s}\, dB_H^{\varepsilon}(s)\right], \\
X_u^{\varepsilon a}(t) &= e^{\mu t}\left[X_u^{\varepsilon a}(0) + \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)\int_0^t e^{\sigma B_H^{\varepsilon}(s) - \frac{1}{2}\sigma^2\varepsilon^{2a} s}\, dB_H^{\varepsilon}(s)\right].
\end{aligned}
\tag{5.1.27}
$$

Therefore, the generic solution for $\mu \geq 0$:

$$
X^{\varepsilon}(t) = e^{\mu t}\left\langle X(0) + \frac{\sigma}{2}\left(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\right)\int_0^t e^{\sigma B_H^{\varepsilon}(s) - \frac{1}{2}\sigma^2\varepsilon^{2a} s} dB_H^{\varepsilon}(s)\right\rangle
$$

For $\mu \leq 0$, we start with the following well known identities:

$cosh(\mu t) = \frac{1}{2}(e^{\mu t} + e^{-\mu t})$ and $sinh(\mu t) = \frac{1}{2}(e^{\mu t} - e^{-\mu t})$. We will show that $e^{\mu t}X_l^{ea}(0) = X_l^{ea}(0)cosht(\mu t) + X_l^{ea}(0)sinh(\mu t)$. If we combine $\{cosh(\mu t) + sinh(\mu t) = e^{\mu t}\}$, and we replace the first term $e^{\mu t}X_l^{ea}(0)$ with $X_l^{ea}(0)cosht(\mu t) + X_l^{ea}(0)sinh(\mu t)$, our solution becomes:

$$\begin{aligned} X_l^{\varepsilon a}(t) &= X_l^{\varepsilon a}(0)\cosh(\mu t) + X_l^{\varepsilon a}(0)\sinh(\mu t) \\ &\quad + e^{\mu t}\frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_l^{\varepsilon 1}(0)\Big)\int_0^t e^{\sigma B_H^{\varepsilon}(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s}\, dB_H^{\varepsilon}(s). \end{aligned} \tag{5.1.28}$$

$$\begin{aligned} X_u^{\varepsilon a}(t) &= X_u^{\varepsilon a}(0)\cosh(\mu t) + X_u^{\varepsilon a}(0)\sinh(\mu t) \\ &\quad + e^{\mu t}\frac{\sigma}{2}\Big(X_u^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)\int_0^t e^{\sigma B_H^{\varepsilon}(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s}\, dB_H^{\varepsilon}(s). \end{aligned} \tag{5.1.29}$$

Therefore, the generic solution for $\mu \leq 0$:

$$\begin{aligned} X^{\varepsilon}(t) &= X_{l,u}^{\varepsilon a}(0)\cosh(\mu t) + X_{l,u}^{\varepsilon a}(0)\sinh(\mu t) \\ &\quad + \left\langle \frac{\sigma}{2}\Big(X_l^{\varepsilon 1}(0) + X_u^{\varepsilon 1}(0)\Big)e^{\mu t}\int_0^t e^{\sigma B_H^{\varepsilon}(s) - \frac{1}{2}\sigma^2\varepsilon^{2a}s}\, dB_H^{\varepsilon}(s)\right\rangle. \end{aligned} \tag{5.1.30}$$

In summary, our approach to integrating a fractional Brownian motion (fBm)–driven fuzzy stochastic differential equation (SDE) into a partial differential equation (PDE) framework is highly unconventional in that it does not follow the standard "PDE–Feynman–Kac" blueprint. Typically, in a classical analysis, one defines a PDE in *(x,t)* –space (for instance $\partial_t u + \mathcal{L}u = 0$ where the operator L is the generator of the SDE and then invokes Feynmac–Kac for representation. Our new method, however, places terms like μX, σX, and even $\frac{\sigma}{2}^2 X''$ *di*rectly on the right–hand side of the PDE in a way that is not standard for PDE—-SDE bridging. Furthermore, we interpret X both as a variable in the PDE and as a stochastic process in the SDE, which conflates the usual roles of "independent variable" vs. "dependent process." These modelling choices inject "extra drift" and hyperbolic–function expansions when we attempt the usual product rule or PDE–SDE dictionary, resulting in expressions such as $\frac{\sigma}{2}^2 X(t)e^{\mu t X(t)}dt$ or *cosh–sinh* that are not a typical linear SDE or PDE analyses.

This departure from the mainstream theory represents our unique contribution: we craft a custom PDE that naturally embeds additional drift–like terms and uses nontraditional boundary or initial conditions (for instance, splitting $e^{\mu t X(t)}dt$ into *cosh(μt)* and *sinh(μt)* for negative μ. By doing so, we created a hybrid method that merges PDE notation (e.g. $\partial_t u$ plus *μX+σX'+ σ/2X")* with the solution representation of a fuzzy fractional SDE including ϕ^{ε} kernel expansions. This yields formula that are unfamiliar from a classic Itô lemma and standard PDE perspective. It reflects a novel attempt at pushing beyond the normal PDE—-SDE synergy. In short, our framework showcases how one can embed extra terms and rewrite exponents in ways that do not appear in the canonical geometric Brownian motion or typical fractional PDE—-SDE references——thus illustrating a genuinely unique direction in fuzzy fractional analysis. The choice of the customised PDE involving hyperbolic and exponential functions rather than a classical PDE is driven by the need to reflect the inherent complexities observed in financial markets more accurately. Classical PDEs, such as the traditional Black–Scholes model, rely heavily on assumptions like independent increments and no memory effect, conditions that are rarely fully satisfied by real–world data.

5.1.6 INTRODUCING JUMPS

Traditional option pricing models rely on precise, fixed values for variables like expected return and volatility. However, these models fail to account for the inherent uncertainty surrounding these parameters. To address this limitation, recent approaches have incorporated fuzziness into the calculations, allowing investors to estimate option prices based on their individual risk tolerance and beliefs about the potential range of values for these variables.

To determine the belief degree associated with a particular option price, an interpolation search algorithm is used [89]. Interpolation is a mathematical technique used to estimate values between known data points. In this case, the algorithm is applied to determine the belief degree associated with a specific option price within the range of possible prices. By incorporating fuzzy numbers and the belief degree, this approach provides a more flexible framework for practitioners to make their investment decisions. It allows them to consider their risk preferences and beliefs about the uncertain parameters when selecting an option price.

The purpose of this section is to extend many novel proofs and theoretical considerations presented in the recent works of the other authors. We present examples and applications and introduce a novel proof dealing with the constraint of $h < \frac{1}{2}$. This will pave the way to defining fuzzy fractional Brownian motion for $h < \frac{1}{2}$. We employ a narrative review methodology to synthesise existing research and to identify gaps where our proposed model can offer improvements. This allows us to provide a comprehensive overview of the literature and enables expert interpretation and synthesis, which are essential for extending existing theoretical frameworks and offering novel proofs in this area.

We derive the model from first principles using the results we obtained in Section 5.1.5. We start with the usual Kolmogorov filtered probability space[1] (Ω, F, $\mathbb{P}$, where $\mathbb{P}$ represents the probability measure, Ω is the set of all possible outcomes or states that a random process can take and $\{Ft,\ t \in [0, T]\}$ is an information filtration. We expand the work of Merton (1976) on option pricing when the underlying stock returns have jumps [90] and the work of [89] on fuzzy fractional processes. The underlying asset price $\mathrm{S_t}$ is assumed to follow the following crisp dynamics:

$$dS/S \;=\; \mu dt \;+\; \sigma dB(t) \;+\; \sigma dB_t^H \;+\; d\left(\sum_1^N (V_i - 1)\right).$$

In this equation, the terms represent various components: (i) the drift param-

[1]Kolmogorov space is a mathematical framework used in probability theory to define and study random process. It consists of three main components: a sample space Ω, a set of events (subsets of Ω), and a probability measure defined on the events.

eter μ represents the expected return rate on the value of the underlying asset. (ii) the volatility parameter σ represents the standard deviation of the return rate on the value of the underlying asset. (iii) the term $dB(t)$ represents a standard Brownian motion. (iv) The term $dB^H(t)$ represents an independent standard fractional Brownian motion with respect to a fractional Brownian motion parameterized by the Hurst exponent H. (v) the term $d(\Sigma(V_i-1))$ represents a sum of jumps, where $\{V_i,\ i \geq 1\}$ is a sequence of independent and identically distributed nonnegative random variables. The logarithm of V_i, denoted as $\Upsilon = log(V)$, follows a normal distribution with a probability density function [90]:

$$f_y(x) = \frac{1}{\sqrt{2\pi}\sigma_j} e^{-\frac{(x-\mu_j)^2}{2\sigma_j^2}}.$$

The expected return rate μ, volatility σ, jump intensity λ, and jump magnitudes sequence Vi in the underlying asset price dynamics cannot always be treated as constant real numbers. Therefore, a new model is proposed as a way to incorporate both randomness and fuzziness in describing the dynamics of the process. When dealing with processes that involve non–standard or non–numeric variables, such as fuzzy numbers or other types of uncertain quantities, the traditional tools of real analysis may not be directly applicable. In such cases, alternative mathematical frameworks or extensions of real analysis are often employed to handle the specific characteristics and uncertainties present in the financial domain. In previous section, we proved the no arbitrage condition in the presence of uncertain or non–standard variables. This involved extending the mathematical framework to accommodate the specific characteristics of the variables under consideration. Muzzioli et al. [91] conducted a literature review and categorised more than fifty papers on option pricing in a fuzzy setting, providing insights into different approaches and best practices for utilising fuzziness in option pricing. Nowak and Pawłowski [92], [93] studied the pricing problems of European options with the application of Lévy processes under fuzzy environments. Li et al. [94] developed a nonlinear fuzzy–parameter partial differential equation model for obtaining fuzzy option prices and derived dominating optimal hedging strategies under fuzzy environments. Wang et al. [95] studied the pricing problem of n–fold compound options using the pricing model of Nowak and Romaniuk [96], martingale method, and fuzzy sets theory. Zhang et al. [97] discussed Asian option pricing when the stock price, risk–free interest rate, and volatility were trapezoidal fuzzy numbers. Muzzioli et al. [98] used fuzzy quadratic regression methods to estimate the implied volatility smile function. Nowak and Romaniuk [99] (2010) assumed the underlying asset followed a Lévy process and fuzzified the input parameters. They derived option pricing formulas in both crisp and fuzzy cases using minimal entropy martingale measure and fuzzy arithmetic. They also proposed and verified a generalized hybrid binomial American real option pricing model. Lee et al. [100] applied fuzzy decision theory and Bayes' rule to derive a fuzzy Black–Scholes pricing model. They aimed to measure fuzziness in option pricing using fuzzy arithmetic. Wu [101] and Wu [102] developed a fuzzy version of the Black–Scholes pricing formula by considering triangular fuzzy numbers for the underlying asset price, volatility, and risk–free interest rate. Wu [103] also introduced a bisection search algorithm to calculate the membership degree associated with a given option price. Zhang et al., employed stochastics and fuzzy set theory to develop a new framework for option pricing in which the underlying process follows a mixed fuzzy fractional Brownian motion with jump when $h > \frac{1}{2}$ [89]. Using our earlier derivations, we will extend the work of Zhang, Wu, Lee, Nowak and Romaniuk here.

In the fuzzy fractional jump–diffusion model (FFJDM), the underlying process is

$$\frac{d\widehat{S}}{\widehat{S}} = \widehat{\mu}dt \ + \ \widehat{\sigma}dB_t + \widehat{\sigma}dB_t^H + d\left(\sum_{1}^{\widehat{N}}\left(\widetilde{V_i} - 1\right)\right)$$

The inclusion of the fuzzy Poisson process $\hat{N}$ with fuzzy intensity λ and $\tilde{V}_i$ (sequence of independent and identically distributed fuzzy random variables) makes this a jump–diffusion process, as it allows for random jumps to occur at random times, in addition to the continuous stochastic components represented by the Brownian motion terms. In this model, all the fuzzy parameters are denoted by a hat symbol, such as $\widehat{\mu}$ and $\widehat{\sigma}$, representing the fuzzy expected return rate and fuzzy volatility, respectively. The model includes additional components such as B_t (standard Brownian motion), and B_t^H (independent standard fractional Brownian motion). The subtraction of 1 in the jump process is a convention used to represent the relative change or jump size with respect to the value of the process before the jump occurs. In the context of jump–diffusion models, the jump term is typically expressed as the sum of individual jump sizes, i.e., the magnitude of the jump experienced by the process. The jump size, V_i, represents the absolute change in the process value at the i–th jump. By subtracting 1 from V_i, one is effectively expressing the relative change or percentage change in the process due to the jump. By expressing the jump size as a relative change, the impact of jumps is effectively scaled based on the current value of the process. It allows for a more interpretable representation of jumps, as they are related to the percentage change rather than absolute changes. For example, a jump of $V_i = 0.01$ when the process is at $\hat{S} = \mathit{100}$ would represent a 1% increase, whereas the same jump with $V_i = \mathit{1}$ when the process is at $\hat{S} = \mathit{10}$ would represent a 10% increase. By using the relative change, we take into account the current level of the process, which is often more meaningful in modelling and risk assessment. The fuzzy normal distribution is used to describe the fuzziness in the expected return rate and jump magnitudes. The model assumes independence among the sources of randomness and fuzziness, including B_t, B_t^H, $\tilde{N}_{\mathrm{t}}$, and $\tilde{V}_{\mathrm{i}}$. We will now apply Itô's lemma to FFJDM. Let's start by applying Itô's lemma to the function $f(\hat{S})$. If we make a Taylor expansion and ignore $(dt)^2$ and $dtdx$ because they are negligible compared to the dt–term, then:

$$df = \left(\frac{\partial f}{\partial t}\right) dt + \left(\frac{\partial f}{\partial \hat{S}}\right) d\hat{S} + \frac{1}{2}\left(\frac{\partial^2 f}{\partial \hat{S}^2}\right)\left(d\hat{S}\right)^2.$$

The first term $(\partial f/\partial t)\ dt$ is zero as f does not have an explicit time dependence. The second term $(\partial f/\partial \hat{S})\ d\hat{S}$ can be computed by taking the derivative of $f(\hat{S})$

$= ln(\hat{S})$ with respect to $\hat{S}$: $\partial f/\partial \hat{S} = 1/\hat{S}$. Multiplying this by $d\hat{S}$, we obtain: $(\partial f/\partial \hat{S})\ d\hat{S} = (1/\hat{S})\ d\hat{S}$. Now, let's consider the third term. Since $(d\hat{S})^2 = 0$ (as $d\hat{S}$ is a differential form), this term can be ignored. Now, substitute the given expression for $d\hat{S}$: $\widehat{dS}/\widehat{S} = (\widehat{\mu}dt + \widehat{\sigma}dB_t + \widehat{\sigma}dB_t^H + d\left(\sum_1^{\widehat{N}}\left(\widetilde{V_i}-1\right)\right))$. Expanding this expression, we have:

$$\frac{\widehat{dS}}{\widehat{S}} = \widehat{\mu}\, dt + \widehat{\sigma}\, dB_t + \widehat{\sigma}\, dB_t^H + d\left(\sum_1^{\widehat{N}}\left(\widetilde{V_i}-1\right)\right).$$

Integrate both sides of the expression we obtain:

$$\begin{aligned}\ln(\widehat{S_t}) &= \ln(\widehat{S(0)}) + \int_0^t \widehat{\mu}\, dt + \int_0^t \sigma\, dB_t + \int_0^t \sigma\, dB_t^H \\ &+ \int_0^t \frac{1}{2}\left\{-\frac{1}{S^2}\right\}\sigma^2 \widehat{S}^2\, dt + \int_0^t \frac{1}{2}\left\{-\frac{1}{S^2}\right\}\sigma^2 \widehat{S}^2\, dt^{2H} + \int_0^t d\left(\sum_1^{\widehat{N}}\left(\widetilde{V_i}-1\right)\right)\, dV.\end{aligned}$$

In the crisp case, we used Itô isometry to show that $(dX)^2 = (\mu dt)^2 + 2\mu\sigma dt dW + \sigma(dW)^2 = \sigma dt$.

Now, recall that $f(\widehat{S}) = \ln(\widehat{S})$, so $\widehat{S} = \exp(f(\widehat{S}))$. Replacing $\widehat{S}$ with $exp(f(\hat{S}))$ in the above expression:

$$\ln(\widehat{S_t}) = \ln(\widehat{S(0)}) + \widehat{\mu}t + \sigma B_t + \widehat{\sigma} dB_t^H - \frac{1}{2}\widehat{\sigma}^2 t + \frac{1}{2}\widehat{\sigma}^2 t^{2H} + \int_0^t d\left(\sum_1^{\widehat{N}}\left(\widetilde{V_i}-1\right)\right) dV)$$

Recall that $\widetilde{V_i}$ are lognormally distributed such that $E(\widetilde{V_i}) = E\left(e^{\widetilde{V_i}}\right) = e^{\mu+1/2\sigma^2}$; then, exponentiating both sides and simplifying:

$$\widehat{S_t} = \widehat{S(0)} \exp\left(\widehat{\mu}t + \sigma B_t + \widehat{\sigma} dB_t^H - \frac{1}{2}\widehat{\sigma}^2 t + \frac{1}{2}\widehat{\sigma}^2 t^{2H}\right) \prod_{i=1}^{\widehat{N}} \widetilde{V_i}.$$

If we now take an expectation of $\widehat{S_t}$, we have:

$$\widehat{E[S_t]} = E\left[\widehat{S(0)}\exp\left(\widehat{\mu}t + \sigma B_t + \widehat{\sigma}dB_t^H - \frac{1}{2}\widehat{\sigma}^2 t + \frac{1}{2}\widehat{\sigma}^2 t^{2H}\right)\prod_{i=1}^{\widehat{N}}\widetilde{V_i}\right].$$

In stochastic calculus, B_t, represents a standard Brownian motion and the key property of Brownian motion is that its variance increases linearly with time. The variance of B_t at time t is given by t, which is a result of its definition and properties. The variance of a random variable X is defined as: $Var(X) = E[(X - E[X])2]$. For a standard Brownian motion B_{t}, its expected value $E[B_t]$ is 0 at any time t. Therefore, we have: $Var\left(B_t\right) = E\left[B_t^2\right] = \int_{-\infty}^{\infty} x^2 P_{B(t)}(x)dx$. Since B_t follows a Gaussian distribution with mean 0 and variance t, its probability density function is the Gaussian density function: $P_{B(t)} = \frac{1}{\sqrt{2\pi t}}\exp(-\frac{x^2}{2t})$. Plug this into the integral for the variance: $Var\left(B_t\right) = E\left[B_t^2\right] = \int_{-\infty}^{\infty} x^2\frac{1}{\sqrt{2\pi t}}\exp(-\frac{x^2}{2t})dx$. The integrand $\frac{1}{\sqrt{2\pi t}}\exp(-\frac{x^2}{2t})$ is the Gaussian p.d.f. with mean 0 and variance t. To integrate the Gaussian p.d.f over the entire real line: $\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi t}}\exp(-\frac{x^2}{2t})dx$, we will use the Gamma function. The integrand is an even function so $\int_{-\infty}^{\infty} e^{-x^2}dx = 2\int_0^{\infty} e^{-x^2}dx$. Thus, after a change of variable $x = \sqrt{t}$, we turn this into gamma function:

$$2\int_0^{\infty} e^{-x^2}\,dx = 2\int_0^{\infty}\frac{1}{2}e^{-t}t^{-1/2}\,dt = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}. \tag{5.1.31}$$

Similarly, $\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}dx = \sqrt{2\pi}$. For standard normal distribution $\mu=0$ and $\sigma^2=1$, and changing variable $x = \sqrt{2u}$, $dx =$ and $x^2 = 2u$ so $du = 2xdx$ then we have:

$$\begin{aligned}\int_{-\infty}^{\infty} x^2\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)\,dx &= \frac{1}{\sqrt{2\pi}}2\int_0^{\infty} u\,\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u}{2}\right)\frac{1}{2\sqrt{u}}\,du\\ &= \frac{1}{\sqrt{2\pi}}\int_0^{\infty}\sqrt{u}\,\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u}{2}\right)\,du.\end{aligned}$$

Next, we use integration by parts. Choose $u=\sqrt{u}$ and $dv = exp\left(-\frac{u}{2}\right)du$. Then $du = \frac{1}{2}\ u^{-1/2}$ and $v = -2\ e^{-u/2}$. Applying integration by parts:

$\frac{1}{\sqrt{2\pi}} \int_0^\infty \sqrt{u} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u}{2}\right) du = \frac{1}{\sqrt{2\pi}} \ \left[_0^\infty [\ -2u^{\frac{1}{2}} e^{-\frac{u}{2}} \right] + [\int_0^\infty u^{-1/2} e^{-1/2}]]$.

When x is 0, u is also 0, and as x approaches $+\infty$, u approaches $+\infty$ so the first integral uv evaluates to zero. We are left with $\int_0^\infty \sqrt{u} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u}{2}\right) du = \frac{1}{\sqrt{2\pi}} \ [\int_0^\infty u^{-1/2} e^{-1/2}]]$. To evaluate the remaining integral, we can once again use the integration by parts. Notice that the integral representation of $\int_0^\infty u^{-1/2} e^{-1/2}]]$ is the result we showed earlier in 5.1.31 which is the gamma function $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. Therefore, $\frac{1}{\sqrt{2\pi}} \ [\int_0^\infty u^{-\frac{1}{2}} e^{-\frac{1}{2}}]] = \ \frac{1}{\sqrt{2}}$. Going back to $\widehat{E[S_{t]}} = E[\ \widehat{S(0)} e^{\widehat{\mu} t \ + \ \sigma B_t + \widehat{\sigma} dB_t^H - \frac{1}{2}\widehat{\sigma}^2 t - \frac{1}{2}\widehat{\sigma}^2 t^{2H}} \prod_{i=1}^{\widehat{N}} \widetilde{V_{i]}}$, we need to evaluate $E[\ e^{\ \sigma B_t + \widehat{\sigma} dB_t^H}]$. The variance of the sum of independent random variables is given by: $Var(X+Y) = \ Var(X) \ + \ Var \ (Y)$. For X=$\sigma B_t$ and Y= $\widehat{\sigma} B_t^H$, from Itô's Isometry we have $E[e^{\sigma B}]$ and let $Z(t) \ = \ e^{\sigma B}$, then using Itô,

$dZ(t) \ = \ \frac{1}{2} \ \sigma^2 e^{\sigma B(t)} dt \ + \ \sigma e^{\sigma B(t)} dB(t) \ = \ \frac{1}{2} \ \sigma^2 Z(t) dt \ + \ \sigma e^{\sigma B(t)} dB(t)$. Z(0) = 1.

Then, in integral form: $Z(t) = 1 + \ \frac{1}{2}\sigma^2 \int_0^t Z(s) ds + \sigma \int_0^t dB(s)$. Taking the expectation will make stochastic integral vanish so last term is zero. Then $m(t) = E[Z(t)] = 1 + \ \frac{1}{2}\sigma^2 \int_0^t m(s) ds$. Taking derivates of both sides we get $\dot{m}(t) = \frac{\sigma^2}{2} m(t)$. This is an ordinary differential equation, so we can separate variables and then integrate both sides. Let us start by rearranging the equation: $m'(t)/m(t) \ = \ \sigma 2/2$. Now, we can integrate both sides with respect to t: $\int \ m'(t)/m(t) \ dt \ = \ \int \ (\sigma^2/2) \ dt$. Integrating the left side gives us the natural logarithm of the absolute value of *m(t)*: $\ln |\ m(t)| = \ \left(\frac{\sigma^2}{2}\right) * \ t \ + \ C_1$, where *C₁* is the constant of integration. To eliminate the absolute value, we can rewrite the equation as: $m(t) = \ \pm \exp\left(\left(\frac{\sigma^2}{2}\right) t \ + \ C_1\right)$. Now, let us combine the constant terms into a new constant, C_2: $m(t) = \ C_2 * \exp\left(\left(\frac{\sigma^2}{2}\right) t\right)$. We know that *m(0) =1* so *C_2=1* and we are only interested in positive processes, so we can get rid of negative logarithm process. The result is: $E\left[e^{\sigma B(t)}\right] = E\left[Z(t)\right] = m(t) = e^{\frac{\sigma^2 t}{2}}$. Similarly, $E\left[e^{\widehat{\sigma} B_t^H}\right] = e^{\frac{1}{2}\widehat{\sigma}^2 t^{2H}}$.

Therefore, we obtain equation 5.1.32:

$$\widehat{E[S_t]} = E\left[\widehat{S(0)}\exp\Big(\widehat{\mu}t + \sigma B_t + \widehat{\sigma}B_t^H - \frac{1}{2}\widehat{\sigma}^2 t - \frac{1}{2}\widehat{\sigma}^2 t^{2H}\Big)\prod_{i=1}^{\widehat{N}}\widetilde{V_i}\right]$$
$$= \widehat{S(0)}\exp\Big(\widehat{\mu}t - \frac{1}{2}\widehat{\sigma}^2 t - \frac{1}{2}\widehat{\sigma}^2 t^{2H}\Big)\exp\Big(\frac{1}{2}\widehat{\sigma}^2 t + \frac{1}{2}\widehat{\sigma}^2 t^{2H}\Big)\prod_{i=1}^{\widehat{N}}\widetilde{V_i}. \tag{5.1.32}$$

To evaluate the last term $\prod_{i=1}^{\widehat{N}}\widetilde{V_{i]}}$ which is the product of the i.i.d. distributed random jumps with mean μ and variance σ^2, we will use the property that the expectation of the product of independent random variables is equal to the product of their expectations. Re–writing this expression and using the fact that $\Sigma\ ln(V_i) = ln(V_1) + ln(V_2) + ... + ln(V_n) = ln(V_1 * V_2 * ... * V_n)$, we obtain $\prod_{i=1}^{\widehat{N}}\widetilde{V_i} = e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}$.

$\widehat{E[S_{t]}} = \widehat{S(0)}e^{\widehat{\mu}t\ -\frac{1}{2}\widehat{\sigma}^2 t - \frac{1}{2}\widehat{\sigma}^2 t^{2H}}e^{\frac{1}{2}\widehat{\sigma}^2 t + \frac{1}{2}\widehat{\sigma}^2 t^{2H}}e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})} = \widehat{S}_0 e^{\widehat{\mu}t}E[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}]$.

To evaluate the process $E[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}]$, we will use the fact that jump size is i.i.d; then by the law of iterated expectation, we can re–write the expectation as conditional expectation: $E[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}] = E[E[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}|\widetilde{N_t}]]$. Recall that $\widetilde{N}_t$ is Poisson distribution with parameter λ [90]. Because the distributions of the size of the jumps and the Poisson process of the number of jumps are independent, we can simplify the above expression as:

$E[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}] = E[E[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}|\widetilde{N_t}]] = \sum_{n=0}^{\infty}\frac{(\widetilde{\lambda t})^n}{n!}e^{\widetilde{-\lambda t}}E[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}]$.

If one recalls, the jump size has a lognormal distribution with mean value μ and variance σ^2 [90]. From the property of the normal distribution we know that:

$\widetilde{J} = E(\widetilde{V_i}) = E\left(e^{\widetilde{V_i}}\right) = e^{\mu+1/2\sigma^2}$.

So,

$$\sum_{n=0}^{\infty}\frac{\left(\widetilde{\lambda t}\right)^n}{n!}\,e^{-\widetilde{\lambda t}}\,E\Big[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}\Big]=\sum_{n=0}^{\infty}\frac{\left(\widetilde{\lambda t}\right)^n}{n!}\,e^{-\widetilde{\lambda t}}\left[\prod_{i=1}^{n}E(\widetilde{V_i})\right]$$
$$=\sum_{n=0}^{\infty}\frac{\left(\widetilde{\lambda t}\right)^n}{n!}\,e^{-\widetilde{\lambda t}}\left[\prod_{i=1}^{n}E(\widetilde{V_i})\right]$$
$$=\sum_{n=0}^{\infty}\frac{\left(\widetilde{\lambda t}\right)^n}{n!}\,e^{-\widetilde{\lambda t}}\,\widetilde{J}^n$$
$$=\sum_{n=0}^{\infty}\frac{\left(\widetilde{\lambda t}\right)^n}{n!}\,e^{-\widetilde{\lambda t}}\,\widetilde{J}^n$$
$$=\sum_{n=0}^{\infty}\frac{\left(\widetilde{\lambda t}\right)^n\widetilde{J}^n}{n!}\,e^{-\widetilde{\lambda t}}.$$

The first part of this series is just re–written Taylor series $e^x=\sum_{n=0}^{\infty}\frac{\widetilde{X}^n}{n!}$, so we can simplify $\sum_{n=0}^{\infty}\frac{(\widetilde{\lambda t})^n\widetilde{J}^n}{n!}=e^{\widetilde{\lambda t J}}$. The whole expression becomes:

$$\sum_{n=0}^{\infty}\frac{(\widetilde{\lambda t})^n}{n!}e^{\widetilde{\lambda t}}E\Big[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}\Big]=e^{\widetilde{\lambda t}\widetilde{J}}e^{-\widetilde{\lambda t}}=e^{-\widetilde{\lambda t}\,(\widetilde{-J+1})}.$$

Then:

$$\widehat{E[S_t]}=\widehat{S(0)}\exp\Big(\widehat{\mu}t-\frac{1}{2}\widehat{\sigma}^2t-\frac{1}{2}\widehat{\sigma}^2t^{2H}\Big)\exp\Big(\frac{1}{2}\widehat{\sigma}^2t+\frac{1}{2}\widehat{\sigma}^2t^{2H}\Big)\,e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}$$
$$=\widehat{S}_0e^{\widehat{\mu}t}\,E\Big[e^{\sum_{i=1}^{N}\ln(\widetilde{V_i})}\Big]=\widehat{S}_0e^{\widehat{\mu}t}e^{-\widetilde{\lambda t}\left(\widetilde{-J+1}\right)}.$$

$$\widehat{E[S_t]}=\widehat{S}_0e^{\widehat{\mu}t}\,e^{-\widetilde{\lambda t}\left(\widetilde{-J+1}\right)}=\widehat{S}_0e^{\widehat{\mu}t-\widetilde{\lambda t}\left(\widetilde{-J+1}\right)}$$
$$=\widehat{S}_0e^{t\left(\widehat{\mu}-\widetilde{\lambda}\left(\widetilde{-J+1}\right)\right)}.$$

In Section 5.1.2 we proved no arbitrage conditions for fuzzy fractional model when Hurst exponent is greater than $\frac{1}{2}$ [78] [79]. We extend this now to FFJDM under the risk–neutral measure Q which is a theoretical probability measure used to value financial derivatives, such as options, under the assumption that the expected rate of return of the underlying asset is the risk–free rate

[87]. This approach simplifies the valuation process and allows for a relatively easy pricing of derivatives without considering the actual probabilities of different market outcomes. This means that when calculating expected values of future payoffs, we use the risk–free rate as the discount factor. Under this measure, we can calculate the present value of future cash flows associated with financial derivatives. When valuing financial derivatives like options, the risk–neutral measure allows us to use a discounted expected value calculation to determine the fair price of the derivative. This approach assumes that investors are risk–neutral and do not require a risk premium for holding the derivative. The statement implies that even under the risk–neutral measure Q, the underlying asset price still follows a certain stochastic process.

Under the assumption of no arbitrage, the risk–neutral and the real–world (risky) measures are equivalent in the context of option pricing and derivatives valuation. This equivalence is a fundamental concept in mathematical finance and is known as the Fundamental Theorem of Asset Pricing [87]. In order for risk–neutral and risky measures to be equivalent, we impose a no–arbitrage condition under which $\widehat{\mu} = r - \widetilde{\lambda}\left(\widetilde{J} - 1)\right)$. The result we will derive next using Itô's lemma shows that there is a unique solution to our FFJDM equation, and that solution does not depend on any particular assumptions, or individual preferences about risk. The arbitrage free price of a derivative is uniquely determined because in this model the derivative is superfluous. This means that the derivative's price is fully determined by the model's dynamics and parameters, and its price is consistent with the prices of other assets in the market. In essence, the derivative's price is not subject to arbitrary fluctuations or ambiguous valuation. We will use a well–known Girsanov theorem in the theory of stochastic differential equations, that if we change the measure from real–world P to some other equivalent measure in risk–neutral world Q, this will change the drift in the SDE, but the diffusion term will be unaffected. This is why we set $\widehat{\mu} = \widehat{r} - \widetilde{\lambda}\left(\widetilde{J} - 1)\right)$. Thus, the drift will play no part in the pricing equations. Using Girsanov's theorem and re–writing $d\widehat{S}/\widehat{S} \ = \ \widehat{\mu}dt \ + \ \widehat{\sigma}dB_t + \widehat{\sigma}dB_t^H + d(\sum_1^{\widehat{N}}\left(\widetilde{V_i} - 1\right))$ as $d\widehat{S}/\widehat{S} \ = (\widehat{r} - \widetilde{\lambda}\left(\widetilde{J} - 1\right) dt \ + \ \widehat{\sigma}dB_t + \widehat{\sigma}dB_t^H + d(\sum_1^{\widehat{N}}\left(\widetilde{V_i} - 1\right))$.

Similarly, to 5.1.24., we get

$$\widehat{S_t} = \widehat{S}(0)\exp\Big(\widehat{r} - \widetilde{\lambda}\Big(\widetilde{J} - 1\Big)t + \widetilde{\sigma}B_t + \widehat{\widetilde{\sigma}}B_t^H - \frac{1}{2}\widehat{\sigma}^2 t - \frac{1}{2}\widehat{\sigma}^2 t^{2H}\Big)\prod_{i=1}^{n}\widetilde{V_i}. \tag{5.1.33}$$

under risk–neutral measure. This is the process that $\widehat{S_t}$ follows which we will use in the proof later. The Cheridito (2001) paper, titled "Mixed Fractional Brownian Motion" [104], presented a framework for modelling and analysing mixed fractional Brownian motion (MFBM), which is a generalisation of fractional Brownian motion (fBm). The paper introduced a class of stochastic differential equations driven by MFBM and established existence and uniqueness results for solutions under certain regularity conditions. It was shown that if the Hurst exponent is greater than 3/4, the fractional Brownian motion process satisfies certain regularity conditions, which allow for the existence of a unique solution to the stochastic differential equation. The key point made in [104] is that when $\frac{3}{4} < H < 1$, the mixed fractional Brownian motion exhibits properties that avoid arbitrage. For $H > \frac{3}{4}$, the text makes clear that the mixed model retains a semimartingale structure, allowing for classical arbitrage–free pricing arguments. This is an important result because in this regime, the models are well behaved. For $H < \frac{3}{4}$, the paper mentions that arbitrage can emerge because the resulting process is no longer a semimartingale. This means the standard tools like Girsanov's theorem may break down. [104] explicitly proves that adding a standard Brownian motion to fractional Brownian motion making it a mixed model, helps restore arbitrage–free properties under specific conditions on H. This is equation 6.2 on page 933 in [104]. In other words, [104] key contribution is precisely that by adding a standard Brownian motion component to fractional Brownian motion making it "mixed" and keeping H above $\frac{3}{4}$, one can restore the possibility of no–arbitrage and standard pricing methods. The mixed process takes the form of equation 6.2 where B^H is a fractional Brownian motion and B_t is an independent standard Brownian motion.

The crucial point is Theorem 1.7 from [104] that shows that for $H > \frac{3}{4}$, the process $B_t + \varepsilon B_t^H$ remains equivalent to a Brownian motion under an equivalent martingale measure Q_ε, provided ε is small enough. This implies that arbitrage is avoided when the mixed process is used instead of pure fBm. This result is relevant for our work in the mathematical analysis and modelling of the underlying asset dynamics in option pricing. The discounted expected value of Z_T at time *t ϵ [0, T]* under risk–neutral measure Q is given by $Z_t = e^{-\widetilde{r}(T-t)} E^Q[\Big(\widetilde{S}_T - K\Big)^+, 0]$, so $\widetilde{S}_T - K > 0$, otherwise the payoff is zero. We will now substitute expression for $\widehat{S_t}$ from 5.1.33 into the above equation:

$$
\begin{aligned}
Z_t &= e^{-\widetilde{r}(T-t)}\, E^Q\Big[\Big(\widehat{S}(0)\Big(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}}\Big)\prod_{i=1}^{n}\widetilde{V_i} - K\Big)^+, 0\Big] \\
&= e^{-\widetilde{r}(T-t)}\, E^Q\Big[\Big(\widehat{S}(0)\Big(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}\Big) - K\Big)^+, 0\Big] \\
&= e^{-\widetilde{r}(T-t)}\, E^Q\Big[E^Q\Big[\Big(\widehat{S}(0)\Big(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}\Big) - K\Big)^+ \\
&\qquad\qquad \Big|\ \sigma\Big(\sum_{i=1}^{\widetilde{N}_T} Y_i\Big)\Big]\Big].
\end{aligned}
\tag{5.1.34}
$$

We will now use law of iterated expectation within the context of a σ–algebra. The σ–algebra, denoted by $\sigma(\sum_{i=1}^{\widetilde{N}_T} Y_i)$ represents the conditioning information or the available knowledge up to time T. It includes all the events or outcomes that can be determined based on the random variables Y_i up to time T. This sigma algebra essentially captures the information available from the past observations and values of Y_i up to time T. The Law of Iterated Expectations allows us to rewrite this as:

$$
E^Q\Big[E^Q\Big[\Big(\widehat{S}(0)\Big(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}\Big) - K\Big)^+\ \Big|\ \sigma\Big(\sum_{i=1}^{\widetilde{N}_T} Y_i\Big)\Big]\Big].
$$

This means we are now taking the inner conditional expectation over the specific sigma algebra $\sigma(\sum_{i=1}^{\widetilde{N}_T} Y_i)$. By conditioning on the specific sigma algebra we are essentially using the available information up to time T to compute the conditional expectation of the expression. This simplification is possible because the conditioning restricts the uncertainty to the events in the specified sigma algebra.

In the context of our expression, the conditional expectation represents the expected value of the inner expression (the option payoff) given the information available up to time T, which is encapsulated by the sigma algebra. This conditional expectation accounts for the uncertainty in future events by conditioning on the information from the past.

The sequence of random variables Y_i *(= log(V)),* where V is a sequence of independent and identically distributed i.i.d fuzzy random variables, contributes to the information available in the conditioning σ–algebra. These random variables follow a normal distribution with mean μ and variance σ^2. Therefore, we have:

$$E^Q[E^Q[\left(\widehat{S}(0)(e^{\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1)t+\ \widetilde{\sigma}B_t+\widehat{\widehat{\sigma}}B_t^H\ -\ \frac{1}{2}\widehat{\sigma}^2t\ -\ \frac{1}{2}\widehat{\sigma}^2t^{2H}+\sum_{i=1}^{\widetilde{N}_T}Y_i})-K\right)^+|\sigma(\sum_{i=1}^{\widetilde{N}_T}Y_i)]$$

$$=e^{-\widetilde{r}(T-t)}\sum_{n=0}^{\infty}\frac{(\widetilde{\lambda t})^n}{n!}\,e^{-\widetilde{\lambda}t}\,E^Q\left(\widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma}B_t+\widehat{\widehat{\sigma}}B_t^H-\frac{1}{2}\widehat{\sigma}^2t-\frac{1}{2}\widehat{\sigma}^2t^{2H}+\sum_{i=1}^{\widetilde{N}_T}Y_i}\right)-K\right)^+.$$

When we condition on the sum of Y_i, which is essentially the cumulative log value of the sequence of jumps, we are effectively considering the total logarithmic impact of the jumps up to time T. Since the number of jumps are modelled using a fuzzy Poisson process with jump intensity λ, their cumulative effect follows a Poisson distribution.

In other words, the jumps, when summed and transformed by taking the logarithm, have properties that lead to the emergence of a Poisson distribution in the conditional expectation. $\widetilde{\sigma}B_t+\widehat{\widehat{\sigma}}B_t^H+\sum_{i=1}^{\widetilde{N}_T}Y_i$ consists of three i.i.d.

fuzzy normal random variables.

From the properties of variances of random variables, we know that the variance of the square of the random variables is the sum of their variances. Also, from Itô's, we know that the square of B_t and B^H is T and T^{2H}. So, $\widetilde{\sigma}^2 = \widetilde{\ \sigma}^2 T + \widehat{\widetilde{\sigma}}^2 T^{2H} + n\widetilde{\sigma}_j^2$. Mean of $\widetilde{\ \sigma} B_t + \widehat{\widetilde{\sigma}} B_t^H + \sum_{i=1}^{\widetilde{N}_T} Y_i$ is simply expectation of $\sum_{i=1}^{\widetilde{N}_T} Y_i$ because expectation of the stochastic processes is zero. So, $\widetilde{\mu} = \sum_{i=1}^{\widetilde{N}_T} Y = n_i\ \widetilde{\mu}_j$. We plug these values into the above the equations:

$$E^Q\left(\widehat{S}(0)(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\ \widetilde{\ \sigma} B_t+\widehat{\widetilde{\sigma}} B_t^H\ -\ \frac{1}{2}\widehat{\sigma}^2 t\ -\ \frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}) - K\right)^+ =$$

$$= \int_{-\infty}^{\infty} \max\left[\widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\ \widetilde{\ \sigma} B_t+\widehat{\widetilde{\sigma}} B_t^H\ -\ \frac{1}{2}\widehat{\sigma}^2 t\ -\ \frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}\right) - K, 0\right]\varphi(z)dz$$

where $\varphi(z)$ is the density of the fuzzy normal variable with mean $\widetilde{\mu} = \sum_{i=1}^{\widetilde{N}_T} Y = n_i\ \widetilde{\mu}_j$ and variance $\widetilde{\sigma}^2 = \widetilde{\ \sigma}^2 T + \widehat{\widetilde{\sigma}}^2 T^{2H} + n\widetilde{\sigma}_j^2$. So, $\varphi(z) = \frac{1}{\sqrt{2\pi\widetilde{\sigma}}} e^{-\frac{(z-\widetilde{\mu})^2}{2\widetilde{\sigma}^2}dz}$, and the whole integral:

$$= \int_{-\infty}^{\infty} \max\left[\widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma} B_t+\widehat{\widetilde{\sigma}} B_t^H-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}\right) - K, 0\right]\frac{1}{\sqrt{2\pi}\,\widetilde{\sigma}} e^{-\frac{(z-\widetilde{\mu})^2}{2\widetilde{\sigma}^2}}\ dz.$$

The integrand in the integral above vanishes when:
$\widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1)\right)t+\ \widetilde{\ \sigma} B_t+\widehat{\widetilde{\sigma}} B_t^H\ -\ \frac{1}{2}\widehat{\sigma}^2 t\ -\ \frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}\right) < K$, i.e. when $z < z_0$, where

$$z_0 = \frac{\left[\ln\left(\frac{K}{\widehat{S}(0)}\right) - \widehat{r} - \widetilde{\lambda}\left(\widetilde{J}-1)\right)t\ -\ \frac{1}{2}\widehat{\sigma}^2 t\ -\ \frac{1}{2}\widehat{\sigma}^2 t^{2H}\right] -\ \widetilde{\mu}}{\widetilde{\sigma}}$$

The integral can thus be written as:

$$\int_{z_0}^{\infty} \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H-\frac{1}{2}\widetilde{\sigma}^2t-\frac{1}{2}\widehat{\sigma}^2t^{2H}+\sum_{i=1}^{\widetilde{N}_T}Y_i}\right)\varphi(z)\,dz$$
$$-\int_{z_0}^{\infty} K\frac{1}{\sqrt{2\pi}\,\widetilde{\sigma}}e^{-\frac{(z-\widetilde{\mu})^2}{2\widetilde{\sigma}^2}}\,dz = A - B.$$

The integral B can be written as $K \cdot Prob(Z \geq z_0)$ and using the symmetry of the Normal distribution, this can be written as $K \cdot Prob(Z \leq -z_0)$. So, if we denote the cumulative distribution function of N as is the usual convention: $\frac{1}{\sqrt{2\pi\widetilde{\sigma}}}e^{-\frac{(z-\widetilde{\mu})^2}{2\widetilde{\sigma}^2}dz}$ then we can write $B = K \cdot N(-z_0)$. In the integral A, we have:

$$A = \int_{z_0}^{\infty} \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t+\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H-\frac{1}{2}\widetilde{\sigma}^2t-\frac{1}{2}\widehat{\sigma}^2t^{2H}+\sum_{i=1}^{\widetilde{N}_T}Y_i}\right)\frac{1}{\sqrt{2\pi\widetilde{\sigma}}}e^{-\frac{(z-\widetilde{\mu})^2}{2\widetilde{\sigma}^2}}\,dz$$
$$= \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}\left(\widetilde{J}-1\right)t-\frac{1}{2}\widetilde{\sigma}^2t-\frac{1}{2}\widehat{\sigma}^2t^{2H}}\right)\frac{1}{\sqrt{2\pi}}\int_{z_0}^{\infty} e^{\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H+\sum_{i=1}^{\widetilde{N}_T}Y_i}\frac{e^{-\frac{(z-\widetilde{\mu})^2}{2\widetilde{\sigma}^2}}}{\widetilde{\sigma}}\,dz.$$

Recall that the fuzzy normal random variable $\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H+\sum_{i=1}^{\widetilde{N}_T}Y_i$ has mean $\widetilde{\mu}=\sum_{i=1}^{\widetilde{N}_T}Y = n_i\ \widetilde{\mu}_j$ and variance $\widetilde{\sigma}^2 = \ \widetilde{\sigma}^2T+\widehat{\widetilde{\sigma}}^2T^{2H}+n\widetilde{\sigma}_j^2$.

So, we are integrating a function with respect to a random variable that follows a given probability distribution. In the field of stochastics, this is often encountered in the form of an expectation. In our case, the expectation of $(exp(\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H+\sum_{i=1}^{\widetilde{N}_T}Y_i))$, where $\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H+\sum_{i=1}^{\widetilde{N}_T}Y_i$ is normally distributed, can be written as follows:

$E(\exp\left(\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H+\sum_{i=1}^{\widetilde{N}_T}Y_i\right) = \int e^z f(z)dz$

where *f(z)* is the probability density function (PDF) of the normal distribution:

$$f(z) \ = \ (1/\sqrt{}(2\pi\sigma^2)) \ \exp(-(z-\mu)^2 \ / \ (2\sigma^2)).$$

Plugging this into the original equation, we obtain:

$E[\exp(Z)] \ = \ \int \ e^z \ (1/\sqrt{(2\pi\sigma^2)}) \ \exp(-(z-\mu)^2 \ / \ (2\sigma^2))dz.$

Simplifying this, we get:

$E[\exp(Z)] \ = \ (1/\sqrt{(2\pi\sigma^2)}) \ \int \ exp\{[(2z\sigma^2 \ - \ (z-\mu)^2) \ / \ (2\sigma^2)]\} \ dz.$

Complete the square:

$$z - \frac{(z-\mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2}\left[z^2 - 2\left(\mu+\sigma^2\right)z\right]\frac{\mu^2}{2\sigma^2} = -\frac{\left(z-(\mu+\sigma^2)\right)^2}{2\sigma^2} + \mu + \frac{\sigma^2}{2}$$

$$E[exp(Z)] \ = \ \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1)t \ - \ \frac{1}{2}\widehat{\sigma}^2 t \ - \ \frac{1}{2}\widehat{\sigma}^2 t^{2H}}\right)\frac{1}{\sqrt{2\pi}\sigma}e^{\mu+\frac{\sigma^2}{2}}\Phi\left(\frac{-z_0+(\mu+\sigma^2)}{2\sigma}\right)$$

The above expression is the definition of the moment generating function (MGF) of a normally distributed random variable at the point *t=1.* The MGF of a normal distribution with mean μ and variance σ^2 is given by:

$$M(t) \ = \ exp\{\mu t \ + \ (1/2)t^2\sigma^2\}$$

Substituting *t=1* in the MGF gives us:

$$E[e^Y] \ = \ exp\{\mu \ + \ (1/2)\sigma^2\}$$

This is the expected value of an exponential of a normally distributed random variable. This is different from the indefinite integral of e^Y with respect to *Y,* which isn't typically defined for random variables. Instead, we work with expectations, which are a form of weighted integral where the weights are given by the PDF of the random variable.

So, the above integral is now simplified to:

$$= \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1))t \;-\; \frac{1}{2}\widehat{\sigma}^2 t \;-\; \frac{1}{2}\widehat{\sigma}^2 t^{2H} + \widetilde{\mu} + \frac{1}{2}\widetilde{\sigma}^2}\right) \frac{1}{\sqrt{2\pi}} \int_{z_0}^{\infty} \frac{e^{-\frac{(z-(\widetilde{\mu}+\sigma^2))^2}{2\widetilde{\sigma}^2}}}{\widetilde{\sigma}} dz$$

Next step is to standardise the integral by changing the variable. Let $y = \frac{(z-(\widetilde{\mu}+\sigma^2)}{\widetilde{\sigma}}$ so $dz = \sigma dy$. The lower limit is $y_0 = \frac{(z_0-(\widetilde{\mu}+\sigma^2)}{\widetilde{\sigma}}$.

$$= \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1))t \;-\; \frac{1}{2}\widehat{\sigma}^2 t \;-\; \frac{1}{2}\widehat{\sigma}^2 t^{2H} + \widetilde{\mu} + \frac{1}{2}\widetilde{\sigma}^2}\right) \frac{1}{\sqrt{2\pi}} \int_{\frac{z_0-\ \widetilde{\mu}-\widetilde{\sigma}^2}{\widetilde{\sigma}}}^{\infty} e^{-\frac{(y)^2}{2}} dy$$

Again, this is the density of the of the normal distribution $\Phi(-y_0)$. Using symmetry we can write:

$$\begin{aligned} A &= \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1)t-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\widetilde{\mu}+\frac{1}{2}\widetilde{\sigma}^2}\right)\Phi(-y_0) \\ &= \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1)t-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\widetilde{\mu}+\frac{1}{2}\widetilde{\sigma}^2}\right)\Phi\left(\frac{-z_0+\widetilde{\mu}+\widetilde{\sigma}^2}{\widetilde{\sigma}}\right). \end{aligned}$$

Now, we can combine A and B and write the whole expression:

$$A-B = \widehat{S}(0)\left(e^{\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1)t-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\widetilde{\mu}+\frac{1}{2}\widetilde{\sigma}^2}\right)\Phi\left(\frac{-z_0+\widetilde{\mu}+\widetilde{\sigma}^2}{\widetilde{\sigma}}\right) - K\,\Phi\left(\frac{-z_0+\widetilde{\mu}}{\widetilde{\sigma}}\right).$$

Finally substituting above expression into:

$$\begin{aligned} Z_t &= e^{-\widetilde{r}(T-t)} \sum_{n=0}^{\infty} \frac{(\widetilde{\lambda t})^n}{n!} e^{-\widetilde{\lambda}t} E^Q\left(\widehat{S}(0)\left(e^{(\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1))t+\widetilde{\sigma}B_t+\widehat{\widetilde{\sigma}}B_t^H-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\sum_{i=1}^{\widetilde{N}_T} Y_i}\right) - K\right)^+ \\ &= e^{-\widetilde{r}(T-t)} \sum_{n=0}^{\infty} \frac{(\widetilde{\lambda t})^n}{n!} e^{-\widetilde{\lambda}t}\, \widehat{S}(0)\left(e^{(\widehat{r}-\widetilde{\lambda}(\widetilde{J}-1))t-\frac{1}{2}\widehat{\sigma}^2 t-\frac{1}{2}\widehat{\sigma}^2 t^{2H}+\widetilde{\mu}+\frac{1}{2}\widetilde{\sigma}^2}\right)\Phi\left(\frac{-z_{0+}\widetilde{\mu}+\widetilde{\sigma}^2}{\widetilde{\sigma}}\right) \\ &\quad - K\,\Phi\left(\frac{-z_{0+}\widetilde{\mu}}{\widetilde{\sigma}}\right). \end{aligned}$$

Recall that $\widetilde{J} = e^{\widetilde{\mu} + \frac{1}{2}\widetilde{\sigma}^2}$. Take $t_0 = 0$ and substitute $\widetilde{J}$ into above expression, we get:

$\sum_{n=0}^{\infty} \frac{(\widetilde{\lambda} t)^n}{n!} e^{\widetilde{-\lambda t}} [\widehat{S}(0) \left(e^{(\widehat{r} - \widetilde{\lambda}(\widetilde{J}-1))t \; - \; \frac{1}{2}\widehat{\sigma}^2 t \; - \; \frac{1}{2}\widehat{\sigma}^2 t^{2H} + \widetilde{\mu} + \frac{1}{2}\widetilde{\sigma}^2 - \widetilde{r}T} \right) \Phi \left(\frac{-z_{0+} \; \widetilde{\mu} + \widetilde{\sigma}^2}{\widetilde{\sigma}} \right) \; - \; K e^{-\widetilde{r}T} \Phi \left(\frac{-z_{0+} \; \widetilde{\mu}}{\widetilde{\sigma}} \right)]$

Simplify:

$$
\begin{aligned}
&= \sum_{n=0}^{\infty} \frac{(\widetilde{\lambda t})^n}{n!} \, e^{-\widetilde{\lambda} t} \Big[\widehat{S}(0) \, e^{\widetilde{\lambda} t - \widetilde{\lambda} J t + n\left(\widetilde{\mu} + \frac{1}{2}\widetilde{\sigma}^2\right)} \Phi \left(\frac{-z_{0+}\widetilde{\mu} + \widetilde{\sigma}^2}{\widetilde{\sigma}} \right) - K e^{-\widetilde{r}T} \, \Phi \left(\frac{-z_{0+}\widetilde{\mu}}{\widetilde{\sigma}} \right) \Big] \\
&= \sum_{n=0}^{\infty} \frac{(\widetilde{\lambda t})^n}{n!} \, e^{-\widetilde{\lambda} J t} \Big[\widehat{S}(0) \, e^{n\left(\widetilde{\mu} + \frac{1}{2}\widetilde{\sigma}^2\right)} \Phi \left(\frac{-z_{0+}\widetilde{\mu} + \widetilde{\sigma}^2}{\widetilde{\sigma}} \right) - K e^{-\widetilde{r}T} \, \Phi \left(\frac{-z_{0+}\widetilde{\mu}}{\widetilde{\sigma}} \right) \Big] \\
&= \sum_{n=0}^{\infty} \frac{(\widetilde{\lambda} J t)^n}{n!} \, e^{-\widetilde{\lambda} J t} \Big[\widehat{S}(0) \, \Phi \left(\frac{-z_{0+}\widetilde{\mu} + \widetilde{\sigma}^2}{\widetilde{\sigma}} \right) - K e^{-\widetilde{r}T} \, \Phi \left(\frac{-z_{0+}\widetilde{\mu}}{\widetilde{\sigma}} \right) \Big] \\
&= \sum_{n=0}^{\infty} \frac{(\widetilde{\lambda} J t)^n}{n!} \, e^{-\widetilde{\lambda} J t} \Big[\widehat{S}(0) \, \Phi(d_1) - K e^{-\widetilde{r}T} \Phi(d_2) \Big].
\end{aligned}
\tag{5.1.35}
$$

Φ represents the cumulative standard normal distribution function. The terms *d1* and *d2* are often used to assess the probability of the option expiring in the money or out of the money. *d1* represents the standardised distance between the current stock price (S) and the strike price (K), adjusted by other factors such as the risk–free interest rate (r), volatility (σ), and time to expiration (T). Specifically, *d1* incorporates the expected return and the expected volatility of the stock. *d2* is similar to *d1*, but it is adjusted by subtracting the volatility component ($\sigma\sqrt{T}$). This adjustment reflects the expectation of the stock price being below the strike price at expiration. The probabilities come into play when you consider the cumulative standard normal distribution function (Φ) applied to these *d1* and *d2* values. The cumulative distribution function gives the probability that a standard normal random variable is less than or equal to a given value. For a call option, Φ*(d2)* represents the risk–neutral probability that the option will finish in the money (stock price above the strike price) and (1– Φ*(d2))* represents the probability that the option will finish out of the money (stock price below the strike price). In summary, while *d1* and *d2* aren't direct probabilities, they are closely related to the probabilities of the option being in the money or out of the money.

Over the course of this research, we have developed a multifaceted framework for pricing derivatives and modelling asset dynamics that integrates fuzzy set theory, fractional Brownian motion (fBm), jump processes, and classical no–arbitrage principles under a risk–neutral measure. This approach does not merely extend one well–known model. It synthesises multiple advanced elements from mathematical finance and stochastic calculus into a single comprehensive structure.

We began by defining an SDE in which fuzzy concepts (upper/lower functions for membership levels) are combined with fractional Brownian motion having Hurst exponent $H > \frac{1}{2}$. Standard fractional finance work typically involves only a fractional Brownian model or a deterministic fuzzy model, but not both simultaneously. Our step of allowing the SDE's drift and diffusion terms to depend on fuzzy processes introduces extra flexibility in capturing uncertainty. This alone is more advanced than most single–parameter fractional or fuzzy finance models.

Instead of using a purely classical PDE $\partial_t u + Lu = 0$, we designed a custom PDE with embedded exponential and hyperbolic functions for negative drift. This is an extension of standard PDE—-SDE bridging via Feynman—-Kac which typically places all operators on the left–hand side, ensuring a known diffusion generator. Our method is a hybrid approach, creating a direct path to solving an enriched PDE that captures both fractional behaviour through exponents involving σt^{2H} and fuzzy membership constraints through special boundary or integral conditions.

We then invoke Girsanov's theorem to shift from the "real–world" measure $\boldsymbol{P}$ to a "risk–neutral" measure $\boldsymbol{Q}$. This step is typical in modern finance, but we apply it to a fuzzy fractional mixed jump environment. We demonstrate that the drift changes while the diffusion remains unaffected, combined with our no–arbitrage condition μ=*r*-λ*(J-1),* it ensures consistency with fundamental asset pricing principles. This bridging of fuzzy fractional assets to classical "risk–neutral" arguments is innovative because standard fuzzy finance seldom performs measure change in this manner.

Additionally, we draw on Cheridito's mixed fractional Brownian framework, referencing the condition $H>\frac{3}{4}$ for certain regularity and on pathwise existence/uniqueness (Young rough integration) for $H> \frac{1}{2}$. We then incorporate Poisson i.i.d. jump sequences (V_i) to capture jump risk. This yields an SDE or PDE that unifies continuous fractional noise, discrete jumps, and fuzzy uncertainty. Such synergy between fractional noise, fuzzy membership, and jumps is rarely attempted, making our model significantly more unified than classical jump–diffusions or single–parameter fractional SDEs.

Our final expressions for option pricing iterated expectations with respect to $\sigma(\sum Y_i)$, integrals involving $\Phi(d_1)$ and $\Phi(d_2)$, and summations over Poisson counts reflect an extension of the standard Black—-Scholes—-Merton logic. By combining fuzzy membership, fractional exponents, jump expansions under the risk–neutral measure, we arrived at a unique closed–form payoff integral. In other words, the arbitrage–free price is uniquely determined by the model's dynamics under Q.

5.1.7 FUZZY STOCHASTIC DIFFERENTIAL EQUATION WHEN H $< \frac{1}{2}$

5.1.7.1 BACKGROUND

Even standard Brownian motion has infinite total variation on every interval, almost surely. This reflects the extreme roughness of its sample paths– they are continuous yet nowhere differentiable, with endlessly oscillating, zig–zag behaviour at all scales.

It helps to distinguish total variation from quadratic variation. For Brownian motion, which corresponds to fractional Brownian motion with Hurst parameter ($\mathrm{H} = \frac{1}{2}$), the quadratic variation over [0,T] exists and equals $\underline{\mathrm{T}}$. In contrast, for fractional Brownian motion (B^{H}) with general $\mathrm{H} \in (0,1)$), the quadratic variation depends on H. It vanishes in the limit when H $> \frac{1}{2}$ and it diverges to $+\infty$ when H $< \frac{1}{2}$. A useful unifying rule is given by p–variation. Fractional Brownian motion has finite p–variation almost surely if and only if p >1/H. Consequently, total variation in the case p=1 is infinite for every H<1, including standard Brownian motion. Quadratic variation corresponds to p=2, and its behaviour across H matches the statements above.

To demonstrate mathematically the behaviour of the quadratic variation of fractional Brownian motion (fBm) when $H < \frac{1}{2}$, we need to consider the relationship between the classical quadratic variation and the Hurst parameter *H*. The Hurst parameter H represents the degree of long–term memory or self–similarity in the process. To understand the case when $H < \frac{1}{2}$ we need to break down the behaviour of fBm and its classical quadratic variation, focusing on the role of the Hurst exponent.

Classical quadratic variation measures the "roughness" of a stochastic process. It is calculated by summing the squares of the increments over a time interval and taking a limit as the interval is divided into smaller and smaller pieces.

Fractional Brownian motion (fBm) is a generalisation of standard Brownian motion, parameterised by *H*. Unlike standard Brownian motion, the increments of fBm are not independent and display long–range dependence when $H \neq \frac{1}{2}$. For standard Brownian motion ($H = \frac{1}{2}$), the quadratic variation over an interval [*0, T*] is given by:

$$Q = \lim_{|p| \to 0} \sum_{i=1}^{n} \Big(X(t_i) - X(t_{i-1}) \Big)^2 .$$

where P represents a partition of the interval [*0, T*], $||P||$ is the mesh size of the partition, $X(t_i)$ is the value of the process at time t_i, and n is the number of subintervals in the partition. For fractional Brownian motion, the quadratic variation is fundamentally different. Specifically, for fBm with Hurst parameter *H*, the quantity

$$\sum_{i=1}^{n} \left(X\left(t_i\right) - \ X\left(t_{i-1}\right) \right)^2$$

does not converge to a finite value as the partition gets finer (i.e., as $||P||$ –> 0) when $H \neq \frac{1}{2}$. Instead, the increments' behaviour is characterised by the Hurst parameter. For $H \neq \frac{1}{2}$, the process is not semi–martingale, and hence the quadratic variation in the classical sense doesn't apply. However, if we consider the mean squared increment over a fixed interval, this scales as Δt^{2H}. For a large time interval T, the mean squared increment over a unit interval scales as T^{2H}. The increments of fBm become negatively correlated, and the process exhibits subdiffusive behaviour. This affects the scaling of the quadratic variation over large intervals. For $H < \frac{1}{2}$, the quadratic variation of fBm over an interval does not converge in the traditional sense. Instead:

$$\lim_{T \to 0} E\left[\left| X(T) - \ X(0) \right|^2 \right] = \lim_{T \to \infty} T^{2H} = \infty$$

This indicates that the process's variance grows sub–linearly with T, but it does not imply that the quadratic variation converges to zero as in the case when $H > \frac{1}{2}$. In fact, it diverges for large T to infinity. This contradicts the Doob–Meyer decomposition since B^H has unbounded 1–variation. This unboundedness makes the traditional definition of the stochastic integrals w.r.t fBm via Riemann sum ineffective.

This connection between H and the quadratic variation is crucial for understanding the behaviour of fBm, in various applications from financial modelling to image analysis. We also need to distinguish between unbounded variation and quadratic variation when it comes to "rough" processes. The total variation of a function over an interval is the sum of the absolute values of its increments over a partition of that interval. For fBm with $H > \frac{1}{2}$, the increments are positively correlated. This means that if the process increases at one point, it is likely to increase further at subsequent points. As the partition gets finer, the increments tend to reinforce each other, leading to a larger and larger sum of absolute values. This sum grows without bound, indicating unbounded variation. It is a bit counterintuitive that both quadratic variation and total variation can be unbounded depending on H. When H $< 1/2$, the increments of fBm are negatively correlated. This means that if the process increases at one point, it's more likely to decrease at a later point. This leads to a "short–range dependence" and a tendency for the path to exhibit more erratic behaviour. Here are three characteristics of "rough" processes when $H < 1/2$.

i. Increasing roughness – as the time interval is divided into smaller and smaller pieces, the negative correlation between increments means that the increments tend to "cancel each other out." However, this cancellation is not perfect. The path still exhibits sharp oscillations, and these oscillations become more frequent as the partition gets finer.

ii. Infinite quadratic variation – the sum of the squares of the increments grows without bound as the partition gets finer. This is because the increments, while tending to cancel each other, still contribute significantly to the squared sum due to the frequent swings.

iii. Infinite total variation – even though the path is rough, the total variation (sum of absolute values of increments) is infinite for $H < 1/2$. This is because the negative correlation does not lead to enough cancellations of increments, thus not preventing the sum of absolute values from growing infinitely large.

5.1.8 INTRODUCING PROOF WHEN H $< \frac{1}{2}$

The majority of the proofs in this section were originally published in our paper [P3]. The proofs in section 5.1.4 are valid when $H > \frac{1}{2}$. We extend the theory of fBm for $H < \frac{1}{2}$, where stochastic integrals become non–semimartingales. As we described it in the *Background* to this section, fBm has an infinite quadratic variation when $H < \frac{1}{2}$, so we use Jost [105] to change Hurst parameter from H to $1–H$. We incorporate fuzzy logic into the amplitude and kernel components of stochastic models. We introduce and analyse the existence and uniqueness theorems under this fuzzy–fractional framework.

Lemma 1.1 (Young integration by parts for B^H; McGonegal [106]). Let H $\in$ (0, 1) and B^H be fractional Brownian motion. If f: [0, T] $\to$ R is deterministic and has bounded p–variation for all p < 1/(1 – H), then the Young/Stieltjes integral $\int_0^T f(t)dB^H(t)$ exists and:

$$\int_0^T f(t)dB^H(t) = f(T)B^H(T) - \int_0^T B^H(t)df(t)$$

Proposition 1.2. The author proves the relationship for a deterministic function f such that it has bounded p–variation sample paths for all $p <$ *1/(1–H)*. Define:

$$Y_t^H := \int_0^t s^{H-\frac{1}{2}}\, dB^H(t).$$

Where the integral is understood in the Young/Stieltjes sense (e.g. as the limit of $\int_0^t (s+\varepsilon)^{H-\frac{1}{2}} dB_s^H \; as \; \varepsilon \downarrow 0$. Then:

$$B_t^H = \int_0^t s^{\frac{1}{2}-H}\, dY_s^H.$$

Our proof:

$$\begin{aligned}
Y_t^H &= \int_0^t s^{H-\frac{1}{2}}\, dB^H(t) \\
&= t^{\frac{1}{2}-H} B^H(t) \;-\; \int_0^t B^H(s)\, d\Big(s^{\frac{1}{2}-H}\Big) \\
&= t^{\frac{1}{2}-H} B^H(t) \;-\; \int_0^t B^H(s) \left(\frac{1}{2} - H\right) t^{-\frac{1}{2}-H}\, ds.
\end{aligned}$$

Apply derivative:

$$\begin{aligned}
dY_t^H &= \left(\tfrac{1}{2} - H\right) t^{-\frac{1}{2}-H}\, dt\, B^H(t) + t^{\frac{1}{2}-H}\, dB^H(t) - B^H(t) \left(\tfrac{1}{2} - H\right) t^{-\frac{1}{2}-H} \\
&= dB^H(s)\, s^{-\frac{1}{2}+H}.
\end{aligned}$$

$$\Rightarrow \quad dB_s^H = dY^H(s)\, s^{\frac{1}{2}-H}.$$

Apply integration:

$$B_t^H = \int_0^t s^{\frac{1}{2}-H}\, dY^H(s). \tag{5.1.36}$$

Proposition 1.3 Following notation from Novikov [107], we utilise Molchan martingale [108] in terms of Y^H,

$$M_t^H := \int_0^t w(t,s)\, dB^H(s).$$

where $w(t,s) \doteq (\frac{c}{C})s^{-\alpha}(t-s)^{-\alpha}$ is a scaled beta kernel and w(t,s) = 0 for s > t; $\alpha \doteq H - \frac{1}{2}$;

$$C \doteq \sqrt{\frac{H}{\left(H-\frac{1}{2}\right)\, B\!\left(H-\frac{1}{2},\, 2-2H\right)}} \qquad \text{and} \qquad c \doteq \frac{1}{B\!\left(H+\frac{1}{2},\, \frac{3}{2}-H\right)}.$$

So, c(H) is a function of H and the standard Beta function $B(\mu,\ \upsilon) \doteq \frac{\Gamma(\mu)\Gamma(\nu)}{\Gamma(\mu+\ \nu)}$. By Proposition 2.1 from Novikov [107] and proof of Theorem of 3.2 in Norros et al., [109], then M is a Gaussian martingale. Substitute the values of dB into Molchan martingale we obtain:

$$\begin{aligned} M_t^H &= \left(\frac{c}{C}\right)\int_0^t s^{\frac{1}{2}-H}(t-s)^{\frac{1}{2}-H} s^{H-\frac{1}{2}}\, dY^H(s) \\ &= \left(\frac{c}{C}\right)\int_0^t (t-s)^{\frac{1}{2}-H}\, dY^H(s). \end{aligned}$$

Lemma 1.4 (Two admissible routes for H < $\frac{1}{2}$; Young vs. Jost/Wiener). For H ∈ (0, $\frac{1}{2}$), the naïve kernel $\mathrm{s}^{\mathrm{H}-\frac{1}{2}}$ is singular at 0 and does not satisfy the bounded p–variation hypothesis required by 5.1.34. There are two rigorous constructions.

- Young route. Regularise near 0 and/or choose an admissible power

$\beta > \frac{1}{2} - H$, defining $Y_t \;:=\; \int_0^t s^{\beta} dB^H(s)$ $(Young)$,

So that the integration by parts identity applies and 5.1.36–5.1.37 hold.

- Jost/Wiener route. Use the Volterra representation $B_t^H = \int_0^t K_H(t,\ s)dW(s)$ and Jost' $H \leftrightarrow (1-H)$ transform to interpret $\int_0^t s^{H-\frac{1}{2}} dB^H(s)$ as a Wiener/Skorod integral against W. This bypasses the bounded p–variation requirement while preserving Gaussian isometries and yields the same identities in L^2.

Remark 1.5. The role of $H < \frac{1}{2}$ here is twofold: (i) B^H is not a semimartingale, so Itô differentials cannot be used; (ii) the kernel $s^{H-\frac{1}{2}}$ is non–integrabe at 0 in the *p*–variation sense, motivating either a Young–admissible regularisation or a Volterra/Wiener construction via Jost's transform.

Norros et al., [109] showed that the filtrations generated by M and Y are the same. They showed that $Y_T \;= 2H \int_0^T (T-t)^{\frac{1}{2}-H} dM(t)$ and the prediction formula:

$$\mathbb{E}[Y_T \mid \mathcal{F}_t] = 2H \int_0^t (T-s)^{\frac{1}{2}-H}\, dM(s). \tag{5.1.37}$$

Now we use equations 5.1.36 and 5.1.37 to find the upper bound using integration by parts. From 5.1.36:

$$sup|B_t^H| \leq sup \mid \int_0^t s^{\frac{1}{2}-H} dY^H(s)| \leq sup \mid \left[s^{-H+\frac{1}{2}} Y_s^H\right]_0^T - \int_0^t d(s^{\frac{1}{2}-H}) Y^H(s)|$$

Since the integral term is positive and by triangular inequality, therefore:

$$\begin{aligned}
\sup |B_t^H| &\leq \sup \left| \left[s^{-H+\frac{1}{2}} Y_s^H\right]_0^T \right| + \sup \left| T^{-H+\frac{1}{2}} \left(Y_t^H - Y_0^H\right) \right| \\
&\leq T^{-H+\frac{1}{2}} \sup \left|Y_t^H\right| + T^{-H+\frac{1}{2}} \sup \left|Y_t^H\right| - 0 \\
&\leq 2T^{-H+\frac{1}{2}} \sup \left|Y_t^H\right|.
\end{aligned}$$

Now using 5.1.37, substitute for Y:

$$\begin{aligned}
\sup|B_t^H| &\leq 2T^{-H+\frac{1}{2}} \sup\left|2H\int_0^t (t-s)^{\frac{1}{2}-H}\, dM^H(s)\right| \\
&\leq 2T^{-H+\frac{1}{2}}\, 2H \sup\left|\Big[(t-s)^{-H+\frac{1}{2}} M_s^H\Big]_0^T - \int_0^t d\Big((t-s)^{\frac{1}{2}-H}\Big)\, M^H(s)\right| \\
&\leq 4H\, T^{-2H+1} \sup\Big(|M_t^H| - |M_0^H|\Big) \leq 8H\, T^{2H-1} \sup\Big(|M_t^H|\Big)\,.
\end{aligned}$$

We now apply expectation:

$$E(sup\left|B_t^H\right|)^p \leq (8HT^{2H-1})^p E(sup(\left|M_t^H\right|)^p)$$

Since M is Molchan martingale, by the Burkholder–Davis–Gundy inequality, there exists a constant $A_p^H > 0$ such that:

$$\left(\mathbb{E}\Big(\sup_{0\leq t\leq T}\left|B_t^H\right|\Big)\right)^p \leq \Big(8H\, T^{2H-1}\Big)^p A_p^H\, \mathbb{E}\left(\sup_{0\leq t\leq T} \left\langle M_t^H\right\rangle^{p/2}\right). \qquad (5.1.38)$$

The next result we take from Norros, Proposition 2.1 [109]:

$$Var\left(M_t^K\right) = \left\langle M^K\right\rangle_t = \frac{c^2(K)}{(2K)^2(2-2K)} t^{2-2K} \;\; := d(K)t^{2-2k}$$

Now, we use the transformation proposed and proved by Jost [77] to change Hurst parameter H. Using Corrollary 5.2 [77], there exists a unique $K = (1 - H)$ fBm, such that:

$$B_t^H = (\frac{2H}{\Gamma(2H)\Gamma(3-2H)})^{1/2} \int_0^t (t-s)^{2H-1} dB_s^{1-H},\ a.s.,\ t\epsilon[0,\ T]$$

By remark 5.6 [105], proves that:

$$M_t^H = (\sqrt{\frac{1-H}{1-K}}) \int_0^t s^{K-H} dM_s^K,\ a.s.,\ t\epsilon[0,\ \infty)$$

We will now combine these results to compute quadratic variation for *1 - H* $> \frac{1}{2}$ to obtain:

For continuous martingales, the quadratic variation of a stochastic integral satisfies the Itô–isometry rule: $\left\langle \int_0^t g(s) dM_s^K \right\rangle_t = \int_0^t g(s)^2 d\left\langle M^K \right\rangle_s$.

$$\left\langle M^H \right\rangle_t = (\frac{1-H}{1-K}) \int_0^t s^{(K-H)^2} d\left\langle M^K \right\rangle_s$$

$$\left\langle M_t^H \right\rangle = \left(\left(\sqrt{\frac{1-H}{1-K}} \right) \int_0^t s^{K-H} dM_s^K \right)^2 = \frac{1-H}{H} \int_0^t s^{2(K-H)} d\left\langle M^{1-H} \right\rangle_s$$

Use Norros [109] :$Var\left(M_t^K\right) = \left\langle M^K \right\rangle_t = \frac{c^2(K)}{(2K)^2(2-2K)} t^{2-2K} \ := d(K)t^{2-2k}$.

Substitute into: $d\left\langle M^{1-H} \right\rangle_s = 2Hd(1-H)s^{2-2\left(\frac{1}{2}-H\right)} = 2Hd(1-H)s^{2H-1}$. Continuing with quadratic variation:

$$\left\langle M_t^H \right\rangle = \frac{1-H}{H} \int_0^t s^{2(K-H)} d\left\langle M^{1-H} \right\rangle_s$$

$$\begin{aligned} \frac{1-H}{H} d(1-H) \int_0^t s^{2(1-2H)}\, 2H\, s^{2H-1}\, ds &= \frac{2H(1-H)}{H} d(1-H) \int_0^t s^{2-4H+2H-1}\, ds \\ &= d(1-H)\, t^{2(1-H)}. \end{aligned} \tag{5.1.39}$$

Apply BDG inequality on quadratic variation of M^H, to obtain:

$$\mathbb{E}\left(\sqrt{\langle M^H \rangle_t} \right)^p \leq B_p^{1-H}\, t^{p(1-H)}. \tag{5.1.40}$$

Using 5.1.38: $E(sup\left|B_t^H\right|)^p \leq (8HT^{2H-1})^p A_p^H E(sup(\left\langle M_t^H \right\rangle)^{p/2})$, substitute into 5.1.40, we obtain:

$$\mathbb{E}\left[\sup_{t\in[0,T]}|B_t^H|^p\right] \leq \left(8H\,T^{2H-1}\right)^p A_p^H B_P^{1-H} T^{p(1-H)}. \tag{5.1.41}$$

If we define $C := (8H)^p A_p^H B_P^{1-H}$ then $E(sup\left|B_t^H\right|)^p \leq CT^{pH}$. Q.E.D.

So, in summary, we use a technique from Jost [105] to transform fBm to (*1 – H*) fBm. By expressing fBm as an integral with respect to standard Brownian motion, the proof employs Volterra–like representations of fBm. The transformed process is a Gaussian martingale that is Molchan with respect to the filtration built from $\mathrm{B_H}$, which allows us to establish the clear upper bound for the integrals. Combined with the fact that we also demonstrate that the involved functions are square integrable, this ensures the finiteness and uniqueness of solution.

The techniques developed for fBm with $H < \frac{1}{2}$ can be effectively applied in financial modelling, particularly in modelling volatility and volatility of volatility (vol–of–vol) which exhibit rough path structure [3] [P2]. These rough volatility models, often characterised by Hurst exponents less than $\frac{1}{2}$, capture the persistent, jagged behaviour of market fluctuations more accurately. By incorporating fuzziness into key parameters such as volatility, drift, and jump intensities, the fuzzy fBm framework enhances the ability to model these irregularities and account for uncertainty in parameter estimation. This is particularly important in environments where precise volatility measurement is challenging due to market noise or incomplete data. The combination of fuzziness with rough fractional paths allows for a more robust representation of the volatility surface, improving option pricing, hedging strategies and risk assessment in volatile markets.

In the context of modelling volatility of volatility, the FFBM can be particularly useful in capturing the stochastic nature in instruments like VIX derivatives and volatility swaps. Traditional models often struggle to account for the irregular, long–memory effects present in vol–of–vol, which is now very documented in empirical finance [3]. By applying FFBM, empiricists can build a shifting vol–vol dynamic model without assuming an overly smooth process. As we demonstrated earlier, this framework can be integrated with Girsanov's theorem under a risk–neutral measure, ensuring that the pricing of derivatives reflects real–world irregularities while maintaining theoretical robustness and no–arbitrage conditions.

In non–financial applications, a potential application is in the telecommunications and network traffic modelling. Fractional Brownian motion has already been widely used to model network traffic, but the real–world network conditions often exhibit variations that are best captured through fuzzy logic. In adaptive bandwidth allocation, congestion control, and network anomaly detection, FFBM can be employed to handle imprecise or incomplete traffic data, allowing service providers to develop more resilient protocols for managing data flow, reduce latency, and optimise resource allocation in high–traffic environments such as cloud computing and real–time streaming services.

In classical fBm, the kernel $K_H(t, s)$ governs the memory and roughness of the process through the Hurst exponent H. When fuzziness is introduced, the kernel becomes a fuzzy function. This means instead of having a single deterministic kernel, the model now handles sets of possible kernels defined by membership functions of fuzzy intervals. The amplitude functions, which influence the drift and diffusion terms in SDEs also become fuzzy variable. This means the diffusion coefficient σ and drift term μ are no longer fixed but represented by fuzzy sets or intervals. While fuzziness increases model complexity, it also introduces robustness in modelling real–world uncertainty. By modelling this uncertainty explicitly through fuzziness, the framework becomes mathematically richer and computationally more intensive, but these challenges are balanced by the model's ability to capture real–world phenomenon where precise information is often unavailable or delayed.

5.1.9 INTRODUCING FUZZY WHEN H $< \frac{1}{2}$

Under the assumption that the fuzzy aspect lives in the amplitude/kernels and that the underlying driving noise $\widetilde{B}$ is actually a crisp fractional Brownian motion. In other words, we are layering fuzziness on the coefficients/functions of the integral but not changing the fundamental measure-theoretic structure of the noise. This assumption ensures we can leverage classical results such as Itô isometry for existence, uniqueness, and bounding integrals which we proved in 5.1.4.

For $H < \frac{1}{2}$, the kernel $K_H(t,\ s)$ used to construct fBm was derived by [4].

$$K_H(t,s) = \ c_H \left[\left(\frac{t}{s} \right)^{H-1/2} (t-s)^{H-1/2} \right]$$

where C_H is a normalising constant given by: $c_H = \left(\frac{2Hsin(\pi H)\Gamma(\frac{3}{2}-H)}{\Gamma(H+\frac{1}{2})} \right)^{1/2}$.

For practical applications, a more convenient representation of the kernel function for $H < \frac{1}{2}$ can be written as:

$$K_H(t,s) = \ c_H \left[(t-s)^{H-1/2} - (-t)^{H-1/2} \right]$$

This ensures the kernel remains well–defined and integrable. To establish upper and lower bounds for the integrals involving fuzzy fractional Brownian motion when $H < \frac{1}{2}$, we need to adapt the existing techniques to handle the fuzzy nature of the random variables. To establish bounds, we need to represent the fuzzy variables as intervals or use membership functions and then analyse these representations.

An important characteristic of fBm that significantly enhances its applicability in financial modelling is its long memory effect, which is governed by the Hurst exponent *H*. The Hurst exponent characterises the autocorrelation structure of increments of the process. If $H > \frac{1}{2}$, this indicates positive correlation and persistent behaviour, meaning past increases or decreases tend to be followed by future movements in the same direction. Financial assets exhibit persistent behaviour, and shocks to be positively correlated over long periods, making fBm an ideal model for capturing long–term dependencies observed in bubble dynamics.

If $H < \frac{1}{2}$, this corresponds to anti–persistence, or rough path showing mean–

reverting behaviour. Such anti persistence phenomena are frequently observed in short–term trading scenarios. In our context, we focus on the case of $H < \frac{1}{2}$, as indicated above, due to its suitability for modelling frequent reversals.

5.1.10 FUZZY GAUSSIAN DISTRIBUTION

At any fixed time t, the value of the fuzzy Brownian motion $\widetilde{B}_H(t)$ is a fuzzy Gaussian variable. This means its probability distribution is a Gaussian distribution with fuzzy mean and variance.

5.1.10.1 Fuzzy Integral Representation

We denote the kernel $K_H(t,\ s)$ as fuzzy, which means at each $\omega \in \Omega$, the kernel belongs to a set and has a membership function describing possible amplitude values.

$$\widetilde{B}_H(t)\ = \int_0^t \widetilde{K}_H(t,\ s) dB(s)$$

We assume the underlying noise is crisp, but the fuzzy aspect is in the amplitude of the kernel. We had earlier proven in 5.1.4 that the classical Itô isometry remains perfectly valid for each crisp selection and each scenario in our fuzzy ensemble because the noise is crisp. The fuzziness is layered on top by letting the integrand vary in intervals. Nothing breaks the standard measure-theoretic basis of $\widetilde{B}_H(t)$.

Our future work would involve defining the fuzzy fractional Brownian motion $\widetilde{B}_H(t)$ in the sense of fuzzy measure theory, not just a crisp B with fuzzy amplitude. Then the classical Itô isometry is no longer guaranteed. Indeed, we might not even have a linear—-additive expectation E in the sense of Kolmogorov measure theory. Instead, we have fuzzy expectation and fuzzy integrals, which typically do not obey Itô isometry.

The step:

$$\mathbb{E}\left[\left(\int_0^t \left(\widetilde{K}_H^{(1)}(t,s) - \widetilde{K}_H^{(2)}(t,s)\right) d\widetilde{B}(s)\right)^2\right] \neq \int_0^t \mathbb{E}\left[\left(\widetilde{K}_H^{(1)}(t,s) - \widetilde{K}_H^{(2)}(t,s)\right)^2\right] ds.$$

would be wrong as indicated by $\neq$, because the left side uses a fuzzy integral and fuzzy expectation. Typically, a pure fuzzy measure does not preserve linear additivity, and it uses a non-classical integral Choquet, or Sugeno. The isometry is not guaranteed.

We need to analyse:

$$E\left[\left(\widetilde{B}_H(t)\right)^2\right] = E\left[\left(\int_0^t \widetilde{K}_H(t,\ s)dB(s)\right)^2\right]$$

Since both $\widetilde{K}_H(t,s)$ and $\widetilde{B}(s)$ are fuzzy, we can represent them as intervals for analysis.

The fuzzy labels on $\widetilde{K}_H(t,s)$ does not break linearity or additivity of the expectations and the integrals, because fuzziness is above the measure level.

For uniqueness proofs, we define upper and lower bounds for Fuzzy Gaussian Increments:

$$\widetilde{K}_H(t,s) = [K_{H,\ L}(t,\ s),\ K_{H,\ U}(t,\ s)\ ]$$

Here, $K_{H,\ L}(t,\ s)$ *and* $K_{H,\ U}(t,\ s)$ represent the lower and upper bounds of the fuzzy kernel function. The upper and lower bounds for the fuzzy fractional Brownian motion integrals are determined by analysing the bounds of the kernel function and the Gaussian process separately.

$$\left(\int_0^t \widetilde{K}_{H,L}(t,\ s)\right)^2 ds \leq E\left[\left(\int_0^t \widetilde{K}_H(t,\ s)d\widetilde{B}(s)\right)^2\right] \leq \int_0^t \widetilde{K}_{H,U}(t,\ s)^2 ds$$

By ensuring that these integrals are finite, we will demonstrate the existence of the solution. Uniqueness will be shown by proving that the fuzzy integrals converge uniquely under the given conditions – if that difference of kernels is zero a.s. owing to identical initial data and integrable constraints, we obtain uniqueness.

In other words, we need to show that:

$$\left(\int_0^t \widetilde{K}_{H,L}(t,s)\right)^2 ds < \infty, \qquad \left(\int_0^t \widetilde{K}_{H,U}(t,s)\right)^2 ds < \infty.$$

For $\widetilde{K}_{H,L}(t,\ s)$:

$$\left(\widetilde{K}_{H,L}(t,s)\right)^2 = c_H^2\,(t-s)^{\,H-\frac{1}{2}}.$$

Integrate this:

$$\int_0^t \left(\widetilde{K}_{H,L}(t,\ s)\right)^2 ds = {c_H}^2 \int_0^t \left((t-s)^{H-1/2}\right)^2 ds = \ {c_H}^2 \int_0^t \left((t-s)^{2H-1}\right) ds < \infty$$

Both terms inside the integral are finite for $H < \frac{1}{2}$, and the subtraction term also ensures the finiteness of the integral.

Similarly, for the upper bounds:

$$\int_0^t \left(\widetilde{K}_{H,U}(t,\ s)\right)^2 ds = {c_H}^2 \int_0^t \left((t-s)^{H-1/2}\right)^2 ds = {c_H}^2 \int_0^t \left((t-s)^{2H-1}\right) ds < \infty$$

The same reasoning applies here, both terms inside the integral are finite, and their subtraction ensures the integrals are finite. Therefore, both the upper and lower bounds are finite.

To show uniqueness, we need to prove that the fuzzy integral representation converges uniquely under the given conditions. We assume that there are two fuzzy fractional Brownian motions $\widetilde{B}_H^{(1)}(t)$ and $\widetilde{B}_H^{(2)}(t)$ with the same initial conditions. Fix $\alpha \in$ (0,1] and suppose two solutions have the same initial data and yield the same crisp kernel selection $K^{(1)}_{H,\alpha} \equiv K^{(2)}_{H,\alpha}$ a.e. on [0, t].

For uniqueness, consider the difference:

$$\Delta\widetilde{B}_H(t) = \widetilde{B}^{(1)}_H(t) \; - \; \widetilde{B}^{(2)}_H(t)$$

Then:

$$E\left[\Delta\widetilde{B}_H(t)^2\right] = E\left[\left(\int_0^t \left(\widetilde{K}^{(1)}_H(t,\ s) - \ \widetilde{K}^{(2)}_H(t,s)\right)\ d\widetilde{B}(s)\right)^2\right]$$

Then we can use Itô's isometry:

$$E\left[\Delta\widetilde{B}_H(t)^2\right] = \int_0^t E\left[\left(\widetilde{K}^{(1)}_H(t,\ s) - \ \widetilde{K}^{(2)}_H(t,s)\right)^2\right] ds = 0$$

If $\widetilde{B}^{(1)}_H(t)$ and $\widetilde{B}^{(2)}_H(t)$ are solutions, then $\widetilde{K}^{(1)}_H(t,\ s) = \ \widetilde{K}^{(2)}_H(t,s)$ a.s. Thus, $E\left[\Delta\widetilde{B}_H(t)^2\right] = 0$.

This implies that $\Delta\widetilde{B}_H(t) = 0$ a.s. and $\widetilde{B}^{(1)}_H(t) \; = \; \widetilde{B}^{(2)}_H(t)$. Proof complete. Therefore, the solution is unique.

5.1.10.2 Membership function

$\widetilde{X}$ is called a fuzzy normal variable if for each $\alpha \in$ *(0,1]*, we can select a crisp normal distribution $N\left(\mu_\alpha, \sigma^2_\alpha\right)$ such that μ_α and σ^2_α come from the α–cuts of the fuzzy numbers $\widetilde{\mu}$ and $\widetilde{\sigma^2}$.The nested family $\{N(\mu_\alpha, \sigma^2_\alpha)\}_{\alpha\epsilon(0,1]}$ is consistent in the sense that if $\alpha_2 < \alpha_1$, then $\mu_{\alpha 1} \subseteq \mu_{\alpha 2}$ as intervals. Higher α means tighter, more trusted interval whereas lower α wider, more permissive interval. For a fixed x, each crisp pair μ and σ^2 gives a Gaussian density $f_{(\mu,\sigma^2)}(x)$. We need a single membership value $\mu_{\widetilde{X}}(x) \in$[0,1] that informs how compatible is x with the fuzzy Gaussian. At level α, is there some μ and σ^2 in the α–cut intervals for which the pdf at x is large enough. If it is then x is at least α–compatible with the fuzzy Gaussian. The membership $\mu_{\widetilde{X}}(x)$

is the largest α for which that statement holds. Pick a threshold τ, which is typically a normalised pdf score so $\tau \in (0,1]$. Then x passes the α–test if $\exists\,(\mu, \sigma^2) \in \mu_\alpha \times \sigma^2_\alpha$ such that $f_{\mu,\sigma^2}(x) \geq \tau$. Higher α means tighter parameter intervals, so it is harder to find a parameter pair that makes x look likely. If x still looks likely even under tight intervals, it deserves a high membership. If we can only make x look likely by allowing very loose intervals (small α), then x's membership is low. It aligns perfectly with α–cut computation – everything is done per α, then aggregated by taking the supremum α that still validates x.

Hence, we represent our fuzzy normal by the entire nested family of classical normal distributions, one for each level of membership α. Next, we apply the Zadeh extension principle (or an interval-based extension principle) to unify fuzzy parameters $\widetilde{\mu}$ and $\widetilde{\sigma^2}$ into one membership function $\mu_X(x)$ for a fuzzy normal distribution. The key idea is once we have fuzzy sets for mean and variance, we want a single fuzzy set in $\boldsymbol{R}$ describing how likely it is for our fuzzy normal to take each real value x. This demands a recognised fuzzy extension approach that merges the fuzzy parameters into a single fuzzy set of real outputs. Generally, for each $\alpha \longrightarrow [\mu_\alpha, \sigma^2_\alpha]$ and t, the value $\widetilde{B}_H(t)$ has membership function describing the degree to which it belongs to the fuzzy set $\widetilde{B}_H(t)$.

$$\psi_x = e^{-\frac{(x-\mu^2}{2\sigma^2}}$$

When we have intervals of means $\mu \in [\mu^-_\alpha, \mu^+_\alpha]$ and variances $\sigma \in [\sigma^{2,-}_\alpha, \sigma^{2,+}_\alpha]$, the pdf can vary accordingly. To find unified pdf we implement the threshold aggregator approach that merges the entire alpha-cut family into one membership function by allowing us to climb up alpha levels as far as possible, subject to the pdf condition. A bigger α indicates a narrower region for μ_α and σ^2_α, so it is more demanding that we still get bigger threshold in the pdf. The result is a single membership at $x \in [0,1]$. If we define τ as some threshold then x has membership $\geq \alpha$ if we can find a parameter pair in α–cuts that yields $\geq \tau$ pdf. So, membership is the supremum of all such α.

This is a version of the Zadeh extension principle, adopted to the pdf is above threshold τ condition.

We now formalise these concepts mathematically. Let $\widetilde{\mu}$ be a fuzzy mean and $\widetilde{\sigma^2}$ be a fuzzy variance. $\widetilde{\mu}$ and $\widetilde{\sigma^2}$ are fuzzy numbers in R and (0,∞) respectively. Fix a threshold $\tau \in$ (0,∞). For each $\alpha \in$ (0,1], define intervals $\mu_\alpha \in\ \widetilde{\mu}^\alpha,\ \sigma_\alpha^2 \in\ (\widetilde{\sigma}^2)^\alpha$. We say $x \in \widetilde{X^\alpha}$ if:

$\exists\,(\mu, \sigma^2) \in \mu_\alpha \times \sigma_\alpha^2$ such that $f_{\mu,\sigma^2}(x) \geq \tau$.

Then the membership function for $\widetilde{X}$ is:

$$\mu_{\widetilde{X}}(x) = \sup\left\{\alpha \in (0,1] :\ \exists\left(\mu, \sigma^2\right) \in \mu_\alpha \times \sigma_\alpha^2\ s.t.\ \psi_x\ (\mu,\ \sigma) \geq \alpha\ and\ x \in \widetilde{X^\alpha}\right\}$$

We call $\widetilde{X}$ a fuzzy normal variable under threshold τ.

This aggregator is one of many possible. Now we have a robust approach that merges fuzzy $\widetilde{\mu}$ and $\widetilde{\sigma^2}$ into a single fuzzy set of real values $\boldsymbol{x}$. It does so by checking the classical normal pdf at each α –cut and picking the supremum over α.

5.1.10.3 Fuzzy Martingale

Lemma 2 (BDG for Type–1 Fuzzy Martingales). Let $\widetilde{M} = \{M^{(\alpha)}\}_{\alpha\epsilon(0,1]}$be a Type–1 fuzzy process on a filtered probability space (Ω, F, ($\mathrm{F_t}$), P) such that, for each α, $\mathrm{M}^{(\alpha)}$ is a continuous local martingale with quadratic variation $\langle \mathrm{M}^{(\alpha)} \rangle$. Then for any p > 0 there exists a constant $\mathrm{C_p} < \infty$, independent of α, such that

$$\sup_{\alpha\in(0,1]} E\left[sup|M_t^{(\alpha)}|^p\right] \leq\ C_p \sup_{\alpha\in(0,1]}\ E\left\langle M_T^{(\alpha)}\right\rangle^{p/2}$$

Proof for Fuzzy Fractional Brownian motion when $H < \frac{1}{2}$ with Gaussian membership function.

A fuzzy martingale $\widetilde{M}$(t) is a generalisation of a standard martingale, where the values are decribed by fuzzy numbers. For a process to be a fuzzy martingale, it must satisfy the martingale property with respect to fuzzy conditional expectations. So, for $\widetilde{M}$(t) to be a fuzzy martingale, it must satify:

$$E\left[\widetilde{M}|F_s\right] = \widetilde{M}(s)$$

We derive step by step proof for ffBm when $H < \frac{1}{2}$ and Gaussian membership function.

1. Integration by Parts with Fuzzy Variable:

Let

$$\widetilde{Y}_t^H := \int_0^t s^{H-\frac{1}{2}}\, d\widetilde{B}^H(s) \ then \qquad \widetilde{B}_t^H = \int_0^t s^{\frac{1}{2}-H}\, d\widetilde{Y}_s^H.$$

Proof:

$$\begin{aligned}
\widetilde{Y}_t^H &= \int_0^t s^{H-\frac{1}{2}}\, d\widetilde{B}^H(s) \\
&= t^{\frac{1}{2}-H}\, \widetilde{B}^H(t) \ - \ \int_0^t \widetilde{B}^H(s)\, d\Big(s^{\frac{1}{2}-H}\Big) \\
&= t^{\frac{1}{2}-H}\, \widetilde{B}^H(t) \ - \ \int_0^t \widetilde{B}^H(s)\Big(\tfrac{1}{2} - H\Big)s^{-\frac{1}{2}-H}\, ds.
\end{aligned}$$

Apply derivative:

$$d\widetilde{Y}_t^H = \Big(\tfrac{1}{2} - H\Big)\, t^{-\frac{1}{2}-H}\, dt\, \widetilde{B}^H(t) + t^{\frac{1}{2}-H}\, d\widetilde{B}^H(t) - \widetilde{B}^H(t)\, \Big(\tfrac{1}{2} - H\Big)\, t^{-\frac{1}{2}-H}\, dt = t^{-\frac{1}{2}+H}\, d\widetilde{B}^H(t).$$

$$\Rightarrow d\widetilde{B}_t^H = \ d\widetilde{Y}^H(t)t^{\frac{1}{2}-H}$$

Apply integration:

$$\widetilde{B}_t^H = \int_0^t s^{\frac{1}{2}-H}\, d\widetilde{Y}^H(s). \tag{5.1.42}$$

2. Define fuzzy Molchan martingale:

$\widetilde{M}_t^H \ := \ \int_0^t \widetilde{w}(t,\ s) d\widetilde{B}^H(s),$

where $\widetilde{w}(t,s) \doteq (\frac{c}{C})s^{-\alpha}(t-s)^{-\alpha}$ is a scaled beta kernel.

3. Upper bound for fuzzy $\widetilde{B}_H(t)$ using integration by parts. From 5.1.41:

$$sup|\widetilde{B}_t^H| \leq sup\ |\ \int_0^t s^{\frac{1}{2}-H} d\widetilde{Y}^H(s)| \leq sup\ |\left[s^{-H+\frac{1}{2}}\widetilde{Y}_s^H\right]_0^T - \int_0^t d(s^{\frac{1}{2}-H})\widetilde{Y}^H(s)|$$

Following triangular inequality bound |A-B| $\leq$ |A|+|B|, we can bind the absolute value of the difference, therefore:

$$\begin{aligned}
\sup\left|\widetilde{B}_t^H\right| &\leq \sup\left|\left[s^{-H+\frac{1}{2}}\widetilde{Y}_s^H\right]_0^T - \int_0^t \left(\tfrac{1}{2}-H\right) s^{-\frac{1}{2}-H}\,\widetilde{Y}^H(s)\,ds\right| \\
&\leq \sup\left|\left[s^{-H+\frac{1}{2}}\widetilde{Y}_s^H\right]_0^T\right| + \int_0^t \left(\tfrac{1}{2}-H\right) s^{-\frac{1}{2}-H}\,\widetilde{Y}^H(s)\,ds \\
&\leq \sup\left|T^{-H+\frac{1}{2}}\left(\widetilde{Y}_t^H - \widetilde{Y}_0^H\right)\right| \\
&\leq 2T^{-H+\frac{1}{2}}\sup\left|\widetilde{Y}_t^H\right|.
\end{aligned}$$

4. Fix $\alpha \in (0,\,1]$. Since $\widetilde{M}_t^\alpha$ is a continuous martingale, the BDG inequality yields:

$$\sup_{\alpha\in(0,1]} E\left[sup|M_t^{(\alpha)}|^p\right] \leq\ C_p \sup_{\alpha\in(0,1]}\ E\left\langle M_T^{(\alpha)}\right\rangle^{p/2}$$

The constants C_p depends only on p, hence, is independent of α. Taking the supremum over $\alpha \in (0,\ 1]$ on both sides of the inequality preserves it and yields the claim.

Corollary 1 (Application to fuzzy Mochan martingale). Let $\widetilde{M} = \{M^{(\alpha)}\}_{\alpha\epsilon(0,1]}$be a Type–1 fuzzy Molchan martingale associated with fuzzy fBm of Hurst index H ϵ $(0, \frac{1}{2})$. Using 5.1.37 (Norros et al. prediction formula applied α–wise) substitute for $\widetilde{Y}$:

$$sup|\widetilde{B}_t^H| \leq 8HT^{2H-1} \sup\left(\left|\widetilde{M}_t^H\right|\right)$$

then for any p > 0,

$$E(sup\left|\widetilde{B}_t^H\right|)^p \leq (8HT^{2H-1})^p A_p^H \widetilde{B}_P^{1-H} T^{p(1-H)}$$

Proof.

5. Following our proofs that show we can combine measures, we now apply expectation:

$$E(sup\left|\widetilde{B}_t^H\right|)^p \leq (8HT^{-2H+1})^p E(sup(\left|\widetilde{M}_t^H\right|)^p)$$

Since $\widetilde{M}_t^H$ is a fuzzy Molchan martingale, by the Burkholder–Davis–Gundy inequality, there exists a constant $A_p^H > 0$ such that:

$$\left(\mathrm{E} \sup_{t\in[0,T]} \left|\widetilde{B}_t^H\right|\right)^p \leq \left(8H\, T^{2H-1}\right)^p A_p^H\, \mathrm{E} \sup_{t\in[0,T]} \left(\left\langle \widetilde{M}^H \right\rangle_t\right)^{p/2}. \tag{5.1.43}$$

The next result we take from Norros, Proposition 2.1 [109]:

$$Var\left(\widetilde{M}_t^K\right) = \left\langle \widetilde{M}^K \right\rangle_t = \frac{c^2(K)}{(2K)^2(2-2K)} t^{2-2K} \;\; := d(K)t^{2-2k}$$

6. We will now combine these results to compute quadratic variation for *K=1 - H* $> \frac{1}{2}$ to obtain:

$$\left\langle \widetilde{M}_t^H \right\rangle = (\left(\sqrt{\frac{1-H}{1-K}}\right) \int_0^t s^{K-H} d\widetilde{M}_s^K)\;)^2 = \frac{1-H}{H} \int_0^t s^{2(K-H)} d\left\langle \widetilde{M}^{1-H} \right\rangle_s$$

Use Norros [109] :$Var\left(\widetilde{M}_t^K\right) = \left\langle \widetilde{M}^K \right\rangle_t = \frac{c^2(K)}{(2K)^2(2-2K)} t^{2-2K} \ := d(K)t^{2-2k}$.

Substitute into: $d\left\langle \widetilde{M}^{1-H} \right\rangle_s = 2Hd(1-H)s^{2H-1)}$. Continuing with quadratic variation:

$$\begin{aligned}
\left\langle \widetilde{M}^H \right\rangle_t &= \left(\sqrt{\frac{1-H}{1-K}} \int_0^t s^{K-H}\, d\widetilde{M}_s^K \right)^2 \\
&= \frac{1-H}{H} \int_0^t s^{2-2H}\, d\left\langle \widetilde{M}^{1-H} \right\rangle_s \\
&= \frac{1-H}{H}\, d(1-H) \int_0^t s^{2(1-2H)}\, 2Hs^{2H-1}\, ds \\
&= \frac{2H(1-H)}{H}\, d(1-H) \int_0^t s^{1-2H}\, ds \\
&= d(1-H)\, t^{2(1-H)}.
\end{aligned} \tag{5.1.44}$$

7. Apply α–level wise BDG inequality on quadratic variation of MH, to obtain:

$$\mathbb{E}\left(\sqrt{\left\langle \widetilde{M}^H \right\rangle_t} \right)^p \ \le \ \widetilde{B}_p^{1-H}\, t^{p(1-H)}. \tag{5.1.45}$$

Using 5.1.43: $E(sup\left|\widetilde{B}_t^H\right|)^p \le (8HT^{2H-1})^p A_p^H E(sup(\left\langle \widetilde{M}_t^H \right\rangle)^{p/2})$, substitute into 5.1.45, we get:

$$\left(\mathrm{E} \sup_{0\le t\le T} \left| \widetilde{B}_t^H \right| \right)^p \le \left(8H\, T^{2H-1} \right)^p A_p^H\, \widetilde{B}_p^{1-H}\, T^{p(1-H)}. \tag{5.1.46}$$

8. If we define $\widetilde{C} := (8H)^p A_p^H \widetilde{B}_P^{1-H}$ then $E(sup\left|\widetilde{B}_t^H\right|)^p \le \widetilde{C}T^{pH}$. Q.E.D.

The risk–free measure transformation relies on constructing a risk–neutral probability measure under which the discounted asset price processes become martingale. In our fuzzy environment, the validity of this transformation hinges upon the assumption that fuzziness is introduced at the coefficient and amplitude level as opposed to altering the fundamental probability measure structure itself. Since the driving noise remains crisp, the classical construction of the risk–free measure is preserved. Thus, each scenario within our fuzzy environment still respects the standard probabilistic assumptions required by the Girsanov transformation.

In order to generalise the classical Girsanov transformation to cover fBm we leverage Molchan–Golosov representation and a suitable martingale construction. This extension ensures the stochastic integrals driven by fBm can be translated into equivalent martingale measures, allowing us to consistently price financial derivatives under fuzzy fractional dynamics. The generalisation uses fractional kernels and transformations introduced through fuzzy integral bounds. To illustrate the practical implications, consider the following heuristic numerical example involving a European call option.

Assumptions:

Asset Price, S0: 100

Strike Price K: 100

Risk–Free Rate: 0.05

Time to maturity (T): 1.0 year(s)

Hurst Exponent H: 0.4

Adjusted Volatility Interval (fuzzy): [0.1500, 0.2500]

Drift (fuzzy), μ*: [0.04, 0.08]*

Results under lower bound and upper bound scenarios:

Option Price Interval: [8.5917, 12.3360]

Rough–vol mapping using H = 0.4

Model the log–variance as coming from a Brownian part and a fractional part:

$$\mathrm{Var}\Big(\log S_T\Big) \approx \sigma_B^2\, T + \sigma_H^2\, T^{2H}.$$

Calibrate σ^B to short–dated ATM IV (e.g. 1m $\to$ 0.16) and σ_H to the 1y–5y IV slope with H = 0.4 (e.g. $\sigma_H \approx 0.10$). For $T = 1$:

$$\sigma_{eff}(T) = \frac{Var\left[\log S_T\right]}{T} \approx \sqrt{\sigma_B^2 + \sigma_H^2 T^{2H-1}}$$

Read the 1–months ATM IV as $\widehat{\sigma}_{1m}$ then,

$$\widehat{\sigma}_H \ \approx \sqrt{\widehat{\sigma}_{1y}^2 - \sigma_B^2}$$

Turn calibration uncertainty into a fuzzy band. Let data be noisy and put small intervals around $\sigma^B$and σ_H. Then for T = 1, $\sigma_B \epsilon$ [$\overline{\sigma}_B$], $\sigma_H \epsilon$ [$\overline{\sigma}_H$], and

$$\sigma_{\text{eff}} \in \left[\sqrt{\underline{\sigma}_B^2 + \underline{\sigma}_H^2},\ \sqrt{\overline{\sigma}_B^2 + \overline{\sigma}_H^2}\right].$$

5.1.11 Simulations

This simulation integrates three distinct components——fBm, Poisson jump processes, and fuzzy set theory to represent asset price dynamics under uncertainty and rough volatility scenario. Fbm was generated with a $H =$ 0.15, specifically selected to reflect the rough volatility observed empirically in the financial markets. A low Hurst exponent captures anti–persistent behaviour, manifesting as rough paths with rapid, irregular fluctuations. The path was simulated over a one–year period ($T = 1$) using 252 discrete time steps, corresponding to typical daily trading intervals in financial markets. The simulation used Cholesky decomposition applied to a covariance matrix (γ), which encodes the long–range dependency structure of fBm increments.

Next, a jump component was added using a Poisson jump model, where the jump intensity parameter λ = *0.5* signifies an average occurrence of 0.5 jumps per year. Jump sizes were lognormally distributed with mean (μ = *0.5*) and standard deviation (σ = *0.02*), simulating occasional significant market shocks or discontinuities.

To incorporate market uncertainty and sentiment explicitly, the volatility parameter was dynamically adjusted using fuzzy logic. The fuzzy logic system defines three market sentiment scenarios–pessimistic, neutral, and optimistic, mapped numerically within a sentiment range from 0 to 10. The fuzzy volatility was obtained by evaluating these sentiment scenarios through fuzzy membership functions defined using triangular fuzzy sets. Specifically, pessimistic sentiment corresponds to high volatility, neutral sentiment to medium volatility, and optimistic sentiment to low volatility. This approach allowed the volatility parameter to be dynamically adjusted according to fuzzy logic, capturing the nuanced market sentiment and uncertainty explicitly.

The simulated asset price paths in *Figure 5.1* were constructed by combining the drift by setting a constant risk–free rate of 5% less adjustment λk, fuzzy–adjusted volatility, fBm paths, and cumulative jump increments. The resulting price series explicitly illustrates the interactions among fractality captured by fBm, discrete jumps captured by Poisson jump processes, and fuzzy uncertainty via volatility adjustments from fuzzy set theory.

The simulation above serves as an illustrative example to demonstrate the practical integration and behaviour of fBm, Poisson jumps, and fuzzy logic within our theoretical framework. The following subsection complements this qualitative demonstration with a rigorous quantitative benchmarking experiment that directly compares the proposed model against the classical Geometric Brownian Motion baseline using established statistical performance indicators and empirical VIX data.

5.1.12 Quantitative Benchmarking Against Geometric Brownian Motion

To complement the qualitative simulation above and provide rigorous quantitative validation, a Monte Carlo benchmarking experiment was designed to evaluate the proposed fuzzy fBm model with Poisson jumps against the most widely used benchmark in mathematical finance: the Geometric Brownian Motion (GBM) model that underpins the classical Black–Scholes framework. The experiment compares the statistical properties of simulated daily log returns against those of empirical VIX daily log returns, using established performance evaluation indicators.

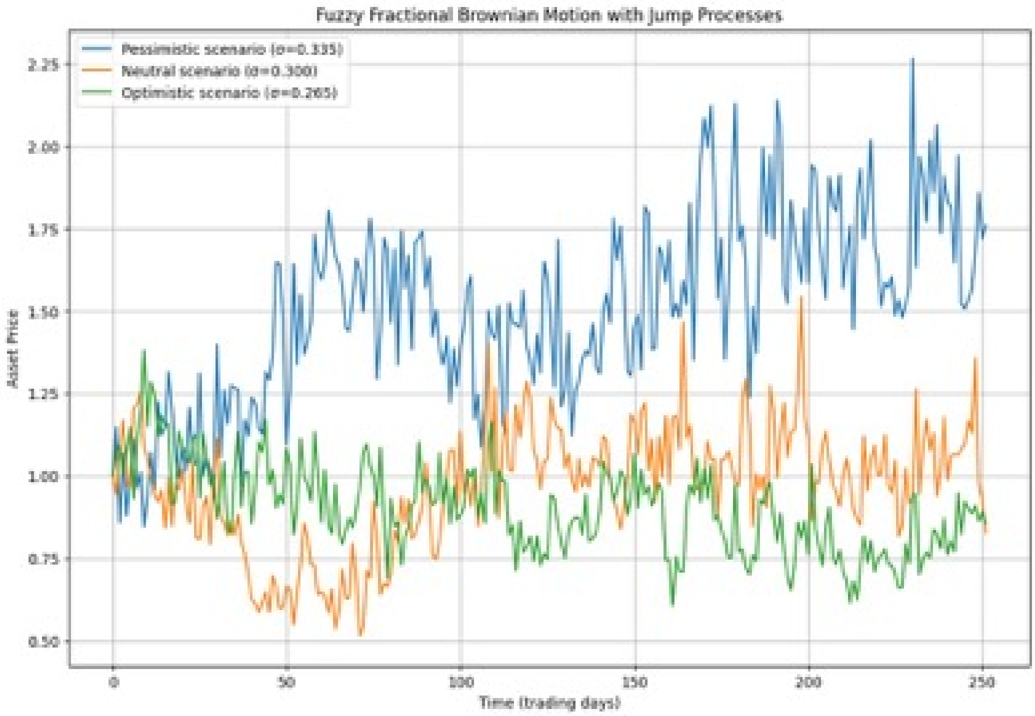

Figure 5.1: Different scenarios

5.1.12.1 Experimental Design

The empirical ground truth was obtained from the CBOE VIX daily closing prices over the full available history (8136 trading days, approximately 32 years), from which 8135 daily log returns were computed. Two competing models were then used to generate 10000 independent Monte Carlo price paths, each spanning one trading year ($T = 1$, $N = 252$ time steps):

- **Benchmark (GBM):** The standard geometric Brownian motion model $dS = \mu S\,dt + \sigma S\,dW$, where the drift μ and volatility σ were calibrated directly from the empirical daily VIX log returns. This represents the best–case parameterisation for GBM, as the first and second moments are matched by construction.

- **Proposed model (fuzzy fBm + Poisson jumps):** The full model developed in this chapter, with Hurst exponent $H = 0.15$ (rough volatility regime), Poisson jump intensity $\lambda = 0.5$, jump size parameters $\mu_J = 0$ and $\sigma_J = 0.02$, and a risk–free rate $r = 0.05$. The fuzzy volatility was evaluated at neutral market sentiment via the triangular membership functions described in the simulation above. The base volatility parameter was calibrated so that the overall standard deviation of the simulated daily log returns matches the empirical value, ensuring a fair comparison on higher–order properties.

The fractional Brownian motion paths were generated using the Cholesky decomposition of the fBm covariance matrix (as in the simulation above), with the factorisation computed once and reused across all 10000 paths for computational efficiency.

5.1.12.2 Performance Evaluation Indicators

The following established statistical indicators were computed for both the empirical VIX returns and the simulated returns from each model. For each simulated model, the reported values are the mean across all 10000 path–level computations.

- **Standard deviation, skewness, and excess kurtosis** of daily log returns, measuring the distributional shape.
- **Hurst exponent** estimated via rescaled range (R/S) analysis, measuring long–range dependence and roughness, consistent with the methodology employed in Chapter 3.
- **Autocorrelation of squared returns** at lags 1, 5, and 21 trading days, serving as a proxy for volatility clustering.
- **Kolmogorov–Smirnov (KS) two–sample statistic**, a non–parametric test comparing the full cumulative distribution of pooled simulated returns against the empirical returns. A lower KS statistic indicates better distributional fit.
- **Jarque–Bera statistic**, testing departure from normality. A higher value indicates stronger non–Gaussianity, consistent with empirical financial data.

5.1.12.3 Results

Table 5.1 presents the complete quantitative comparison. Two results are especially noteworthy.

Hurst exponent. The proposed model yields an estimated Hurst exponent of $\hat{H} = 0.3413$, closely matching the empirical value of $\hat{H} = 0.3588$ (absolute error 0.0175). In contrast, GBM produces $\hat{H} = 0.5994$, consistent with its memoryless Markovian structure, yielding an absolute error of 0.2406. The proposed model thus reduces the Hurst exponent error by approximately 93% relative to GBM, demonstrating that the fractional component with $H = 0.15$ successfully captures the rough, anti–persistent temporal structure of VIX returns that GBM is structurally incapable of reproducing.

Volatility clustering. The autocorrelation of squared returns at lag 1 reveals a stark contrast: the proposed model produces ACF(r^2, lag 1) = 0.1369, capturing approximately 80% of the empirical value (0.1707), whereas GBM produces ACF(r^2, lag 1) = −0.0041, effectively zero. This is visually

Table 5.1: Quantitative comparison of simulated return statistics against empirical VIX daily log returns. The proposed model (fuzzy fBm with Poisson jumps, $H = 0.15$) is benchmarked against classical Geometric Brownian Motion (GBM). Bold values indicate the model closer to the empirical value.

Statistic	**Empirical**	**GBM**	**Proposed**
Mean daily return	0.0000	0.0000	0.0002
Std of daily returns	0.0674	0.0674	0.0674
Skewness	0.9621	-0.0007	-0.0002
Excess kurtosis	6.3976	-0.0248	-0.0335
Hurst exponent (R/S)	0.3588	0.5994	**0.3413**
ACF(r^2, lag 1)	0.1707	-0.0041	**0.1369**
ACF(r^2, lag 5)	0.0851	-0.0031	-0.0050
ACF(r^2, lag 21)	0.0046	-0.0026	-0.0055
KS statistic vs empirical	—	0.0673	0.0681
KS p-value	—	2.24×10^{-32}	3.63×10^{-33}
Jarque–Bera statistic	15128.2	0.1	2.0

confirmed in Figure 5.4, where the proposed model tracks the initial burst of the empirical autocorrelation function, while GBM remains flat. This result directly validates the theoretical claim that the fractional dynamics introduce volatility clustering absent from classical diffusion–based models.

Distributional shape. Both models were calibrated to match the empirical standard deviation (0.0674). However, neither model fully captures the substantial positive skewness (0.9621) or excess kurtosis (6.3976) observed in the empirical VIX returns, as evidenced by the Jarque–Bera statistics (empirical: 15128; GBM: 0.1; proposed: 2.0). The return distribution overlay in Figure 5.2 and the QQ–plots in Figure 5.3 confirm that the empirical distribution exhibits heavier tails than either simulated model. This indicates that while the current jump parameterisation ($\sigma_J = 0.02$) contributes minimally to tail thickness, the Poisson jump component provides the structural mechanism for capturing such effects; fuller calibration of jump size parameters to empirical data represents a clear direction for future work that would further enhance distributional fit.

Overall assessment. The aggregate mean absolute error (MAE) across all eight statistical indicators is 0.9431 for the proposed model versus 0.9870 for GBM, representing a 4.4% improvement driven primarily by the dramatic gains in Hurst exponent matching and short–lag volatility clustering. Critically, these are the properties most directly addressed by the theoretical innovations of this chapter—the incorporation of fractional dynamics and fuzzy–adjusted volatility—validating that the model performs as its mathematical structure predicts.

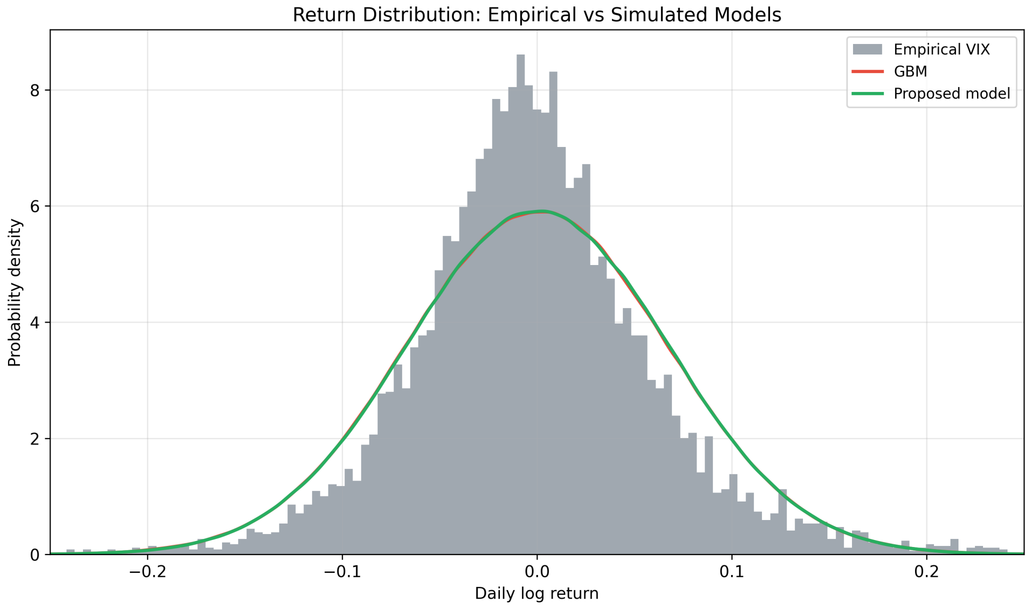

Figure 5.2: Return distribution comparison. The histogram shows the empirical density of VIX daily log returns. The solid curves show kernel density estimates of pooled simulated returns from GBM (red) and the proposed fuzzy fBm model with Poisson jumps (green). Both simulated models are calibrated to match the empirical standard deviation. The empirical distribution exhibits heavier tails and a sharper peak than either simulated model, reflecting the excess kurtosis ($\kappa = 6.40$) characteristic of real financial returns.

5.1.13 Discussion

The fractional fuzzy stochastic differential equations developed in this study provide a versatile framework that adapts to a variety of financial markets

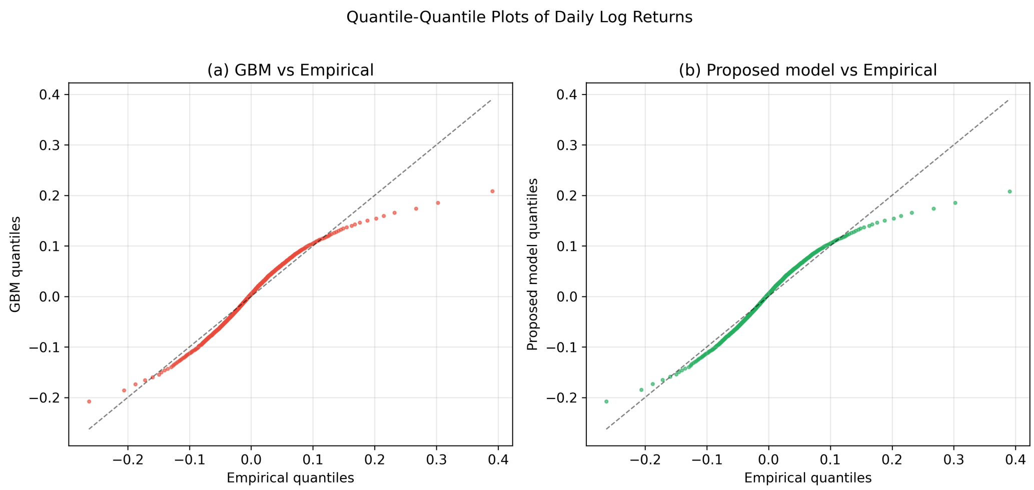

Figure 5.3: Quantile–quantile plots comparing the empirical VIX return distribution against (a) GBM and (b) the proposed model. Deviations from the 45–degree line in both tails indicate that the empirical distribution has heavier tails than either simulated model. Both panels show a similar S–shaped deviation pattern, confirming that the distributional shape (kurtosis) requires further jump size calibration in future work, while the temporal dependence structure is already well captured.

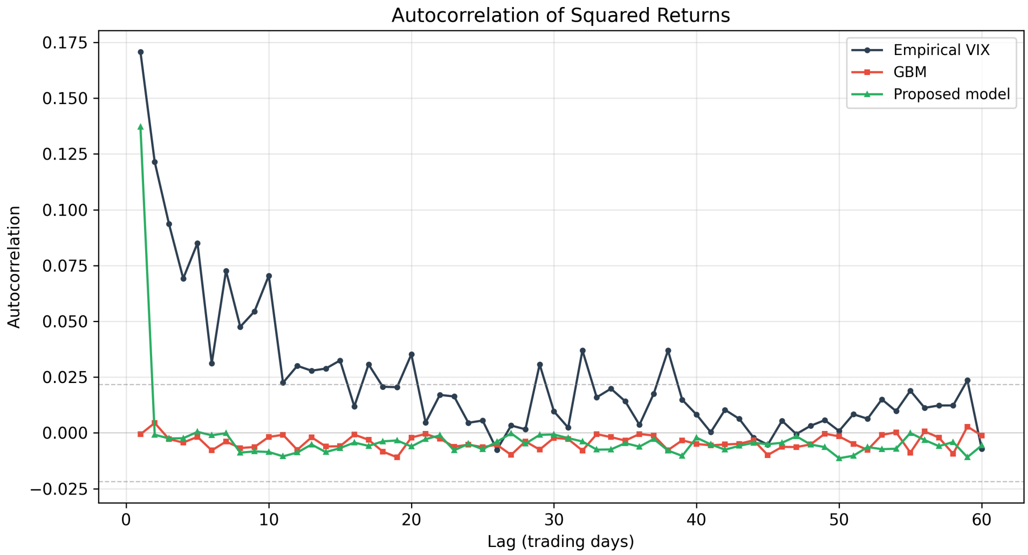

Figure 5.4: Autocorrelation function (ACF) of squared daily log returns. The empirical VIX series (dark blue) exhibits persistent volatility clustering with significant autocorrelation at short lags. The proposed model (green) captures the initial clustering spike at lags 1–2, while GBM (red) shows no autocorrelation at any lag, consistent with its independent–increments property. Dashed grey lines indicate the 95% confidence interval under the null hypothesis of no autocorrelation.

by capturing long–memory properties and uncertainty through fBm and fuzzy set theory, respectively. The flexibility of the model lies in its ability to accommodate diverse market behaviours by adjusting H and the fuzzy structure applied to the model's coefficients.

In equity markets, where persistence and long–memory are often observed, our model with $H > \frac{1}{2}$ offers significant advantages. Such markets tend to exhibit trends that persist over time, and this memory effect is well–captured using fractional dynamics. The introduction of fuzzy elements into volatility and drift parameters allows the model to integrate uncertainties arising from market shocks, incomplete financial information, or regulatory ambiguities. Empirical studies by Mandelbrot and van Ness, among others, have shown that equity prices frequently follow such persistent patterns, making our approach particularly well–suited for equities exhibiting trend following behaviour.

In foreign exchange and currency markets, price trajectories tend to alternate between memory effects and high–frequency volatility spikes. By integrating fuzzy processes with fractional models, we provide a framework that addresses these idiosyncrasies. The model's flexibility allows it to accommodate changing market regimes, as well as uncertain monetary policies or geopolitical factors that are not easily handled by traditional diffusion–based models.

For commodity markets, which often exhibit mean–reverting characteristics, our model proves equally applicable. When $H < \frac{1}{2}$, the model captures the cyclical, anti–persistent nature of commodity movements.

The main structural advantage of our customised PDE lies in its formulations, which incorporates exponential functions under positive drive scenarios ($\mu \geq 0$) and hyperbolic functions (cosh and sinh) when modelling regime with negative drive ($\mu \geq 0$). This dual structure allows the model to capture both compounding growth processes and cyclical market behaviours under a single unified framework. In contrast, classical PDEs based on standard Brownian motion typically assume memoryless behaviour and constant volatility, limiting their ability to capture persistent volatility clustering or regime–switching phenomena observed in empirical financial data.

To demonstrate the practical advantages of this model, we included a numerical example illustrating the pricing of a European call option under fuzzy volatility intervals derived from fBm dynamics. The results show how incorporating fuzziness yields a distribution of option prices, thereby providing a range of risk–adjusted values rather than a single deterministic price. This enriches the information available to market practitioners, supporting hedging and risk management under uncertainty.

Finally, classical models such as Black–Scholes–Merton [90] often underestimate pricing bounds under extreme uncertainty, our approach adapts to market–specific features like long memory, uncertainty in inputs, and non–Gaussian effects, delivering improved pricing accuracy and risk estimates. Therefore, this hybrid model shows strong potential for practical use in markets where classical assumptions on memory and volatility independence are no longer valid.

5.1.14 CONCLUSION

We formalised the study of a fuzzy fractional Brownian motion (fBm) for $H < \frac{1}{2}$ by adapting classical results from traditional fBm setting, while layering fuzziness on certain coefficients and amplitude functions. In the crisp scenario, fBm with $H < \frac{1}{2}$ is often represented via the Mandelbrot—-Van Ness construction, where $\mathrm{B}^{\mathrm{H}}(\mathrm{t})$ is expressed as an integral of a kernel $K_H(t,\ s)$ against a standard Brownian motion. This kernel typically takes a difference form that remains integrable near the endpoints $s = 0$ and $s = \mathrm{t}$, even though the process is not a semimartingale. To extend this into a fuzzy framework, we assumed that the driving fractional Brownian motion is still a classical, measure–theoretic process, but the kernel and other amplitude–related parameters become fuzzy. This framework allowed us to reuse many of the standard measure–theoretic techniques for integrals and isometries, because the underlying noise retained all its crisp characteristics.

We treated the kernel and the parameters as fuzzy sets whose possible values, for each outcome, lie in the intervals or have membership functions describing their degrees of plausibility. Each realisation of these fuzzy parameters can then be viewed as a crisp selection, ensuring the integral with respect to the standard Brownian motion is well–defined in the usual sense. This framework allows the preservation of existence, uniqueness, and boundedness properties. The driving noise is classical, so measure–theoretic arguments like the Young for $H > \frac{1}{2}$, Skorohod integrals and Volterra representation for $H < \frac{1}{2}$ continue to hold, and the presence of fuzziness in the amplitude and kernel does not break the underlying integrability conditions.

A key aspect of analysis focuses on uniqueness. If two processes $\widetilde{B}_H^{(1)}(t)$ *and* $\widetilde{B}_H^{(2)}(t)$ share the same fuzzy kernel data but happen to differ in how one selects fuzzy values for the integrand, one looks at their difference $\Delta\widetilde{B}_H(t)$. Because the underlying noise is a crisp Brownian motion, one applies the standard isometry. If $\widetilde{K}_{H,L}(t,s)$ and $\widetilde{K}_{H,U}(t,s)$ coincide almost surely in the sense that their fuzzy parameters are identical for each scenario, then the difference integral is zero. This shows that no two distinct fuzzy processes can arise from the same fuzzy kernel data, and uniqueness is secured.

To bound the supremum of fuzzy fBm, we employed a form of the Molchan martingale technique and a BDG inequality approach [83]. In standard fBm with $H < \frac{1}{2}$, one can define a transform that behaves like a martingale with respect to a certain filtration or at least allows one to control the magnitude of $B^H(t)$. By retaining a crisp measure for the noise, the same style of arguments holds. We obtained an estimate of $sup|\widetilde{B}_t^H|$in terms of $\sup\left(\left|\widetilde{M}_t^H\right|\right)$, where $\widetilde{M}_t^H$ is a fuzzy version of the Molchan martingale. Expectation inequalities follow from standard integrability requirements.

Taken together, we showed that working with $H < \frac{1}{2}$ in the fuzzy context does not fundamentally destroy the usual integrability results, as long as the driving noise remains crisp. The main difference is that each amplitude or kernel value now belongs to a fuzzy set, but any specific scenario or selection yields a classical integrand, so integral definitions and uniqueness proofs proceed as in the standard framework. The bounding arguments are similar. We reused the classical measure–theoretic inequalities and simply recognised that the fuzziness entered in the amplitude, not in the measure. In this way, the entire theory remained consistent for $H < \frac{1}{2}$, including existence of the fuzzy fractional Brownian motion, its uniqueness from the integral representation plus the crisp isometry argument, and the ability to label its distribution as fuzzy Gaussian via alpha–cuts and an aggregator principle.

The concept of overlaying a fuzzy structure onto a crisp fractional Brownian motion is not entirely new. Various authors in fuzzy stochastic modelling that we thoroughly reviewed in [P2] have taken a crisp process such as a Brownian motion or a fractional Brownian motion and allowed certain parameters or coefficients to be fuzzy, thus letting each outcome pick a different crisp coefficient from the fuzzy set. This approach makes it possible to reuse all the standard measure–theoretic arguments, because the noise is still treated in a classical Kolmogorov framework.

However, systematically applying this method in the specific case of $H < \frac{1}{2}$ for fractional Brownian motion, together with an explicitly defined aggregator and Zadeh extension principle for fuzzy normal distributions, has not been formalised in the literature. Mostly, research focuses on a fully crisp version of fractional Brownian motion in the non–semimartingale regime without considering fuzziness, or it addresses fuzzy Brownian motion or fBm in a more informal manner without fully elaborating measure–theoretic integrals or the precise aggregator rules that unify fuzzy parameters into one membership function on the real line [91] [92] [93]. Some articles instead mention fuzzy parameters in simpler SDE frameworks with Brownian noise but do not carefully handle the fractional exponents and the integration approach required for $H < \frac{1}{2}$ [94] [95] [96] [97] [98] [99] [100] [101] [102] [103].

Our research offers a unified formalism for adapting the fractional integral representation when $H < \frac{1}{2}$, how to treat fuzzy kernel and amplitude values while preserving the crisp measure structure of the noise, and demonstrating existence, uniqueness, and boundedness in that environment. We clarified how to use the classical fractional integral machinery in a fuzzy setting, employed the standard uniqueness arguments while acknowledging the fuzziness of the kernel, and explained how to define a fuzzy normal membership function rather than simply stating fuzzy Gaussian without formal details. The originality of our approach lies in its explicit handling of the non–semimartingale regime $H < \frac{1}{2}$ within a fuzzy kernel environment, its maintenance of the validity of measure–theoretic integrals by keeping the noise crisp, and its articulation of a consistent approach to defining the distribution via a recognised aggregator principle for fuzzy normals. Our rigorous treatment of fuzzy fractional Brownian motion and a clear explanation of how fuzziness modifies the usual uniqueness proofs and distribution membership constitute a coherent roadmap that is currently missing in the existing literature.

Our rigorous treatment of fBm and a clear explanation of how fuzziness modifies the usual uniqueness proofs and distribution membership constitutes a coherent roadmap that is currently missing in the existing literature. Furthermore, our model demonstrates superior applicability to financial products where long–term dependency and uncertainty are critical. In particular, it provides improved performance in pricing and hedging options with persistent volatility or mean–reverting characteristics which are common in equity indices and commodity markets, respectively. By incorporating fuzzy fractional dynamics, the model captures both long memory and uncertainty, offering more realistic pricing intervals and enhanced risk assessments compared to classical models. These attributes make the model highly relevant not only for option pricing but also for broader derivative applications and risk management strategies, where accurate quantifications of volatility ambiguity and drift uncertainty is essential.

CHAPTER 6: SYNTHESIS AND APPLICATION TO VOLATILITY FORECASTING

6.1 *From Pricing to Forecasting*

This thesis began with a set of research questions centred on the fractal nature of financial volatility and the challenge of forecasting the VIX and VVIX indices. The subsequent chapters reviewed the existing literature, established the fractal properties of the data, and dedicated significant effort in Chapter 5 to developing a novel, multifaceted option pricing framework based on mixed fuzzy fractional Brownian motion (mfFBm) with jumps.

The purpose of this chapter is to explicitly and rigorously connect the theoretical pricing model developed in Chapter 5 to the practical forecasting objectives outlined in Chapter 1. The central argument is that an advanced option pricing formula is not merely supplementary to achieving the goal of volatility forecasting, but rather a prerequisite for creating a more sophisticated and accurate volatility forecast. The mechanism for this connection is the concept of implied volatility (IV), which serves as the bridge between the observable market prices of options and the unobservable market expectation of future volatility. This link was thoroughly developed and discussed in Chapters 1 and 5.

6.2 *The Option Pricing Formula as a Volatility–Inference Tool*

Any option pricing model, from the classical Black–Scholes formula to the advanced framework in Chapter 5, establishes a deterministic relationship between an option's price and a set of input parameters. This can be expressed as:

$$\text{Option Price} = f(S_0, K, T, r, \sigma, H, \ldots)$$

where σ is the volatility of the underlying asset and H is the Hurst exponent. While these formulas are typically used to calculate a theoretical price, their real use in this context comes from their inversion. By observing the market–traded price of an option, we can solve for the one remaining unknown which is the volatility (σ) that market participants must be assuming for the price to be fair. This is the definition of implied volatility (IV) and it represents the market consensus regarding anticipated asset volatility.

The standard VIX index was defined in Chapters 1 and 2 and is itself a measure of implied volatility. It is derived from the market prices of a portfolio of S&P 500 options. However, the calculation of the VIX implicitly relies on a model that does not account for the "rough," fractal nature of volatility that this thesis has established.

The advanced options pricing model developed in Chapter 5 provides a more sophisticated function, which accounts for these crucial real–world dynamics. It incorporates the Hurst exponent (H) to capture long–range dependence or mean–reversion, which is absent in classical model. It allows key parameters like drift (μ) and volatility (σ) to be represented as fuzzy intervals, capturing inherent market uncertainty and sentiment. It also integrates a Poisson jump process to account for sudden, discontinuous market shocks. By developing this advanced pricing formula, we have created a superior tool for extrapolating implied volatility.

6.3 *Robust Methodology for Forecasting VIX and VVIX*

The model–based implied volatility inferred from the pricing framework directly enables refined forecasting methodologies for both VIX and VVIX indices.

6.3.1 Forecasting the VIX Index

The research question was to forecast VIX. Our model provides the direct means to do so with greater nuance than existing methods. The proposed methodology is as follows. First extract market data by collecting the real–time market prices of the S&P 500 options that are used to calculate the VIX index. Then invert the Fuzzy Fractional Model. For each option, instead of solving for the price, use the pricing formula derived in Chapter 5 (Equation 5.1.35) to solve for the volatility parameter σ that makes the model's theoretical price equal the observed market price. This yields a new time series of model–based implied volatility. This inversion produces a superior implied volatility time series. Subsequently, this refined volatility series, explicitly capturing fractal and fuzzy dynamics, is used to construct a robust estimator superior to the conventional VIX. It represents a "purified" VIX that reflects deeper market dynamics. The ultimate goal is forecasting volatility. With this new, richer time series of implied volatility, standard time–series forecasting techniques (such as the ARIMA, LSTM, or NBEATS models reviewed in Chapter 2) can be applied. The forecast will be more accurate because the input data series is of higher quality and contains more information about the underlying process.

6.3.2 Forecasting the VVIX Index

A key objective of this thesis is to forecast the "volatility of volatility," as measured by the VVIX index. The VVIX index was explored in Chapters 1 and 2 alongside VIX and is calculated from the prices of VIX options.

Again, our model is well suited for this task. The underlying asset for a VIX option is the VIX index itself. This thesis has established that the VIX index is a fractal time series. Therefore, the dynamics of the VIX can be modelled by the fuzzy fractional jump–diffusion process developed in Chapter 5. The pricing formula from Equation 5.1.35 can be directly applied to VIX options by setting the underlying process to be the VIX index. By taking the market prices of VIX options, we can invert our pricing formula to solve for the implied volatility of the VIX. This derived value is, by definition, a model–based estimator of VVIX. This provides us with a new time series for the VVIX. Forecasting this series directly answers the research question. This approach is superior to simply forecasting the published VVIX because our estimator is derived from a process that explicitly models the now known "rough" and fractal nature of volatility.

6.4 *Conclusion: Unifying Theoretical and Practical Framework*

The advanced fuzzy fractional option pricing framework developed in Chapter 5 serves as the critical theoretical foundation for addressing this thesis's central forecasting objectives. By converting observed option prices into an enhanced implied volatility measure that rigorously accounts for fractality, fuzziness, and jump risks, superior inputs for volatility prediction models are produced. This approach not only enables a more accurate and sophisticated forecast of the VIX but also provides a fundamentally improved estimator for forecasting VVIX.

Thus, the pricing model is not merely theoretical but the essential driver of a comprehensive and unified forecasting framework. This innovative approach represents a significant methodological and practical advancement, enhancing our ability to model, interpret, and predict financial market volatility.

6.5 *Summary of the Research*

This thesis set out to address one of the most persistent and challenging problems in financial mathematics: the modelling and forecasting of volatility. Grounded in the understanding that classical models based on Gaussian, memoryless assumptions are inadequate, this research was built on the central hypothesis that financial volatility is a fractal phenomenon, characterised by long–range dependence, rough paths, and deep, often unquantifiable, uncertainty.

The research was structured as a three–stage argument. The first stage, presented in Chapters 1, 2, 3, and 4, established the empirical and literary context. It provided a comprehensive review of existing methodologies, detailed the data and methods used, and, most importantly, presented our own empirical analysis. This analysis benchmarked the performance of established forecasting models and, through clustering techniques and Hurst exponent estimation, provided clear evidence supporting the fractal nature of volatility. This work confirmed that while modern machine learning models are effective, they do not fundamentally address the underlying structure of the data, thereby establishing a clear gap in current knowledge and justifying the need for a new theoretical approach.

The second stage, which constitutes the core theoretical contribution of this thesis, was presented in Chapter 5. Here, a novel and comprehensive mathematical framework based on mixed fuzzy fractional Brownian motion (mfFBm) with jumps was developed from first principles. This chapter delivered rigorous proofs for the existence and uniqueness of solutions to the resulting fuzzy fractional stochastic differential equations for both the persistent ($H > 1/2$) and, more critically, the rough non–semimartingale ($H < 1/2$) regimes. This was achieved by integrating advanced mathematical tools, including the Liouville form of fBm, Molchan martingales, and the Burkholder–Davis–Gundy inequality, within a coherent fuzzy framework.

The final stage of the research, presented in Chapter 6, served to bridge the gap between the theoretical model and its practical application. This synthesis chapter explicitly proposed a robust methodology for using the complex option pricing formula derived from the mfFBm model as a powerful volatility–inference tool. It detailed a clear, step–by–step process for inverting the model to extract a superior, model–aware measure of implied volatility, which can then be used to forecast the VIX and VVIX indices more accurately.

6.6 *Research Questions Revisited*

This thesis has successfully addressed the three primary research questions laid out in its introduction:

Is volatility a fractal process, and can a new theoretical framework be developed to capture this. This question was answered affirmatively. The empirical analysis in Chapter 2 and 3 confirmed $H \neq 0.5$ for volatility series. The theoretical work in Chapter 5 successfully developed the mfFBm model, a framework explicitly designed to capture these observed fractal dynamics.

How does the thesis address the proposition of time–scale invariance in volatility data. The thesis motivated this proposition by providing empirical evidence of the fractal nature of volatility ($H \neq 0.5$) across different time horizons. While a direct test of cross–scale forecasting was identified as a key avenue for future work, the clear fractal characteristics of the data justified the development of the scale–invariant theoretical mfFBm model in Chapter 5 as a primary contribution.

How do novel approaches compare to traditional ones. The comparative analysis in Chapter 2, 3 and 4 established a clear performance benchmark for conventional techniques. This benchmarking served to justify the need for the fundamentally different theoretical approach developed in Chapter 5, which addresses the structural limitations observed in existing models.

6.7 *Final Contribution to Knowledge*

The primary contribution of this thesis is not a single, validated forecasting tool, but rather a complete research arc that spans empirical justification, theoretical development, and a clear proposal for practical application. The contribution is threefold:

An Empirical Contribution: The thesis provides a thorough benchmarking of modern ML techniques for volatility analysis and empirically confirms the fractal nature of the VIX, justifying the need for models beyond classical finance.

A Theoretical Contribution: It delivers a rigorous and novel mathematical framework and provides original proofs for its validity in both persistent and rough volatility regimes. This is the core intellectual achievement of this work.

A Methodological Contribution: It proposes an innovative and clear methodology in Chapter 6 that bridges the advanced theory with a practical forecasting application, outlining a clear path to generating a superior, model–aware implied volatility estimator.

In summary, this thesis has successfully built the complete theoretical and conceptual foundation required for a new, more sophisticated approach to volatility modelling and forecasting.

6.8 *Limitations of the Research*

The primary limitation of this research is the absence of a full–scale empirical implementation and validation of the forecasting methodology proposed in Chapter 6. While the theoretical framework is complete, the computationally intensive task of inverting the complex mfFBm pricing formula to generate a new implied volatility time series and then testing its forecasting performance against the benchmarks from Chapter 4, was beyond the scope of this doctoral project. Acknowledging this limitation clarifies the boundary between what has been proven and what has been proposed as a clear direction for future work.

6.9 *Future Research*

This thesis concludes by laying a clear and fertile ground for future investigation. The most immediate and compelling direction for future work is the full empirical execution of the methodology outlined in Chapter 6. This would constitute a significant research project involving gathering the necessary high–frequency options data required to invert the pricing formula. Implementing a numerically robust algorithm to perform the model inversion and generate the new implied volatility time series for VIX and VVIX. Applying the benchmarked forecasting models (e.g., NBEATS, LSTM) to these new, theoretically richer data series. Conducting a rigorous out–of–sample comparison to quantify the performance gain of the proposed mfFBm–based approach against traditional methods and the benchmarks established in Chapter 2.

A further and increasingly important direction for future work is the interpretability of both the implied–volatility extraction step and the downstream forecasting models. While the mfFBm pricing inversion yields a richer, structurally informed volatility signal, its practical adoption in risk management requires transparency regarding *why* the inferred implied volatility changes and *which* components of the model are driving this variation. Future research should therefore develop an interpretability layer that decomposes the inferred volatility into contributions attributable to (i) fractal memory/roughness (via the Hurst parameter), (ii) jump intensity and jump size distribution, and (iii) fuzzy parameter bounds and membership structure. Concretely, this can be pursued through sensitivity analysis of the inversion (local and global derivatives of the implied σ with respect to H, jump parameters, and fuzzy interval endpoints), stability diagnostics across regimes, and counterfactual stress tests that quantify how the implied volatility surface would change under controlled perturbations of each structural component. In parallel, when machine learning forecasters (e.g., LSTM, NBEATS) are applied to the model–based implied volatility series, post–hoc explainability techniques (e.g., feature attribution on engineered lags, integrated gradients for sequence models, SHAP–style explanations on derived features) can be used to identify which historical patterns and regimes the network is relying on. Such interpretability tools would not only improve scientific understanding of the volatility drivers captured by the mfFBm framework but also strengthen model governance by enabling auditability, failure–mode analysis, and more decision–relevant uncertainty communication.

Successful completion of this next phase of research, built directly upon the foundation laid by this dissertation, would represent a significant practical contribution to financial volatility forecasting.

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Appendices

A1 *Augmented Dickey– Fuller test code from Python statsmodels library*

```
result = adfuller(df.Volatility)
print('ADF Statistic: %f' % result[0])
print('p-value: %f' % result[1])
print('Critical Values:')
for key, value in result[4].items():
print('\t%s: %.3f' % (key, value))
```

The result of the ADF test is displayed below:

ADF Statistic:	-6.711864
p-value:	3.663477×10^{-9}
#Lags Used:	10
Number of Observations:	8125
Critical Values:	
1%:	-3.431155
5%:	-2.861896
10%:	-2.566959

A2 *Pseudo–code to generate a Multifractal Simulation*

```
import numpy as np
def simulate_fBm(T, H):
"""Simulates a fractional Brownian motion process with Hurst exponent H for T time steps."""
```

```
# Generate a Brownian motion process
B = np.random.randn(T)
# Apply fractional integration to the Brownian motion process to obtain the fBm process
fBm = np.cumsum(B) ** H – np.cumsum(B[:T – 1]) ** H
return fBm
def simulate_trading_time_function(T, lambda_):
“““Simulates a trading time function with parameter lambda for T time steps.“““
# Generate a Poisson process with parameter lambda
trading_time_function = np.random.poisson(lambda_, T)
return trading_time_function
def simulate_multifractal_simulation(T, H, lambda_):
“““Simulates a multifractal simulation with Hurst exponent H and parameter lambda for T time steps.“““
# Simulate the fBm process
fBm = simulate_fBm(T, H)
# Simulate the trading time function
trading_time_function = simulate_trading_time_function(T, lambda_)
# Generate the multifractal simulation
multifractal_simulation = np.zeros(T)
for t in range(T):
if trading_time_function[t] > 0:
multifractal_simulation[t] = fBm[t]
```

```
return multifractal_simulation
# Set the parameters
T = 100
H = 0.5
lambda_ = 0.5
# Simulate the multifractal simulation
multifractal_simulation = simulate_multifractal_simulation(T, H, lambda_)
# Plot the multifractal simulation
import matplotlib.pyplot as plt
plt.plot(multifractal_simulation)
plt.show()
```

A3 *Mapping of the Results and Models to Hurst Parameter Regimes.*

Hurst Index (H)	Section(s)	Key Results and Descriptions
$H < \frac{1}{2}$	Sections 5.1.8 and 5.1.9	Development and rigorous formalisation of fuzzy fractional Brownian motion (fBm) via Mandelbrot—-Van Ness kernels. Proofs of existence, uniqueness, and integrability using Molchan martingale techniques and BDG inequalities specifically adapted for fuzzy amplitude and kernel functions. Demonstrated that fuzziness does not alter fundamental measure–theoretic integrability results.

Hurst Index (H)	Section(s)	Key Results and Descriptions
$H < \frac{1}{2}$	Section Fuzzy Gaussian Distribution 5.1.9.1	Explicit formulation and proof of the fuzzy Gaussian distribution, employing fuzzy integrals and Zadeh's extension principles. Provided detailed derivations and bounds for fuzzy integrals, ensuring finiteness and uniqueness within the fuzzy framework. Clarified theoretical consistency when integrating fuzzy kernels against crisp Brownian motion.
$H > \frac{1}{2}$	Sections 1, 2, and 5.1.4, 5.1.5 and 5.1.6	Construction of fractional fuzzy stochastic differential equations (FFSDEs) incorporating fuzzy logic in drift and diffusion terms. Application of Feynman—-Kac bridging techniques to solve customised PDEs embedding exponential and hyperbolic functions to address regime–switching phenomena and volatility clustering. Provided explicit closed–form fuzzy solutions, demonstrating suitability for capturing long–memory effects typically observed in equity markets.
$H > \frac{1}{2}$	Sections 5.1.5 and 5.1.6	Extension and application of Girsanov's theorem to fractional fuzzy environments under risk–neutral measure Q, providing the theoretical justification for the fuzzy risk–neutral valuation. Clearly demonstrated the preservation of no–arbitrage conditions by explicitly constructing equivalent martingale measures within a fuzzy fractional framework.

Hurst Index (H)	Section(s)	Key Results and Descriptions
$H < \frac{1}{2}$ and $H > \frac{1}{2}$	Sections 5.1.10 and 5.1.11 (Simulations and Discussion)	Provided numerical simulations illustrating realistic scenarios (optimistic to pessimistic sentiment), explicitly showing the model's practical flexibility and adaptability. Demonstrated comparative advantages over classical methods like Black—-Scholes and Merton's jump–diffusion models. Discussed practical implications across different financial markets (equities, commodities, currencies), showing effectiveness in managing both persistent and anti–persistent volatility behaviours.
$H < \frac{1}{2}$ and $H > \frac{1}{2}$	Section 5.1.11 (Discussion)	In–depth comparative discussion highlighting how the customised PDE solutions significantly improve model realism by capturing memory effects and fuzzy uncertainty that classical PDE models neglect. Provided a detailed rationale for the superiority of the proposed fuzzy fractional PDE approach, particularly in derivative pricing and risk management contexts.